Supplementary Table 01 - r5_orthogonal_aging_mr

ST01_r5_orthogonal_aging_mr.csv

n_steiger_remov	n_snp	mean_F	ivw_beta	ivw_se	ivw_p	ivw_or	ivw_re_beta	ivw_re_p	cochran_q	cochran_q_p	egger_beta	egger_p	egger_intercep	egger_int_p	wmed_beta	wmed_p	n_presso_outlie	trait	outcome	n_out	presso_beta	presso_p	ivw_q	ivw_fdr05
0	5	35.83903726	0.110916693	0.031934086	0.000514098	1.117301824	0.110916693	0.378516047	62.22345414	9.885425817	0.137548255	0.320889402	0.018989221	0.513890790	0.010852798	0.942422034	3	GrimAge	ALLERG_AST	283740			0.001101639	True
0	5	35.83903726	-0.02671488	0.021375474	0.211375743	0.973638801	-0.02671488	0.648258410	30.02319770	4.841502442	-0.00407733	0.934456857	0.015936014	0.181806203	0.025576298	0.763869024	3	GrimAge	L12_ATOPIC	464119			0.226474010	False
0	5	35.83903726	-0.00014842	0.029331400	0.995962467	0.999851584	-0.00014842	0.998543968	30.75770202	3.430309447	0.012291483	0.895221084	0.008841451	0.645006546	-0.09365758	0.381370191	2	GrimAge	ALLERG_RHI	490219	-0.10386421	0.003462496	0.995962467	False
0	56	54.07377010	0.019310097	0.007373032	0.008818357	1.019497743	0.019310097	0.029285205	79.41003455	0.017294776	0.018420637	0.039031561	-0.00339253	0.363792010	0.029771519	0.0	1	Hannum	ALLERG_AST	283740	0.015151995	0.040681428	0.016534420	True
0	56	54.07377010	0.009267154	0.004910751	0.059144789	1.009310221	0.009267154	0.059144789	27.18445290	0.999404608	0.010015960	0.002986786	0.002900774	0.043396124	0.009896096	3.227296208	1	Hannum	L12_ATOPIC	464119	0.007405416	0.133810069	0.087533196	False
0	56	54.07377010	-0.02777058	0.006756517	3.953296210	0.972611472	-0.02777058	0.001698004	94.32694763	0.000766845	-0.02707025	0.002462529	0.002675669	0.474376019	-0.03690094	0.0	2	Hannum	ALLERG_RHI	490219	-0.03409756	5.828993132	9.883240520	True
0	133	130.8930925	-0.01400459	0.003369517	3.234981420	0.986093013	-0.01400459	0.015053383	385.8128563	0.0	-0.01542495	0.006745482	0.007809141	0.019075596	-0.05689229	0.0	17	IEAA	ALLERG_AST	283740	-0.04594733	0.0	9.704944262	True
0	133	130.8930925	-0.00419210	0.002264960	0.064191010	0.995816670	-0.00419210	0.064191010	97.55761211	0.989178241	-0.00394730	0.043286749	-0.00139599	0.217053295	-0.00770925	0.0	1	IEAA	L12_ATOPIC	464119	-0.00460923	0.042048260	0.087533196	False
0	133	130.8930925	0.005509707	0.003112267	0.076673958	1.005524913	0.005509707	0.076673958	108.9906003	0.928713443	0.005315807	0.062018741	0.001104350	0.502190153	-0.00325581	0.0	0	IEAA	ALLERG_RHI	490219	0.005509707	0.076673958	0.095842448	False
0	89	69.24580254	-0.05699032	0.005137566	0.0	0.944603209	-0.05699032	0.0	96.61151316	0.248582313	-0.05682583	0.0	0.000941603	0.742046602	-0.07779067	0.0	0	PhenoAge	ALLERG_AST	283740	-0.05699032	0.0	0.0	True
0	89	69.24580254	-0.00733228	0.003430238	0.032553900	0.992694534	-0.00733228	0.032553900	32.91353005	0.999999986	-0.00792004	7.742581676	-0.00336925	0.001904884	-0.01048486	0.0	0	PhenoAge	L12_ATOPIC	464119	-0.00733228	0.032553900	0.054256501	False
0	89	69.24580254	-0.00602499	0.004713361	0.201150713	0.993993117	-0.00602499	0.201150713	16.78900321	1.0	-0.00610739	0.003277870	-0.00047189	0.666284797	-0.00446909	0.0	0	PhenoAge	ALLERG_RHI	490219	-0.00602499	0.201150713	0.226474010	False
5	66	46.45580341	1.234571241	0.084643040	0.0	3.436904603	1.234571241	0.0	145.7227443	3.963074735	1.232983649	0.0	0.000991252	0.802913482	1.433223776	2.075375238	3	Frailty	ALLERG_AST	283740	1.218054947	0.0	0.0	True
0	71	45.62663311	0.376283863	0.053993265	3.189892794	1.456860624	0.376283863	0.000671819	293.9548193	0.0	0.366775030	0.000904181	0.004031120	0.223401289	0.331849214	0.036368643	7	Frailty	L12_ATOPIC	464119	0.667967399	0.0	1.196209797	True
1	70	45.83208167	0.525667623	0.074844494	2.164268764	1.691587815	0.525667623	6.815935593	101.3410265	0.006812566	0.530399920	5.076566722	-0.00284881	0.297651522	0.632032064	8.541078067	1	Frailty	ALLERG_RHI	490219	0.560132205	1.192379528	1.082134382	True

Supplementary Table 02 - r5_aging_instruments_clumped

ST02_r5_aging_instruments_clumped.csv

trait	snp	beta	se	p	eaf	n_snp
GrimAge	rs9386796	0.1983	0.0294	1.64e-11	0.4597	32418
GrimAge	rs887466	-0.1928	0.031	5.092e-10	0.3815	30674
GrimAge	rs4065321	-0.1703	0.0296	8.776e-09	0.5333	32420
GrimAge	rs17094148	-0.18	0.0323	2.547e-08	0.7067	31838
GrimAge	rs3134745	0.185	0.0333	2.866e-08	0.3016	30188
Hannum	rs10786282	-0.36	0.0362	2.6089999999999998e-23	0.2124	31412
Hannum	rs941997	0.3481	0.0363	9.98e-22	0.7807	28541
Hannum	rs1005277	0.3006	0.033	8.919000000000001e-20	0.2982	28556
Hannum	rs117932856	0.7878	0.0912	5.595e-18	0.04	26835
Hannum	rs144646268	-0.6295	0.0743	2.357e-17	0.0439	28449
Hannum	rs77125544	0.6263	0.0739	2.41e-17	0.9557	28449
Hannum	rs80065902	-0.6246	0.074	3.114e-17	0.0442	28449
Hannum	rs17386635	0.6274	0.0743	3.127e-17	0.9556	28054
Hannum	rs117090076	-0.7294	0.0868	4.495e-17	0.9578	27475
Hannum	rs117699776	0.6208	0.0746	9.047e-17	0.9563	28165
Hannum	rs117219667	-0.6202	0.0746	9.382e-17	0.0437	28165
Hannum	rs143370018	-0.6186	0.0746	1.092e-16	0.0437	28165
Hannum	rs117774156	-0.6183	0.0747	1.275e-16	0.0436	28165
Hannum	rs76715209	-0.6172	0.0746	1.285e-16	0.0438	28165
Hannum	rs117148433	0.6115	0.0747	2.749e-16	0.9564	28171
Hannum	rs80106405	0.6145	0.0751	2.89e-16	0.957	28449
Hannum	rs4142016	-0.6126	0.0751	3.467e-16	0.0433	28165
Hannum	rs79663593	-0.6116	0.0751	3.888e-16	0.0432	28165
Hannum	rs2505248	-0.249	0.0307	5.525e-16	0.574	28161
Hannum	rs117480177	0.6004	0.0747	9.537e-16	0.9565	28171
Hannum	rs55668627	0.5948	0.0747	1.733e-15	0.9563	28167
Hannum	rs117315625	-0.5948	0.0748	1.783e-15	0.0437	28171
Hannum	rs149538538	-0.5956	0.0756	3.375e-15	0.0428	28171
Hannum	rs117567726	0.5995	0.0761	3.379e-15	0.9572	28171
Hannum	rs111716638	0.5927	0.0756	4.59e-15	0.9571	28171
Hannum	rs55769428	0.5945	0.0763	6.416e-15	0.9572	28171
Hannum	rs111925661	0.5886	0.0757	7.481e-15	0.9567	28171
Hannum	rs141225626	-0.6137	0.0792	9.099e-15	0.0395	27308
Hannum	rs185532555	0.7067	0.0915	1.173e-14	0.0421	22713
Hannum	rs150082435	0.6264	0.0823	2.786e-14	0.0531	22713
Hannum	rs148942702	0.6792	0.0893	2.804e-14	0.0434	23351
Hannum	rs112231354	-0.5412	0.0715	3.642e-14	0.9392	27756
Hannum	rs73318809	0.5673	0.0757	6.663e-14	0.9564	28166
Hannum	rs144205159	0.7377	0.101	2.756e-13	0.0334	25316
Hannum	rs12417758	-0.2093	0.0304	6.215e-12	0.5461	28548

(showing first 40 of 384 rows; full data in the accompanying CSV)

Supplementary Table 03 - r5_presso_summary

ST03_r5_presso_summary.csv

outcome	n_snps	n_outliers	ivw_beta_full	ivw_beta_corrected	ivw_se_corrected	ivw_p_corrected	or_corrected	iterations
ALLERG_ASTHMA	93	17	0.8414810043551096	0.8512854316797401	0.09016669694066765	0.0	2.342656241877074	2
ALLERG_RHINITIS	93	12	0.4850807357802657	0.5831192789218208	0.06647935929489704	0.0	1.7916182808107328	2
L12_ATOPIC	93	16	0.6402973800129684	0.49179665798823735	0.06539921513548222	5.484501741648273e-14	1.6352515700517865	2

Supplementary Table 04 - r5_presso_outliers

ST04_r5_presso_outliers.csv

outcome	snp	beta_exp	se_exp	beta_out	se_out	wald_ratio	resid_z	p_raw	p_bonferroni	is_outlier	cochran_Q	cochran_Q_p	ivw_full_beta	ivw_presso_corrected_b	distortion_p	n_snps
ALLERG_ASTHMA	rs3828037	-0.0097723999204	0.00023817937259	-0.00624693	0.0127093	0.63924215656601	0.15550493733360	0.87642326186140	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs6698842	0.0131579691813	0.00024385248869	-0.0122091	0.0127633	-0.9278863502218	-1.8240800672226	0.06813996850273	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs4845620	-0.0092050883146	0.00023827087440	0.0675241	0.0136788	-7.3355189751695	5.50267618213310	3.74069104580598	3.4788426725995	True	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs1811472	-0.0121880493390	0.00024220545499	-0.0322656	0.0129051	2.64731452117222	-1.7054953468827	0.08810207936829	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs10437340	0.01211484784150	0.00024732955978	-0.0041461	0.0127985	-0.3422329404579	-1.1204839886924	0.26250757186750	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs2701185	-0.0094063924328	0.00023515981081	-0.0297543	0.012735	3.16319994221663	-1.7148802079515	0.08636719205854	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs147636938	-0.0296191559387	0.00075315190771	0.0130989	0.0434532	-0.4422442026055	0.87503008035134	0.38155753890054	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs12972156	-0.0180533193287	0.00033206029317	-0.0448471	0.0170091	2.48414705258815	-1.7435122798550	0.08124417117410	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs77301115	-0.0294270020077	0.00073887761571	0.019007	0.0430586	-0.6459033779581	1.01650455901160	0.30938917761524	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs112849259	-0.0294361521949	0.00073805409880	0.0189872	0.0430549	-0.6450299575245	1.01631087085011	0.30948137352599	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs116881820	-0.0294178518205	0.00073777959324	0.0191222	0.0430459	-0.6500202705700	1.01930180332960	0.30805970813380	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs79398853	-0.0295459544412	0.00073805409880	0.0192067	0.0430247	-0.6500619243215	1.02427348518083	0.30570611799451	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs75687619	-0.0295551046284	0.00073823710260	0.0192203	0.0430125	-0.6503208241563	1.02505920666241	0.30533525212150	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs76366838	-0.0294819031309	0.00073787109511	0.0192683	0.0429715	-0.6535636425653	1.02572080232024	0.30502320603251	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs114536010	-0.0296283061259	0.00073796259699	0.0205363	0.0430029	-0.6931310859513	1.05732303626577	0.29036418330184	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs115881343	-0.0292805990127	0.00071362309900	0.0207631	0.0417897	-0.7091077607732	1.08644397699260	0.27728261406591	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs112019714	-0.0293171997614	0.00071344009531	0.0190436	0.0418334	-0.6495709056438	1.04494176184961	0.29604988676931	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs1081105	-0.0300309143624	0.00071435511401	0.0258954	0.044507	-0.8622914270108	1.14961340866334	0.25030313302474	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs150966173	-0.0294361521949	0.00071353159719	-0.0041759	0.043899	0.14186297082384	0.46912373660821	0.63898119336961	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs4239225	0.00319221100400	8.05047487060650	0.0813649	0.0127299	25.488571994872	6.18062318464279	6.38490593729555	5.9379625216848	True	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs2095044	-0.0619	0.0014725	-0.10999	0.0145434	1.77689822294021	-3.9813472661426	6.85257477277545	0.0063728945386	True	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs56177354	-0.0111966146899	0.00025960545020	0.0377952	0.0252682	-3.3755917343451	1.86863087100250	0.06167418771519	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs1401884	-0.0061976163725	0.00014963700000	0.0319182	0.0143705	-5.1500767523071	2.58400030964701	0.00976617066711	0.9082538720429	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs76428106	-0.0247857090693	0.00055174481904	0.0148091	0.0604311	-0.5974854283394	0.59018954414754	0.55506358121262	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs391855	0.00600735953839	0.00012219245171	-0.0207023	0.0127352	-3.4461563133806	-2.0225343094646	0.04312118283539	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs139795227	-0.0255181978808	0.00050751010510	0.185889	0.0364273	-7.2845661307267	5.69249103782342	1.25199173250933	1.1643523112336	True	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs141094656	-0.0252470818921	0.00051844987300	0.184033	0.0363963	-7.2892780554182	5.64007714596444	1.69973992658611	1.5807581317250	True	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs1002607	0.00503229326336	0.00012276322221	-0.00675064	0.0129332	-1.3414639502727	-0.8493813742510	0.39566911388261	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs1610009	-0.0057409999705	0.00013698492059	0.00628296	0.0148286	-1.0944016777901	0.74949101204761	0.45356131390936	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs6740847	-0.0064259245735	0.00012157411701	0.000569017	0.0129691	-0.0885502146015	0.46081150303791	0.64493384814577	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs150649461	-0.0256038134562	0.00050703446301	0.184539	0.0364698	-7.2074810385403	5.65081581644141	1.59688124945489	1.4850995619930	True	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs4857909	0.08425000000000	0.0019965	0.00214183	0.018804	0.02542231454005	-3.6562935873706	0.00025588827665	0.0237976097293	True	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs17575497	-0.0224170114841	0.00058266155459	0.192889	0.0364032	-8.6045813973213	5.81686470800147	5.99616334362457	5.5764319095708	True	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs9807989	-0.05405	0.001422	-0.082897	0.0129277	1.53370952821461	-2.8941692423715	0.00380163195250	0.3535517715884	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs13277237	0.00439112773220	0.00012185950227	0.00722604	0.0128836	1.64560004640850	0.27406853873162	0.78403195090797	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs34856868	-0.0124570662162	0.00034945424011	0.165396	0.0332944	-13.277283521608	5.28252152286660	1.27417860706911	1.1849861045742	True	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs876036	0.00457139608262	0.00012889900511	-0.0262035	0.0138329	-5.7320563622995	-2.1723747707936	0.02982740639561	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs3093068	0.01793436689529	0.00051103795464	0.00935736	0.0253979	0.52175580295849	-0.2257694166649	0.82138076491669	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs2524079	0.0028269925194	8.15119906963631	-0.0756414	0.013	-26.756844766981	-6.0015585003507	1.95432514615841	1.8175223859273	True	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93
ALLERG_ASTHMA	rs11065394	0.00974494935891	0.00028786488904	0.0411541	0.0147244	4.22312097110562	2.23804774565142	0.02521794170822	1.0	False	646.693040555310	0.0	0.84148100435510	0.85128543167974	0.93214737606711	93

(showing first 40 of 279 rows; full data in the accompanying CSV)

Supplementary Table 05 - r5_raps_vs_ivw

ST05_r5_raps_vs_ivw.csv

method	n_snps	beta	se	p	or	or_lower	or_upper	outcome	egger_intercept	egger_intercept_p	cochran_Q	cochran_Q_p	presso_outliers_removed	phi
IVW_fixed	93	0.8414810043551096	0.071623534168863	0.0	2.319800067945715	2.015966791058354	2.669425101202063	ALLERG_ASTHMA						
IVW_random	93	0.8414810043551096	0.189893861883129	9.365675015082786	2.319800067945715	1.59885705772809	3.365824561507579	ALLERG_ASTHMA						
MR_Egger	93	0.8438130605464101	0.191218105132045	1.020279377073052	2.325216285063411	1.598435877410440	3.38245083755449	ALLERG_ASTHMA	0.0010177688206276	0.835718711009366	646.6930405553165	0.0		
weighted_median	93	0.8497319963536911	0.448384240766096	0.0580794436718115	2.339019902054779	0.971318244124488	5.632565984735239	ALLERG_ASTHMA						
IVW_PRESSO_corrected	93	0.8512854316797400	0.090166696940667	0.0	2.342656241877074	1.963166450575896	2.795503288065918	ALLERG_ASTHMA					17.0	
MR_RAPS	93	0.7176065441879759	0.143595210081240	5.810039680476109	2.049521895388071	1.546756151436648	2.715709257579866	ALLERG_ASTHMA						2.47391647156698
IVW_fixed	93	0.4850807357802651	0.065801519187213	1.683098105331737	1.624306143174401	1.427764248459977	1.847903426353414	ALLERG_RHINITIS						
IVW_random	93	0.4850807357802651	0.142565327489666	0.000667685691276	1.624306143174401	1.228326524611274	2.147938999842944	ALLERG_RHINITIS						
MR_Egger	93	0.4917610476981791	0.143135868099976	0.000591190480991	1.635193339305872	1.235177576301034	2.164755342238031	ALLERG_RHINITIS	0.0028351121817074	0.440445498713412	431.8608333754709	0.0		
weighted_median	93	0.5013254329990884	0.167186686643433	0.002712261441546	1.650907989132867	1.189626660441175	2.29105254548681	ALLERG_RHINITIS						
IVW_PRESSO_corrected	93	0.5831192789218208	0.066479359294897	0.0	1.791618280810732	1.572740650399927	2.040957015588661	ALLERG_RHINITIS					12.0	
MR_RAPS	93	0.4917642833244204	0.101743383805713	1.342352217159614	1.635198630188910	1.339568103334690	1.996072132141252	ALLERG_RHINITIS						2.0840928504524276
IVW_fixed	93	0.6402973800129684	0.047985857096085	0.0	1.897044938679268	1.72675702981679	2.084126160905488	L12_ATOPIC						
IVW_random	93	0.6402973800129684	0.114238198035745	2.083362327098825	1.897044938679268	1.516477955886154	2.373116922273809	L12_ATOPIC						
MR_Egger	93	0.6366771176540485	0.114869978310341	2.980312507894211	1.890189554925859	1.509127939016803	2.367470948737791	L12_ATOPIC	-0.001598513180670	0.587375942608924	521.4155787102993	0.0		
weighted_median	93	0.4602136498516311	0.312105559937411	0.140334948047244	1.584412458322449	0.859406497101920	2.921042424688202	L12_ATOPIC						
IVW_PRESSO_corrected	93	0.4917966579882377	0.065399215135482	5.484501741648273	1.635251570051786	1.438519121119658	1.858889227190468	L12_ATOPIC					16.0	
MR_RAPS	93	0.5467416894551396	0.088762700331596	7.293430304144977	1.727614731913933	1.451745748607434	2.055905908310071	L12_ATOPIC						2.1722187260745107

Supplementary Table 06 - r5_bidirectional_matrix

ST06_r5_bidirectional_matrix.csv

direction	outcome	trait	method	n_snps	beta	se	p	or	or_lower	or_upper
forward_aging_to_allergy	ALLERG_ASTHMA	GrimAge	IVW_fixed	5	0.110916693868455 26	0.0319340862517817	0.0005140984354838896	1.1173018248845636		
forward_aging_to_allergy	L12_ATOPIC	GrimAge	IVW_fixed	5	-0.026714884528899016	0.02137547404867976	0.21137574322038066	0.9736388014396982		
forward_aging_to_allergy	ALLERG_RHINITIS	GrimAge	IVW_fixed	5	-0.00014842620940936372	0.02933140056574105	0.9959624677580692	0.9998515848052155		
forward_aging_to_allergy	ALLERG_ASTHMA	Hannum	IVW_fixed	56	0.0193100976482815	0.007373032824284832	0.00881835759313443	1.0194977434574162		
forward_aging_to_allergy	L12_ATOPIC	Hannum	IVW_fixed	56	0.009267154804618745	0.004910751195691883	0.059144789438002165	1.009310227835704		
forward_aging_to_allergy	ALLERG_RHINITIS	Hannum	IVW_fixed	56	-0.027770585160008195	0.006756517118176393	3.95329621052376e-05	0.9726114727137516		
forward_aging_to_allergy	ALLERG_ASTHMA	IEAA	IVW_fixed	133	-0.014004594978069052	0.0033695176342471737	3.234981420852101e-05	0.9860930131766822		
forward_aging_to_allergy	L12_ATOPIC	IEAA	IVW_fixed	133	-0.004192104543932167	0.0022649607363169977	0.06419101073995459	0.995816670060686		
forward_aging_to_allergy	ALLERG_RHINITIS	IEAA	IVW_fixed	133	0.005509707319350415	0.0031122674936183243	0.07667395891445117	1.005524913671412		
forward_aging_to_allergy	ALLERG_ASTHMA	PhenoAge	IVW_fixed	89	-0.056990324120883	0.005137566080018194	0.0	0.944603209188771		
forward_aging_to_allergy	L12_ATOPIC	PhenoAge	IVW_fixed	89	-0.007332280929135882	0.003430238824326783	0.03255390094357802	0.9926945346628314		
forward_aging_to_allergy	ALLERG_RHINITIS	PhenoAge	IVW_fixed	89	-0.006024996006940149	0.004713361667395943	0.2011507130563026	0.9939931178845359		
forward_aging_to_allergy	ALLERG_ASTHMA	Frailty	IVW_fixed	66	1.2345712417627024	0.08464304000004685	0.0	3.4369046035910564		
forward_aging_to_allergy	L12_ATOPIC	Frailty	IVW_fixed	71	0.37628386365900324	0.053993265705615495	3.1898927943529998e-12	1.4568606248683944		
forward_aging_to_allergy	ALLERG_RHINITIS	Frailty	IVW_fixed	70	0.5256676234344337	0.07484449468671586	2.16426876420428e-12	1.691587815141615		
reverse_allergy_to_trait	ALLERG_ASTHMA	crp	IVW_fixed	165.0	0.0097758146293003	0.003361005835153	0.0036305344425144	1.0098237539932349	1.0031933310453949	1.016497999509573
reverse_allergy_to_trait	ALLERG_ASTHMA	wbc	IVW_fixed	137.0	0.0232273875206738	0.0031125810949168	8.504308368628699e-14	1.0234992440446804	1.0172742318702264	1.0297623489726495
reverse_allergy_to_trait	ALLERG_ASTHMA	neutrophil_pct	IVW_fixed	161.0	-0.0045086274241062	0.0034161242301993	0.1868991948375125	0.9955015211787	0.9888583027149304	1.002189369243532
reverse_allergy_to_trait	ALLERG_ASTHMA	lymphocyte_pct	IVW_fixed	160.0	-0.0332339982371797	0.0033827966922168	0.0	0.9673121837584904	0.9609198465806956	0.973747044747964
reverse_allergy_to_trait	ALLERG_ASTHMA	monocyte_pct	IVW_fixed	162.0	-0.0198879430997359	0.0033888816184483	4.395275032820223e-09	0.9803085174860132	0.9738186817158028	0.9868416035749072
reverse_allergy_to_trait	ALLERG_ASTHMA	eosinophil_pct	IVW_fixed	121.0	0.0197423667415969	0.0044696688813819	1.0009687265055687e-05	1.019938536086269	1.0110423372148514	1.0289130129401671
reverse_allergy_to_trait	ALLERG_ASTHMA	eosinophil_pct_astle	IVW_fixed	120.0	-0.0082259943878753	0.0070186575997368	0.2411897525412607	0.9918077465230588	0.9782572924977074	1.0055458963680082
reverse_allergy_to_trait	ALLERG_ASTHMA	wbc_astle	IVW_fixed	138.0	0.037265993023517	0.0056279036765275	3.552180771748681e-11	1.0379690766550853	1.0265824767789264	1.049481973891342
reverse_allergy_to_trait	ALLERG_RHINITIS	crp	IVW_fixed	19.0	-0.014339967884148	0.0085946613753207	0.095221695819579	0.985762359746389	0.9692957479846976	1.0025087099712626
reverse_allergy_to_trait	ALLERG_RHINITIS	wbc	IVW_fixed	15.0	0.0490894983247357	0.0080451000088732	1.048881426157777e-09	1.0503143479082546	1.0338824679929046	1.0670073858235816
reverse_allergy_to_trait	ALLERG_RHINITIS	neutrophil_pct	IVW_fixed	15.0	-0.050542027024444	0.0098795840236235	3.1240540221411095e-07	0.9507139721533592	0.9324814581244106	0.9693029807430512
reverse_allergy_to_trait	ALLERG_RHINITIS	lymphocyte_pct	IVW_fixed	16.0	-0.0089543881792539	0.0100059451774785	0.3708369221301972	0.9910855829599072	0.9718381111313192	1.0107142552863444
reverse_allergy_to_trait	ALLERG_RHINITIS	monocyte_pct	IVW_fixed	20.0	-0.0543139796319624	0.0082766104266501	5.296785232644652e-11	0.947134678813251	0.9318940639299058	0.9626245455708304
reverse_allergy_to_trait	ALLERG_RHINITIS	eosinophil_pct	IVW_fixed	8.0	-0.0088886467239901	0.0172126811892559	0.605574329155008	0.9911507405101746	0.9582701932871271	1.0251594981202927
reverse_allergy_to_trait	ALLERG_RHINITIS	eosinophil_pct_astle	IVW_fixed	4.0	-0.0222297853024585	0.0605546021431981	0.7135420763915716	0.9780154756471146	0.8685615682346257	1.101262484534507
reverse_allergy_to_trait	ALLERG_RHINITIS	wbc_astle	IVW_fixed	10.0	0.036458899959508	0.0181505148510935	0.0445691880376641	1.0371316769878496	1.0008842815980006	1.0746917852424172
reverse_allergy_to_trait	L12_ATOPIC	crp	IVW_fixed	159.0	0.0243782198183607	0.0032147837571088	3.3750779948604765e-14	1.0246777980603234	1.0182416260564475	1.0311546522647728
reverse_allergy_to_trait	L12_ATOPIC	wbc	IVW_fixed	151.0	0.0230607105895003	0.0027381570831941	0.0	1.023328664547853	1.0178513875091504	1.0288354159912936
reverse_allergy_to_trait	L12_ATOPIC	neutrophil_pct	IVW_fixed	163.0	-0.000499339585369	0.0030988909379545	0.8719871267407546	0.9995007850638934	0.99344839017447	1.0055900530151285
reverse_allergy_to_trait	L12_ATOPIC	lymphocyte_pct	IVW_fixed	160.0	-0.016520407574854	0.0031019852596657	1.0053562338185885e-07	0.9836153059828638	0.9776531749490024	0.9896137965431666
reverse_allergy_to_trait	L12_ATOPIC	monocyte_pct	IVW_fixed	161.0	0.0068643624119932	0.0031402695444717	0.0288218220329579	1.0068879761478136	1.0007096858127231	1.0131044106839704
reverse_allergy_to_trait	L12_ATOPIC	eosinophil_pct	IVW_fixed	144.0	0.0220532934283435	0.0034350165648384	1.3614753768820262e-10	1.0222982647980243	1.0154386239411	1.029204244913252
reverse_allergy_to_trait	L12_ATOPIC	eosinophil_pct_astle	IVW_fixed	100.0	0.0595455691200307	0.0064370222649362	0.0	1.061354124851089	1.0480476008028328	1.0748295950255624
reverse_allergy_to_trait	L12_ATOPIC	wbc_astle	IVW_fixed	121.0	0.0067239494020526	0.0051545079046725	0.1920701772076141	1.0067466059017527	0.9966268159205456	1.016969152649713

Supplementary Table 07 - r5_reverse_mr_components

ST07_r5_reverse_mr_components.csv

method	n_snps	beta	se	p	or	or_lower	or_upper	outcome	exposure	trait	source	n_harmonised	steiger_applied	egger_intercept	egger_intercept_p	cochran_Q	cochran_Q_ppresso_outliers_removed
IVW_fixed	165	0.009775814629	0.003361005835	0.003630534442	1.009823753993	1.003193331045	1.016497999509	ALLERG_ASTHMA	ALLERG_ASTHMA	crp	panukbb	168	True				
IVW_random	165	0.009775814629	0.004313902962	0.023444138491	1.009823753993	1.001321437153	1.018398264825	ALLERG_ASTHMA	ALLERG_ASTHMA	crp	panukbb	168	True				
MR_Egger	165	0.007477057698	0.004194578264	0.074659110836	1.007505080693	0.999255961233	1.015822298793	ALLERG_ASTHMA	ALLERG_ASTHMA	crp	panukbb	168	True	-0.002214104061	0.000228346583	270.1755589587	3.430704784701
weighted_media	165	0.016561207037	0.004736617224	0.000471544429	1.016699104022	1.007303982177	1.026181854145	ALLERG_ASTHMA	ALLERG_ASTHMA	crp	panukbb	168	True				
IVW_PRESSO_cor	165	0.005974256387	0.003402758706	0.079137993736	1.005992137849	0.999305090883	1.012723932507	ALLERG_ASTHMA	ALLERG_ASTHMA	crp	panukbb	168	True				1.0
IVW_fixed	137	0.023227387520	0.003112581094	8.504308368628	1.023499244044	1.017274231870	1.029762348972	ALLERG_ASTHMA	ALLERG_ASTHMA	wbc	panukbb	168	True				
IVW_random	137	0.023227387520	0.004149630744	2.175164026674	1.023499244044	1.015208602682	1.031857590441	ALLERG_ASTHMA	ALLERG_ASTHMA	wbc	panukbb	168	True				
MR_Egger	137	0.025132273699	0.004363390171	8.421148889325	1.025450751718	1.016718240437	1.034258265837	ALLERG_ASTHMA	ALLERG_ASTHMA	wbc	panukbb	168	True	0.000856964229	0.172642629599	241.7221581277	6.364209192977
weighted_media	137	0.016694176662	0.004613626336	0.000296372631	1.016834303109	1.007680816335	1.026070937560	ALLERG_ASTHMA	ALLERG_ASTHMA	wbc	panukbb	168	True				
IVW_PRESSO_cor	137	0.029138600402	0.003235630535	0.0	1.029567283026	1.023058596645	1.036117377589	ALLERG_ASTHMA	ALLERG_ASTHMA	wbc	panukbb	168	True				4.0
IVW_fixed	161	-0.00450862742	0.003416124230	0.186899194837	0.995501521178	0.988858302714	1.002189369243	ALLERG_ASTHMA	ALLERG_ASTHMA	neutrophil_pct	panukbb	168	True				
IVW_random	161	-0.00450862742	0.004521900581	0.318733108390	0.995501521178	0.986717449505	1.004363791443	ALLERG_ASTHMA	ALLERG_ASTHMA	neutrophil_pct	panukbb	168	True				
MR_Egger	161	-0.00582814866	0.004579187297	0.203107696454	0.994188802045	0.985305675229	1.003152015624	ALLERG_ASTHMA	ALLERG_ASTHMA	neutrophil_pct	panukbb	168	True	-0.001019755451	0.118495002431	280.3462107645	1.288504070107
weighted_media	161	-0.00089613519	0.012477386797	0.942744564708	0.999104266215	0.974966845364	1.023839261320	ALLERG_ASTHMA	ALLERG_ASTHMA	neutrophil_pct	panukbb	168	True				
IVW_PRESSO_cor	161	0.003030323027	0.003598761968	0.399762220057	1.003034919097	0.995984831999	1.010134910297	ALLERG_ASTHMA	ALLERG_ASTHMA	neutrophil_pct	panukbb	168	True				6.0
IVW_fixed	160	-0.03323399823	0.003382796692	0.0	0.967312183758	0.960919846580	0.973747044747	ALLERG_ASTHMA	ALLERG_ASTHMA	lymphocyte_pct	panukbb	168	True				
IVW_random	160	-0.03323399823	0.003740823546	0.0	0.967312183758	0.960245774302	0.974430594632	ALLERG_ASTHMA	ALLERG_ASTHMA	lymphocyte_pct	panukbb	168	True				
MR_Egger	160	-0.03274689913	0.003806834251	0.0	0.967783475433	0.960589333065	0.975031496896	ALLERG_ASTHMA	ALLERG_ASTHMA	lymphocyte_pct	panukbb	168	True	0.000389135554	0.471719266444	194.4373869760	0.029179710038
weighted_media	160	-0.03315197027	0.002007942809	0.0	0.967391533656	0.963591780581	0.971206270383	ALLERG_ASTHMA	ALLERG_ASTHMA	lymphocyte_pct	panukbb	168	True				
IVW_PRESSO_cor	160	-0.03352273223	0.003494505341	0.0	0.967032928157	0.960432127766	0.973679094133	ALLERG_ASTHMA	ALLERG_ASTHMA	lymphocyte_pct	panukbb	168	True				3.0
IVW_fixed	162	-0.01988794309	0.003388881618	4.395275032820	0.980308517486	0.973818681715	0.986841603574	ALLERG_ASTHMA	ALLERG_ASTHMA	monocyte_pct	panukbb	168	True				
IVW_random	162	-0.01988794309	0.005109869845	9.939480454002	0.980308517486	0.970539391818	0.990175975913	ALLERG_ASTHMA	ALLERG_ASTHMA	monocyte_pct	panukbb	168	True				
MR_Egger	162	-0.02417569183	0.004808746913	4.970681097837	0.976114199398	0.966957402407	0.985357708514	ALLERG_ASTHMA	ALLERG_ASTHMA	monocyte_pct	panukbb	168	True	-0.003573276411	5.466564200561	366.0435480032	0.0
weighted_media	162	-0.00329475577	0.006190965962	0.594596018583	0.996710665971	0.984689368902	1.008878721590	ALLERG_ASTHMA	ALLERG_ASTHMA	monocyte_pct	panukbb	168	True				
IVW_PRESSO_cor	162	-0.02516223363	0.003560816774	1.589839371263	0.975151696791	0.968369611510	0.981981281168	ALLERG_ASTHMA	ALLERG_ASTHMA	monocyte_pct	panukbb	168	True				5.0
IVW_fixed	121	0.019742366741	0.004469668881	1.000968726505	1.019938536086	1.011042337214	1.028913012940	ALLERG_ASTHMA	ALLERG_ASTHMA	eosinophil_pct	panukbb	168	True				
IVW_random	121	0.019742366741	0.005255037660	0.000172068372	1.019938536086	1.009487213911	1.030498061845	ALLERG_ASTHMA	ALLERG_ASTHMA	eosinophil_pct	panukbb	168	True				
MR_Egger	121	0.020280221098	0.006082867478	0.000856096468	1.020487262025	1.008392824764	1.032726757254	ALLERG_ASTHMA	ALLERG_ASTHMA	eosinophil_pct	panukbb	168	True	0.000174610980	0.859260225501	165.8754862907	0.003546118467
weighted_media	121	0.020829238463	0.001948435389	0.0	1.021047681079	1.017155804163	1.024954449229	ALLERG_ASTHMA	ALLERG_ASTHMA	eosinophil_pct	panukbb	168	True				
IVW_PRESSO_cor	121	0.030662100817	0.004714316573	7.819145331211	1.031137024656	1.021653139493	1.040708947602	ALLERG_ASTHMA	ALLERG_ASTHMA	eosinophil_pct	panukbb	168	True				3.0
IVW_fixed	120	-0.00822599438	0.007018657599	0.241189752541	0.991807746523	0.978257292497	1.005545896368	ALLERG_ASTHMA	ALLERG_ASTHMA	eosinophil_pct_a	astle	173	True				
IVW_random	120	-0.00822599438	0.007018657599	0.241189752541	0.991807746523	0.978257292497	1.005545896368	ALLERG_ASTHMA	ALLERG_ASTHMA	eosinophil_pct_a	astle	173	True				
MR_Egger	120	0.000709831508	0.004402947407	0.871922217458	1.000710083498	0.992111334724	1.009383358666	ALLERG_ASTHMA	ALLERG_ASTHMA	eosinophil_pct_a	astle	173	True	0.003125630314	5.233431235208	40.90379262426	0.999999999998
weighted_media	120	-0.01002215355	0.007709511362	0.193610359281	0.990027900869	0.975180403544	1.005101457061	ALLERG_ASTHMA	ALLERG_ASTHMA	eosinophil_pct_a	astle	173	True				
IVW_PRESSO_cor	120	-0.01002215355	0.007709511362	0.193610359281	0.990027900869	0.975180403544	1.005101457061	ALLERG_ASTHMA	ALLERG_ASTHMA	eosinophil_pct_a	astle	173	True				0.0
IVW_fixed	138	0.037265993023	0.005627903676	3.552180771748	1.037969076655	1.026582476778	1.049481973891	ALLERG_ASTHMA	ALLERG_ASTHMA	wbc_astle	astle	173	True				
IVW_random	138	0.037265993023	0.005627903676	3.552180771748	1.037969076655	1.026582476778	1.049481973891	ALLERG_ASTHMA	ALLERG_ASTHMA	wbc_astle	astle	173	True				
MR_Egger	138	0.036738538492	0.004069415044	0.0	1.037421739522	1.029180119732	1.045729357767	ALLERG_ASTHMA	ALLERG_ASTHMA	wbc_astle	astle	173	True	-0.000284407347	0.630582157294	66.07730886373	0.999999953561
weighted_media	138	0.042499788468	0.002272313855	0.0	1.043415835658	1.038779070992	1.048073297301	ALLERG_ASTHMA	ALLERG_ASTHMA	wbc_astle	astle	173	True				
IVW_PRESSO_cor	138	0.043134371346	0.005728736399	5.084821452783	1.044078179616	1.032420502780	1.055867490247	ALLERG_ASTHMA	ALLERG_ASTHMA	wbc_astle	astle	173	True				1.0

(showing first 40 of 120 rows; full data in the accompanying CSV)

Supplementary Table 08 - r5_winners_curse_corrected

ST08_r5_winners_curse_corrected.csv

trait	component	outcome	naive_n	naive_beta	naive_se	naive_p	naive_or	naive_or_lower	naive_or_upper	corrected_n	corrected_beta	corrected_se	corrected_p	corrected_or	corrected_or_low	corrected_or_upp	r4_index_or	r4_index_beta	r4_index_direction_corrected	corrected_direction_consi	naive_corrected_ci_overlap
astle_eosinop	eosinophil_pct	ALLERG_ASTH	46	0.517310602	0.043335972	0.0	1.677510086	1.540908375	1.826221554	88	0.497853152	0.035036094	0.0	1.645185514	1.536001186	1.762131046	2.319800067	0.841481004	0.0	True	True	True
astle_eosinop	eosinophil_pct	ALLERG_RHIN	46	0.207105016	0.039774508	1.919401546	1.230111746	1.137857354	1.329845876	88	0.197702249	0.032147923	7.759670683	1.218599500	1.144184685	1.297854062	1.624306143	0.485080735	1.683098105	True	True	True
astle_eosinop	eosinophil_pct	L12_ATOPIC	35	0.210712568	0.031092351	1.227040691	1.234557453	1.161568862	1.312132371	69	0.199011480	0.025220654	3.108624468	1.220195974	1.161345210	1.282028979	1.897044938	0.640297380	0.0	True	True	True
astle_wbc	wbc	ALLERG_ASTH	43	0.087619267	0.051690587	0.090061763	1.091572446	0.986399055	1.207959798	88	0.143349774	0.053278439	0.007132845	1.154133417	1.039691498	1.281172297	2.319800067	0.841481004	0.0	True	True	True
astle_wbc	wbc	ALLERG_RHIN	43	-0.055024090	0.047386208	0.245567221	0.946462347	0.862516449	1.038578423	88	-0.118438694	0.048842283	0.015311704	0.888306271	0.807211494	0.977548062	1.624306143	0.485080735	1.683098105	False	False	True
astle_wbc	wbc	L12_ATOPIC	34	0.056893488	0.036318863	0.117231683	1.058543057	0.985810153	1.136642181	69	0.043907235	0.036913048	0.234251878	1.044885421	0.971958335	1.123284306	1.897044938	0.640297380	0.0	True	True	True

Supplementary Table 09 - r5_astle_selection_sensitivity

ST09_r5_astle_selection_sensitivity.csv

trait	category	item	n_raw_gws	n_clumped	median_abs_beta_naive	median_abs_beta_corrected	inflation_ratio_naive_over_c	naive_beta	corrected_beta	naive_over_corrected_beta_	naive_p	corrected_p	naive_n	corrected_n
astle_eosinophil	instrument_selection	counts	137.0	46.0	0.05308625	0.059125	0.891011726161324							
astle_wbc	instrument_selection	counts	167.0	46.0	0.03221085	0.0298300000000000	1.245005599404404							
astle_eosinophil	mr_effect	ALLERG_ASTHMA						0.517310602644196	0.497853152837773	1.039082708817882	0.0	0.0	46.0	88.0
astle_eosinophil	mr_effect	ALLERG_RHINITIS						0.207105016436274	0.197702249068754	1.047560244821747	1.919401546590649	7.75967068378236e-	46.0	88.0
astle_eosinophil	mr_effect	L12_ATOPIC						0.210712568800275	0.199011480580993	1.058796046263877	1.227040691276215	3.108624468950438	35.0	69.0
astle_wbc	mr_effect	ALLERG_ASTHMA						0.087619267832370	0.143349774459945	0.611227106303214	0.090061763890442	0.007132845035221	43.0	88.0
astle_wbc	mr_effect	ALLERG_RHINITIS						-0.055024090166128	-0.118438694603318	0.464578661141260	0.245567221129214	0.015311704661854	43.0	88.0
astle_wbc	mr_effect	L12_ATOPIC						0.056893488494465	0.043907235271338	1.295765678318725	0.117231683412557	0.234251878734631	34.0	69.0

Supplementary Table 10 - r5_heterogeneity_snp_clusters

ST10_r5_heterogeneity_snp_clusters.csv

rsid	chrom	pos	ref	alt	crp_beta	eosinophil_pct_b	lymphocyte_pct_b	monocyte_pct_b	neutrophil_pct_b	wbc_beta	crp_se	eosinophil_pct_	lymphocyte_pct_	monocyte_pct_s	neutrophil_pct_	wbc_se	cluster	crp_z	wbc_z	neutrophil_pct_	lymphocyte_pct_	monocyte_pct_s	eosinophil_pct_	cluster_metho	quartile
rs1002607	9	113858933	T	C	-0.0005239	0.003551	0.007752	-0.1058	0.01989	-0.02108	0.002624	0.002634	0.002538	0.002581	0.002558	0.002203	0	0.34356906	-0.81272812	0.48763987	0.32943265	-0.94103798	0.19895956	eosinophil_s	Q1_low
rs10437340	1	159710809	G	C	0.1324	-0.0009136	-0.002186	0.002207	0.0007677	0.00123	0.002703	0.002713	0.002615	0.002659	0.002636	0.00227	1	1.38595633	-0.22597050	0.01114825	0.14507971	-0.23715160	0.10464945	eosinophil_s	Q1_low
rs1081105	19	45412955	A	C	-0.3282	-0.002148	-0.01344	0.0008059	0.01307	-0.01988	0.007807	0.007841	0.007557	0.007684	0.007617	0.00656	1	-2.22606208	-0.78116787	0.31769835	-0.06368543	-0.24628263	0.07857401	eosinophil_s	Q1_low
rs10849448	12	6493351	A	G	-0.004738	-0.02719	0.03335	-0.07971	-0.003327	0.003588	0.003041	0.003051	0.00294	0.002989	0.002963	0.002552	1	0.31052215	-0.16395461	-0.09088393	0.80428340	-0.77100830	-0.45041265	eosinophil_s	Q3
rs11065394	12	121446922	A	G	0.1065	0.009083	0.0008694	-0.005701	-0.001281	0.01188	0.003146	0.003156	0.003042	0.003093	0.003066	0.00264	0	1.18284891	0.05412670	-0.03990148	0.20175831	-0.28868839	0.31581741	eosinophil_s	Q2
rs11201971	19	45404857	T	C	-0.3204	-0.003199	-0.01311	-5.569e-05	0.01302	-0.01736	0.007797	0.007829	0.007545	0.007672	0.007605	0.006549	1	-2.16489460	-0.71489135	0.31645244	-0.05756383	-0.25189765	0.05637271	eosinophil_s	Q1_low
rs11284925	19	45397307	C	T	-0.3217	0.0002972	-0.01181	-0.0006855	0.01163	-0.0169	0.008066	0.008098	0.007804	0.007935	0.007866	0.006774	0	-2.17508918	-0.70279325	0.28181626	-0.03344843	-0.25600215	0.13022636	eosinophil_s	Q1_low
rs11307182	4	38665818	G	A	-0.001152	0.0894	-0.02454	-0.002513	0.005111	-0.004768	0.003121	0.003131	0.003018	0.003068	0.003042	0.002619	0	0.33864351	-0.38371914	0.11937509	-0.26953938	-0.26791205	2.01243200	eosinophil_s	Q4_high
rs11354238	2	43464818	G	A	0.01056	-0.03879	-0.1039	-0.01976	0.102	0.003098	0.004901	0.004922	0.004744	0.004823	0.004781	0.004118	1	0.43048885	-0.17684171	2.53366609	-1.74174615	-0.38031151	-0.69545080	eosinophil_s	Q4_high
rs11440397	1	65924032	A	G	0.1104	-0.02312	-0.02624	-0.01582	0.03182	0.03215	0.00349	0.003504	0.003377	0.003433	0.003404	0.002931	1	1.21343266	0.58723190	0.78491296	-0.30112935	-0.35463436	-0.36443806	eosinophil_s	Q3
rs11442573	1	92171673	G	C	0.05175	-0.0003094	-0.102	0.3457	0.00672	0.04602	0.01121	0.01127	0.01086	0.01104	0.01094	0.009421	1	0.75350022	0.95201577	0.15946833	-1.70650057	2.00140741	0.11741256	eosinophil_s	Q1_low
rs11453601	19	45400725	C	T	-0.3238	0.0002111	-0.01176	-0.0002716	0.01157	-0.01713	0.008065	0.008097	0.007804	0.007935	0.007865	0.006773	0	-2.19155735	-0.70884230	0.28032118	-0.03252092	-0.25330475	0.12840759	eosinophil_s	Q1_low
rs11466597	1	92185351	T	C	0.05342	0.004585	-0.1098	0.3731	0.005054	0.04841	0.01172	0.01178	0.01135	0.01154	0.01144	0.009847	0	0.76659634	1.01487326	0.11795476	-1.85119296	2.17997442	0.22080176	eosinophil_s	Q2
rs11588134	19	45403216	C	T	-0.32	-0.003772	-0.0129	4.082e-05	0.01284	-0.01726	0.007799	0.007832	0.007548	0.007675	0.007607	0.006552	1	-2.16175780	-0.71226133	0.31196718	-0.05366827	-0.25126869	0.04426867	eosinophil_s	Q1_low
rs11635924	1	93035991	G	A	0.04006	-0.01077	-0.06853	0.2156	0.009291	0.03334	0.006963	0.00699	0.006737	0.00685	0.00679	0.005847	1	0.66182741	0.61852914	0.22353280	-1.08562181	1.15353998	-0.10355692	eosinophil_s	Q2
rs11688182	19	45397952	T	C	-0.3215	-0.0001532	-0.01129	-0.0003994	0.01125	-0.01687	0.008063	0.008095	0.007802	0.007933	0.007864	0.006772	1	-2.17352078	-0.70200425	0.27234738	-0.02380227	-0.25413763	0.12071212	eosinophil_s	Q1_low
rs11731044	19	45393516	C	T	-0.3693	0.02184	-0.00759	0.01042	0.0009031	-0.01326	0.0118	0.01186	0.01143	0.01163	0.01152	0.009923	0	-2.54836768	-0.60706050	0.01452216	0.04483385	-0.18362712	0.58529601	eosinophil_s	Q3
rs11752086	19	16580644	C	A	-0.004778	-0.02118	0.1019	-0.03855	-0.07534	0.04164	0.005963	0.005979	0.005763	0.005859	0.005808	0.005001	1	0.31020847	0.83682086	-1.88531197	2.07590690	-0.50276677	-0.32345754	eosinophil_s	Q3
rs11786638	19	16582942	C	A	-0.004745	-0.02129	0.1017	-0.03854	-0.07516	0.04178	0.005959	0.005976	0.005759	0.005856	0.005805	0.004998	1	0.31046726	0.84050289	-1.88082671	2.07219684	-0.50270160	-0.32578118	eosinophil_s	Q3
rs11805144	19	16589807	A	G	-0.004168	-0.02219	0.101	-0.03951	-0.07415	0.04228	0.005938	0.005955	0.00574	0.005836	0.005785	0.004981	1	0.31499209	0.85365299	-1.85565942	2.05921163	-0.50902313	-0.34479276	eosinophil_s	Q3
rs11817034	19	45401868	T	C	-0.2144	-0.001377	-0.01482	-0.004628	0.01553	-0.01512	0.00654	0.006571	0.006333	0.006439	0.006383	0.005496	1	-1.33364416	-0.65597888	0.37899690	-0.08928485	-0.28169560	0.09486060	eosinophil_s	Q1_low
rs11864884	16	85901039	C	T	-0.002108	-0.005487	-0.004497	-0.06845	0.02185	0.0006576	0.002876	0.002888	0.002783	0.00283	0.002805	0.002415	1	0.33114658	-0.24102474	0.53647937	0.10220995	-0.69762639	0.00804105	eosinophil_s	Q2
rs12515180	5	131785684	C	T	0.006039	-0.0748	0.01047	0.003008	0.00516	-0.02673	0.002595	0.002605	0.00251	0.002552	0.00253	0.002179	1	0.39503523	-0.96132429	0.12059608	0.37985239	-0.23193145	-1.45612527	eosinophil_s	Q4_high
rs1260326	2	27730940	T	C	-0.0851	0.0191	0.005805	0.05027	-0.02235	-0.03276	0.002624	0.002634	0.002539	0.002581	0.002559	0.002204	0	-0.31967546	-1.11991454	-0.56490119	0.29331521	0.07607708	0.52741631	eosinophil_s	Q3
rs12619285	2	213824045	A	G	-0.003383	-0.07749	0.005682	-0.0006747	0.008612	-0.00685	0.002909	0.00292	0.002814	0.002861	0.002836	0.002442	1	0.32114804	-0.43847617	0.20661340	0.29103352	-0.25593177	-1.51294877	eosinophil_s	Q4_high
rs12972156	19	45387459	C	G	-0.1973	-0.006393	0.003378	-0.01268	0.001468	-0.008993	0.003629	0.003643	0.003511	0.00357	0.003539	0.003047	1	-1.19954623	-0.49483751	0.02859841	0.24829361	-0.33417084	-0.01109727	eosinophil_s	Q2
rs13277237	8	130604563	G	A	0.004256	0.03298	0.04566	-0.09232	-0.02376	-0.03236	0.002605	0.002614	0.00252	0.002562	0.002539	0.002187	0	0.38105297	-1.10939445	-0.60003573	1.03263767	-0.85318823	0.82061713	eosinophil_s	Q3
rs13979522	1	92842367	A	C	0.07748	-0.0163	-0.1471	0.5365	0.0002698	0.06408	0.01084	0.01089	0.0105	0.01067	0.01058	0.009112	1	0.95527450	1.42699751	-0.00125847	-2.54311939	3.24485942	-0.22037252	eosinophil_s	Q2
rs1401884	16	86013551	G	C	0.0002157	0.03918	0.0127	0.1303	-0.05082	0.00281	0.003198	0.003211	0.003094	0.003146	0.003119	0.002685	0	0.34936900	-0.18441617	-1.27431985	0.42121957	0.59763612	0.95158580	eosinophil_s	Q4_high
rs14109465	1	92753336	T	C	0.08537	-0.009281	-0.1497	0.5308	0.001553	0.06197	0.01108	0.01112	0.01072	0.0109	0.01081	0.009305	1	1.01714777	1.37150407	0.03071645	-2.59135019	3.20771227	-0.07210332	eosinophil_s	Q2
rs14493724	13	28697891	G	C	-0.0008097	0.000706	-0.03131	0.2342	-0.03468	0.07016	0.009847	0.00988	0.009523	0.009682	0.009598	0.008267	0	0.34132783	1.58690277	-0.87214152	-0.39517941	1.27475700	0.13886185	eosinophil_s	Q1_low
rs14509671	13	28761592	G	A	0.005747	-0.01097	-0.05877	0.4051	-0.05194	0.1116	0.01233	0.01237	0.01193	0.01213	0.01202	0.01035	1	0.39274537	2.67678335	-1.30222814	-0.90457082	2.38851983	-0.10778172	eosinophil_s	Q2
rs14763693	19	45386634	C	T	-0.3237	-0.001005	-0.01561	0.000362	0.01523	-0.01735	0.008231	0.008262	0.007963	0.008096	0.008026	0.006911	1	-2.19077315	-0.71462835	0.37152147	-0.10393960	-0.24917555	0.10271872	eosinophil_s	Q1_low
rs14777508	19	16577203	G	A	-0.004703	-0.02155	0.1017	-0.03861	-0.07508	0.04186	0.005966	0.005983	0.005766	0.005863	0.005812	0.005004	1	0.31079662	0.84260691	-1.87883326	2.07219684	-0.50315779	-0.33127341	eosinophil_s	Q3
rs14900776	7	50370254	C	T	0.001637	-0.02023	-0.02459	-0.09546	0.04726	0.009736	0.003594	0.003609	0.003478	0.003536	0.003506	0.003019	1	0.36051481	-0.00226094	1.16964861	-0.27052135	-0.87365175	-0.30338976	eosinophil_s	Q3
rs15064946	1	92925654	G	C	0.0786	-0.01649	-0.1507	0.5383	0.003366	0.06564	0.01083	0.01088	0.01049	0.01066	0.01057	0.009103	1	0.96405752	1.46802583	0.07589298	-2.60990049	3.25659010	-0.22438608	eosinophil_s	Q2
rs15096617	19	45421204	C	T	-0.3217	-0.003241	-0.0132	9.738e-05	0.0132	-0.02137	0.007798	0.007831	0.007547	0.007674	0.007607	0.006551	1	-2.17508918	-0.82035518	0.32093770	-0.05923336	-0.25090009	0.05548550	eosinophil_s	Q1_low
rs1610009	9	113999400	G	A	-0.005148	-0.004598	-0.01454	0.1207	-0.01704	0.02325	0.002929	0.002939	0.002832	0.00288	0.002855	0.002458	1	0.30730694	0.35316005	-0.43258601	-0.08409077	0.53507249	0.02682026	eosinophil_s	Q2
rs16850073	4	74703999	C	T	-0.001608	-0.01707	-0.04323	-0.03126	0.05022	0.03887	0.002667	0.002676	0.00258	0.002623	0.0026	0.002239	1	0.33506757	0.76396929	1.24340623	-0.61629906	-0.45525752	-0.23663799	eosinophil_s	Q2
rs16939945	16	85945249	C	T	-0.001025	-0.01182	0.001202	-0.2024	0.04931	-0.006617	0.007646	0.007676	0.007399	0.007522	0.007457	0.006421	1	0.33963945	-0.43234822	1.22073074	0.20792815	-1.57058444	-0.12573710	eosinophil_s	Q2

(showing first 40 of 93 rows; full data in the accompanying CSV)

Supplementary Table 11 - r5_heterogeneity_cluster_enrichment

ST11_r5_heterogeneity_cluster_enrichment.csv

cluster	category	item	value	p	fdr	se	or	n_snps
0	component_mean_z	crp	0.07484236992809633					
0	component_mean_z	wbc	-0.30268980061458317					
0	component_mean_z	neutrophil_pct	-0.3355502795096762					
0	component_mean_z	lymphocyte_pct	0.1649230316947806					
0	component_mean_z	monocyte_pct	-0.04575466309370691					
0	component_mean_z	eosinophil_pct	0.825828694876237					
1	component_mean_z	crp	-0.041163303460452855					
1	component_mean_z	wbc	0.16647939033802073					
1	component_mean_z	neutrophil_pct	0.18455265373032187					
1	component_mean_z	lymphocyte_pct	-0.09070766743212928					
1	component_mean_z	monocyte_pct	0.025165064701538817					
1	component_mean_z	eosinophil_pct	-0.45420578218193036					
0	nearest_gene_top10	TOMM40	3.0					
0	nearest_gene_top10	RNY4P18	1.0					
0	nearest_gene_top10	C12orf43	1.0					
0	nearest_gene_top10	KLF3	1.0					
0	nearest_gene_top10	KLF3-AS1	1.0					
0	nearest_gene_top10	TGFBR3	1.0					
0	nearest_gene_top10	CTB-129P6.4	1.0					
0	nearest_gene_top10	GCKR	1.0					
0	nearest_gene_top10	CCDC26	1.0					
0	nearest_gene_top10	RP11-542M13.6	1.0					
1	nearest_gene_top10	TOMM40	7.0					
1	nearest_gene_top10	EPS15L1	3.0					
1	nearest_gene_top10	TGFBR3	2.0					
1	nearest_gene_top10	CTB-129P6.4	2.0					
1	nearest_gene_top10	NECTIN2	2.0					
1	nearest_gene_top10	IRF8	2.0					
1	nearest_gene_top10	RP11-419N10.5	1.0					
1	nearest_gene_top10	CTB-129P6.7	1.0					
1	nearest_gene_top10	LTBR	1.0					
1	nearest_gene_top10	THADA	1.0					
0	dice_eqtl_enrichment	B_CELL_NAIVE	2.0	0.6132320227691896	0.8127245447872012			
0	dice_eqtl_enrichment	CD4_NAIVE	3.0	0.662219999456238	0.8127245447872012			
0	dice_eqtl_enrichment	CD4_STIM	1.0	1.0	1.0			
0	dice_eqtl_enrichment	CD8_NAIVE	2.0	0.6132320227691896	0.8127245447872012			
0	dice_eqtl_enrichment	CD8_STIM	2.0	0.28617665644313606	0.8127245447872012			
0	dice_eqtl_enrichment	M2	2.0	0.28617665644313606	0.8127245447872012			
0	dice_eqtl_enrichment	MONOCYTES	2.0	0.28617665644313606	0.8127245447872012			
0	dice_eqtl_enrichment	NK	2.0	0.28617665644313606	0.8127245447872012			

(showing first 40 of 92 rows; full data in the accompanying CSV)

Supplementary Table 12 - r5_cluster_annotation_enrichment

ST12_r5_cluster_annotation_enrichment.csv

cluster	n_snps	mean_z_crp	mean_z_wbc	mean_z_neutrophil	mean_z_lymphocyte	mean_z_monocyte	mean_z_eosinophil	dominant_compon	top_genes	nsnps_ALLERG_GwAS	beta_ALLERG_GwAS	se_ALLERG_GwAS	p_ALLERG_GwAS	or_ALLERG_GwAS	nsnps_ALLERG_GwRe	beta_ALLERG_GwRe	se_ALLERG_GwRe	p_ALLERG_GwRe	or_ALLERG_GwRe	nsnps_ALLERG_RHo	beta_ALLERG_RHo	se_ALLERG_RHo	p_ALLERG_RHo	or_ALLERG_RHo	nsnps_ALLERG_RHtr	beta_ALLERG_RHtr	se_ALLERG_RHtr	p_ALLERG_RHtr	or_ALLERG_RHtr	nsnps_L12_ATOPIC	beta_L12_ATOPIC	se_L12_ATOPIC	p_L12_ATOPIC	or_L12_ATOPIC
0	44	0.37141626	-0.06112839	-0.06613776	0.335750470	-0.31460569	-0.12158149	crp(+); lymp	EPS15L1(4);	44	1.01718967	0.07439420	0.0	2.76541212	44	0.65330941	0.06836107	0.0	1.92189065	44	0.63329118	0.04982846	0.0	1.88380032	44	0.63329118	0.04982846	0.0	1.88380032	44	0.63329118	0.04982846	0.0	1.88380032
1	16	-2.01315218	-0.68971383	0.269561386	-0.02358431	-0.25497873	0.112102958	crp(-); wbc(-	TOMM40(10)	16	0.08297145	0.34729935	0.81117947	1.08651079	16	-0.89967320	0.31698443	0.00453651	0.40670254	16	1.44733081	0.23165519	4.16326306	4.25175063	16	1.44733081	0.23165519	4.16326306	4.25175063	16	1.44733081	0.23165519	4.16326306	4.25175063
2	33	0.48085209	0.41591123	-0.04251274	-0.43623247	0.543100310	0.107755703	monocyte_pc	TGFBR3(3); I	33	-3.43107431	0.40965710	0.0	0.03235216	33	-2.68032433	0.37746242	1.23945298	0.06854092	33	-0.30744343	0.27852627	0.26967035	0.73532446	33	-0.30744343	0.27852627	0.26967035	0.73532446	33	-0.30744343	0.27852627	0.26967035	0.73532446

Supplementary Table 13 - r5_snp_clusters

ST13_r5_snp_clusters.csv

rsid	cluster	crp_beta	wbc_beta	neutrophil_pct_beta	lymphocyte_pct_beta	monocyte_pct_beta	eosinophil_pct_beta
rs1002607	2	-0.0005239	-0.02108	0.01989	0.007752	-0.1058	0.003551
rs10437340	0	0.1324	0.00123	0.0007677	-0.002186	0.002207	-0.0009136
rs1081105	1	-0.3282	-0.01988	0.01307	-0.01344	0.0008059	-0.002148
rs10849448	2	-0.004738	0.003588	-0.003327	0.03335	-0.07971	-0.02719
rs11065394	0	0.1065	0.01188	-0.001281	0.0008694	-0.005701	0.009083
rs112019714	1	-0.3204	-0.01736	0.01302	-0.01311	-5.569e-05	-0.003199
rs112849259	1	-0.3217	-0.0169	0.01163	-0.01181	-0.0006855	0.0002972
rs113071828	0	-0.001152	-0.004768	0.005111	-0.02454	-0.002513	0.0894
rs113542380	0	0.01056	0.003098	0.102	-0.1039	-0.01976	-0.03879
rs114403972	0	0.1104	0.03215	0.03182	-0.02624	-0.01582	-0.02312
rs114425738	2	0.05175	0.04602	0.00672	-0.102	0.3457	-0.0003094
rs114536010	1	-0.3238	-0.01713	0.01157	-0.01176	-0.0002716	0.0002111
rs11466597	2	0.05342	0.04841	0.005054	-0.1098	0.3731	0.004585
rs115881343	1	-0.32	-0.01726	0.01284	-0.0129	4.082e-05	-0.003772
rs116359246	2	0.04006	0.03334	0.009291	-0.06853	0.2156	-0.01077
rs116881820	1	-0.3215	-0.01687	0.01125	-0.01129	-0.0003994	-0.0001532
rs117310449	1	-0.3693	-0.01326	0.0009031	-0.00759	0.01042	0.02184
rs117520868	0	-0.004778	0.04164	-0.07534	0.1019	-0.03855	-0.02118
rs117866384	0	-0.004745	0.04178	-0.07516	0.1017	-0.03854	-0.02129
rs118051449	0	-0.004168	0.04228	-0.07415	0.101	-0.03951	-0.02219
rs118170342	1	-0.2144	-0.01512	0.01553	-0.01482	-0.004628	-0.001377
rs11864884	2	-0.002108	0.0006576	0.02185	-0.004497	-0.06845	-0.005487
rs12515180	0	0.006039	-0.02673	0.00516	0.01047	0.003008	-0.0748
rs1260326	0	-0.0851	-0.03276	-0.02235	0.005805	0.05027	0.0191
rs12619285	0	-0.003383	-0.00685	0.008612	0.005682	-0.0006747	-0.07749
rs12972156	1	-0.1973	-0.008993	0.001468	0.003378	-0.01268	-0.006393
rs13277237	2	0.004256	-0.03236	-0.02376	0.04566	-0.09232	0.03298
rs139795227	2	0.07748	0.06408	0.0002698	-0.1471	0.5365	-0.0163
rs1401884	2	0.0002157	0.00281	-0.05082	0.0127	0.1303	0.03918
rs141094656	2	0.08537	0.06197	0.001553	-0.1497	0.5308	-0.009281
rs144937240	2	-0.0008097	0.07016	-0.03468	-0.03131	0.2342	0.000706
rs145096717	2	0.005747	0.1116	-0.05194	-0.05877	0.4051	-0.01097
rs147636938	1	-0.3237	-0.01735	0.01523	-0.01561	0.000362	-0.001005
rs147775088	0	-0.004703	0.04186	-0.07508	0.1017	-0.03861	-0.02155
rs149007767	2	0.001637	0.009736	0.04726	-0.02459	-0.09546	-0.02023
rs150649461	2	0.0786	0.06564	0.003366	-0.1507	0.5383	-0.01649
rs150966173	1	-0.3217	-0.02137	0.0132	-0.0132	9.738e-05	-0.003241
rs1610009	2	-0.005148	0.02325	-0.01704	-0.01454	0.1207	-0.004598
rs16850073	0	-0.001608	0.03887	0.05022	-0.04323	-0.03126	-0.01707
rs16939945	2	-0.001025	-0.006617	0.04931	0.001202	-0.2024	-0.01182

(showing first 40 of 93 rows; full data in the accompanying CSV)

Supplementary Table 14 - r5_nhanes_weighted_results

ST14_r5_nhanes_weighted_results.csv

analysis	outcome	exposure	model	beta	se	p	n	n_events	effect	ci_lower	ci_upper	effect_scale	weight	note
main_logistic	allergy_composite	l_z (per SD)	unweighted	0.300493340175733	0.032498345953473	2.321386321184097	7021	1092.0	1.350524911451739	1.267183323794352	1.439347805643728	or	none	
main_logistic	allergy_composite	l_z (per SD)	weighted	0.351013222092375	0.050449324729286	3.457776269560892	7021	1092.0	1.420506107851005	1.286766581260296	1.568145794139044	or	WTMEC2YR	
main_logistic	allergy_composite	l (per unit, R4 scale)	unweighted	0.485184983453456	0.052472741773509	2.321386321183063	7021	1092.0	1.624475482136820	1.465708010843730	1.800440860348828	or	none	
main_logistic	allergy_composite	l (per unit, R4 scale)	weighted	0.566755803151031	0.081456896081965	3.457776269562833	7021	1092.0	1.762539740322908	1.502454774533899	2.067647152428449	or	WTMEC2YR	
main_logistic	allergy_composite	l_z (per SD)	weighted_strataFE	0.358390699603224	0.050130883844079	8.734870969432758	7021	1092.0	1.431024612022021	1.297104103906808	1.578771845717562	or	WTMEC2YR	SDMVSTRA fixed effects a
ige_validation	total_lgE_continuous	l_z (per SD)	unweighted	0.305338137681337	0.016213125783983	4.068765243134736	7854		0.305338137681337	0.273560411144730	0.337115864217944	beta	none	
ige_validation	total_lgE_continuous	l_z (per SD)	weighted	0.309753984957546	0.024058426899381	6.218085231425764	7854		0.309753984957546	0.262599468234758	0.356908501680333	beta	WTMEC2YR	
ige_validation	high_total_lgE_gt100	l_z (per SD)	unweighted	0.410430230202944	0.026681341207481	2.139447917019810	7854		1.507466203108844	1.430658462297472	1.588397520024517	or	none	
ige_validation	high_total_lgE_gt100	l_z (per SD)	weighted	0.427007599426146	0.035827213735181	9.476971055745822	7854		1.532664309048985	1.428730461089111	1.644158886653773	or	WTMEC2YR	
ige_validation	any_specific_lgE_posi	l_z (per SD)	unweighted	0.354234373402286	0.025673787320405	2.638032206542825	7869		1.425089150334962	1.355152169519009	1.498635453702041	or	none	
ige_validation	any_specific_lgE_posi	l_z (per SD)	weighted	0.328583544846272	0.027389091895497	3.689606296796308	7869		1.388999279212064	1.316400245698251	1.465602125156302	or	WTMEC2YR	
ige_validation	self_reported_allergy_	l_z (per SD)	unweighted	0.237237572573222	0.036345429236792	6.696519256982458	3747	1282.0	1.267742262793523	1.180573728925461	1.361346949788271	or	none	
ige_validation	self_reported_allergy_	l_z (per SD)	weighted	0.309461290666990	0.062113795529538	6.287619460129222	3747	1282.0	1.362690822101078	1.206493539627420	1.539110004030318	or	WTMEC2YR	
mortality_cox	all_cause_mortality	l_z	unweighted	0.201732264632388	0.119805153762326	0.092213313293697	4981	54.0	1.223520384575166	0.967455512795822	1.547360174882686	hr	none	
mortality_cox	all_cause_mortality	age	unweighted	0.076172399664333	0.011569138895701	4.576404496704487	4981	54.0	1.079148602978089	1.054953708143909	1.103898396981321	hr	none	
mortality_cox	all_cause_mortality	sex_female	unweighted	-0.218134217280837	0.276675659194799	0.430456160823372	4981	54.0	0.804017521378872	0.467470745209854	1.382854823127013	hr	none	
mortality_cox	all_cause_mortality	bmi	unweighted	-0.030545738037338	0.023853729682988	0.200354062996662	4981	54.0	0.969916068995299	0.925613009294527	1.016339627305253	hr	none	
mortality_cox	all_cause_mortality	l_z	weighted	0.069218283305390	0.133803592536271	0.604937965187784	4981	54.0	1.071670111313134	0.824451668333528	1.393018986550578	hr	WTMEC2YR	
mortality_cox	all_cause_mortality	age	weighted	0.102159246838376	0.022920826787590	8.309563617912403	4981	54.0	1.107559833086798	1.058904016288701	1.158451346862027	hr	WTMEC2YR	
mortality_cox	all_cause_mortality	sex_female	weighted	-0.038096967304428	0.366157377927967	0.917133371097167	4981	54.0	0.962619593738267	0.469650993075208	1.973032093856207	hr	WTMEC2YR	
mortality_cox	all_cause_mortality	bmi	weighted	0.029190160608748	0.043455680292226	0.501760339443378	4981	54.0	1.029620369096397	0.945555235945465	1.121159361354690	hr	WTMEC2YR	
threshold_hinge	allergy_composite	l_z hinge knot	weighted	-1.75	1.007653061224489		7021	1092.0	-1.75	-3.725	0.225000000000000	beta	WTMEC2YR	knot on l_z scale; boots

Supplementary Table 15 - r5_nhanes_weighted_vs_unweighted

ST15_r5_nhanes_weighted_vs_unweighted.csv

model	predictor	n	beta	se	p	or_	or_lower	or_upper
unweighted|l_z (per SD)	l_z	7021	0.300493340175733	0.032498345953473576	2.321386321184097e-20	1.350524911451739	1.267183323794352	1.4393478056437288
unweighted|l_z (per SD)	age	7021	-0.011120180051377408	0.0015913896870298422	2.793961356043248e-12	0.9889414206025838	0.985861595568761	0.9920308669892229
unweighted|l_z (per SD)	sex_female	7021	0.10549821977218848	0.06723278469082565	0.11661335303573511	1.1112641262177878	0.9740642248700883	1.267789049929725
unweighted|l_z (per SD)	bmi	7021	0.02862697909808704	0.004138414129414575	4.600759655729551e-12	1.0290406691949772	1.020727580683972	1.0374214618044115
weighted|l_z (per SD)	l_z	7021	0.35101322209237584	0.050449324729286885	3.4577762695608926e-12	1.420506107851005	1.2867665812602969	1.5681457941390444
weighted|l_z (per SD)	age	7021	-0.011887628518819004	0.0021463634114457487	3.050774956404705e-08	0.9881827501824151	0.9840343236252567	0.9923486654007787
weighted|l_z (per SD)	sex_female	7021	0.26296595262603284	0.1052050435638247	0.012434886157649145	1.3007824299719237	1.0584060682715875	1.5986632927065623
weighted|l_z (per SD)	bmi	7021	0.02349747527814673	0.00621003142863049	0.00015445611635447266	1.0237757159944434	1.01139019352759	1.0363129120366967
unweighted|l (per unit,	l	7021	0.4851849834534569	0.05247274177350993	2.3213863211830637e-20	1.6244754821368206	1.4657080108437304	1.8004408603488284
unweighted|l (per unit,	age	7021	-0.011120180051380395	0.0015913896870298886	2.793961356009938e-12	0.9889414206025808	0.9858615955687579	0.9920308669892202
unweighted|l (per unit,	sex_female	7021	0.10549821977218815	0.06723278469082526	0.11661335303573418	1.1112641262177874	0.9740642248700886	1.2677890499297237
unweighted|l (per unit,	bmi	7021	0.028626979098091684	0.004138414129414566	4.6007596556926025e-12	1.029040669194982	1.0207275806839768	1.0374214618044162
weighted|l (per unit, R4	l	7021	0.5667558031510318	0.08145689608196546	3.4577762695628333e-12	1.762539740322908	1.502454774533899	2.0676471524284494
weighted|l (per unit, R4	age	7021	-0.011887628518820182	0.002146363411445811	3.050774956397947e-08	0.988182750182414	0.9840343236252554	0.9923486654007777
weighted|l (per unit, R4	sex_female	7021	0.2629659526260328	0.10520504356382576	0.012434886157650047	1.3007824299719235	1.0584060682715852	1.5986632927065654
weighted|l (per unit, R4	bmi	7021	0.023497475278148407	0.0062100314286305325	0.00015445611635432141	1.023775715994445	1.0113901935275915	1.0363129120366985
weighted_strataFE|l_z (p	l_z	7021	0.3583906996032241	0.05013088384407967	8.734870969432758e-13	1.4310246120220214	1.2971041039068085	1.5787718457175623
weighted_strataFE|l_z (p	age	7021	-0.011848201188247228	0.002152439194239111	3.7012019952177394e-08	0.9882217123584541	0.9840614034379578	0.9923996097853724
weighted_strataFE|l_z (p	sex_female	7021	0.2727316838322891	0.10513216015651884	0.009481699824041389	1.3135477514382454	1.0689455015057399	1.6141212932539817
weighted_strataFE|l_z (p	bmi	7021	0.023397962477642156	0.006211630242881479	0.00016535403302026284	1.0236738422748097	1.0112863832158945	1.0362130379184187

Supplementary Table 16 - r5_age_interaction

ST16_r5_age_interaction.csv

model	group	predictor	n	beta	se	or	or_lower	or_upper	p
unweighted		l_z	14878	0.2302546760897763	0.020996024510620027	1.2589205860784065	1.2081647444246189	1.3118087159601635	5.532292448476026e-28
unweighted		age_c	14878	-0.001201413334977347	0.000989229635016575	0.9987993080730914	0.99686462217587	1.0007377487524944	0.22455918646146011
unweighted		l_z:age_c	14878	-0.0013944977937510518	0.0008541895016727033	0.9986064740664925	0.9969359944584655	1.0002797527530325	0.10256596035420314
unweighted		sex_female	14878	0.042298235757660695	0.04155020690485863	1.0432055535603038	0.9616159102126193	1.1317177840146537	0.3086765249187793
unweighted		bmi	14878	0.0821944986645252	0.0029039489417625026	1.0856669500675435	1.0795051684057109	1.0918639029859476	3.0517744352211987e-176
unweighted		cycle_D	14878	1.058486500079604	0.04431599419471427	2.8820057712173313	2.642240288561787	3.1435283540586187	4.387999177062446e-126
weighted		l_z	14878	0.2668958788487068	0.03851197378318662	1.3059044669750333	1.2109587608997259	1.4082944291168653	4.2023077831370565e-12
weighted		age_c	14878	-0.003022094681857345	0.001730255677816448	0.996982467249589	0.993607126117347	1.0003692746117518	0.0807037260559217
weighted		l_z:age_c	14878	-0.0009581009287722066	0.0011244706852774045	0.9990423579033751	0.9968429305855553	1.0012466380223515	0.3941884729571111
weighted		sex_female	14878	0.18845668750757674	0.060761493547509635	1.2073847870096495	1.0718264545753662	1.3600877433836762	0.001924902706965574
weighted		bmi	14878	0.08751123759328612	0.0107908040483275	1.0914545296961307	1.0686126962370812	1.1147846124129404	5.070411745825354e-16
weighted		cycle_D	14878	1.3410892924628426	0.08569697217320915	3.8232058272533593	3.2320713838537376	4.52245667300692	3.364074831000522e-55

Supplementary Table 17 - r5_age_interaction_results

ST17_r5_age_interaction_results.csv

analysis	group	predictor	model	beta	se	p	n	n_events	effect	ci_lower	ci_upper	effect_scale	weight	note
age_interaction_poole	all	l_z	unweighted	0.230254676089776	0.020996024510620	5.532292448476026	14878		1.258920586078406	1.208164744424618	1.311808715960163	or	none	
age_interaction_poole	all	age_c	unweighted	-0.001201413334977	0.000989229635016	0.224559186461460	14878		0.998799308073091	0.99686462217587	1.000737748752494	or	none	
age_interaction_poole	all	l_z:age_c	unweighted	-0.001394497793751	0.000854189501672	0.102565960354203	14878		0.998606474066492	0.996935994458465	1.000279752753032	or	none	
age_interaction_poole	all	sex_female	unweighted	0.042298235757660	0.041550206904858	0.308676524918779	14878		1.043205553560303	0.961615910212619	1.131717784014653	or	none	
age_interaction_poole	all	bmi	unweighted	0.082194498664525	0.002903948941762	3.051774435221198	14878		1.085666950067543	1.079505168405710	1.091863902985947	or	none	
age_interaction_poole	all	cycle_D	unweighted	1.058486500079604	0.044315994194714	4.387999177062446	14878		2.882005771217331	2.642240288561787	3.143528354058618	or	none	
age_interaction_poole	all	l_z	weighted	0.266895878848706	0.038511973783186	4.202307783137056	14878		1.305904466975033	1.210958760899725	1.408294429116865	or	WTMEC4YR	
age_interaction_poole	all	age_c	weighted	-0.003022094681857	0.001730255677816	0.080703726055921	14878		0.996982467249589	0.993607126117347	1.000369274611751	or	WTMEC4YR	
age_interaction_poole	all	l_z:age_c	weighted	-0.000958100928772	0.001124470685277	0.394188472957111	14878		0.999042357903375	0.996842930585555	1.001246638022351	or	WTMEC4YR	
age_interaction_poole	all	sex_female	weighted	0.188456687507576	0.060761493547509	0.001924902706965	14878		1.207384787009649	1.071826454575366	1.360087743383676	or	WTMEC4YR	
age_interaction_poole	all	bmi	weighted	0.087511237593286	0.010790804048327	5.070411745825354	14878		1.091454529696130	1.068612696237081	1.114784612412940	or	WTMEC4YR	
age_interaction_poole	all	cycle_D	weighted	1.341089292462842	0.085696972173209	3.364074831000522	14878		3.823205827253359	3.232071383853737	4.52245667300692	or	WTMEC4YR	
age_stratified	<40	l_z	unweighted	0.265429747039065	0.026662827830418	2.397057125654843	8803	1832.0	1.303991241758536	1.237595772631364	1.373948744966711	or	none	scheme=broad
age_stratified	<40	l_z	weighted	0.288797469081978	0.038320004164977	4.828729439037965	8803	1832.0	1.334821358555150	1.238239064359724	1.438937044177599	or	WTMEC4YR	scheme=broad
age_stratified	40-60	l_z	unweighted	0.230681518481090	0.050844229108005	5.705345234780984	2882	831.0	1.259458061452319	1.139998366407386	1.391435861049631	or	none	scheme=broad
age_stratified	40-60	l_z	weighted	0.263897971729704	0.081158066321298	0.001147354973320	2882	831.0	1.301995349190518	1.110519764174819	1.526485114448516	or	WTMEC4YR	scheme=broad
age_stratified	>60	l_z	unweighted	0.229495620758403	0.048366669065304	2.085963117350597	3193	796.0	1.257965358277010	1.144189984332747	1.383054269215487	or	none	scheme=broad
age_stratified	>60	l_z	weighted	0.258162777280681	0.080877525582880	0.001412809565657	3193	796.0	1.294549524726032	1.104776251502391	1.516921159093876	or	WTMEC4YR	scheme=broad
age_stratified	0-10	l_z	unweighted	0.243246946657009	0.051357746655861	2.176291775422983	2138	294.0	1.275383536934376	1.153251983309998	1.410449051745529	or	none	scheme=decade
age_stratified	0-10	l_z	weighted	0.252568725596732	0.092877591861138	0.006540599785772	2138	294.0	1.287327965497325	1.073075307821655	1.544358796323091	or	WTMEC4YR	scheme=decade
age_stratified	10-20	l_z	unweighted	0.302614980697946	0.041201465951887	2.061248297800757	3539	747.0	1.353393281579318	1.248396724224868	1.467220587078458	or	none	scheme=decade
age_stratified	10-20	l_z	weighted	0.318225027294221	0.066548523893258	1.736896794139635	3539	747.0	1.374685568198817	1.206580010952965	1.566212264631799	or	WTMEC4YR	scheme=decade
age_stratified	20-30	l_z	unweighted	0.122460309968176	0.065331933281187	0.060870814443733	1629	418.0	1.130274258562448	0.994425317271639	1.284681591850423	or	none	scheme=decade
age_stratified	20-30	l_z	weighted	0.182731473992869	0.071963483145352	0.011109835484511	1629	418.0	1.200492001471741	1.042563982624850	1.382343021259168	or	WTMEC4YR	scheme=decade
age_stratified	30-40	l_z	unweighted	0.290635574379615	0.079986146031741	0.000279522314258	1497	373.0	1.337277157082413	1.143235874777781	1.564253042008525	or	none	scheme=decade
age_stratified	30-40	l_z	weighted	0.367783460273139	0.119828071812298	0.002145900302510	1497	373.0	1.444529207144884	1.142159123988978	1.826947389788364	or	WTMEC4YR	scheme=decade
age_stratified	40-50	l_z	unweighted	0.208809380618840	0.078129336051989	0.007526330444975	1469	428.0	1.232210093016805	1.057254952790658	1.436116907586739	or	none	scheme=decade
age_stratified	40-50	l_z	weighted	0.243871841507605	0.105200273322456	0.020440226353726	1469	428.0	1.276180766605557	1.038398171200505	1.568413152318113	or	WTMEC4YR	scheme=decade
age_stratified	50-60	l_z	unweighted	0.240599686094519	0.067054021549638	0.000333036515219	1413	403.0	1.272011729390700	1.115356186703639	1.450670072032731	or	none	scheme=decade
age_stratified	50-60	l_z	weighted	0.283949255832168	0.122163721563310	0.020107615574724	1413	403.0	1.328365522239512	1.045513718808045	1.687739652700457	or	WTMEC4YR	scheme=decade
age_stratified	60-70	l_z	unweighted	0.272806951902916	0.065432809059281	3.055787794601694	1598	452.0	1.313646623364075	1.15552946032692	1.493399787996418	or	none	scheme=decade
age_stratified	60-70	l_z	weighted	0.327745304073226	0.128152458766733	0.010543846090474	1598	452.0	1.387835451234675	1.079573963307889	1.784117906847335	or	WTMEC4YR	scheme=decade
age_stratified	70+	l_z	unweighted	0.178783938835844	0.072406795716336	0.013543077910947	1595	344.0	1.195762358448109	1.037554621776	1.378093825493148	or	none	scheme=decade
age_stratified	70+	l_z	weighted	0.182062261936182	0.098493404260650	0.064534147128392	1595	344.0	1.199688886507481	0.989075302304948	1.455150503763983	or	WTMEC4YR	scheme=decade

Supplementary Table 17b - r21_age_interaction_power

ST17b_age_interaction_power.csv

model	interaction	se	z_alpha	z_power	contrast_years	mde_or_ratio	power	alpha	n	note
weighted	l_z:age_c	0.0011244706852774	1.96	0.84	40	1.1342149263440644	0.8	0.05	14878	Minimum detectable 40-ye
unweighted	l_z:age_c	0.0008541895016727	1.96	0.84	40	1.1003950197318837	0.8	0.05	14878	Minimum detectable 40-ye

Supplementary Table 18 - r5_age_stratified_or

ST18_r5_age_stratified_or.csv

model	group	predictor	n	beta	se	or	or_lower	or_upper	p	scheme	n_events
unweighted	<40	l_z	8803	0.26542974703906597	0.026662827830418696	1.3039912417585369	1.2375957726313649	1.3739487449667112	2.3970571256548433e-23	broad	1832
weighted	<40	l_z	8803	0.2887974690819786	0.0383200041649776	1.3348213585551505	1.2382390643597243	1.438937044177599	4.828729439037965e-14	broad	1832
unweighted	40-60	l_z	2882	0.2306815184810907	0.05084422910800526	1.2594580614523196	1.1399983664073863	1.3914358610496316	5.7053452347809845e-06	broad	831
weighted	40-60	l_z	2882	0.2638979717297044	0.08115806632129889	1.301995349190518	1.1105197641748197	1.5264851144485168	0.001147354973320446	broad	831
unweighted	>60	l_z	3193	0.22949562075840363	0.048366669065304596	1.2579653582770105	1.1441899843327474	1.383054269215487	2.085963117350597e-06	broad	796
weighted	>60	l_z	3193	0.2581627772806819	0.08087752558288006	1.2945495247260324	1.1047762515023918	1.5169211590938765	0.0014128095656578484	broad	796
unweighted	0-10	l_z	2138	0.2432469466570091	0.051357746655861305	1.275383536934376	1.1532519833099988	1.4104490517455295	2.1762917754229834e-06	decade	294
weighted	0-10	l_z	2138	0.25256872559673244	0.0928775918611388	1.2873279654973253	1.0730753078216557	1.544358796323091	0.006540599785772423	decade	294
unweighted	10-20	l_z	3539	0.3026149806979467	0.041201465951887874	1.3533932815793184	1.2483967242248684	1.4672205870784587	2.0612482978007571e-13	decade	747
weighted	10-20	l_z	3539	0.318225027294221	0.06654852389325816	1.374685568198817	1.206580010952965	1.566212264631799	1.7368967941396356e-06	decade	747
unweighted	20-30	l_z	1629	0.12246030996817686	0.06533193328118744	1.1302742585624483	0.9944253172716396	1.2846815918504233	0.06087081444373359	decade	418
weighted	20-30	l_z	1629	0.1827314739928698	0.07196348314535224	1.2004920014717415	1.0425639826248507	1.3823430212591687	0.011109835484511072	decade	418
unweighted	30-40	l_z	1497	0.290635574379615	0.07998614603174183	1.3372771570824138	1.1432358747777818	1.5642530420085254	0.0002795223142584787	decade	373
weighted	30-40	l_z	1497	0.3677834602731392	0.11982807181229888	1.4445292071448845	1.1421591239889783	1.826947389788364	0.002145900302510187	decade	373
unweighted	40-50	l_z	1469	0.20880938061884083	0.0781293360519891	1.2322100930168054	1.0572549527906583	1.4361169075867393	0.007526330444975524	decade	428
weighted	40-50	l_z	1469	0.24387184150760585	0.10520027332245606	1.2761807666055576	1.0383981712005055	1.5684131523181133	0.020440226353726883	decade	428
unweighted	50-60	l_z	1413	0.2405996860945195	0.06705402154963812	1.2720117293907007	1.1153561867036392	1.4506700720327315	0.0003330365152194565	decade	403
weighted	50-60	l_z	1413	0.28394925583216846	0.12216372156331048	1.3283655222395125	1.045513718808045	1.687739652700457	0.020107615574724572	decade	403
unweighted	60-70	l_z	1598	0.2728069519029163	0.06543280905928134	1.3136466233640758	1.15552946032692	1.4933997879964187	3.0557877946016944e-05	decade	452
weighted	60-70	l_z	1598	0.32774530407322605	0.12815245876673356	1.387835451234675	1.0795739633078898	1.7841179068473356	0.010543846090474662	decade	452
unweighted	70+	l_z	1595	0.1787839388358445	0.07240679571633621	1.1957623584481094	1.037554621776	1.3780938254931485	0.01354307791094798	decade	344
weighted	70+	l_z	1595	0.18206226193618286	0.09849340426065016	1.1996888865074813	0.9890753023049488	1.4551505037639836	0.06453414712839214	decade	344

Supplementary Table 19 - r5_ige_validation_weighted

ST19_r5_ige_validation_weighted.csv

model	predictor	n	beta	se	p	analysis	weight	or_	or_lower	or_upper
total_lgE_continuous|unw	l_z	7854	0.30533813768133755	0.016213125783983335	4.06876524313473e-79	total_lgE_continuous	none			
total_lgE_continuous|wei	l_z	7854	0.30975398495754614	0.024058426899381518	6.218085231425764e-38	total_lgE_continuous	WTMEC2YR			
high_total_lgE_gt100|unw	l_z	7854	0.41043023020294434	0.0266813412074818	2.1394479170198107e-53	high_total_lgE_gt100	none	1.5074662031088442	1.4306584622974727	1.588397520024517
high_total_lgE_gt100|wei	l_z	7854	0.42700759942614624	0.035827213735181086	9.476971055745822e-33	high_total_lgE_gt100	WTMEC2YR	1.5326643090489858	1.4287304610891112	1.6441588866537733
any_specific_lgE_positiv	l_z	7869	0.3542343734022866	0.025673787320405136	2.6380322065428256e-43	any_specific_lgE_positiv	none	1.4250891503349628	1.3551521695190096	1.498635453702041
any_specific_lgE_positiv	l_z	7869	0.3285835448462727	0.027238091895497108	3.6896062967963084e-33	any_specific_lgE_positiv	WTMEC2YR	1.3889992792120647	1.316400245698251	1.4656021251563023
self_reported_allergy_in	l_z	3747	0.2372375725732224	0.03634542923679282	6.696519256982458e-11	self_reported_allergy_in	none	1.2677422627935235	1.1805737289254619	1.3613469497882713
self_reported_allergy_in	l_z	3747	0.30946129066699024	0.06211379552953862	6.287619460129222e-07	self_reported_allergy_in	WTMEC2YR	1.362690822101078	1.2064935396274208	1.539110004030318

Supplementary Table 20 - r5_threshold_weighted_bootstrap

ST20_r5_threshold_weighted_bootstrap.csv

model	knot_hat	loglik	ci_lower	ci_upper	boot_median	boot_mean	n_boot_ok	n_boot_fail	hinge_beta	hinge_p	n	n_events
weighted	-1.75	-115234585.98341203	-1.95	2.0	-1.35	-1.063	200	0	1.7338326964095223	8.158718109317754e-0	7021	1092
unweighted	-1.75	-2951.318085388518					0	0			7021	1092

Supplementary Table 21 - r5_mortality_cox_weighted

ST21_r5_mortality_cox_weighted.csv

model	predictor	n	n_events	beta	se	hr	hr_lower	hr_upper	p
unweighted	l_z	4981	54	0.20173226463238825	0.11980515376232645	1.2235203845751665	0.9674555127958225	1.5473601748826866	0.09221331329369735
unweighted	age	4981	54	0.07617239966433378	0.011569138895701016	1.0791486029780897	1.0549537081439093	1.1038983969813219	4.5764044967044873e-11
unweighted	sex_female	4981	54	-0.21813421728083782	0.2766756591947994	0.8040175213788725	0.467470745209854	1.382854823127013	0.4304561608233721
unweighted	bmi	4981	54	-0.030545738037338704	0.023853729682988357	0.9699160689952993	0.9256130092945276	1.0163396273052532	0.2003540629966625
weighted	l_z	4981	54	0.06921828330539037	0.13380359253627128	1.0716701113131342	0.824451668333528	1.3930189865505789	0.6049379651877848
weighted	age	4981	54	0.1021592468383764	0.022920826787590446	1.107559833086798	1.0589040162887013	1.1584513468620272	8.309563617912403e-06
weighted	sex_female	4981	54	-0.038096967304428235	0.3661573779279672	0.9626195937382672	0.4696509930752088	1.9730320938562078	0.9171333710971671
weighted	bmi	4981	54	0.02919016060874849	0.04345568029222617	1.0296203690963974	0.9455552359454656	1.1211593613546904	0.5017603394433789

Supplementary Table 22 - r5_il4_locus_coloc

ST22_r5_il4_locus_coloc.csv

trait1	trait2	n_snps	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4	max_snp_pph4	lead_coloc_snp
IL4_eQTLGen	ALLERG_ASTHMA	175	5.111770592484172e-58	1.2067971612131893e-08	4.231539202521209e-50	0.998989340107187	0.0010106478248413648	0.9999999998519917	rs3756752
IL4_eQTLGen	ALLERG_RHINITIS	175	1.477475235566224e-53	0.0003488053479266227	4.1080279887658338e-50	0.9698017236273498	0.029849471024723528	0.9999955724274586	rs803140
IL4_eQTLGen	L12_ATOPIC	175	1.1074011225819127e-76	2.6143750132704 91e-27	4.2349020195019095e-50	0.999784035438487	0.00021596456151299045	1.0	rs2070874
IL13_DICE_TH1	ALLERG_ASTHMA	6	0.9800197947984679	0.01700421143795713	0.0012595963434353782	2.0158873629726557e-05	0.0016962385465099869	0.0008392764459674579	rs78330488
IL13_DICE_TH1	ALLERG_RHINITIS	6	0.982206147810373	0.017042146599154658	0.00022043000075785006	3.296676833921213e-06	0.0005279789128806393	0.00012994437441124517	rs7725926
IL13_DICE_TH1	L12_ATOPIC	6	0.9733722706981011	0.016888871007139813	0.0035351164241690454	5.518884797010875e-05	0.006148553022620023	0.0019867875045476656	rs78330488
IL4_DICE_TREG_NAIVE	ALLERG_ASTHMA	42	6.413399028834831e-06	2.3326411991652332e-06	0.15091101903617346	0.05409342085073779	0.7949868140728608	0.9411406625346809	rs370314650
IL4_DICE_TREG_NAIVE	ALLERG_RHINITIS	42	0.007313746056900257	0.002660109763302423	0.14973973844675778	0.05367578622851934	0.7866106195045202	0.8334832503448446	rs187681
IL4_DICE_TREG_NAIVE	L12_ATOPIC	42	9.905528460807366e-29	3.602776574995191e-29	0.05849626300796483	0.02035474425452453	0.9211489927375106	0.9543574488768719	rs370314650

Supplementary Table 23 - r5_il4_locus_coloc_IL4_eQTLGen_ALLERG_ASTHMA_persnp

ST23_r5_il4_locus_coloc_IL4_eQTLGen_ALLERG_ASTHMA_persnp.csv

trait1	trait2	snp	pos37	pos38	beta_x	se_x	p_x	beta_y	se_y	p_y	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4
IL4_eQTLGen	ALLERG_ASTHMA	rs2070874	132009710	132674018	-0.13985918132477	0.00884568852854	2.6129e-56	0.0753801	0.0132661	1.32993e-08	0.0	1.25424943418588	0.0	0.0	0.9999874575056581
IL4_eQTLGen	ALLERG_ASTHMA	rs734244	132010726	132675034	-0.14273284580923	0.00903824354007	3.52080000000000	0.074907	0.0135428	3.18171e-08	0.0	2.86107863936124	0.0	0.0	0.9999713892136064
IL4_eQTLGen	ALLERG_ASTHMA	rs2243266	132013789	132678097	-0.13935994880167	0.00884089733629	5.5738e-56	0.0762468	0.0132651	9.03441e-09	0.0	8.63866409747628	0.0	0.0	0.9999913613359025
IL4_eQTLGen	ALLERG_ASTHMA	rs2243267	132013886	132678194	-0.13915115743881	0.00884111273445	8.1467e-56	0.0760255	0.0132648	9.96208e-09	0.0	9.49282660145085	0.0	0.0	0.9999905071733985
IL4_eQTLGen	ALLERG_ASTHMA	rs2243290	132018169	132682477	-0.13895320119135	0.00883785665074	1.05890000000000	0.0758617	0.0132667	1.07656e-08	0.0	1.02289915426250	0.0	0.0	0.9999897710084574
IL4_eQTLGen	ALLERG_ASTHMA	rs2243268	132013963	132678271	-0.13873914505525	0.00884528278782	1.9096e-55	0.0766543	0.0132692	7.60992e-09	0.0	7.31982006928023	0.0	0.0	0.9999926801799307
IL4_eQTLGen	ALLERG_ASTHMA	rs2243289	132018132	132682440	-0.13867773764303	0.00884249527472	1.9707e-55	0.0766239	0.0132708	7.74604e-09	0.0	7.44554124363969	0.0	0.0	0.9999925544587563
IL4_eQTLGen	ALLERG_ASTHMA	rs2243282	132016554	132680862	-0.13857397982739	0.00883954300214	2.1865e-55	0.0756726	0.0132651	1.1659e-08	0.0	1.10473755941680	0.0	0.0	0.9999889526244038
IL4_eQTLGen	ALLERG_ASTHMA	rs2243288	132017944	132682252	-0.13662789645728	0.00879580620069	2.0615e-54	0.0747899	0.0132091	1.49603e-08	0.0	1.40947976650613	0.0	0.0	0.9999859052023349
IL4_eQTLGen	ALLERG_ASTHMA	rs2243291	132018983	132683291	-0.13652911737273	0.00879680917074	2.52470000000000	0.0744352	0.0132115	1.75971e-08	0.0	1.64759717297955	0.0	0.0	0.9999835240282702
IL4_eQTLGen	ALLERG_ASTHMA	rs2243250	132009154	132673462	-0.13754862443102	0.00889556894901	6.2043e-54	0.0751868	0.0132625	1.43529e-08	0.0	1.35003777255400	0.0	0.0	0.9999864996222745
IL4_eQTLGen	ALLERG_ASTHMA	rs2243270	132014109	132678417	-0.13544689954455	0.00879879558942	1.7958e-53	0.0748234	0.0132117	1.48395e-08	0.0	1.39806172480410	0.0	0.0	0.9999860193827519
IL4_eQTLGen	ALLERG_ASTHMA	rs74863864	132023259	132687567	-0.13625277989886	0.00886024059688	2.2923e-53	0.0763814	0.0132723	8.66662e-09	0.0	8.29595369732533	0.0	0.0	0.9999917040463027
IL4_eQTLGen	ALLERG_ASTHMA	rs2299007	132043032	132707340	-0.13623398609206	0.00887130608086	3.1893e-53	0.078389	0.013309	3.86376e-09	0.0	3.79615883492645	0.0	0.0	0.9999962038411615
IL4_eQTLGen	ALLERG_ASTHMA	rs2299010	132042740	132707048	-0.13598313008212	0.00887131925589	4.9249e-53	0.0783993	0.013309	3.84521e-09	0.0	3.77902772820111	0.0	0.0	0.9999962209722718
IL4_eQTLGen	ALLERG_ASTHMA	rs10455052	132034588	132698896	-0.13701388301260	0.00894136384482	5.3188e-53	0.0781874	0.0133094	4.23838e-09	0.0	4.15122922152225	0.0	0.0	0.9999958487707785
IL4_eQTLGen	ALLERG_ASTHMA	rs11242123	132025487	132689795	-0.13568964708945	0.00887336004194	8.6454e-53	0.0746856	0.0133119	2.01804e-08	0.0	1.86967609452478	0.0	0.0	0.9999813032390548
IL4_eQTLGen	ALLERG_ASTHMA	rs2243274	132014832	132679140	-0.15298723753240	0.01000786549957	9.36399999999999	0.074712	0.0132116	1.55822e-08	0.0	1.46550839960140	0.0	0.0	0.9999853449160044
IL4_eQTLGen	ALLERG_ASTHMA	rs2299005	132057589	132721897	-0.13443122685738	0.00887334830741	7.5767e-52	0.0773793	0.0132875	5.76381e-09	0.0	5.59209761919863	0.0	0.0	0.9999944079023808
IL4_eQTLGen	ALLERG_ASTHMA	rs9784675	132069739	132734047	-0.13282091155285	0.00881160929537	2.4158e-51	0.0752934	0.0132299	1.26186e-08	0.0	1.19460884008246	0.0	0.0	0.9999880539115992
IL4_eQTLGen	ALLERG_ASTHMA	rs9784600	132069847	132734155	-0.13264693518056	0.00881170061318	3.26949999999999	0.0753219	0.0132319	1.25233e-08	0.0	1.18577901272507	0.0	0.0	0.9999881422098728
IL4_eQTLGen	ALLERG_ASTHMA	rs10463516	132067476	132731784	-0.13302868050075	0.00887183170500	7.9641e-51	0.0762483	0.0132877	9.56445e-09	0.0	9.11623923801299	0.0	0.0	0.9999908837607623
IL4_eQTLGen	ALLERG_ASTHMA	rs2406411	132064301	132728609	-0.13293767507092	0.00888169613504	1.1956e-50	0.0750894	0.0132301	1.38169e-08	0.0	1.30375651192417	0.0	0.0	0.9999869624348807
IL4_eQTLGen	ALLERG_ASTHMA	rs7732667	132070333	132734641	-0.13150387966124	0.00882843004002	3.52009999999999	0.0753162	0.0132328	1.2584e-08	0.0	1.19120917421108	0.0	0.0	0.9999880879082579
IL4_eQTLGen	ALLERG_ASTHMA	rs60168940	132071212	132735520	-0.13206437001124	0.00887249121656	4.13719999999999	0.0762496	0.0132959	9.76225e-09	0.0	9.29232454951528	0.0	0.0	0.9999907076754505
IL4_eQTLGen	ALLERG_ASTHMA	rs3798129	132063560	132727868	-0.13189618784135	0.00887598085056	5.9925e-50	0.0746329	0.0132225	1.65768e-08	0.0	1.55465899870522	0.0	0.0	0.999984453410013
IL4_eQTLGen	ALLERG_ASTHMA	rs12514976	132031972	132696280	-0.12556001551742	0.00845430899818	6.79279999999999	0.0769659	0.0132891	6.97044e-09	0.0	6.71646145715411	0.0	0.0	0.9999932853385429
IL4_eQTLGen	ALLERG_ASTHMA	rs7737031	132046590	132710898	-0.12505583357651	0.00845434552535	1.6498e-49	0.0770821	0.0132875	6.58749e-09	0.0	6.36200867909113	0.0	0.0	0.9999936379913209
IL4_eQTLGen	ALLERG_ASTHMA	rs12186803	132040069	132704377	-0.12496030335920	0.00845337351827	1.9026e-49	0.0772741	0.0132879	6.04923e-09	0.0	5.85946262187497	0.0	0.0	0.9999941405373781
IL4_eQTLGen	ALLERG_ASTHMA	rs11242126	132030414	132694722	-0.12496482560713	0.00846955021533	2.8654e-49	0.0769442	0.0132912	7.07522e-09	0.0	6.81470635043245	0.0	0.0	0.9999931852936496
IL4_eQTLGen	ALLERG_ASTHMA	rs59354656	132066422	132730730	-0.13104990465554	0.00888214992616	2.8824e-49	0.0751961	0.013229	1.31444e-08	0.0	1.24258517484744	0.0	0.0	0.9999875741482516
IL4_eQTLGen	ALLERG_ASTHMA	rs2406410	132065880	132730188	-0.13065455875079	0.00888214379194	5.5607e-49	0.0751896	0.013229	1.31804e-08	0.0	1.24603338421023	0.0	0.0	0.9999875396615179
IL4_eQTLGen	ALLERG_ASTHMA	rs17166215	132050692	132715000	-0.12422678665316	0.00845609406248	7.3619e-49	0.075606	0.0132278	1.09257e-08	0.0	1.03985544365797	0.0	0.0	0.9999896014455634
IL4_eQTLGen	ALLERG_ASTHMA	rs1080001	132049116	132713424	-0.12419087249004	0.00847054342939	1.1356e-48	0.0772363	0.0132904	6.19455e-09	0.0	5.99325993693338	0.0	0.0	0.9999940067400661
IL4_eQTLGen	ALLERG_ASTHMA	rs7735429	132058845	132723153	-0.13018539387107	0.00887982878636	1.14910000000000	0.0749538	0.0132274	1.45693e-08	0.0	1.37241260361460	0.0	0.0	0.9999862758739638
IL4_eQTLGen	ALLERG_ASTHMA	rs57276537	132066652	132730960	-0.14801109423316	0.01009742563825	1.1905e-48	0.0762429	0.0132876	9.58562e-09	0.0	9.13516071544253	0.0	0.0	0.9999908648392845
IL4_eQTLGen	ALLERG_ASTHMA	rs3798130	132042146	132706454	-0.12451951237352	0.00849568202974	1.2152e-48	0.076419	0.0132837	8.77667e-09	0.0	8.39110294154402	0.0	0.0	0.9999916088970584
IL4_eQTLGen	ALLERG_ASTHMA	rs17166203	132040995	132705303	-0.12352542700427	0.00851200924787	1.0171e-47	0.0763747	0.013283	8.93532e-09	0.0	8.53787828858705	0.0	0.0	0.9999914621217114
IL4_eQTLGen	ALLERG_ASTHMA	rs2299011	132042542	132706850	-0.12349903977560	0.00851200924787	1.0626e-47	0.0763766	0.013283	8.9275e-09	0.0	8.53091374091933	0.0	0.0	0.9999914690862591
IL4_eQTLGen	ALLERG_ASTHMA	rs3798131	132042007	132706315	-0.12349411827973	0.00851167004023	1.0626e-47	0.0764357	0.0132842	8.72168e-09	0.0	8.34117354145839	0.0	0.0	0.9999916588264586

(showing first 40 of 175 rows; full data in the accompanying CSV)

Supplementary Table 24 - r5_il4_locus_coloc_IL4_eQTLGen_ALLERG_RHINITIS_persnp

ST24_r5_il4_locus_coloc_IL4_eQTLGen_ALLERG_RHINITIS_persnp.csv

trait1	trait2	snp	pos37	pos38	beta_x	se_x	p_x	beta_y	se_y	p_y	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4
IL4_eQTLGen	ALLERG_RHINITIS	rs2070874	132009710	132674018	-0.13985918132477	0.00884568852854	2.6129e-56	0.0513078	0.0122244	2.7029e-05	0.0	0.01913381652348	0.0	0.0	0.9808661834765122
IL4_eQTLGen	ALLERG_RHINITIS	rs734244	132010726	132675034	-0.14273284580923	0.00903824354007	3.52080000000000	0.0565054	0.0124852	6.01672e-06	0.0	0.00459172497086	0.0	0.0	0.9954082750291378
IL4_eQTLGen	ALLERG_RHINITIS	rs2243266	132013789	132678097	-0.13935994880167	0.00884089733629	5.5738e-56	0.0508327	0.0122257	3.21225e-05	0.0	0.02244580184486	0.0	0.0	0.9775541981551323
IL4_eQTLGen	ALLERG_RHINITIS	rs2243267	132013886	132678194	-0.13915115743881	0.00884111273445	8.1467e-56	0.0506012	0.0122255	3.48827e-05	0.0	0.02421711228817	0.0	0.0	0.9757828871182776
IL4_eQTLGen	ALLERG_RHINITIS	rs2243290	132018169	132682477	-0.13895320119135	0.00883785665074	1.05890000000000	0.0507102	0.0122273	3.36434e-05	0.0	0.02341939689791	0.0	0.0	0.9765806031020817
IL4_eQTLGen	ALLERG_RHINITIS	rs2243268	132013963	132678271	-0.13873914505525	0.00884528278782	1.9096e-55	0.0510415	0.012231	3.00435e-05	0.0	0.02109255155832	0.0	0.0	0.9789074484416739
IL4_eQTLGen	ALLERG_RHINITIS	rs2243289	132018132	132682440	-0.13867773764303	0.00884249527472	1.9707e-55	0.0512603	0.0122323	2.78266e-05	0.0	0.01964532552821	0.0	0.0	0.9803546744718706
IL4_eQTLGen	ALLERG_RHINITIS	rs2243282	132016554	132680862	-0.13857397982739	0.00883954300214	2.1865e-55	0.0503838	0.0122264	3.77398e-05	0.0	0.02603218035243	0.0	0.0	0.9739678196475681
IL4_eQTLGen	ALLERG_RHINITIS	rs2243288	132017944	132682252	-0.13662789645728	0.00879580620069	2.0615e-54	0.048841	0.0121707	5.99501e-05	0.0	0.03984701269283	0.0	0.0	0.9601529873071621
IL4_eQTLGen	ALLERG_RHINITIS	rs2243291	132018983	132683291	-0.13652911737273	0.00879680917074	2.52470000000000	0.0487839	0.0121728	6.13352e-05	0.0	0.04066970631848	0.0	0.0	0.9593302936815137
IL4_eQTLGen	ALLERG_RHINITIS	rs2243250	132009154	132673462	-0.13754862443102	0.00889556894901	6.2043e-54	0.0501738	0.0122213	4.03534e-05	0.0	0.02769298720401	0.0	0.0	0.9723070127959813
IL4_eQTLGen	ALLERG_RHINITIS	rs2243270	132014109	132678417	-0.13544689954455	0.00879879558942	1.7958e-53	0.0485752	0.0121735	6.60085e-05	0.0	0.04344762608463	0.0	0.0	0.9565523739153647
IL4_eQTLGen	ALLERG_RHINITIS	rs74863864	132023259	132687567	-0.13625277989886	0.00886024059688	2.2923e-53	0.0509715	0.0122346	3.09763e-05	0.0	0.02168967204782	0.0	0.0	0.9783103279521717
IL4_eQTLGen	ALLERG_RHINITIS	rs2299007	132043032	132707340	-0.13623398609206	0.00887130608086	3.1893e-53	0.0512092	0.012273	3.01245e-05	0.0	0.02108315569413	0.0	0.0	0.9789168443058652
IL4_eQTLGen	ALLERG_RHINITIS	rs2299010	132042740	132707048	-0.13598313008212	0.00887131925589	4.9249e-53	0.0512131	0.0122729	3.00822e-05	0.0	0.02105323161895	0.0	0.0	0.9789467683810447
IL4_eQTLGen	ALLERG_RHINITIS	rs10455052	132034588	132698896	-0.13701388301260	0.00894136384482	5.3188e-53	0.0511873	0.0122731	3.03655e-05	0.0	0.02123908457090	0.0	0.0	0.9787609154290916
IL4_eQTLGen	ALLERG_RHINITIS	rs11242123	132025487	132689795	-0.13568964708945	0.00887336004194	8.6454e-53	0.05115	0.0122687	3.05731e-05	0.0	0.02137907588916	0.0	0.0	0.9786209241108335
IL4_eQTLGen	ALLERG_RHINITIS	rs2243274	132014832	132679140	-0.15298723753240	0.01000786549957	9.36399999999999	0.0485704	0.0121729	6.60633e-05	0.0	0.04348202235823	0.0	0.0	0.9565179776417649
IL4_eQTLGen	ALLERG_RHINITIS	rs2299005	132057589	132721897	-0.13443122685738	0.00887334830741	7.5767e-52	0.0501452	0.0122522	4.26265e-05	0.0	0.02905565429778	0.0	0.0	0.9709443457022152
IL4_eQTLGen	ALLERG_RHINITIS	rs9784675	132069739	132734047	-0.13282091155285	0.00881160929537	2.4158e-51	0.0487176	0.0121946	6.46904e-05	0.0	0.04260468155353	0.0	0.0	0.9573953184464686
IL4_eQTLGen	ALLERG_RHINITIS	rs9784600	132069847	132734155	-0.13264693518056	0.00881170061318	3.26949999999999	0.0487249	0.0121964	6.46904e-05	0.0	0.04259788165876	0.0	0.0	0.9574021183412398
IL4_eQTLGen	ALLERG_RHINITIS	rs10463516	132067476	132731784	-0.13302868050075	0.00887183170500	7.9641e-51	0.0503642	0.0122503	3.93514e-05	0.0	0.02700513239996	0.0	0.0	0.9729948676000382
IL4_eQTLGen	ALLERG_RHINITIS	rs2406411	132064301	132728609	-0.13293767507092	0.00888169613504	1.1956e-50	0.04868	0.0121946	6.55375e-05	0.0	0.04310643452563	0.0	0.0	0.9568935654743647
IL4_eQTLGen	ALLERG_RHINITIS	rs7732667	132070333	132734641	-0.13150387966124	0.00882843004002	3.52009999999999	0.0483439	0.0121973	7.38635e-05	0.0	0.04797834580355	0.0	0.0	0.9520216541964496
IL4_eQTLGen	ALLERG_RHINITIS	rs60168940	132071212	132735520	-0.13206437001124	0.00887249121656	4.13719999999999	0.0504061	0.0122582	3.92211e-05	0.0	0.02690788074429	0.0	0.0	0.9730921192557085
IL4_eQTLGen	ALLERG_RHINITIS	rs3798129	132063560	132727868	-0.13189618784135	0.00887598085056	5.9925e-50	0.0476733	0.0121892	9.18671e-05	0.0	0.05828515638649	0.0	0.0	0.9417148436135079
IL4_eQTLGen	ALLERG_RHINITIS	rs12514976	132031972	132696280	-0.12556001551742	0.00845430899818	6.79279999999999	0.0501995	0.0122521	4.18138e-05	0.0	0.02854777831856	0.0	0.0	0.9714522216814391
IL4_eQTLGen	ALLERG_RHINITIS	rs7737031	132046590	132710898	-0.12505583357651	0.00845434552535	1.6498e-49	0.0501603	0.0122509	4.23253e-05	0.0	0.02886766678973	0.0	0.0	0.9711323332102607
IL4_eQTLGen	ALLERG_RHINITIS	rs12186803	132040069	132704377	-0.12496030335920	0.00845337351827	1.9026e-49	0.0504227	0.0122512	3.85967e-05	0.0	0.02652732265814	0.0	0.0	0.9734726773418564
IL4_eQTLGen	ALLERG_RHINITIS	rs11242126	132030414	132694722	-0.12496482560713	0.00846955021533	2.8654e-49	0.0501264	0.0122539	4.3017e-05	0.0	0.02929417311453	0.0	0.0	0.9707058268854699
IL4_eQTLGen	ALLERG_RHINITIS	rs59354656	132066422	132730730	-0.13104990465554	0.00888214992616	2.8824e-49	0.0487244	0.0121935	6.44421e-05	0.0	0.04245955143901	0.0	0.0	0.9575404485609859
IL4_eQTLGen	ALLERG_RHINITIS	rs2406410	132065880	132730188	-0.13065455875079	0.00888214379194	5.5607e-49	0.0485857	0.0121936	6.76192e-05	0.0	0.04433625858103	0.0	0.0	0.955663741418961
IL4_eQTLGen	ALLERG_RHINITIS	rs17166215	132050692	132715000	-0.12422678665316	0.00845609406248	7.3619e-49	0.0482435	0.0121922	7.59242e-05	0.0	0.04919155314714	0.0	0.0	0.950808446852866
IL4_eQTLGen	ALLERG_RHINITIS	rs1080001	132049116	132713424	-0.12419087249004	0.00847054342939	1.1356e-48	0.0503038	0.0122541	4.04194e-05	0.0	0.02767103642764	0.0	0.0	0.972328963572356
IL4_eQTLGen	ALLERG_RHINITIS	rs7735429	132058845	132723153	-0.13018539387107	0.00887982878636	1.14910000000000	0.0479048	0.0121913	8.51491e-05	0.0	0.05448323857601	0.0	0.0	0.9455167614239807
IL4_eQTLGen	ALLERG_RHINITIS	rs57276537	132066652	132730960	-0.14801109423316	0.01009742563825	1.1905e-48	0.0503749	0.0122502	3.91967e-05	0.0	0.02690814762418	0.0	0.0	0.9730918523758114
IL4_eQTLGen	ALLERG_RHINITIS	rs3798130	132042146	132706454	-0.12451951237352	0.00849568202974	1.2152e-48	0.0497531	0.0122461	4.84953e-05	0.0	0.03269866915387	0.0	0.0	0.9673013308461271
IL4_eQTLGen	ALLERG_RHINITIS	rs17166203	132040995	132705303	-0.12352542700427	0.00851200924787	1.0171e-47	0.0496298	0.0122458	5.06046e-05	0.0	0.03399456377459	0.0	0.0	0.9660054362254064
IL4_eQTLGen	ALLERG_RHINITIS	rs2299011	132042542	132706850	-0.12349903977560	0.00851200924787	1.0626e-47	0.049629	0.0122458	5.06186e-05	0.0	0.03400320175164	0.0	0.0	0.9659967982483573
IL4_eQTLGen	ALLERG_RHINITIS	rs3798131	132042007	132706315	-0.12349411827973	0.00851167004023	1.0626e-47	0.049594	0.0122469	5.13216e-05	0.0	0.03442902818775	0.0	0.0	0.9655709718122452

(showing first 40 of 175 rows; full data in the accompanying CSV)

Supplementary Table 25 - r5_il4_locus_coloc_IL4_eQTLGen_L12_ATOPIC_persnp

ST25_r5_il4_locus_coloc_IL4_eQTLGen_L12_ATOPIC_persnp.csv

trait1	trait2	snp	pos37	pos38	beta_x	se_x	p_x	beta_y	se_y	p_y	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4
IL4_eQTLGen	L12_ATOPIC	rs2070874	132009710	132674018	-0.13985918132477	0.00884568852854	2.6129e-56	0.0960295	0.00886117	2.29615e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs734244	132010726	132675034	-0.14273284580923	0.00903824354007	3.52080000000000	0.0887699	0.00905804	1.1246e-22	0.0	2.22044604925031	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2243266	132013789	132678097	-0.13935994880167	0.00884089733629	5.5738e-56	0.095502	0.00886251	4.47095e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2243267	132013886	132678194	-0.13915115743881	0.00884111273445	8.1467e-56	0.0954384	0.00886217	4.81282e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2243290	132018169	132682477	-0.13895320119135	0.00883785665074	1.05890000000000	0.0953336	0.00886385	5.59371e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2243268	132013963	132678271	-0.13873914505525	0.00884528278782	1.9096e-55	0.0957428	0.00886665	3.5148e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2243289	132018132	132682440	-0.13867773764303	0.00884249527472	1.9707e-55	0.0956925	0.00886804	3.80802999999999	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2243282	132016554	132680862	-0.13857397982739	0.00883954300214	2.1865e-55	0.0954148	0.00886264	4.98425e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2243288	132017944	132682252	-0.13662789645728	0.00879580620069	2.0615e-54	0.0920204	0.00882589	1.88322e-25	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2243291	132018983	132683291	-0.13652911737273	0.00879680917074	2.52470000000000	0.092141	0.00882723	1.65806e-25	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2243250	132009154	132673462	-0.13754862443102	0.00889556894901	6.2043e-54	0.0958265	0.00885851	2.84577e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2243270	132014109	132678417	-0.13544689954455	0.00879879558942	1.7958e-53	0.0922409	0.00882726	1.47231e-25	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs74863864	132023259	132687567	-0.13625277989886	0.00886024059688	2.2923e-53	0.0957212	0.00886915	3.73078e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2299007	132043032	132707340	-0.13623398609206	0.00887130608086	3.1893e-53	0.0957891	0.00889535	4.85065e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2299010	132042740	132707048	-0.13598313008212	0.00887131925589	4.9249e-53	0.0957928	0.00889534	4.82836e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs10455052	132034588	132698896	-0.13701388301260	0.00894136384482	5.3188e-53	0.0957128	0.00889571	5.34934e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs11242123	132025487	132689795	-0.13568964708945	0.00887336004194	8.6454e-53	0.0938663	0.00889527	4.95222e-26	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2243274	132014832	132679140	-0.15298723753240	0.01000786549957	9.36399999999999	0.0920497	0.00882698	1.84289e-25	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2299005	132057589	132721897	-0.13443122685738	0.00887334830741	7.5767e-52	0.0952502	0.00887987	7.64188e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs9784675	132069739	132734047	-0.13282091155285	0.00881160929537	2.4158e-51	0.0909502	0.00884227	8.16206e-25	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs9784600	132069847	132734155	-0.13264693518056	0.00881170061318	3.26949999999999	0.0909676	0.00884363	8.13017999999999	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs10463516	132067476	132731784	-0.13302868050075	0.00887183170500	7.9641e-51	0.0946073	0.00887915	1.65272e-26	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2406411	132064301	132728609	-0.13293767507092	0.00888169613504	1.1956e-50	0.0909672	0.00884215	7.99098e-25	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs7732667	132070333	132734641	-0.13150387966124	0.00882843004002	3.52009999999999	0.0908236	0.00884422	9.69393e-25	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs60168940	132071212	132735520	-0.13206437001124	0.00887249121656	4.13719999999999	0.0946936	0.00888485	1.60214000000000	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs3798129	132063560	132727868	-0.13189618784135	0.00887598085056	5.9925e-50	0.0914973	0.00883738	4.03738e-25	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs12514976	132031972	132696280	-0.12556001551742	0.00845430899818	6.79279999999999	0.0953695	0.00888	6.61912e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs7737031	132046590	132710898	-0.12505583357651	0.00845434552535	1.6498e-49	0.0953897	0.00887886	6.36063e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs12186803	132040069	132704377	-0.12496030335920	0.00845337351827	1.9026e-49	0.0956498	0.00887898	4.6366e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs11242126	132030414	132694722	-0.12496482560713	0.00846955021533	2.8654e-49	0.0954105	0.00888122	6.39587999999999	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs59354656	132066422	132730730	-0.13104990465554	0.00888214992616	2.8824e-49	0.090979	0.00884143	7.81268e-25	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2406410	132065880	132730188	-0.13065455875079	0.00888214379194	5.5607e-49	0.0909709	0.00884142	7.88497e-25	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs17166215	132050692	132715000	-0.12422678665316	0.00845609406248	7.3619e-49	0.0915128	0.00883926	4.05602e-25	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs1080001	132049116	132713424	-0.12419087249004	0.00847054342939	1.1356e-48	0.095385	0.00888121	6.59780999999999	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs7735429	132058845	132723153	-0.13018539387107	0.00887982878636	1.14910000000000	0.0911132	0.00883874	6.45803e-25	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs57276537	132066652	132730960	-0.14801109423316	0.01009742563825	1.1905e-48	0.0945943	0.00887909	1.67764000000000	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs3798130	132042146	132706454	-0.12451951237352	0.00849568202974	1.2152e-48	0.0952564	0.00887465	7.08434999999999	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs17166203	132040995	132705303	-0.12352542700427	0.00851200924787	1.0171e-47	0.095162	0.00887442	7.92318999999999	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs2299011	132042542	132706850	-0.12349903977560	0.00851200924787	1.0626e-47	0.0951618	0.00887441	7.92501e-27	0.0	0.0	0.0	0.0	1.0
IL4_eQTLGen	L12_ATOPIC	rs3798131	132042007	132706315	-0.12349411827973	0.00851167004023	1.0626e-47	0.0952039	0.00887516	7.60326e-27	0.0	0.0	0.0	0.0	1.0

(showing first 40 of 175 rows; full data in the accompanying CSV)

Supplementary Table 26 - r5_il4_locus_coloc_IL4_DICE_TREG_NAIVE_ALLERG_ASTHMA_persnp

ST26_r5_il4_locus_coloc_IL4_DICE_TREG_NAIVE_ALLERG_ASTHMA_persnp.csv

trait1	trait2	snp	pos37	pos38	beta_x	se_x	p_x	beta_y	se_y	p_y	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs370314650	132022653	132686961	0.779455135928	0.15897469966967	9.43773827539e-07	-0.0649468	0.0127512	3.51747e-07	0.01356524393231	0.00028364084114	0.04501045269185	0.0	0.9411406625346809
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs6897429	132023263	132687571	0.779455135928	0.15897469966967	9.43773827539e-07	-0.0647532	0.0127526	3.82164e-07	0.01466923136825	0.00030672453403	0.04495901029024	0.0	0.9400650338074702
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs2406539	132023276	132687584	0.779455135928	0.15897469966967	9.43773827539e-07	-0.0647524	0.0127526	3.82279e-07	0.01467380166187	0.00030682009604	0.04495879732868	0.0	0.9400605809133897
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs6879672	132025947	132690255	0.782607228262	0.16068626391390	1.11369579381e-06	-0.0619722	0.0127858	1.25372e-06	0.04989383587449	0.00091306917193	0.04918037775999	0.0	0.900012717193577
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs11242122	132024708	132689016	0.798055047934	0.16472621089683	1.26779970795e-06	-0.0644973	0.0127472	4.19914e-07	0.02168126541985	0.00032765813301	0.06069766021718	0.0	0.9172934162299368
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs11747814	132024166	132688474	0.798055047934	0.16472621089683	1.26779970795e-06	-0.0644783	0.0127501	4.25736e-07	0.02196033653560	0.00033187559537	0.06068007830435	0.0	0.9170277095466648
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs70974054	132026345	132690653	0.748754087505	0.16030083156212	2.9983798042e-06	-0.0640432	0.0127337	4.91994e-07	0.03125853173468	0.00037166984250	0.07512456309816	0.0	0.893245235324636
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs11949166	132027681	132691989	0.760529789278	0.16408187402424	3.56829146016e-06	-0.064232	0.0127561	4.76848e-07	0.03593047005473	0.00035272411484	0.08909401607094	0.0	0.8746227897594813
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs10588133	132022972	132687280	0.73939763246	0.16002219513461	3.82642491347e-06	-0.0649857	0.012752	3.46697e-07	0.02466400826470	0.00026515651462	0.08297947605356	0.0	0.892091359167117
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs2051809	132056874	132721182	0.743160909642	0.16128241563660	4.06905773535e-06	-0.0652207	0.0127071	2.85752e-07	0.02187556604122	0.00022030125043	0.08833300704883	0.0	0.8895711256594986
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs7717027	132058479	132722787	0.743160909642	0.16128241563660	4.06905773535e-06	-0.0648696	0.0127059	3.29997e-07	0.02503219787114	0.00025209059650	0.08804500046333	0.0	0.8866707110690233
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs2074529	132056147	132720455	0.743204919631	0.16165124219534	4.27421434323e-06	-0.0650854	0.0127031	2.99806e-07	0.02356328590253	0.00022963841381	0.09084712484064	0.0	0.8853599508430032
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs6896296	132058698	132723006	0.743204919631	0.16165124219534	4.27421434323e-06	-0.0648689	0.012706	3.30119e-07	0.02578188543727	0.00025126000259	0.09063864689252	0.0	0.8833282076676102
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs2227282	132013179	132677487	-0.706925839928	0.15582597580777	5.71584465754e-06	-0.0648629	0.0127366	3.53102e-07	0.02639791413293	0.00026833079324	0.08717841271083	0.0	0.886155342362988
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs11242125	132025905	132690213	0.733467346697	0.16275735270799	6.59031357226e-06	-0.0614353	0.0127615	1.47846e-06	0.11899461659448	0.00092780057712	0.10004296463140	0.0	0.7800346181969738
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs11242124	132025904	132690212	0.733467346697	0.16275735270799	6.59031357226e-06	-0.0614353	0.0127615	1.4785e-06	0.11899461659448	0.00092780057712	0.10004296463140	0.0	0.7800346181969738
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs1859138	132027049	132691357	0.733235927856	0.16272941458929	6.61047237939e-06	-0.0636434	0.0127365	5.82465e-07	0.05266784850608	0.00041042307776	0.10769446626952	0.0	0.8392272621466257
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs10068370	132027568	132691876	0.733235927856	0.16272941458929	6.61047237939e-06	-0.0636734	0.0127365	5.75427e-07	0.05208795420628	0.00040590415379	0.10776093223848	0.0	0.8397452094014359
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs2299003	132058296	132722604	0.703575724989	0.15656539116479	6.99587848467e-06	-0.0653351	0.0127259	2.83616e-07	0.02381516700833	0.00021586940989	0.09697271373480	0.0	0.8789962498469665
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs4426908	132022718	132687026	0.726703038418	0.16266024111121	7.90971622052e-06	-0.0641512	0.0127317	4.68727e-07	0.04611182840922	0.00033273196925	0.11606415383365	0.0	0.8374912857878702
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs17690965	132030667	132694975	0.69445583213	0.15806745119678	1.11584085551e-05	-0.0646761	0.012712	3.62226e-07	0.03727569539982	0.00026184469807	0.11993985790314	0.0	0.8425226019989569
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs2023823	132032806	132697114	0.69445583213	0.15806745119678	1.11584085551e-05	-0.064735	0.0127107	3.52541e-07	0.03635518939585	0.00025537856473	0.12005537503868	0.0	0.8433340570007218
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs2237057	132043944	132708252	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0641978	0.0127292	4.57457e-07	0.05307911481585	0.00031441008728	0.13672519760924	0.0	0.809881277487612
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs11740584	132046789	132711097	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0651782	0.0127272	3.0362e-07	0.03647773911941	0.00021607310484	0.13913725750541	0.0	0.8241689302703221
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs6865485	132035572	132699880	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0649488	0.0127279	3.34572e-07	0.03988500091788	0.00023625576017	0.13864220699333	0.0	0.8212365363286092
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs3213639	132038199	132702507	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0652434	0.0127299	2.97201e-07	0.03575734578146	0.00021180590986	0.13924192545051	0.0	0.8247889228581531
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs1468215	132036489	132700797	0.702963168517	0.16143648771835	1.33416305877e-05	-0.063946	0.012726	5.03918e-07	0.05793762508142	0.00034318910219	0.13601929118432	0.0	0.805699894632051
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs2285700	132039132	132703440	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0641733	0.012726	4.5905e-07	0.05325494233603	0.00031545158819	0.13669965114142	0.0	0.8097299549343503
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs10062446	132040374	132704682	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0641738	0.012726	4.58965e-07	0.05324502859547	0.00031539286490	0.13670109153627	0.0	0.8097384870033493
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs4705964	132046165	132710473	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0642018	0.0127263	4.53963e-07	0.05272128391525	0.00031229050324	0.13677718785274	0.0	0.810189237728748
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs6864565	132047971	132712279	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0641663	0.012726	4.60341e-07	0.05339390953612	0.00031627474979	0.13667946021168	0.0	0.8096103555023942
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs6864396	132047875	132712183	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0641948	0.0127277	4.56636e-07	0.05299374824969	0.00031390442494	0.13673760075429	0.0	0.8099547465706453
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs10036532	132045815	132710123	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0641685	0.0127265	4.60426e-07	0.05339867817811	0.00031630299648	0.13667876736247	0.0	0.8096062514629232
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs187681	132101903	132766211	-0.718216176316	0.16531697280451	1.3960636769e-05	0.0939126	0.014214	3.92013e-11	6.83140350990168	3.58736474049241	0.15996600988453	0.0	0.840027122838312
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs10477742	132049789	132714097	0.7030535515399	0.16194807168493	1.41688553808e-05	-0.0641397	0.0127262	4.65597e-07	0.05571039680692	0.00031707410190	0.14107100577544	0.0	0.8029015233157295
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs2299004	132057838	132722146	0.677525207284	0.16023790105244	2.35520573784e-05	-0.0645424	0.0127059	3.78016e-07	0.05332325508871	0.00025317570832	0.16465434284053	0.0	0.7817692263624159
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs7734673	132058466	132722774	0.677525207284	0.16023790105244	2.35520573784e-05	-0.0647964	0.0127059	3.40181e-07	0.04844899842416	0.00023003302167	0.16550636945741	0.0	0.7858145990967439
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs10073911	132035399	132699707	0.679669207524	0.16093548480232	2.4081493953e-05	-0.064475	0.0127076	3.90086e-07	0.05603142920718	0.00025891239119	0.16789470848784	0.0	0.7758149499137692
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs2897442	132049027	132713335	0.679669207524	0.16093548480232	2.4081493953e-05	-0.0639019	0.0127057	4.92062e-07	0.06902900088485	0.00031897211857	0.16557163464964	0.0	0.7650803923469257
IL4_DICE_TREG_NAI	ALLERG_ASTHMA	rs10594222	132054753	132719061	0.677386571374	0.16053902356710	2.44897751985e-05	-0.0645501	0.0127052	3.76244e-07	0.05413738132640	0.00025074694350	0.16790942110578	0.0	0.7777024506243009

(showing first 40 of 42 rows; full data in the accompanying CSV)

Supplementary Table 27 - r5_il4_locus_coloc_IL4_DICE_TREG_NAIVE_ALLERG_RHINITIS_persnp

ST27_r5_il4_locus_coloc_IL4_DICE_TREG_NAIVE_ALLERG_RHINITIS_persnp.csv

trait1	trait2	snp	pos37	pos38	beta_x	se_x	p_x	beta_y	se_y	p_y	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs370314650	132022653	132686961	0.779455135928	0.15897469966967	9.43773827539e-07	-0.0328312	0.0117329	0.00513866	0.97155596910949	0.02031463301906	0.00037104645794	0.0	0.007758351413489802
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs6897429	132023263	132687571	0.779455135928	0.15897469966967	9.43773827539e-07	-0.0328174	0.011734	0.00516167	0.97158679688810	0.02031527760880	0.00036960997876	0.0	0.007728315524324484
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs2406539	132023276	132687584	0.779455135928	0.15897469966967	9.43773827539e-07	-0.032816	0.011734	0.00516345	0.97158940583824	0.02031533216033	0.00036948840975	0.0	0.007725773591663881
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs6879672	132025947	132690255	0.782607228262	0.16068626391390	1.11369579381e-06	-0.0342984	0.0117635	0.00354936	0.97218181891485	0.01779116062746	0.00051952827779	0.0	0.009507492179882296
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs11242122	132024708	132689016	0.798055047934	0.16472621089683	1.26779970795e-06	-0.032501	0.011729	0.00558869	0.97960689778392	0.01480430966556	0.00034686066102	0.0	0.0052419318894805 3
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs11747814	132024166	132688474	0.798055047934	0.16472621089683	1.26779970795e-06	-0.0325023	0.0117314	0.00559654	0.97961270968546	0.01480439749793	0.00034649450222	0.0	0.005236398314380374
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs70974054	132026345	132690653	0.748754087505	0.16030083156212	2.9983798042e-06	-0.0302032	0.0117152	0.00993413	0.98559022845617	0.01171885385077	0.00020875704338	0.0	0.0024821606496594994
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs11949166	132027681	132691989	0.760529789278	0.16408187402424	3.56829146016e-06	-0.0310604	0.0117358	0.00812962	0.98759711751200	0.00969509495889	0.00025033045410	0.0	0.0024574570749 96298
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs10588133	132022972	132687280	0.73939763246	0.16002219513461	3.82642491347e-06	-0.0327731	0.0117336	0.00522047	0.98505150505222	0.01059003958309	0.00037090878888	0.0	0.0039875465758000455
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs2051809	132056874	132721182	0.743160909642	0.16128241563660	4.06905773535e-06	-0.0306521	0.0116903	0.00874118	0.98746722107108	0.00994444043905	0.00023380177506	0.0	0.002354536714789199
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs7717027	132058479	132722787	0.743160909642	0.16128241563660	4.06905773535e-06	-0.0308359	0.0116888	0.00833777	0.98735796243067	0.00994334013312	0.00024377037756	0.0	0.0024549270586366233
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs2074529	132056147	132720455	0.743204919631	0.16165124219534	4.27421434323e-06	-0.0313636	0.011686	0.00727763	0.98742077222522	0.00962301017112	0.00027510952991	0.0	0.0026811080737359507
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs6896296	132058698	132723006	0.743204919631	0.16165124219534	4.27421434323e-06	-0.0308347	0.0116888	0.00834046	0.98775393818978	0.00962625707412	0.00024380250240	0.0	0.002376002233691 6393
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs2227282	132013179	132677487	-0.706925839928	0.15582597580777	5.71584465754e-06	-0.0317738	0.0117171	0.00669299	0.98665497434636	0.01002919816269	0.00029698813585	0.0	0.003018839355087 9813
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs11242125	132025905	132690213	0.733467346697	0.16275735270799	6.59031357226e-06	-0.0330977	0.0117402	0.00481482	0.98876643289962	0.00770940815093	0.00040060935082	0.0	0.0031235495986177 32
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs11242124	132025904	132690212	0.733467346697	0.16275735270799	6.59031357226e-06	-0.0330976	0.0117402	0.00481493	0.98876651606410	0.00770940879936	0.00040059982338	0.0	0.00312347531314653
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs1859138	132027049	132691357	0.733235927856	0.16272941458929	6.61047237939e-06	-0.0298808	0.0117364	0.0108968	0.99057791757406	0.00771924521755	0.00019366557847	0.0	0.0015091716299088821
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs10068370	132027568	132691876	0.733235927856	0.16272941458929	6.61047237939e-06	-0.0298698	0.0117365	0.0109265	0.99058199134228	0.00771927696308	0.00019319865380	0.0	0.001505533040825477
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs2299003	132058296	132722604	0.703575724989	0.15656539116479	6.99587848467e-06	-0.0320132	0.0117084	0.00625302	0.98786760071378	0.00895439431289	0.00031576799880	0.0	0.0028622369745169773
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs4426908	132022718	132687026	0.726703038418	0.16266024111121	7.90971622052e-06	-0.0312701	0.0117127	0.00759049	0.99066271331007	0.00714838614807	0.00026642697284	0.0	0.0019224735690145697
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs17690965	132030667	132694975	0.69445583213	0.15806745119678	1.11584085551e-05	-0.0310767	0.0116942	0.00787354	0.99097196460569	0.00696112445624	0.00025757368680	0.0	0.001809337251260949
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs2023823	132032806	132697114	0.69445583213	0.15806745119678	1.11584085551e-05	-0.0314231	0.0116929	0.00720178	0.99080279732712	0.00695993613354	0.00027880300998	0.0	0.0019584635293444404
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs2237057	132043944	132708252	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0314584	0.0117117	0.00722986	0.99219358994649	0.00587718303715	0.00027865216643	0.0	0.0016505748499184502
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs11740584	132046789	132711097	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0321942	0.0117094	0.00597008	0.99183678217241	0.00587506951352	0.00033049375704	0.0	0.0019576545570132325
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs6865485	132035572	132699880	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0321232	0.01171	0.00608373	0.99187478552745	0.00587529462350	0.00032497214424	0.0	0.0019249477048017187
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs3213639	132038199	132702507	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0322907	0.0117115	0.00583042	0.99178776400894	0.00587477915817	0.00033761574190	0.0	0.001999841090979709
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs1468215	132036489	132700797	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0313049	0.0117088	0.00750361	0.99225643933508	0.00587755532071	0.00026952060453	0.0	0.0015964847396666242
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs2285700	132039132	132703440	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0313519	0.011709	0.00741532	0.99223676564459	0.00587743878511	0.00027237904956	0.0	0.0016134165207330356
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs10062446	132040374	132704682	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0313619	0.011709	0.00739656	0.99223250714098	0.00587741356018	0.00027299777936	0.0	0.0016170815194689514
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs4705964	132046165	132710473	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0313946	0.0117093	0.00733652	0.99221880458910	0.00587733239416	0.00027498866110	0.0	0.0016288743556188516
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs6864565	132047971	132712279	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0313636	0.011709	0.00739316	0.99223178210049	0.00587740926546	0.00027310312250	0.0	0.0016177055115340565
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs6864396	132047875	132712183	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0313015	0.0117104	0.00751848	0.99225942195395	0.00587757298803	0.00026908725157	0.0	0.0015939178064397674
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs10036532	132045815	132710123	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0313659	0.0117094	0.00739112	0.99223119699248	0.00587740579962	0.00027318813446	0.0	0.0016182090734219502
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs187681	132101903	132766211	-0.718216176316	0.16531697280451	1.3960636769e-05	0.0713516	0.013123	5.41365e-08	0.00775615685231	4.07297908451870	0.15871986301199	0.0	0.833483250344844 6
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs10477742	132049789	132714097	0.7030535 15399	0.16194807168493	1.41688553808e-05	-0.0312945	0.011709	0.00752454	0.99255133915627	0.00564907705730	0.00026893695198	0.0	0.0015306468344436613
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs2299004	132057838	132722146	0.677525207284	0.16023790105244	2.35520573784e-05	-0.0308926	0.0116886	0.00821807	0.99385263370959	0.00471875439887	0.00024854321027	0.0	0.0011800686812461993
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs7734673	132058466	132722774	0.677525207284	0.16023790105244	2.35520573784e-05	-0.0308097	0.0116889	0.00839363	0.99387902157097	0.00471887968690	0.00024393057663	0.0	0.0011581681654747027
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs10073911	132035399	132699707	0.679669207524	0.16093548480232	2.4081493953e-05	-0.0309549	0.0116901	0.00809804	0.99399107982679	0.00459307590308	0.00025189183863	0.0	0.0011639524314831008
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs2897442	132049027	132713335	0.679669207524	0.16093548480232	2.4081493953e-05	-0.0303872	0.011689	0.00933233	0.99415784048581	0.00459384647776	0.00022208647700	0.0	0.001026226559408 7661
IL4_DICE_TREG_NAI	ALLERG_RHINITIS	rs10594222	132054753	132719061	0.677386571374	0.16053902356710	2.44897751985e-05	-0.0308987	0.0116879	0.00820182	0.99399380195286	0.00460385969169	0.00024900895229	0.0	0.0011533294031409132

(showing first 40 of 42 rows; full data in the accompanying CSV)

Supplementary Table 28 - r5_il4_locus_coloc_IL4_DICE_TREG_NAIVE_L12_ATOPIC_persnp

ST28_r5_il4_locus_coloc_IL4_DICE_TREG_NAIVE_L12_ATOPIC_persnp.csv

trait1	trait2	snp	pos37	pos38	beta_x	se_x	p_x	beta_y	se_y	p_y	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4
IL4_DICE_TREG_NAI	L12_ATOPIC	rs370314650	132022653	132686961	0.779455135928	0.15897469966967	9.43773827539e-07	-0.102153	0.00851645	3.7853e-33	0.0	0.0	0.045642551123122	0.0	0.954357448876871
IL4_DICE_TREG_NAI	L12_ATOPIC	rs6897429	132023263	132687571	0.779455135928	0.15897469966967	9.43773827539e-07	-0.102131	0.00851719	3.95549e-33	0.0	0.0	0.045642551123122	0.0	0.954357448876871
IL4_DICE_TREG_NAI	L12_ATOPIC	rs2406539	132023276	132687584	0.779455135928	0.15897469966967	9.43773827539e-07	-0.10213	0.00851719	3.95640000000000	0.0	0.0	0.045642551123122	0.0	0.954357448876871
IL4_DICE_TREG_NAI	L12_ATOPIC	rs6879672	132025947	132690255	0.782607228262	0.16068626391390	1.11369579381e-06	-0.10188	0.00853439	7.54223e-33	0.0	0.0	0.051812827149151	0.0	0.948187172850847
IL4_DICE_TREG_NAI	L12_ATOPIC	rs11242122	132024708	132689016	0.798055047934	0.16472621089683	1.26779970795e-06	-0.101879	0.00851364	5.32108e-33	0.0	0.0	0.062063613543072	0.0	0.937936386456926
IL4_DICE_TREG_NAI	L12_ATOPIC	rs11747814	132024166	132688474	0.798055047934	0.16472621089683	1.26779970795e-06	-0.101903	0.00851543	5.30152000000000	0.0	0.0	0.062063613543072	0.0	0.937936386456926
IL4_DICE_TREG_NAI	L12_ATOPIC	rs70974054	132026345	132690653	0.748754087505	0.16030083156212	2.9983798042e-06	-0.0993409	0.0085051	1.6099e-31	0.0	0.0	0.077578382990171	0.0	0.922421617009828
IL4_DICE_TREG_NAI	L12_ATOPIC	rs11949166	132027681	132691989	0.760529789278	0.16408187402424	3.56829146016e-06	-0.102366	0.00851658	2.80221e-33	0.0	0.0	0.092448337034201	0.0	0.907551662965799
IL4_DICE_TREG_NAI	L12_ATOPIC	rs10588133	132022972	132687280	0.73939763246	0.16002219513461	3.82642491347e-06	-0.102181	0.00851656	3.64502e-33	0.0	0.0	0.085100972212732	0.0	0.914899027787266
IL4_DICE_TREG_NAI	L12_ATOPIC	rs2051809	132056874	132721182	0.743160909642	0.16128241563660	4.06905773535e-06	-0.0956642	0.0084905	1.90502000000000	0.0	0.0	0.090328902490877	0.0	0.909671097509123
IL4_DICE_TREG_NAI	L12_ATOPIC	rs7717027	132058479	132722787	0.743160909642	0.16128241563660	4.06905773535e-06	-0.0955978	0.00848953	2.05211000000000	0.0	0.0	0.090328902490877	0.0	0.909671097509123
IL4_DICE_TREG_NAI	L12_ATOPIC	rs2074529	132056147	132720455	0.743204919631	0.16165124219534	4.27421434323e-06	-0.0949996	0.00848826	4.46786000000000	0.0	0.0	0.093061325925157	0.0	0.906938674074843
IL4_DICE_TREG_NAI	L12_ATOPIC	rs6896296	132058698	132723006	0.743204919631	0.16165124219534	4.27421434323e-06	-0.0955969	0.00848956	2.05494e-29	0.0	0.0	0.093061325925157	0.0	0.906938674074843
IL4_DICE_TREG_NAI	L12_ATOPIC	rs2227282	132013179	132677487	-0.706925839928	0.15582597580777	5.71584465754e-06	-0.0997931	0.00850808	9.02818e-32	0.0	0.0	0.089566823565284	0.0	0.910433176434716
IL4_DICE_TREG_NAI	L12_ATOPIC	rs11242125	132025905	132690213	0.733467346697	0.16275735270799	6.59031357226e-06	-0.0989872	0.00852119	3.39391e-31	0.0	0.0	0.113675165216551	0.0	0.886324834783440
IL4_DICE_TREG_NAI	L12_ATOPIC	rs11242124	132025904	132690212	0.733467346697	0.16275735270799	6.59031357226e-06	-0.0989871	0.00852119	3.39391e-31	0.0	0.0	0.113675165216551	0.0	0.886324834783440
IL4_DICE_TREG_NAI	L12_ATOPIC	rs1859138	132027049	132691357	0.733235927856	0.16272941458929	6.61047237939e-06	-0.0994882	0.00850618	1.33629e-31	0.0	0.0	0.113731117406771	0.0	0.886268882593227
IL4_DICE_TREG_NAI	L12_ATOPIC	rs10068370	132027568	132691876	0.733235927856	0.16272941458929	6.61047237939e-06	-0.0995578	0.00850621	1.21395e-31	0.0	0.0	0.113731117406771	0.0	0.886268882593227
IL4_DICE_TREG_NAI	L12_ATOPIC	rs2299003	132058296	132722604	0.703575724989	0.15656539116479	6.99587848467e-06	-0.0984927	0.00850036	4.80286e-31	0.0	0.0	0.099360448286104	0.0	0.900639551713896
IL4_DICE_TREG_NAI	L12_ATOPIC	rs4426908	132022718	132687026	0.726703038418	0.16266024111121	7.90971622052e-06	-0.0993693	0.00850511	1.5481e-31	0.0	0.0	0.121717258390051	0.0	0.878282741609949
IL4_DICE_TREG_NAI	L12_ATOPIC	rs17690965	132030667	132694975	0.69445583213	0.15806745119678	1.11584085551e-05	-0.0957207	0.00849375	1.85524e-29	0.0	0.0	0.124617699806541	0.0	0.875382300193457
IL4_DICE_TREG_NAI	L12_ATOPIC	rs2023823	132032806	132697114	0.69445583213	0.15806745119678	1.11584085551e-05	-0.0957297	0.00849296	1.81134000000000	0.0	0.0	0.124617699806541	0.0	0.875382300193457
IL4_DICE_TREG_NAI	L12_ATOPIC	rs2237057	132043944	132708252	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0984898	0.00850305	5.03385e-31	0.0	0.0	0.144437209343256	0.0	0.855562790656744
IL4_DICE_TREG_NAI	L12_ATOPIC	rs11740584	132046789	132711097	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0984963	0.00850134	4.85400000000000	0.0	0.0	0.144437209343256	0.0	0.855562790656744
IL4_DICE_TREG_NAI	L12_ATOPIC	rs6865485	132035572	132699880	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0983561	0.00850186	5.93198e-31	0.0	0.0	0.144437209343256	0.0	0.855562790656744
IL4_DICE_TREG_NAI	L12_ATOPIC	rs3213639	132038199	132702507	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0984279	0.00850296	5.47142e-31	0.0	0.0	0.144437209343256	0.0	0.855562790656744
IL4_DICE_TREG_NAI	L12_ATOPIC	rs1468215	132036489	132700797	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0983918	0.00850101	5.57313999999999	0.0	0.0	0.144437209343256	0.0	0.855562790656744
IL4_DICE_TREG_NAI	L12_ATOPIC	rs2285700	132039132	132703440	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0984916	0.00850099	4.85847999999999	0.0	0.0	0.144437209343256	0.0	0.855562790656744
IL4_DICE_TREG_NAI	L12_ATOPIC	rs10062446	132040374	132704682	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0985006	0.00850101	4.80065e-31	0.0	0.0	0.144437209343256	0.0	0.855562790656744
IL4_DICE_TREG_NAI	L12_ATOPIC	rs4705964	132046165	132710473	0.702963168517	0.16143648771835	1.33416305877e-05	-0.098533	0.0085012	4.60575e-31	0.0	0.0	0.144437209343256	0.0	0.855562790656744
IL4_DICE_TREG_NAI	L12_ATOPIC	rs6864565	132047971	132712279	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0984997	0.00850099	4.80397e-31	0.0	0.0	0.144437209343256	0.0	0.855562790656744
IL4_DICE_TREG_NAI	L12_ATOPIC	rs6864396	132047875	132712183	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0985084	0.008502	4.82392e-31	0.0	0.0	0.144437209343256	0.0	0.855562790656744
IL4_DICE_TREG_NAI	L12_ATOPIC	rs10036532	132045815	132710123	0.702963168517	0.16143648771835	1.33416305877e-05	-0.0985193	0.00850132	4.70111e-31	0.0	0.0	0.144437209343256	0.0	0.855562790656744
IL4_DICE_TREG_NAI	L12_ATOPIC	rs187681	132101903	132766211	-0.718216176316	0.16531697280451	1.3960636769e-05	0.0794765	0.00954181	8.13392e-17	2.49459874716967	1.30998492100517	0.159967108418987	0.0	0.840032891555934
IL4_DICE_TREG_NAI	L12_ATOPIC	rs10477742	132049789	132714097	0.703053515399	0.16194807168493	1.41688553808e-05	-0.0984882	0.00850097	4.87978e-31	0.0	0.0	0.149443973662306	0.0	0.850556026337690
IL4_DICE_TREG_NAI	L12_ATOPIC	rs2299004	132057838	132722146	0.677525207284	0.16023790105244	2.35520573784e-05	-0.0955165	0.00848956	2.28876e-29	0.0	0.0	0.173975319506461	0.0	0.826024680493530
IL4_DICE_TREG_NAI	L12_ATOPIC	rs7734673	132058466	132722774	0.677525207284	0.16023790105244	2.35520573784e-05	-0.0956615	0.00848957	1.88538e-29	0.0	0.0	0.173975319506461	0.0	0.826024680493530
IL4_DICE_TREG_NAI	L12_ATOPIC	rs10073911	132035399	132699707	0.679669207524	0.16093548480232	2.4081493953e-05	-0.0955687	0.0084908	2.17470000000000	0.0	0.0	0.177909282789591	0.0	0.822090717210404
IL4_DICE_TREG_NAI	L12_ATOPIC	rs2897442	132049027	132713335	0.679669207524	0.16093548480232	2.4081493953e-05	-0.0956841	0.00848999	1.84120000000000	0.0	0.0	0.177909282789591	0.0	0.822090717210404
IL4_DICE_TREG_NAI	L12_ATOPIC	rs10594222	132054753	132719061	0.677386571374	0.16053902356710	2.44897751985e-05	-0.095602	0.00848907	2.02628e-29	0.0	0.0	0.177566955455601	0.0	0.822433044544398

(showing first 40 of 42 rows; full data in the accompanying CSV)

Supplementary Table 29 - r5_il4_locus_coloc_IL13_DICE_TH1_ALLERG_ASTHMA_persnp

ST29_r5_il4_locus_coloc_IL13_DICE_TH1_ALLERG_ASTHMA_persnp.csv

trait1	trait2	snp	pos37	pos38	beta_x	se_x	p_x	beta_y	se_y	p_y	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4
IL13_DICE_TH1	ALLERG_ASTHMA	rs78330488	131813917	132478225	1.35032520591	0.27559087562378	9.59607815202e-07	0.162853	0.0638199	0.0107179	0.99718472544942	0.00136113237298	0.00061486573162	0.0	0.000839276445967 4579
IL13_DICE_TH1	ALLERG_ASTHMA	rs75893502	131813827	132478135	1.25718354702	0.26974830498392	3.15319617726e-06	0.162894	0.0638209	0.0106997	0.99755054201621	0.00113364653715	0.00061589255850	0.0	0.000699918888111 99992
IL13_DICE_TH1	ALLERG_ASTHMA	rs7725926	132938624	133602933	-0.814568045843	0.17822089348961	4.86438790641e-06	0.0101586	0.0133659	0.447234	0.99413720142759	0.00578268464644	1.17524461953213	0.0	6.836147975782054e-05
IL13_DICE_TH1	ALLERG_ASTHMA	rs7727007	132935260	133599569	-0.802503321888	0.18542146702155	1.50472370011e-05	0.0116162	0.013309	0.382767	0.99679677384584	0.00314974386067	1.28567358438756	0.0	4.062555764129978e-05
IL13_DICE_TH1	ALLERG_ASTHMA	rs55674913	132927135	133591444	-0.776087652761	0.18262243817141	2.14072303946e-05	0.0121405	0.0132896	0.360963	0.99701905732677	0.00292852548524	1.33130409775410	0.0	3.910414700930776e-05
IL13_DICE_TH1	ALLERG_ASTHMA	rs80082968	132925277	133589586	-0.776087652761	0.18262243817141	2.14072303946e-05	0.0121097	0.0132955	0.362394	0.99701916310077	0.00292852579592	1.32860973367704	0.0	3.902500595570196e-05

Supplementary Table 30 - r5_il4_locus_coloc_IL13_DICE_TH1_ALLERG_RHINITIS_persnp

ST30_r5_il4_locus_coloc_IL13_DICE_TH1_ALLERG_RHINITIS_persnp.csv

trait1	trait2	snp	pos37	pos38	beta_x	se_x	p_x	beta_y	se_y	p_y	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4
IL13_DICE_TH1	ALLERG_RHINITIS	rs78330488	131813917	132478225	1.35032520591	0.27559087562378	9.59607815202e-07	0.0599944	0.0593883	0.312397	0.99850193077610	0.00136293032552	5.71417835848920	0.0	7.799711477976120e-05
IL13_DICE_TH1	ALLERG_RHINITIS	rs75893502	131813827	132478135	1.25718354702	0.26974830498392	3.15319617726e-06	0.0598624	0.0593884	0.313463	0.99874312581419	0.00113500182537	5.70448600918997	0.0	6.482750034408396e-05
IL13_DICE_TH1	ALLERG_RHINITIS	rs7725926	132938624	133602933	-0.814568045843	0.178220893489611	4.86438790641e-06	0.0175312	0.0122709	0.153096	0.99406544880512	0.00578226727670	2.23395437615310	0.0	0.000129944374411.24517
IL13_DICE_TH1	ALLERG_RHINITIS	rs7727007	132935260	133599569	-0.802503321888	0.185421467021551	1.50472370011e-05	0.0196376	0.012221	0.108083	0.99672905722073	0.00314952988523	2.91867346148463	0.0	9.222615941211768e-05
IL13_DICE_TH1	ALLERG_RHINITIS	rs55674913	132927135	133591444	-0.776087652761	0.182622438171411	2.14072303946e-05	0.0196068	0.0122047	0.108165	0.99695693319862	0.00292834300919	2.91378191956672	0.0	8.558597297793721e-05
IL13_DICE_TH1	ALLERG_RHINITIS	rs80082968	132925277	133589586	-0.776087652761	0.182622438171411	2.14072303946e-05	0.0195844	0.0122105	0.108735	0.99695735311867	0.00292834424262	2.90308536440720	0.0	8.527178505786952e-05

Supplementary Table 31 - r5_il4_locus_coloc_IL13_DICE_TH1_L12_ATOPIC_persnp

ST31_r5_il4_locus_coloc_IL13_DICE_TH1_L12_ATOPIC_persnp.csv

trait1	trait2	snp	pos37	pos38	beta_x	se_x	p_x	beta_y	se_y	p_y	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4
IL13_DICE_TH1	L12_ATOPIC	rs78330488	131813917	132478225	1.35032520591	0.27559087562378	9.59607815202e-07	-0.130009	0.0443974	0.00340824	0.99519924178152	0.00135842223711	0.00145554847681	0.0	0.00198678750454766656
IL13_DICE_TH1	L12_ATOPIC	rs75893502	131813827	132478135	1.25718354702	0.26974830498392	3.15319617726e-06	-0.130041	0.0443978	0.00340048	0.99575116501677	0.00113160167083	0.00145908504227	0.0	0.0016581482701 1.792
IL13_DICE_TH1	L12_ATOPIC	rs7725926	132938624	133602933	-0.814568045843	0.17822089348961	4.86438790641e-06	0.0235279	0.00890754	0.00825791	0.99292689956791	0.00577564457796	0.00019033245387	0.0	0.00110712340024 2554
IL13_DICE_TH1	L12_ATOPIC	rs7727007	132935260	133599569	-0.802503321888	0.18542146702155	1.50472370011e-05	0.0228394	0.0088719	0.0100427	0.99618734937036	0.00314781816122	0.00015982065969	0.0	0.00050501180872 0973
IL13_DICE_TH1	L12_ATOPIC	rs55674913	132927135	133591444	-0.776087652761	0.18262243817141	2.14072303946e-05	0.0230577	0.00886071	0.00926168	0.99639779698747	0.00292670066883	0.00017156567773	0.0	0.00050393666595 37334
IL13_DICE_TH1	L12_ATOPIC	rs80082968	132925277	133589586	-0.776087652761	0.18262243817141	2.14072303946e-05	0.0232181	0.00886483	0.00881556	0.99636725496854	0.00292661095832	0.00017934559659	0.0	0.00052678847653 80645

Supplementary Table 32 - r5_il4_il13_direction

ST32_r5_il4_il13_direction.csv

gene	lead_snp	eqtl_source	eqtl_beta	eqtl_se	eqtl_p	outcome	gwas_beta	gwas_se	gwas_p	wald_beta	wald_se	wald_p	direction
IL4	rs2070874	eQTLGen	-0.1398591813247735	0.0088456885285417	2.6129e-56	ALLERG_ASTHMA	0.0753801	0.0132661	1.32993e-08	-0.5389714088555713	0.0948532650794956	1.330043053471286e-	protective
IL4	rs2070874	eQTLGen	-0.1398591813247735	0.0088456885285417	2.6129e-56	ALLERG_RHINITIS	0.0513078	0.0122244	2.7029e-05	-0.3668532842392074	0.0874050590330079	2.7027934915935248	protective
IL4	rs2070874	eQTLGen	-0.1398591813247735	0.0088456885285417	2.6129e-56	L12_ATOPIC	0.0960295	0.00886117	2.29615e-27	-0.6866156307393606	0.0633577997244461	0.0	protective
IL13	rs78330488	DICE_TH1	1.35032520591	0.2755908756237814	9.59607815202e-07	ALLERG_ASTHMA	0.162853	0.0638199	0.0107179	0.1206027994495233	0.0472626147543406	0.0107180648710294	risk
IL13	rs78330488	DICE_TH1	1.35032520591	0.2755908756237814	9.59607815202e-07	ALLERG_RHINITIS	0.0599944	0.0593883	0.312397	0.0444295935063798	0.0439807386695248	0.3123967422904969	risk
IL13	rs78330488	DICE_TH1	1.35032520591	0.2755908756237814	9.59607815202e-07	L12_ATOPIC	-0.130009	0.0443974	0.00340824	-0.0962797698146984	0.0328790426229806	0.0034018710702417	protective

Supplementary Table 33 - r5_il4_celltype_eqtl

ST33_r5_il4_celltype_eqtl.csv

gene	source	cell_or_tissue	is_focus	any_eqtl_for_gene	lead_snp_present	top_snp	top_p	top_beta	direction	note
IL13	DICE	TH1	False	True	True	rs78330488	9.59607815202e-07	1.35032520591	up	
IL4/IL13	DICE	TH2	True	False	False					
IL4/IL13	DICE	TH17	False	False	False					
IL4/IL13	DICE	TREG	False	False	False					vcf not on disk
IL4/IL13	DICE	CD4_NAIVE	False	False	False					
IL4/IL13	DICE	CD8_NAIVE	False	False	False					
IL4/IL13	DICE	B_CELL_NAIVE	False	False	False					
IL4/IL13	DICE	MONOCYTES	True	False	False					
IL4/IL13	DICE	M2	False	False	False					
IL4/IL13	DICE	NK	False	False	False					
IL4/IL13	DICE	TFH	False	False	False					
IL4/IL13	DICE	THSTAR	False	False	False					
IL4/IL13	DICE	TREG_MEM	False	False	False					
IL4	DICE	TREG_NAIVE	True	True	False	rs370314650	9.43773827539e-07	0.779455135928	up	
IL4/IL13	DICE	CD4_STIM	False	False	False					
IL4/IL13	DICE	CD8_STIM	False	False	False					
IL4/IL13	OneK1K	cd4nc	False	False	False					absent in top-eSNP table
IL4/IL13	OneK1K	cd4et	False	False	False					absent in top-eSNP table
IL4/IL13	OneK1K	cd8et	False	False	False					absent in top-eSNP table
IL4/IL13	OneK1K	monoc	False	False	False					absent in top-eSNP table
IL4/IL13	OneK1K	nk	False	False	False					absent in top-eSNP table
IL4/IL13	OneK1K	bin	False	False	False					absent in top-eSNP table
IL4	GTEx	Adipose_Subcutaneous	False	False	False					
IL13	GTEx	Adipose_Subcutaneous	False	True	False	chr5_133249836_C_T_b38	0.00072946	-0.181795	down	
IL4	GTEx	Adipose_Visceral_Omentum	False	False	False					
IL13	GTEx	Adipose_Visceral_Omentum	False	True	False	chr5_133312780_G_A_b38	0.00230274	0.554313	up	
IL4	GTEx	Adrenal_Gland	False	False	False					
IL13	GTEx	Adrenal_Gland	False	False	False					
IL4	GTEx	Artery_Aorta	False	True	False	chr5_132721182_A_C_b38	3.22056e-05	0.287292	up	
IL13	GTEx	Artery_Aorta	False	True	False	chr5_132488235_T_C_b38	0.000239396	0.245077	up	

Supplementary Table 34 - r5_celltype_mr_matrix

ST34_r5_celltype_mr_matrix.csv

cell	gene	snp_lead	instrument	beta	se	p	ea	oa	fstat	snp_mr	outcome	beta_outcome	se_outcome	p_outcome	wald_beta	wald_se	wald_p	or	or_lower	or_upper	wald_q	significant
B_CELL_NAIVE	IL4		no instrument																			
B_CELL_NAIVE	IL5		no instrument																			
B_CELL_NAIVE	IL13		no instrument																			
B_CELL_NAIVE	IL33		no instrument																			
B_CELL_NAIVE	TSLP	rs6898145	pass	0.8081025900	0.1959097088	3.7094002634	T	G	17.01457397	rs6898145	ALLERG_ASTH	-0.0617131	0.0198496	0.00187702	-0.076367902	0.0245632178	0.0018769951	0.9264752910	0.8829278341	0.9721705804	0.0054603496	True
B_CELL_NAIVE	TSLP	rs6898145	pass	0.8081025900	0.1959097088	3.7094002634	T	G	17.01457397	rs6898145	ALLERG_RHIN	-0.0852463	0.0183992	3.60139e-06	-0.105489452	0.0227683962	3.6013899329	0.8998839640	0.8606085622	0.9409517687	2.3048895576	True
B_CELL_NAIVE	TSLP	rs6898145	pass	0.8081025900	0.1959097088	3.7094002634	T	G	17.01457397	rs6898145	L12_ATOPIC	0.00784719	0.0131511	0.550711	0.009710635	0.0162740475	0.5507113165	1.0097579370	0.9780578128	1.0424855035	0.7049104851	False
B_CELL_NAIVE	GATA3	rs3781093	pass	0.7956097554	0.1704566380	3.0485718650	C	T	21.78574229	rs3781093	ALLERG_ASTH	0.0399063	0.0155041	0.0100554	0.050158133	0.0194870662	0.0100553765	1.0514373502	1.0120353315	1.0923734247	0.0244984335	True
B_CELL_NAIVE	GATA3	rs3781093	pass	0.7956097554	0.1704566380	3.0485718650	C	T	21.78574229	rs3781093	ALLERG_RHIN	0.0340449	0.0143677	0.0178098	0.042790953	0.0180587277	0.0178100268	1.0437196860	1.0074232631	1.0813238305	0.0474934049	True
B_CELL_NAIVE	GATA3	rs3781093	pass	0.7956097554	0.1704566380	3.0485718650	C	T	21.78574229	rs3781093	L12_ATOPIC	-0.044194	0.0105156	2.63736e-05	-0.055547332	0.0132170325	2.6374029708	0.9459672473	0.9217762934	0.9707930645	0.0004219844	True
B_CELL_NAIVE	STAT6		no instrument																			
B_CELL_NAIVE	FOXP3		no instrument																			
B_CELL_NAIVE	RORC		no instrument																			
B_CELL_NAIVE	IL6		no instrument																			
B_CELL_NAIVE	TNF		no instrument																			
B_CELL_NAIVE	IL1R1		no instrument																			
B_CELL_NAIVE	STAT5A		no instrument																			
B_CELL_NAIVE	ARMS2		no instrument																			
B_CELL_NAIVE	OCA2		no instrument																			
B_CELL_NAIVE	COL1A1		no instrument																			
CD4_NAIVE	IL4		no instrument																			
CD4_NAIVE	IL5	rs13165038	pass	0.7290291592	0.1708154151	1.9729043078	C	T	18.21527265	rs13165038	ALLERG_ASTH	0.0782268	0.0136517	1.00323e-08	0.107302703	0.0187258627	1.0031971253	1.1132711949	1.0731518930	1.1548903390	6.4204616023	True
CD4_NAIVE	IL5	rs13165038	pass	0.7290291592	0.1708154151	1.9729043078	C	T	18.21527265	rs13165038	ALLERG_RHIN	0.0371228	0.0125764	0.00315937	0.050920871	0.0172508874	0.0031594518	1.0522396276	1.0172563311	1.0884259944	0.0144382043	True
CD4_NAIVE	IL5	rs13165038	pass	0.7290291592	0.1708154151	1.9729043078	C	T	18.21527265	rs13165038	L12_ATOPIC	0.0292695	0.00916547	0.00140582	0.040148599	0.0125721581	0.0014058209	1.0409654500	1.0156280296	1.0669349760	0.0074977118	True
CD4_NAIVE	IL13		no instrument																			
CD4_NAIVE	IL33		no instrument																			
CD4_NAIVE	TSLP		no instrument																			
CD4_NAIVE	GATA3		no instrument																			
CD4_NAIVE	STAT6		no instrument																			
CD4_NAIVE	FOXP3		no instrument																			
CD4_NAIVE	RORC		no instrument																			
CD4_NAIVE	IL6		no instrument																			
CD4_NAIVE	TNF		no instrument																			
CD4_NAIVE	IL1R1		no instrument																			
CD4_NAIVE	STAT5A		no instrument																			
CD4_NAIVE	ARMS2		no instrument																			
CD4_NAIVE	OCA2		no instrument																			
CD4_NAIVE	COL1A1		no instrument																			
CD4_STIM	IL4		no instrument																			
CD4_STIM	IL5		no instrument																			

(showing first 40 of 304 rows; full data in the accompanying CSV)

Supplementary Table 35 - r5_celltype_replication

ST35_r5_celltype_replication.csv

cell	gene	snp	ea	oa	wald_beta	outcome	onek1k_cell	onek1k_snp	onek1k_beta	onek1k_p	allele_aligned	same_direction	exact_snp	note	onek1k_beta_aligned
B_CELL_NAIVE	TSLP	rs6898145	T	G	-0.07636790273446	ALLERG_ASTHMA	bin						False	SNP absent in OneK	
B_CELL_NAIVE	GATA3	rs3781093	C	T	0.05015813308608	ALLERG_ASTHMA	bin	rs3781093	0.04141063332964	0.19477456770371	True	True	True		0.0414106333296487
B_CELL_NAIVE	TSLP	rs6898145	T	G	-0.10548945275595	ALLERG_RHINITIS	bin						False	SNP absent in OneK	
B_CELL_NAIVE	GATA3	rs3781093	C	T	0.04279095343598	ALLERG_RHINITIS	bin	rs3781093	0.04141063332964	0.19477456770371	True	True	True		0.0414106333296487
B_CELL_NAIVE	GATA3	rs3781093	C	T	-0.05554733296763	L12_ATOPIC	bin	rs3781093	0.04141063332964	0.19477456770371	True	False	True		0.0414106333296487
CD4_STIM	TSLP	rs2289277	G	C	0.11760327680559	ALLERG_ASTHMA	cd4et						False	SNP absent in OneK	
TFH	IL1R1	rs4851546	G	C	0.06195442341180	ALLERG_ASTHMA	cd4et	rs4851546	0.00904447186593	0.77712292898700	True	True	True		0.009044471865937
TH1	IL13	rs78330488	C	T	0.12060279944952	ALLERG_ASTHMA	cd4et						False	SNP absent in OneK	
TH2	COL1A1	rs2214615	T	C	0.06150835579396	ALLERG_ASTHMA	cd4et						False	SNP absent in OneK	
THSTAR	IL5	rs7734334	A	C	0.08386682214462	ALLERG_ASTHMA	cd4et						False	SNP absent in OneK	
CD4_STIM	TSLP	rs2289277	G	C	0.11370329344407	ALLERG_RHINITIS	cd4et						False	SNP absent in OneK	
TFH	IL1R1	rs4851546	G	C	0.06038727845414	ALLERG_RHINITIS	cd4et	rs4851546	0.00904447186593	0.77712292898700	True	True	True		0.009044471865937
THSTAR	IL5	rs7734334	A	C	0.04102541287176	ALLERG_RHINITIS	cd4et						False	SNP absent in OneK	
TFH	STAT6	rs3024949	TGGGGAGGCAA	T	-0.04774453910001	L12_ATOPIC	cd4et						False	SNP absent in OneK	
TH1	IL13	rs78330488	C	T	-0.09627976981469	L12_ATOPIC	cd4et						False	SNP absent in OneK	
CD4_NAIVE	IL5	rs13165038	C	T	0.10730270388053	ALLERG_ASTHMA	cd4nc						False	SNP absent in OneK	
TREG_NAIVE	IL4	rs370314650	G	GTTCC	-0.08332333319309	ALLERG_ASTHMA	cd4nc						False	SNP absent in OneK	
TREG_NAIVE	TSLP	rs3806932	G	A	0.17897158360883	ALLERG_ASTHMA	cd4nc	rs3806932	-0.06007158604159	0.05986888192437	True	False	True		-0.0600715860415964
CD4_NAIVE	IL5	rs13165038	C	T	0.05092087130774	ALLERG_RHINITIS	cd4nc						False	SNP absent in OneK	
TREG_NAIVE	IL4	rs370314650	G	GTTCC	-0.04212070520378	ALLERG_RHINITIS	cd4nc						False	SNP absent in OneK	
TREG_NAIVE	TSLP	rs3806932	G	A	0.17253735970923	ALLERG_RHINITIS	cd4nc	rs3806932	-0.06007158604159	0.05986888192437	True	False	True		-0.0600715860415964
CD4_NAIVE	IL5	rs13165038	C	T	0.04014859985620	L12_ATOPIC	cd4nc						False	SNP absent in OneK	
TREG_NAIVE	IL4	rs370314650	G	GTTCC	-0.13105693360834	L12_ATOPIC	cd4nc						False	SNP absent in OneK	
CD8_NAIVE	TSLP	rs3806933	T	C	0.16863329596118	ALLERG_ASTHMA	cd8et						False	SNP absent in OneK	
CD8_STIM	TSLP	rs2289277	G	C	0.11905368261893	ALLERG_ASTHMA	cd8et						False	SNP absent in OneK	
CD8_NAIVE	TSLP	rs3806933	T	C	0.16215880764411	ALLERG_RHINITIS	cd8et						False	SNP absent in OneK	
CD8_STIM	TSLP	rs2289277	G	C	0.11510560061005	ALLERG_RHINITIS	cd8et						False	SNP absent in OneK	
CD8_NAIVE	STAT6	rs3024949	TGGGGAGGCAA	T	-0.04583603877425	L12_ATOPIC	cd8et						False	SNP absent in OneK	
MONOCYTES	IL1R1	rs549445698	A	ATAT	-0.05838304994698	ALLERG_ASTHMA	monoc						False	SNP absent in OneK	
MONOCYTES	IL1R1	rs549445698	A	ATAT	-0.04259303946398	ALLERG_RHINITIS	monoc						False	SNP absent in OneK	
NK	IL5	rs368020909	C	CAGG	0.11147788586455	ALLERG_ASTHMA	nk						False	SNP absent in OneK	
NK	IL5	rs368020909	C	CAGG	0.05174412034184	ALLERG_RHINITIS	nk						False	SNP absent in OneK	
NK	IL5	rs368020909	C	CAGG	0.04253550378624	L12_ATOPIC	nk						False	SNP absent in OneK	

Supplementary Table 36 - r5_ewas_convergence

ST36_r5_ewas_convergence.csv

cpg	gene	in_type2_panel	type2_match
cg00010445	RFX1	False	
cg00014561	NUP93	False	
cg00019841	MAPK4	False	
cg00021513	-	False	
cg00028211	-	False	
cg00040588	MXRA8	False	
cg00052385	ST3GAL1	False	
cg00055764	RIMBP2	False	
cg00057667	-	False	
cg00058708	UBXN2A	False	
cg00068153	CACNA1S	False	
cg00068428	-	False	
cg00073543	-	False	
cg00085732	SLC13A2	False	
cg00095018	MAPK13	False	
cg00095523	-	False	
cg00095688	PRKCZ	False	
cg00109344	RGS3	False	
cg00110695	FBXL4	False	
cg00113315	SLC2A5	False	
cg00114012	SLC2A8	False	
cg00115178	CAMK1D	False	
cg00115288	EPS8L2	False	
cg00127505	-	False	
cg00138126	PMEPA1	False	
cg00144186	FJX1	False	
cg00145969	FAM5C	False	
cg00152644	SPRR2E	False	
cg00153082	-	False	
cg00154119	-	False	
cg00170714	MLX;PSMC3IP	False	
cg00178661	IGSF3	False	
cg00186499	RBM38	False	
cg00192031	-	False	
cg00193959	DOCK1	False	
cg00199059	TP53RK	False	
cg00204465	MYO15B	False	
cg00207280	TRIB1	False	
cg00217953	SEMA4C	False	
cg00220721	PRR5L	False	

(showing first 40 of 4716 rows; full data in the accompanying CSV)

Supplementary Table 37 - r5_ewas_overlap_enrichment

ST37_r5_ewas_overlap_enrichment.csv

analysis	n_age	n_allergy	n_background	overlap	expected_overlap	perm_p	perm_n	fisher_p	fisher_or	type2_cpgs_in_overlap	type2_bg_cpgs	type2_genes_hit
all_tissues	301823.0	13924.0	724456	4716	5801.02	1.0	1000	8.263804790235691e-8	0.7126763330403711			
all_tissues | type2 enri			724456	4716			0	1.978249637410501e-05	5.641451675885224	10.0	281.0	GATA3;IL1RL1;IL4;IL5RA;R
blood_stratified	301684.0	434.0	713949	241	183.39	0.0009990009990009991	1000	3.313773986304761e-05	1.7069768678631603			
blood_stratified | type2			713949	241			0	5.203826584861716e-06	55.36684360836903	5.0	278.0	IL1RL1;IL4;IL5RA

Supplementary Table 38 - r5_ewas_overlap_genes

ST38_r5_ewas_overlap_genes.csv

scope	cpg	gene	in_type2_panel	type2_match
all_tissues	cg00010445	RFX1	False	
all_tissues	cg00014561	NUP93	False	
all_tissues	cg00019841	MAPK4	False	
all_tissues	cg00021513	-	False	
all_tissues	cg00028211	-	False	
all_tissues	cg00040588	MXRA8	False	
all_tissues	cg00052385	ST3GAL1	False	
all_tissues	cg00055764	RIMBP2	False	
all_tissues	cg00057667	-	False	
all_tissues	cg00058708	UBXN2A	False	
all_tissues	cg00068153	CACNA1S	False	
all_tissues	cg00068428	-	False	
all_tissues	cg00073543	-	False	
all_tissues	cg00085732	SLC13A2	False	
all_tissues	cg00095018	MAPK13	False	
all_tissues	cg00095523	-	False	
all_tissues	cg00095688	PRKCZ	False	
all_tissues	cg00109344	RGS3	False	
all_tissues	cg00110695	FBXL4	False	
all_tissues	cg00113315	SLC2A5	False	
all_tissues	cg00114012	SLC2A8	False	
all_tissues	cg00115178	CAMK1D	False	
all_tissues	cg00115288	EPS8L2	False	
all_tissues	cg00127505	-	False	
all_tissues	cg00138126	PMEPA1	False	
all_tissues	cg00144186	FJX1	False	
all_tissues	cg00145969	FAM5C	False	
all_tissues	cg00152644	SPRR2E	False	
all_tissues	cg00153082	-	False	
all_tissues	cg00154119	-	False	
all_tissues	cg00170714	MLX;PSMC3IP	False	
all_tissues	cg00178661	IGSF3	False	
all_tissues	cg00186499	RBM38	False	
all_tissues	cg00192031	-	False	
all_tissues	cg00193959	DOCK1	False	
all_tissues	cg00199059	TP53RK	False	
all_tissues	cg00204465	MYO15B	False	
all_tissues	cg00207280	TRIB1	False	
all_tissues	cg00217953	SEMA4C	False	
all_tissues	cg00220721	PRR5L	False	

(showing first 40 of 4957 rows; full data in the accompanying CSV)

Supplementary Table 39 - r5_ewas_study_classification

ST39_r5_ewas_study_classification.csv

trait	tissue	study_id	is_age	is_allergy	is_blood
Smoking	Lymphoblasts	22232023_smoking_current	False	False	False
Smoking	Whole blood	23070629_smoking_females	False	False	True
Smoking	Whole blood	23070629_smoking_males_o	False	False	True
Smoking	Peripheral blood	23175441_smoking_breast_	False	False	True
Smoking	Peripheral blood	23175441_smoking_colon_c	False	False	True
Age	Whole blood	23177740_Battram-T_age_e	True	False	True
Child abuse	Saliva	23332324_child_abuse	False	False	False
Rheumatoid arthritis	Whole blood	23334450_Battram-T_rheum	False	False	True
Rheumatoid arthritis	Whole blood	23334450_rheumatoid_arth	False	False	True
Clear cell renal carcino	Clear cell renal carcino	23526956_clear_cell_rena	False	False	False
Smoking	Whole blood	23691101_smoking_current	False	False	True
Smoking	Whole blood	23691101_smoking_current	False	False	True
Smoking	Whole blood	23691101_smoking_former_	False	False	True
Ethnicity	Lymphoblasts	23792949_Battram-T_ethni	False	False	False
Right vs left colon aden	Colon adenoma	23868178_right_vs_left_c	False	False	False
Right vs left colon aden	Colon adenomas	23868178_right_vs_left_c	False	False	False
M00054_tryptophan	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M00054_tryptophan	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M00060_leucine	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M00542_3-hydroxybutyrate	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M00542_3-hydroxybutyrate	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M00555_adenosine	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M00555_adenosine	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M00584_mannose	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01105_linoleate_18:2n6	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01105_linoleate_18:2n6	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01125_isoleucine	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01125_isoleucine	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01299_tyrosine	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01299_tyrosine	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01302_methionine	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01302_methionine	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01336_palmitate_16:0	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01358_stearate_18:0	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01444_pipecolate	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01494_5-oxoproline	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01604_urate	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M01898_proline	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M02137_biliverdin	Whole blood	24014485_Petersen-A-K_m0	False	False	True
M02730_gamma-glutamylglu	Whole blood	24014485_Petersen-A-K_m0	False	False	True

(showing first 40 of 7639 rows; full data in the accompanying CSV)

Supplementary Table 40 - r5_gene_coords_extended

ST40_r5_gene_coords_extended.csv

gene	ensembl_id	chr	start	end	strand	source
ADA	ENSG00000196839	20	44584896	44652252	-1	ensembl_rest
ARTN	ENSG00000117407	1	43933272	43937244	1	ensembl_rest
CASP8	ENSG00000064012	2	201233443	201361836	1	ensembl_rest
CCL2	ENSG00000108691	17	34255274	34257208	1	ensembl_rest
CCL3	ENSG00000277632	17	36088256	36090169	-1	ensembl_rest
CD274	ENSG00000120217	9	5450503	5470566	1	ensembl_rest
CST5	ENSG00000170367	20	23875934	23880086	-1	ensembl_rest
CX3CL1	ENSG00000006210	16	57372457	57385044	1	ensembl_rest
CXCL8	ENSG00000169429	4	73740519	73747379	1	ensembl_rest
GDNF	ENSG00000168621	5	37812677	37840063	-1	ensembl_rest
HGF	ENSG00000019991	7	81699010	81770438	-1	ensembl_rest
IL10	ENSG00000136634	1	206767602	206774541	-1	ensembl_rest
IL17A	ENSG00000112115	6	52186375	52190638	1	ensembl_rest
IL17C	ENSG00000124391	16	88638571	88640468	1	ensembl_rest
IL18	ENSG00000150782	11	112142701	112164391	-1	ensembl_rest
IL1A	ENSG00000115008	2	112773916	112784500	-1	ensembl_rest
IL20	ENSG00000162891	1	206865621	206869223	1	ensembl_rest
IL22RA1	ENSG00000142677	1	24119769	24143301	-1	ensembl_rest
IL24	ENSG00000162892	1	206897443	206904139	1	ensembl_rest
IL7	ENSG00000104432	8	78675743	78805527	-1	ensembl_rest
KITLG	ENSG00000049130	12	88492789	88580851	-1	ensembl_rest
LTA	ENSG00000226979	6	31572054	31574324	1	ensembl_rest
MMP1	ENSG00000196611	11	102789401	102798167	-1	ensembl_rest
MMP10	ENSG00000166670	11	102770502	102780628	-1	ensembl_rest
NGF	ENSG00000134259	1	115285904	115338770	-1	ensembl_rest
NRTN	ENSG00000171119	19	5805053	5828324	1	ensembl_rest
NTF3	ENSG00000185652	12	5432088	5521536	1	ensembl_rest
OSM	ENSG00000099985	22	30262829	30266858	-1	ensembl_rest
PLAU	ENSG00000122861	10	73909177	73917501	1	ensembl_rest
S100A12	ENSG00000163221	1	153373703	153375660	-1	ensembl_rest
SLAMF1	ENSG00000117090	1	160608100	160647056	-1	ensembl_rest
TGFA	ENSG00000163235	2	70447284	70554193	-1	ensembl_rest
TGFB1	ENSG00000105329	19	41288203	41353961	-1	ensembl_rest
TNFRSF11B	ENSG00000164761	8	118923557	118952251	-1	ensembl_rest
TNFSF10	ENSG00000121858	3	172505503	172523529	-1	ensembl_rest
TNFSF12	ENSG00000239697	17	7548508	7557890	1	ensembl_rest
TNFSF14	ENSG00000125735	19	6658085	6670621	-1	ensembl_rest
VEGFA	ENSG00000112715	6	43770184	43786487	1	ensembl_rest

Supplementary Table 41 - r5_pqtl_instruments

ST41_r5_pqtl_instruments.csv

chrom	pos	ea	oa	beta	se	eaf	p	variant_id	rsid	n	fstat	snp	protein	tss	instrument_note	build_flag	family	f_pass
20	44935764	A	G	-0.1703	0.0516	0.0214	0.000966	20_44935764_A_	rs138477745	14288	10.89255828976	rs138477745	ADA	44652252	ok	grch38_ok	exploratory	True
1	44346226	T	C	-0.3087	0.0908	0.01	0.000674	1_44346226_T_C	rs143259274	7837	11.55850225115	rs143259274	ARTN	43933272	ok	grch38_ok	exploratory	True
2	201677213	T	C	0.2275	0.0481	0.0195	2.25e-06	2_201677213_T_	rs544253385	14089	22.37034331628	rs544253385	CASP8	201233443	ok	grch38_ok	exploratory	True
17	34188197	T	C	-0.4149	0.1306	0.9931	0.00149	17_34188197_T_	rs191123319	7285	10.09254084693	rs191123319	CCL2_MCP1	34255274	ok	grch38_ok	hypothesis	True
17	36223556	A	G	0.4067	0.0957	0.037	2.14e-05	17_36223556_A_	rs184164134	8717	18.06027958757	rs184164134	CCL3_MIP1A	36090169	ok	grch38_ok	exploratory	True
20	23847165	T	G	0.6649	0.0137	0.19	0.0	20_23847165_T_	rs6049191	14733	2355.437210293	rs6049191	CST5	23880086	ok	grch38_ok	exploratory	True
20	23854714	T	G	0.5303	0.0108	0.4548	0.0	20_23854714_T_	rs6036565	14708	2410.991855281	rs6036565	CST5	23880086	ok	grch38_ok	exploratory	True
20	23862056	T	C	-0.3478	0.012	0.6442	1.070000000000	20_23862056_T_	rs4075883	14288	840.0336111111	rs4075883	CST5	23880086	ok	grch38_ok	exploratory	True
16	57412802	C	G	0.2041	0.0118	0.424	4.989999999999	16_57412802_C_	rs671623	14295	299.1727233553	rs671623	CX3CL1	57372457	ok	grch38_ok	exploratory	True
16	57374418	T	C	0.192	0.012	0.3762	1.28e-57	16_57374418_T_	rs9921681	14295	256.0	rs9921681	CX3CL1	57372457	ok	grch38_ok	exploratory	True
16	57444576	T	G	0.1859	0.0132	0.2491	4.81e-45	16_57444576_T_	rs223899	14726	198.3402777777	rs223899	CX3CL1	57372457	ok	grch38_ok	exploratory	True
5	37868088	A	G	-0.3888	0.0198	0.0906	7.560000000000	5_37868088_A_G	rs11740708	14710	385.5867768595	rs11740708	GDNF	37840063	ok	grch38_ok	exploratory	True
5	37772566	T	G	-0.1953	0.0142	0.2147	4.850000000000	5_37772566_T_G	rs10472290	14288	189.1593433842	rs10472290	GDNF	37840063	ok	grch38_ok	exploratory	True
5	37893013	A	G	-0.2331	0.018	0.1301	2.35e-38	5_37893013_A_G	rs62360399	14288	167.7025000000	rs62360399	GDNF	37840063	ok	grch38_ok	exploratory	True
7	81359051	T	C	-0.1943	0.0242	0.0608	9.83e-16	7_81359051_T_C	rs5745687	14733	64.46364660883	rs5745687	HGF	81770438	ok	grch38_ok	exploratory	True
7	81377916	A	G	0.0935	0.0138	0.7651	1.24e-11	7_81377916_A_G	rs2214827	14286	45.90553455156	rs2214827	HGF	81770438	ok	grch38_ok	exploratory	True
7	81431551	A	G	-0.2035	0.0311	0.0394	6.01e-11	7_81431551_A_G	rs112490145	14727	42.81619296740	rs112490145	HGF	81770438	ok	grch38_ok	exploratory	True
12	68210754	A	G	-0.058	0.0152	0.2225	0.000136	12_68210754_A_	rs2160584	11784	14.56024930747	rs2160584	IFNG	68159748	ok	grch38_ok	hypothesis	True
1	206954566	A	G	-0.0996	0.0151	0.2042	4.22e-11	1_206954566_A_	rs12123181	14296	43.50756545765	rs12123181	IL10	206774541	ok	grch38_ok	exploratory	True
1	206955041	A	G	0.1049	0.0169	0.8554	5.4e-10	1_206955041_A_	rs3122605	14735	38.52809775568	rs3122605	IL10	206774541	ok	grch38_ok	exploratory	True
1	206926898	A	G	0.1591	0.0275	0.9451	7.23e-09	1_206926898_A_	rs74630871	14724	33.47148429752	rs74630871	IL10	206774541	ok	grch38_ok	exploratory	True
5	132457166	A	G	0.2699	0.0789	0.01	0.000624	5_132457166_A_	rs60312712	8844	11.70177552243	rs60312712	IL13	132656263	ok	grch38_ok	positive_control	True
6	52453340	A	T	-0.0501	0.0134	0.5578	0.000185	6_52453340_A_T	rs2239618	11336	13.97867008242	rs2239618	IL17A	52186375	ok	grch38_ok	exploratory	True
16	88684495	T	G	-0.105	0.0136	0.3783	1.16e-14	16_88684495_T_	rs17700884	11775	59.60748269896	rs17700884	IL17C	88638571	ok	grch38_ok	exploratory	True
16	88702487	T	C	-0.0958	0.0142	0.5254	1.51e-11	16_88702487_T_	rs8059449	11345	45.51497718706	rs8059449	IL17C	88638571	ok	grch38_ok	exploratory	True
16	88634563	A	G	-0.088	0.0151	0.7087	5.62e-09	16_88634563_A_	rs72809412	10725	33.96342265690	rs72809412	IL17C	88638571	ok	grch38_ok	exploratory	True
11	112025306	A	C	0.2039	0.0135	0.7618	1.53e-51	11_112025306_A	rs5744249	14742	228.1218655692	rs5744249	IL18	112164391	ok	grch38_ok	hypothesis	True
11	112080384	T	C	0.1577	0.0115	0.4761	8.480000000000	11_112080384_T	rs10891343	14743	188.0475614366	rs10891343	IL18	112164391	ok	grch38_ok	hypothesis	True
11	112033636	A	G	0.1398	0.012	0.3639	2.29e-31	11_112033636_A	rs7106524	14731	135.7225	rs7106524	IL18	112164391	ok	grch38_ok	hypothesis	True
2	112637801	T	C	-0.2183	0.0543	0.9772	5.81e-05	2_112637801_T_	rs114374674	11340	16.16247299465	rs114374674	IL1A	112784500	ok	grch38_ok	hypothesis	True
1	207010900	A	G	0.2598	0.0807	0.9863	0.00128	1_207010900_A_	rs2243195	9867	10.36409115407	rs2243195	IL20	206865621	ok	grch38_ok	exploratory	True
1	24522881	T	C	0.4005	0.1034	0.9894	0.000107	1_24522881_T_C	rs114879793	8631	15.00251132669	rs114879793	IL22RA1	24143301	ok	grch38_ok	exploratory	True
1	206896342	A	G	0.1674	0.0524	0.0213	0.0014	1_206896342_A_	rs4240846	10850	10.20583007983	rs4240846	IL24	206897443	ok	grch38_ok	exploratory	True
4	122406746	A	G	-0.5035	0.1392	0.0044	0.000298	4_122406746_A_	rs143837636	8140	13.08339577966	rs143837636	IL2	122456725	ok	grch38_ok	exploratory	True
9	5806889	A	G	-0.1354	0.0428	0.0259	0.00156	9_5806889_A_G	rs117980606	11793	10.00805747226	rs117980606	IL33	6215136	ok	grch38_ok	positive_control	True
5	132428808	T	C	0.6895	0.1991	0.003	0.000534	5_132428808_T_	rs575237428	7844	11.99294994224	rs575237428	IL4	132673986	ok	grch38_ok	positive_control	True
5	132613677	A	G	0.0702	0.0197	0.1672	0.000366	5_132613677_A_	rs55650607	11784	12.69818856450	rs55650607	IL5	132556838	ok	grch38_ok	positive_control	True
7	22304702	A	T	0.2074	0.057	0.0232	0.000274	7_22304702_A_T	rs118147027	13808	13.23938442597	rs118147027	IL6	22725884	ok	grch38_ok	hypothesis	True
8	79299642	A	G	0.0613	0.0169	0.154	0.000286	8_79299642_A_G	rs78574099	14736	13.15671720177	rs78574099	IL7	78805527	ok	grch38_ok	exploratory	True
4	74215450	A	G	0.2139	0.0416	0.9785	2.72e-07	4_74215450_A_G	rs74696018	14742	26.43838410687	rs74696018	IL8	73740519	ok	grch38_ok	hypothesis	True

(showing first 40 of 93 rows; full data in the accompanying CSV)

Supplementary Table 42 - r5_pqtl_instrument_qc

ST42_r5_pqtl_instrument_qc.csv

protein	gene	status	chromosome	base_pair_location	variant_id	rsid	effect_allele	other_allele	beta	standard_error	effect_allele_frequen	p_value	n	fstat	build_flag	n_snps_cis	n_snps_5mb	instrument_pass	protein_category
ADA	ADA	ok	20	44935764	20_44935764_A	rs138477745	A	G	-0.1703	0.0516	0.0214	0.000966	14288	10.89	grch38_cis	4751	46680	True	exploratory
ARTN	ARTN	ok	1	44346226	1_44346226_T_	rs143259274	T	C	-0.3087	0.0908	0.01	0.000674	7837	11.56	grch38_cis	3787	40722	True	exploratory
CASP8	CASP8	ok	2	201677213	2_201677213_T	rs544253385	T	C	0.2275	0.0481	0.0195	2.25e-06	14089	22.37	grch38_cis	3754	37040	True	exploratory
CCL2_MCP1	CCL2	ok	17	34188197	17_34188197_T	rs191123319	T	C	-0.4149	0.1306	0.9931	0.00149	7285	10.09	grch38_cis	3774	40761	True	hypothesis
CCL3_MIP1A	CCL3	ok	17	36223556	17_36223556_A	rs184164134	A	G	0.4067	0.0957	0.037	2.14e-05	8717	18.06	grch38_cis	3426	40772	True	exploratory
CST5	CST5	ok	20	23810508	20_23810508_T	rs11696133	T	C	-0.6554	0.0149	0.8166	0.0	14288	1934.82	grch38_cis	6200	33588	True	exploratory
CX3CL1	CX3CL1	ok	16	57412802	16_57412802_C	rs671623	C	G	0.2041	0.0118	0.424	4.98999999999	14295	299.17	grch38_cis	5247	49644	True	exploratory
GDNF	GDNF	ok	5	37868088	5_37868088_A_	rs11740708	A	G	-0.3888	0.0198	0.0906	7.56000000000	14710	385.59	grch38_cis	4968	47374	True	exploratory
HGF	HGF	ok	7	81359051	7_81359051_T_	rs5745687	T	C	-0.1943	0.0242	0.0608	9.83e-16	14733	64.46	grch38_cis	4720	46935	True	exploratory
IFNG	IFNG	ok	12	68210754	12_68210754_A	rs2160584	A	G	-0.058	0.0152	0.2225	0.000136	11784	14.56	grch38_cis	4716	44326	True	hypothesis
IL10	IL10	ok	1	206954566	1_206954566_A	rs12123181	A	G	-0.0996	0.0151	0.2042	4.22e-11	14296	43.51	grch38_cis	3046	43681	True	exploratory
IL13	IL13	ok	5	132457166	5_132457166_A	rs60312712	A	G	0.2699	0.0789	0.01	0.000624	8844	11.7	grch38_cis	4585	40971	True	positive_control
IL17A	IL17A	ok	6	52453340	6_52453340_A_	rs2239618	A	T	-0.0501	0.0134	0.5578	0.000185	11336	13.98	grch38_cis	4892	46950	True	exploratory
IL17C	IL17C	ok	16	88684495	16_88684495_T	rs17700884	T	G	-0.105	0.0136	0.3783	1.16e-14	11775	59.61	grch38_cis	6242	45437	True	exploratory
IL18	IL18	ok	11	112025306	11_112025306_	rs5744249	A	C	0.2039	0.0135	0.7618	1.53e-51	14742	228.12	grch38_cis	4063	44007	True	hypothesis
IL1A	IL1A	ok	2	112637801	2_112637801_T	rs114374674	T	C	-0.2183	0.0543	0.9772	5.81e-05	11340	16.16	grch38_cis	3785	39221	True	hypothesis
IL20	IL20	ok	1	207010900	1_207010900_A	rs2243195	A	G	0.2598	0.0807	0.9863	0.00128	9867	10.36	grch38_cis	3080	43023	True	exploratory
IL22RA1	IL22RA1	ok	1	24522881	1_24522881_T_	rs114879793	T	C	0.4005	0.1034	0.9894	0.000107	8631	15.0	grch38_cis	3747	40905	True	exploratory
IL24	IL24	ok	1	206896342	1_206896342_A	rs4240846	A	G	0.1674	0.0524	0.0213	0.0014	10850	10.21	grch38_cis	3190	43046	True	exploratory
IL2	IL2	ok	4	122406746	4_122406746_A	rs143837636	A	G	-0.5035	0.1392	0.0044	0.000298	8140	13.08	grch38_cis	5264	49583	True	exploratory
IL33	IL33	ok	9	5806889	9_5806889_A_G	rs117980606	A	G	-0.1354	0.0428	0.0259	0.00156	11793	10.01	grch38_cis	4974	62098	True	positive_control
IL4	IL4	ok	5	132428808	5_132428808_T	rs575237428	T	C	0.6895	0.1991	0.003	0.000534	7844	11.99	grch38_cis	4626	40981	True	positive_control
IL5	IL5	ok	5	132613677	5_132613677_A	rs55650607	A	G	0.0702	0.0197	0.1672	0.000366	11784	12.7	grch38_cis	4405	40949	True	positive_control
IL6	IL6	ok	7	22304702	7_22304702_A_	rs118147027	A	T	0.2074	0.057	0.0232	0.000274	13808	13.24	grch38_cis	5103	52575	True	hypothesis
IL7	IL7	ok	8	79299642	8_79299642_A_	rs78574099	A	G	0.0613	0.0169	0.154	0.000286	14736	13.16	grch38_cis	4877	46347	True	exploratory
IL8	CXCL8	ok	4	74215450	4_74215450_A_	rs74696018	A	G	0.2139	0.0416	0.9785	2.72e-07	14742	26.44	grch38_cis	3935	48559	True	hypothesis
LAP_TGFB1	TGFB1	ok	19	41847860	19_41847860_A	rs1800472	A	G	-0.4623	0.0335	0.0345	2.55000000000	14736	190.44	grch38_cis	5093	46057	True	exploratory
MMP10	MMP10	ok	11	102649482	11_102649482_	rs17860955	T	C	1.1371	0.047	0.983	2.6e-129	14256	585.33	grch38_cis	5512	52871	True	exploratory
MMP1	MMP1	ok	11	102673248	11_102673248_	rs484915	A	T	-0.3517	0.0119	0.5613	5.73000000000	14295	873.48	grch38_cis	5641	52878	True	exploratory
NGF	NGF	ok	1	115829943	1_115829943_A	rs6328	A	C	-0.0797	0.0122	0.3498	6.46e-11	13224	42.68	grch38_cis	4526	43912	True	exploratory
NRTN	NRTN	ok	19	5306000	19_5306000_T_	rs117764349	T	C	-0.0885	0.0256	0.9288	0.000546	11770	11.95	grch38_cis	5437	53906	True	exploratory
NTF3	NTF3	ok	12	5680104	12_5680104_A_	rs58399805	A	T	0.1106	0.022	0.0799	4.98e-07	14737	25.27	grch38_cis	4973	50303	True	exploratory
OPG	TNFRSF11B	ok	8	119443927	8_119443927_C	rs9643126	C	G	0.0837	0.0146	0.4689	9.87e-09	10893	32.87	grch38_cis	4419	44708	True	exploratory
OSM	OSM	ok	22	30600105	22_30600105_C	rs4823082	C	G	-0.159	0.0346	0.0363	4.32e-06	14724	21.12	grch38_cis	3986	51264	True	hypothesis
PDL1	CD274	ok	9	5448218	9_5448218_T_C	rs822335	T	C	-0.1295	0.012	0.3471	3.77000000000	14736	116.46	grch38_cis	4687	64071	True	exploratory
PLAU	PLAU	ok	10	73958362	10_73958362_A	rs80073349	A	G	0.6049	0.1661	0.9944	0.000271	6861	13.26	grch38_cis	4141	43771	True	exploratory
S100A12	S100A12	ok	1	153337943	1_153337943_A	rs3014874	A	G	-0.1726	0.0137	0.2623	2.15e-36	14294	158.72	grch38_cis	4190	33049	True	hypothesis
SCF	KITLG	ok	12	88342657	12_88342657_T	rs150334106	T	C	0.1397	0.0379	0.0338	0.000228	14726	13.59	grch38_cis	3359	43017	True	exploratory
SLAMF1	SLAMF1	ok	1	160636559	1_160636559_T	rs60094514	T	C	0.2209	0.0162	0.8506	2.45e-42	14732	185.94	grch38_cis	4525	48229	True	exploratory
TGFA	TGFA	ok	2	70774295	2_70774295_A_	rs72912115	A	T	0.137	0.0184	0.114	9.65e-14	14728	55.44	grch38_cis	4410	43988	True	exploratory

(showing first 40 of 47 rows; full data in the accompanying CSV)

Supplementary Table 43 - r5_pqtl_mr_results

ST43_r5_pqtl_mr_results.csv

protein	gene	lead_snp	variant_id	chromosome	pos	fstat	n_exp	outcome	n_outcome	outcome_cla	protein_catego	build_flag	match_metho	strument_pa	beta	se	p	raw_p	or	or_lower	or_upper	steiger_pass	steiger_p	beta_outcome_ali	p_outcome	fdr
ADA	ADA	rs13847774	20_4493576	20	44935764	10.89	14288	ALLERG_AS	283740	allergic	exploratory	grch38_cis	rsid	True	-0.5399213	0.25759248	0.03607961	0.03607961	0.58279410	0.35176125	0.96556674	True	0.00576417	0.0919486	0.0360795	0.49274383621549905
ADA	ADA	rs13847774	20_4493576	20	44935764	10.89	14288	ALLERG_RH	490219	allergic	exploratory	grch38_cis	rsid	True	-0.1208391	0.23395772	0.60550540	0.60550540	0.88617652	0.56023651	1.40174519	True	0.00154610	0.0205789	0.605505	0.8687686251800483
ADA	ADA	rs13847774	20_4493576	20	44935764	10.89	14288	L12_ATOPIC	464119	allergic	exploratory	grch38_cis	rsid	True	0.06375455	0.17067997	0.70875201	0.70875201	1.06583075	0.76278566	1.48927184	True	0.00144421	-0.0108574	0.708752	0.8970860396244189
ADA	ADA	rs13847774	20_4493576	20	44935764	10.89	14288	I9_MI_STRIC	447837	negative_co	exploratory	grch38_cis	rsid	True	0.34092953	0.18238520	0.06158346	0.06158346	1.40625414	0.98359066	2.01054239	True	0.00350259	-0.0580603	0.0615843	
ADA	ADA	rs13847774	20_4493576	20	44935764	10.89	14288	M13_RHEUN	331429	negative_co	exploratory	grch38_cis	rsid	True	0.45799060	0.23408396	0.05040360	0.05040360	1.58089415	0.99918635	2.50126145	True	0.00460777	-0.0779958	0.0504034	
ARTN	ARTN	rs14325927	1_44346226	1	44346226	11.56	7837	ALLERG_AS	283740	allergic	exploratory	grch38_cis	rsid	True	-0.2115140	0.26648040	0.42735131	0.42735131	0.80935787	0.48007368	1.36449922	True	0.00127009	0.0652944	0.427351	0.7988161269291195
ARTN	ARTN	rs14325927	1_44346226	1	44346226	11.56	7837	ALLERG_RH	490219	allergic	exploratory	grch38_cis	rsid	True	-0.1561959	0.24960479	0.53146373	0.53146373	0.85539153	0.52444150	1.39518833	True	0.00098980	0.0482177	0.531464	0.8224587078778532
ARTN	ARTN	rs14325927	1_44346226	1	44346226	11.56	7837	L12_ATOPIC	464119	allergic	exploratory	grch38_cis	rsid	True	-0.1340858	0.18262163	0.46281004	0.46281004	0.87451498	0.61138753	1.25088658	True	0.00105361	0.0413923	0.46281	0.8185996149915309
ARTN	ARTN	rs14325927	1_44346226	1	44346226	11.56	7837	I9_MI_STRIC	447837	negative_co	exploratory	grch38_cis	rsid	True	0.00700851	0.19174829	0.97084331	0.97084331	1.00703313	0.69155115	1.46643632	True	0.00076656	-0.0021635	0.970843	
ARTN	ARTN	rs14325927	1_44346226	1	44346226	11.56	7837	M13_RHEUN	331429	negative_co	exploratory	grch38_cis	rsid	True	-0.0119643	0.24739488	0.96142841	0.96142841	0.98810698	0.60843921	1.60468851	True	0.00080256	0.00369338	0.961428	
CASP8	CASP8	rs54425338	2_2016772	2	201677213	22.37	14089	ALLERG_AS	283740	allergic	exploratory	grch38_cis	rsid	True	0.28661230	0.25111692	0.25372429	0.25372429	1.33190774	0.81417751	2.17885928	True	1.26009309	0.0652043	0.253724	0.7966196045700854
CASP8	CASP8	rs54425338	2_2016772	2	201677213	22.37	14089	ALLERG_RH	490219	allergic	exploratory	grch38_cis	rsid	True	-0.1055986	0.22924879	0.64506467	0.64506467	0.89978567	0.57411456	1.41019634	True	4.54367568	-0.0240237	0.645064	0.8778199728895998
CASP8	CASP8	rs54425338	2_2016772	2	201677213	22.37	14089	L12_ATOPIC	464119	allergic	exploratory	grch38_cis	rsid	True	0.16067208	0.16684659	0.33555107	0.33555107	1.17429983	0.84675216	1.62855221	True	7.02679769	0.0365529	0.335551	0.7966196045700854
CASP8	CASP8	rs54425338	2_2016772	2	201677213	22.37	14089	I9_MI_STRIC	447837	negative_co	exploratory	grch38_cis	rsid	True	-0.0913283	0.17987516	0.61164140	0.61164140	0.91271796	0.64154004	1.29852235	True	4.94763952	-0.0207772	0.611642	
CASP8	CASP8	rs54425338	2_2016772	2	201677213	22.37	14089	M13_RHEUN	331429	negative_co	exploratory	grch38_cis	rsid	True	-0.0201967	0.22721186	0.92916988	0.92916988	0.98000588	0.62780104	1.52980237	True	3.96941758	-0.0045947	0.92917	
CCL2_MCP1	CCL2	rs19112331	17_3418819	17	34188197	10.09	7285	ALLERG_AS	283740	allergic	hypothesis	grch38_cis	rsid	True	0.28715594	0.25541817	0.26090356	0.26090356	1.33263200	0.80778149	2.19850055	True	0.00309543	-0.119141	0.260905	0.7966196045700854
CCL2_MCP1	CCL2	rs19112331	17_3418819	17	34188197	10.09	7285	ALLERG_RH	490219	allergic	hypothesis	grch38_cis	rsid	True	0.30780911	0.22641166	0.17398477	0.17398477	1.36044127	0.87287950	2.12033898	True	0.00280702	-0.12771	0.173983	0.7445028621239614
CCL2_MCP1	CCL2	rs19112331	17_3418819	17	34188197	10.09	7285	L12_ATOPIC	464119	allergic	hypothesis	grch38_cis	rsid	True	0.05637020	0.16410243	0.73121764	0.73121764	1.05798928	0.76699840	1.45937896	True	0.00187972	-0.023388	0.731218	0.9020628947786848
CCL2_MCP1	CCL2	rs19112331	17_3418819	17	34188197	10.09	7285	I9_MI_STRIC	447837	negative_co	hypothesis	grch38_cis	rsid	True	-0.0468524	0.17739527	0.79169242	0.79169242	0.95422814	0.67398512	1.35099620	True	0.00182697	0.0194391	0.791692	
CCL2_MCP1	CCL2	rs19112331	17_3418819	17	34188197	10.09	7285	M13_RHEUN	331429	negative_co	hypothesis	grch38_cis	rsid	True	-0.1537146	0.22814822	0.50047093	0.50047093	0.85751669	0.54832606	1.34105403	True	0.00234380	0.0637762	0.500471	
CCL3_MIP1A	CCL3	rs18416413	17_3622355	17	36223556	18.06	8717	ALLERG_AS	283740	allergic	exploratory	grch38_cis	rsid	True	-0.2943594	0.36427096	0.41904547	0.41904547	0.74500862	0.36482752	1.52137054	True	5.23734912	-0.119716	0.419045	0.7988161269291195
CCL3_MIP1A	CCL3	rs18416413	17_3622355	17	36223556	18.06	8717	ALLERG_RH	490219	allergic	exploratory	grch38_cis	rsid	True	0.29155151	0.32970985	0.37655215	0.37655215	1.33850258	0.70139798	2.55431181	True	4.24164702	0.118574	0.376554	0.7966196045700854
CCL3_MIP1A	CCL3	rs18416413	17_3622355	17	36223556	18.06	8717	L12_ATOPIC	464119	allergic	exploratory	grch38_cis	rsid	True	-0.1186378	0.24033808	0.62156831	0.62156831	0.88812940	0.55449334	1.42251272	True	3.44756782	-0.04825	0.621569	0.8727045108074245
CCL3_MIP1A	CCL3	rs18416413	17_3622355	17	36223556	18.06	8717	I9_MI_STRIC	447837	negative_co	exploratory	grch38_cis	rsid	True	-0.5012294	0.26842389	0.06185894	0.06185894	0.60578544	0.35795777	1.02519356	True	7.83746688	-0.20385	0.0618586	
CCL3_MIP1A	CCL3	rs18416413	17_3622355	17	36223556	18.06	8717	M13_RHEUN	331429	negative_co	exploratory	grch38_cis	rsid	True	-0.1433481	0.34243914	0.67550178	0.67550178	0.86645234	0.44284790	1.69525395	True	3.68712521	-0.0582997	0.675501	
CST5	CST5	rs11696133	20_2381050	20	23810508	1934.82	14288	ALLERG_AS	283740	allergic	exploratory	grch38_cis	rsid	True	-0.0279647	0.02691943	0.29888321	0.29888321	0.97242263	0.92244561	1.02510735	True	0.0	0.0183281	0.298884	0.7966196045700854
CST5	CST5	rs11696133	20_2381050	20	23810508	1934.82	14288	ALLERG_RH	490219	allergic	exploratory	grch38_cis	rsid	True	-0.0332171	0.02480256	0.18048554	0.18048554	0.96732851	0.92142845	1.01551503	True	0.0	0.0217705	0.180486	0.7445028621239614
CST5	CST5	rs11696133	20_2381050	20	23810508	1934.82	14288	L12_ATOPIC	464119	allergic	exploratory	grch38_cis	rsid	True	-0.0142103	0.01806789	0.43157496	0.43157496	0.98589010	0.95158765	1.02142906	True	0.0	0.0093134	0.431576	0.7988161269291195
CST5	CST5	rs11696133	20_2381050	20	23810508	1934.82	14288	I9_MI_STRIC	447837	negative_co	exploratory	grch38_cis	rsid	True	-0.0035120	0.01961779	0.85791770	0.85791770	0.99649407	0.95890530	1.03555631	True	0.0	0.0023018	0.857917	
CST5	CST5	rs11696133	20_2381050	20	23810508	1934.82	14288	M13_RHEUN	331429	negative_co	exploratory	grch38_cis	rsid	True	-0.0266036	0.02502212	0.28768983	0.28768983	0.97374715	0.92714346	1.02269342	True	0.0	0.017436	0.28769	
CX3CL1	CX3CL1	rs671623	16_5741280	16	57412802	299.17	14295	ALLERG_AS	283740	allergic	exploratory	grch38_cis	rsid	True	0.02940137	0.06369818	0.64438746	0.64438746	1.02983785	0.90896646	1.16678233	True	0.0	0.00600082	0.644386	0.8778199728895998
CX3CL1	CX3CL1	rs671623	16_5741280	16	57412802	299.17	14295	ALLERG_RH	490219	allergic	exploratory	grch38_cis	rsid	True	0.02372689	0.05845075	0.68479426	0.68479426	1.02401062	0.91316691	1.14830895	True	0.0	0.0048426	0.684793	0.8949786410152447
CX3CL1	CX3CL1	rs671623	16_5741280	16	57412802	299.17	14295	L12_ATOPIC	464119	allergic	exploratory	grch38_cis	rsid	True	-0.0033844	0.04256246	0.93662078	0.93662078	0.99662124	0.91685411	1.08332819	True	0.0	-0.0006907	0.936621	0.947142572094392
CX3CL1	CX3CL1	rs671623	16_5741280	16	57412802	299.17	14295	I9_MI_STRIC	447837	negative_co	exploratory	grch38_cis	rsid	True	0.01091386	0.04604365	0.81263118	0.81263118	1.01097363	0.92373347	1.10645302	True	0.0	0.0022275	0.812631	
CX3CL1	CX3CL1	rs671623	16_5741280	16	57412802	299.17	14295	M13_RHEUN	331429	negative_co	exploratory	grch38_cis	rsid	True	0.20411807	0.05853307	0.00048805	0.00048805	1.22644296	1.09351060	1.37553521	True	0.0	0.0416605	0.00048807	
GDNF	GDNF	rs11740708	5_37868088	5	37868088	385.59	14710	ALLERG_AS	283740	allergic	exploratory	grch38_cis	rsid	True	-0.0234717	0.06144470	0.70246308	0.70246308	0.97680158	0.86597144	1.10181616	True	0.0	0.00912581	0.702462	0.8970860396244189
GDNF	GDNF	rs11740708	5_37868088	5	37868088	385.59	14710	ALLERG_RH	490219	allergic	exploratory	grch38_cis	rsid	True	0.08965843	0.05654603	0.11283394	0.11283394	1.09380061	0.97905074	1.22199977	True	0.0	-0.0348592	0.112833	0.7092419306189307
GDNF	GDNF	rs11740708	5_37868088	5	37868088	385.59	14710	L12_ATOPIC	464119	allergic	exploratory	grch38_cis	rsid	True	0.02792052	0.04129886	0.49900237	0.49900237	1.02831395	0.94835605	1.11501327	True	0.0	-0.0108555	0.498999	0.8185996149915309
GDNF	GDNF	rs11740708	5_37868088	5	37868088	385.59	14710	I9_MI_STRIC	447837	negative_co	exploratory	grch38_cis	rsid	True	-0.0700995	0.04444984	0.11478449	0.11478449	0.93230101	0.85451498	1.01716786	True	0.0	0.0272547	0.114784	
GDNF	GDNF	rs11740708	5_37868088	5	37868088	385.59	14710	M13_RHEUN	331429	negative_co	exploratory	grch38_cis	rsid	True	-0.0054799	0.05639043	0.92258463	0.92258463	0.99453504	0.89047060	1.11076093	True	0.0	0.0021306	0.922584	

(showing first 40 of 235 rows; full data in the accompanying CSV)

Supplementary Table 44 - r5_pqtl_drug_target_mr

ST44_r5_pqtl_drug_target_mr.csv

protein	gene	lead_snp	variant_id	chromosome	pos	fstat	n_exp	outcome	n_outcome	outcome_cla	protein_catego	build_flag	match_metho	strument_pa	beta	se	p	raw_p	or	or_lower	or_upper	steiger_pass	steiger_p	beta_outcome_ali	p_outcome	fdr
IL18	IL18	rs5744249	11_1120253	11	112025306	228.12	14742	L12_ATOPIC	464119	allergic	hypothesis	grch38_cis	rsid	True	0.21226826	0.04875188	1.33643826	1.33643826	1.23647954	1.12379861	1.36045876	True	0.0	0.0432815	1.33644e-0	0.00088204
TNFSF14	TNFSF14	rs344560	19_6665020	19	6665020	321.37	11793	L12_ATOPIC	464119	allergic	exploratory	grch38_cis	rsid	True	0.15989272	0.03788046	2.43241467	2.43241467	1.17338499	1.08942170	1.26381944	True	0.0	-0.0782515	2.43232e-0	0.00107026
TNF	TNF	rs6908066	6_3127050	6	31270509	12.05	11785	L12_ATOPIC	464119	allergic	hypothesis	grch38_cis	rsid	True	0.81585924	0.16653682	9.63393556	9.63393556	2.26111769	1.63141391	3.13387865	True	0.00788867	0.049849	9.6343e-07	0.00012716
TRAIL	TNFSF10	rs57404467	3_1722742	3	172274232	212.62	13173	L12_ATOPIC	464119	allergic	exploratory	grch38_cis	rsid	True	0.15085080	0.04510072	0.00082355	0.00082355	1.16282316	1.06444484	1.27029381	True	0.0	0.087765	0.00082356	0.02717736

Supplementary Table 45 - r5_pqtl_coloc

ST45_r5_pqtl_coloc.csv

protein	outcome	lead_snp	n_snps	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4	max_snp_pph4	lead_coloc_snp
IL18	L12_ATOPIC	rs5744249	1171	4.623908731437817e-47	0.012329288320098844	7.666999596148045e-47	0.019475254892544404	0.9681954567873569	0.9960838711943519	rs71478719
TNFSF14	L12_ATOPIC	rs344560	2512	1.1285945871389405e-66	0.0015489481551575614	1.737492075903447e-65	0.022870760779860277	0.9755802910649822	0.9996026639246643	rs344563
TNF	L12_ATOPIC	rs6908066	9300	4.273527014211859e-11	1.8683486334017644e-11	0.6888504882637377	0.30114940852716104	0.010000103147682338	0.6936924390269964	rs6908066
TRAIL	L12_ATOPIC	rs574044675	1020	7.24276008630716e-37	0.6828446097297751	3.737842886263705e-38	0.03495804030096916	0.2821973499692557	0.292440518152626	rs142552223

Supplementary Table 46 - r5_pqtl_coloc_IL18_L12_ATOPIC_persnp

ST46_r5_pqtl_coloc_IL18_L12_ATOPIC_persnp.csv

snp	pos	beta_x	se_x	p_x	beta_y	se_y	p_y	match	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4
rs12785809	111776055	0.001	0.0133	0.94	-0.0133581	0.00994321	0.179129	rsid	0.9999747538048162	8.857483851616701e-	1.6244819473325276e	0.0	1.438918588793618e-07
rs12808601	111776066	-0.002	0.0131	0.879	-0.0135978	0.00988805	0.169078	rsid	0.9999741846422078	8.80208250667892e-0	1.68648258646627156	0.0	1.484494209964509 6e-07
rs7950926	111776437	-0.0056	0.0131	0.669	-0.0139182	0.00988938	0.159312	rsid	0.999972668506867	9.52670647329364e-0	1.763676181441417e-	0.0	1.680248453152235e-07
rs1940392	111776846	0.0156	0.0122	0.201	0.00927309	0.00874605	0.289026	rsid	0.9999713576946471	1.8263330525838 77e-	1.0192814750383083e	0.0	1.8616007682865612e-07
rs1940391	111776891	-0.0039	0.0131	0.766	-0.0135971	0.00988805	0.1691	rsid	0.999973891423169	9.092065317960603e-	1.68631863184651876	0.0	1.533251945790479 2e-07
rs182641993	111777491	-0.0735	0.0641	0.252	-0.0326628	0.0673354	0.627622	rsid	0.9998832296070282	6.851035458312739e-	4.516538077247981e-	0.0	3.094657615990488e-06
rs185533636	111777868	-0.075	0.0643	0.243	-0.0326129	0.0673355	0.628148	rsid	0.9998816950344914	6.99924512391363e-0	4.515185157743991e-	0.0	3.160662691981441 8e-06
rs117843740	111778393	-0.0738	0.0648	0.255	-0.032579	0.0673356	0.628507	rsid	0.9998832666353166	6.849800595581216e-	4.514280555937379e-	0.0	3.092553168204843 3e-06
rs41302377	111779861	-0.0348	0.0349	0.319	-0.00190688	0.019695	0.922869	rsid	0.9999501276539541	3.63182721126171 65e	1.3079042034790506e	0.0	4.750318985471127 6e-07
rs80015283	111780693	0.0476	0.1297	0.714	0.00662442	0.0299083	0.82471	rsid	0.9999106410092035	6.79769427056268e-0	2.00209626358881 62e	0.0	1.361085455233932 1e-06
rs2070894	111780837	-0.0004	0.0132	0.976	-0.0131514	0.00994416	0.185992	rsid	0.9999752890067602	8.77077822137347e-0	1.58016190968513 73e	0.0	1.385959214799819e-07
rs11603779	111781047	0.0005	0.0133	0.97	0.0131723	0.00994263	0.185228	rsid	0.9999751737542468	8.838876092292824e-	1.58472939159570 86e	0.0	1.40075744876157e-07
rs117627181	111781358	-0.0725	0.0644	0.26	-0.0322714	0.067337	0.63176	rsid	0.9998845396561339	6.736366863182029e-	4.506085988453685e-	0.0	3.035815349811028e-06
rs117865217	111782139	-0.0722	0.0645	0.263	-0.0321064	0.0673379	0.633508	rsid	0.9998849378196923	6.702711269315284e-	4.501733818816561 5e	0.0	3.017729426411331 6e-06
rs2228387	111782284	0.0332	0.0459	0.469	0.00515367	0.0344206	0.880981	rsid	0.9999393844996448	3.716911517840691e-	2.260608602643219e-	0.0	8.402991504028829e-07
rs14133	111782697	-0.0026	0.0133	0.845	-0.0133262	0.00994003	0.180031	rsid	0.9999746731350106	9.001731585118266e-	1.61794863224059 12e	0.0	1.456470818433682 2e-07
rs2234702	111783098	-0.0047	0.066	0.943	0.0793568	0.0846037	0.348253	rsid	0.9998882834313162	4.03590338332454e-0	6.858903824831785e-	0.0	2.768496602194433e-06
rs762550	111783100	-0.0177	0.0123	0.15	-0.0145387	0.008967	0.10494	rsid	0.9999545251914489	2.2858385578043137e	2.211097871126424e-	0.0	5.054442618710233e-07
rs4252588	111784229	0.0728	0.0249	0.00346	0.0160924	0.0181516	0.375318	rsid	0.998916235034539	0.0010475469559512 6	1.7678675023028287e	0.0	1.853933448682476 7e-05
rs4252589	111784401	0.014	0.0746	0.851	-0.0231019	0.0274362	0.399774	rsid	0.9999283345493476	4.5164645368521316e	2.535554883722273e-	0.0	1.145256446676751 2e-06
rs4252590	111784816	0.0158	0.0391	0.686	-0.0213718	0.0187863	0.255277	rsid	0.999948630147142	2.72291335716090 65e	2.350078052112818 4e	0.0	6.39938765407085e-07
rs117765764	111784827	0.068	0.0643	0.29	0.00806392	0.0676925	0.905176	rsid	0.9998928192324251	6.318930818788571e-	4.137661918490656e-	0.0	2.614840202027785e-06
rs4252591	111784923	0.0154	0.0124	0.214	0.0153874	0.00860518	0.0737497	rsid	0.9999535920078029	1.77227143888092 75e	2.818572703526253e-	0.0	4.995507732352844e-07
rs11214040	111786628	-0.0274	0.0189	0.147	-0.00126921	0.0172464	0.941334	rsid	0.9999529692142863	3.517437203285834e-	1.1453524210957576e	0.0	4.028894698918969e-07
rs148419924	111786863	-0.0688	0.0642	0.284	-0.00824137	0.0676499	0.903039	rsid	0.9998920621426898	6.392698211440553e-	4.13661750030739e-0	0.0	2.644700192834917e-06
rs12284944	111787383	-0.0151	0.0123	0.22	-0.0160942	0.00858469	0.0608261	rsid	0.999949218992336	1.727658924694529e-	3.293537851446351e-	0.0	5.690399027067138e-07
rs10891299	111787962	0.0122	0.0123	0.321	0.0156928	0.00858821	0.0676628	rsid	0.9999560913754827	1.33225355822970 15e	3.018394460153218e-	0.0	4.02144333572539e-07
rs186661052	111788462	-0.0554	0.0824	0.501	-0.211525	0.197611	0.284434	rsid	0.9998389001507576	5.72723934027613 05e	9.820227055109156e-	0.0	5.625185288548463e-06
rs10789852	111788464	-0.0153	0.0124	0.217	-0.015985	0.0085991	0.0630391	rsid	0.9999498584126926	1.75477906048177e-0	3.203168324696575e-	0.0	5.621134555985128e-07
rs79613093	111788492	-0.0671	0.0642	0.296	-0.00826803	0.0676165	0.902679	rsid	0.9998936363133574	6.243061473743867e-	4.13512154865386 24e	0.0	2.581856418731979e-06
rs139772777	111789512	0.0617	0.0468	0.187	-0.0404713	0.0457661	0.376529	rsid	0.9998897694032532	6.57543758298017e-0	4.17318656284214 45e	0.0	2.744355288531147 6e-06
rs188029389	111790429	-0.1434	0.1347	0.287	-0.0474805	0.14693	0.746581	rsid	0.9998301406274511	9.142772069725291e-	7.186049446195212e-	0.0	6.571157389504949e-06
rs191284997	111790850	-0.0329	0.0612	0.591	-0.0158813	0.059536	0.789662	rsid	0.9999175705976655	4.275932518766614e-	3.804323978629421e-	0.0	1.626837360445829 7e-06
rs57377042	111791075	0.0183	0.0131	0.162	0.0156424	0.00860245	0.0690081	rsid	0.9999466555553108	2.291378923275228e-	2.97489577441755 83e	0.0	6.816977124300086e-07
rs57527690	111791248	0.0149	0.0124	0.23	0.0156406	0.00860244	0.0690399	rsid	0.9999528837717321	1.68763916330202 4e	2.97379438013427 33e	0.0	5.018928332578841e-07
rs76851368	111791654	-0.0099	0.0892	0.912	0.0174977	0.0552352	0.751407	rsid	0.9999106830698111	5.134575432513580 6e	3.611657731109304e-	0.0	1.854598552729619 8e-06
rs10502152	111791695	0.0151	0.0124	0.223	0.0156442	0.00860237	0.0689747	rsid	0.9999525202185423	1.72066138267168 33e	2.976105631309145e-	0.0	5.121113179879977e-07
rs11214043	111791781	-0.0154	0.0124	0.214	-0.0160307	0.00859828	0.0622644	rsid	0.9999493471451306	1.77226391548290 63e	3.235673984178 94e-0	0.0	5.734758726328619e-07
rs117502493	111793027	-0.066	0.0642	0.304	-0.00823532	0.0675934	0.903029	rsid	0.9998946192445621	6.150050926803859e-	4.133768929003731e-	0.0	2.542556879865982 7e-06
rs185492252	111793591	0.0666	0.0642	0.3	0.00826008	0.0675932	0.902738	rsid	0.9998940932250866	6.200420183937812e-	4.133910346982409e-	0.0	2.563469604200286e-05

(showing first 20 of 1171 rows; full data in the accompanying CSV)

Supplementary Table 47 - r5_pqtl_coloc_TNFSF14_L12_ATOPIC_persnp

ST47_r5_pqtl_coloc_TNFSF14_L12_ATOPIC_persnp.csv

snp	pos	beta_x	se_x	p_x	beta_y	se_y	p_y	match	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4
rs114436343	6415176	-0.079	0.0746	0.29	0.00342595	0.0278915	0.902241	rsid	0.9999104940085143	6.980544541159248e-	1.8414966096527146e	0.0	1.28557997771799 18e-06
rs1616305	6415443	0.0873	0.0541	0.107	0.010575	0.0411827	0.797347	rsid	0.9998623979156132	0.00010737193128497	2.729864182549537e-	0.0	2.9315112763222414e-06
rs140018828	6415657	0.0708	0.0548	0.196	0.0152613	0.0452562	0.735951	rsid	0.9998957318070338	7.165755389254661e-	3.042988086213373e-	0.0	2.1807582115401344e-06
rs62107578	6415711	-0.0017	0.0332	0.959	0.0157232	0.0172051	0.360787	rsid	0.9999607801211642	2.1638673187513158e	1.7208815116858774e	0.0	3.723905313695631e-07
rs145698454	6415735	0.013	0.1003	0.897	-0.115832	0.0573779	0.0435131	rsid	0.999720976191107	5.589862915320615e-	0.00021130994666338	0.0	1.18152330762560 78e-05
rs8103311	6415795	0.0263	0.0144	0.0678	-0.00184566	0.00912885	0.839777	rsid	0.9999435999264847	4.989084246904957e-	6.199895585524877e-	0.0	3.09335460524691 95e-07
rs189946046	6416113	0.0442	0.0524	0.399	0.0105833	0.0411831	0.797193	rsid	0.9999261773725378	4.5284261898286426e	2.7301926688368323e	0.0	1.2364388756503839e-06
rs182213090	6416117	0.0483	0.0526	0.358	0.0105833	0.0411831	0.797193	rsid	0.9999232140209912	4.816893150341258e-	2.7301845777188565e	0.0	1.315201728210425e-06
rs2305058	6416123	0.0288	0.0138	0.0369	0.00309365	0.00860755	0.719287	rsid	0.999914011140417	7.939375928912284e-	6.1099654513794 11e-	0.0	4.851348425035247e-07
rs146253337	6416443	0.0162	0.0445	0.716	-0.00704336	0.0239644	0.768828	rsid	0.9999528177322216	3.022910927637956e-	1.6455694056630372e	0.0	4.974644453572205e-07
rs80332810	6416696	0.0894	0.0828	0.28	0.0202772	0.0362989	0.576423	rsid	0.99989514023398	7.55449643233933 6e-	2.7255565074505524e	0.0	2.0592366222378343e-06
rs201923567	6416724	-0.046	0.0552	0.405	-0.0105893	0.0411834	0.797081	rsid	0.9999245255394564	4.6891091961746956e	2.730300426331398e-	0.0	1.2803643185492668e-06
rs74489818	6416725	-0.0468	0.0523	0.371	-0.0105893	0.0411834	0.797081	rsid	0.9999243646460165	4.704771324067567e-	2.7302999870108134e	0.0	1.2846408727662059e-06
rs77809861	6416916	0.0411	0.0534	0.441	0.0105927	0.0411835	0.797019	rsid	0.9999278820939793	4.362305733905418e-	2.730369224557278e-	0.0	1.1911564361044255e-06
rs3786689	6417350	-0.0287	0.0146	0.0493	0.00174266	0.0092145	0.849997	rsid	0.9999276658042097	6.568235505891699e-	6.241832795909663e-	0.0	4.100079355136142e-07
rs181107021	6417551	0.2302	0.199	0.247	0.0284314	0.0451726	0.529089	rsid	0.9998601470825677	0.00010175733067923	3.457666544263056e-	0.0	3.518921310642581e-06
rs183882319	6417568	-0.0364	0.0523	0.486	-0.0106022	0.0411839	0.796843	rsid	0.9999307190675136	4.085965564747667e-	2.730550596612154e-	0.0	1.115770872702348e-06
rs2258496	6417855	-0.0269	0.0143	0.06	0.00134776	0.00921906	0.883769	rsid	0.9999386605042889	5.479933859342392e-	6.200360613992924e-	0.0	3.397965036337054e-07
rs150345802	6417860	0.0428	0.0524	0.414	0.0106043	0.041184	0.796804	rsid	0.9999270838521216	4.439794097530417e-	2.7305797321132597e	0.0	1.2124095819836328e-06
rs2860167	6418142	0.0164	0.0205	0.424	-0.00129041	0.0116582	0.911864	rsid	0.9999735191048856	1.85396791840799 25e	7.796664444986066e-	0.0	1.4455148536868 69e-07
rs72983829	6418237	-0.005	0.0304	0.869	-0.0115266	0.0155377	0.45818	rsid	0.9999660755547437	2.0123925086838798e	1.3528269060354968e	0.0	2.7225110909302885e-07
rs144144696	6418342	0.0478	0.0521	0.359	0.0106218	0.0411846	0.796478	rsid	0.9999236194972482	4.776711239195437e-	2.730882691211654e-	0.0	1.3045634476154997e-06
rs117689481	6418350	0.0483	0.0526	0.358	0.0106241	0.0411847	0.796435	rsid	0.9999232062723076	4.816893113013812e-	2.730923869759905e-	0.0	1.3155578646335715e-06
rs2860166	6419098	-0.0415	0.0526	0.43	-0.0108032	0.0411865	0.793091	rsid	0.999927806368079	4.3660761297203194e	2.7339136913039154e	0.0	1.193733710803889e-06
rs2305059	6419275	0.0245	0.0141	0.0823	-0.00195919	0.0091308	0.830103	rsid	0.9999517291109634	4.179379696551841e-	6.217237563135967e-	0.0	2.598545078082112e-07
rs151235276	6419286	-0.0438	0.043	0.308	0.0379845	0.0208993	0.0691401	rsid	0.999882646753872	4.450550983176890 6e	6.974340614510771e-	0.0	3.10433015111243e-05
rs2860163	6419306	-0.0001	0.0311	0.997	0.000207389	0.0177137	0.990659	rsid	0.9999677295764685	2.0303047386945874e	1.172922935981646e-	0.0	2.3814678460231222e-07
rs5826930	6419751	-0.0092	0.0663	0.89	0.0362587	0.031136	0.244211	rsid	0.99991872394501	4.075474481810309e-	3.89344188891139 04e	0.0	1.5868912827301284e-06
rs8105183	6419755	-0.0367	0.0571	0.52	-0.0109934	0.041238	0.789789	rsid	0.9999288216044994	4.261113852742822 6e	2.7399643845021 19e-	0.0	1.1676131283114327e-06
rs184558016	6419761	0.0477	0.0525	0.364	0.0109947	0.041238	0.789765	rsid	0.9999235751819878	4.771755649396416e-	2.7399714179955334e	0.0	1.3075473383678 67e-06
rs117786289	6420068	0.0495	0.0527	0.348	0.0111604	0.0412688	0.786827	rsid	0.9999221190754227	4.908885303835646e-	2.7444735995087148e	0.0	1.3473355436771 82e-06
rs117024124	6420301	0.0482	0.0527	0.36	0.0110563	0.0412996	0.788924	rsid	0.9999231349120035	4.809987210665772 5e	2.7445012806287805e	0.0	1.3202030834742115e-06
rs74790427	6420497	0.0477	0.0527	0.365	0.0110172	0.0413155	0.789732	rsid	0.9999235104313222	4.773170526606518e-	2.7447640506318275e	0.0	1.3102229052813 93e-06
rs117808562	6421150	-0.285	0.2278	0.211	0.0773019	0.0659345	0.241035	rsid	0.9998150327533478	0.00010581568020313	7.157627322961426e-	0.0	7.575293219326092e-06
rs181452781	6421373	0.0384	0.0526	0.465	0.0107954	0.0414106	0.794331	rsid	0.9999294268352038	4.195472935953188e-	2.7466024483671337e	0.0	1.1524109530828125e-06
rs75599457	6421529	0.0415	0.0527	0.431	0.01122	0.0414361	0.786562	rsid	0.9999275626004444	4.368368495833414 7e	2.7550135953622 48e-	0.0	1.2035786436644268e-06
rs182620609	6422206	0.0467	0.0528	0.376	0.0114692	0.0414991	0.782262	rsid	0.9999240472854954	4.702540402044835 4e	2.7627994207671717e	0.0	1.2993162764885688e-06
rs149581684	6422306	-0.1492	0.1168	0.201	0.0223992	0.270004	0.933884	rsid	0.9998015009150459	0.00010208839697221	8.747838666406 94e-05	0.0	8.932301317888895e-06
rs61751242	6422423	0.0579	0.0605	0.339	0.00147937	0.0472589	0.975027	rsid	0.99991280666357	5.546221083957651 6e	3.006358718140277 7e	0.0	1.667538409086464e-06
rs191130575	6422644	-0.0466	0.0524	0.374	-0.0115538	0.0415399	0.780908	rsid	0.9999241219303546	4.691439354369585e-	2.7665659952138345e	0.0	1.2980161496004845e-06

(showing first 20 of 2512 rows; full data in the accompanying CSV)

Supplementary Table 48 - r5_pqtl_coloc_TNF_L12_ATOPIC_persnp

ST48_r5_pqtl_coloc_TNF_L12_ATOPIC_persnp.csv

snp	pos	beta_x	se_x	p_x	beta_y	se_y	p_y	match	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4
rs77203474	31020528	0.0974	0.0563	0.0836	0.067051	0.0364502	0.0658385	rsid	0.999737691721192	0.00013043366278922	0.00011665489811707	0.0	1.52197179017202286-05
rs9262603	31020543	-0.0341	0.0252	0.176	0.0478748	0.0207791	0.0212236	rsid	0.9997666569059872	4.035594930781903e-	0.00018549939337419	0.0	7.487751330702925e-06
rs116127591	31020544	-0.0045	0.0303	0.882	-0.000943371	0.0197441	0.961892	rsid	0.9999666604996144	2.00123282498501166	1.30656885697417756	0.0	2.6148356619944333e-07
rs9262614	31020716	0.0394	0.0248	0.112	-0.0520841	0.0200463	0.00937152	rsid	0.9995591303829706	5.569964148270628e-	0.00036483957875704	0.0	2.03303967897159966-05
rs3131789	31020821	-0.019	0.0131	0.147	0.0154272	0.00852676	0.0704093	rsid	0.9999455626887847	2.471068179283645e-	2.9009739944348848e	0.0	7.16889478193373e-07
rs553442784	31020895	-0.0352	0.1326	0.791	-0.0742134	0.0792565	0.349082	rsid	0.9998621904993832	6.755196437049864e-	6.581124465120663e-	0.0	4.446291595080793e-06
rs4713427	31020928	0.0082	0.0182	0.652	-0.0549	0.0100936	5.35538e-08	rsid	0.05587360652533367	7.438699582155622e-	0.931721234022432	0.0	0.012404415582276219
rs4713428	31020933	-0.0083	0.0182	0.648	0.0548976	0.0100936	5.36253e-08	rsid	0.05593985365048168	7.465821791412203e-	0.9316257740930165	0.0	0.012433625674322833
rs4713429	31021017	0.0105	0.0156	0.501	0.0371052	0.00950473	9.46695e-05	rsid	0.9874770801140584	1.27817368538803526	0.012350278229720059	0.0	0.00015985991936770378
rs2517528	31021078	-0.0245	0.0136	0.0716	-0.016458	0.00942524	0.080783	rsid	0.9999249267877337	4.514837111841716e-	2.86320535162189426	0.0	1.292787631752928e-06
rs9262615	31021161	0.0115	0.0153	0.452	0.0378223	0.00938365	5.56212e-05	rsid	0.9797359514887266	1.3149827183513548e	0.01998269480432151	0.0	0.0002682038797682577
rs2428516	31021325	-0.0128	0.0133	0.336	0.0164926	0.00863831	0.0562302	rsid	0.9999501566286029	1.398427872713452e-	3.53645207042688356	0.0	4.945719657141847e-07
rs2523857	31021504	-0.0046	0.0132	0.727	0.0382756	0.00851446	6.94496e-06	rsid	0.8807637004453295	8.20122039185458e-0	0.11812814951917774	0.0	0.0010999488151009036
rs117803081	31021542	-0.108	0.0593	0.0686	-0.0488536	0.0209321	0.0196006	rsid	0.9996150959026611	0.00015424743256135	0.00019982266378752	0.0	3.08340009901157146-05
rs2517527	31021547	0.0	0.0133	1.0	0.0458427	0.0086384	1.1154e-07	rsid	0.12161042884062645	1.07417278707966826	0.8706977109594205	0.0	0.007690786027166107
rs2523856	31021630	0.0057	0.0132	0.666	-0.0380642	0.00849555	7.44749e-06	rsid	0.8877539053629684	8.537352544099987e-	0.11116847222226817	0.0	0.0010690850622194056
rs17190134	31021860	0.0324	0.0409	0.428	0.0823815	0.025489	0.00122911	rsid	0.9971939676880157	3.5133538949600284e	0.00267659587076603	0.0	9.430290226877787e-05
rs17190141	31021915	0.0701	0.1487	0.637	0.106604	0.082382	0.19566	rsid	0.999827142208263	7.445501544789052e-	9.158279871016817e-	0.0	6.819977578990619e-06
rs117264051	31021957	0.1454	0.1686	0.388	0.0606261	0.0894837	0.498082	rsid	0.999845931061749	8.804755816443606e-	6.067800652943002e-	0.0	5.34337355709264e-05
rs2523855	31022158	0.006	0.0137	0.661	-0.00530397	0.00870706	0.54242	rsid	0.9999829538341992	1.0003845123012105e	6.972567008284759e-	0.0	6.975366949331999e-08
rs11756181	31022169	-0.0125	0.0294	0.671	0.00192925	0.0195057	0.921212	rsid	0.9999657852254787	2.09845854842452e-0	1.2958256100017758e	0.0	2.719329369816882e-07
rs9262616	31022195	-0.0081	0.0155	0.601	-0.0358153	0.00950537	0.00016463	rsid	0.9925170140849088	1.1678573983912848e	0.00738441767879813	0.0	8.68896623091847e-05
rs9262617	31022206	-0.0032	0.0145	0.825	-0.0328172	0.00906336	0.000293616	rsid	0.9958264607456044	9.816623842474736e-	0.00412307829114983	0.0	4.06443394032599046-05
rs9262618	31022237	-0.0052	0.0157	0.74	-0.0352241	0.00950688	0.000211295	rsid	0.9940689530540534	1.092610469831628e-	0.00585575847484742	0.0	6.436236640091196-05
rs9262619	31022266	-0.0011	0.0147	0.94	-0.0329227	0.0090631	0.000280563	rsid	0.9956472589298241	9.738763780932e-06	0.00430093341682754	0.0	4.20688895673969e-05
rs3915966	31022274	0.0096	0.0139	0.49	-0.00620466	0.00870648	0.476064	rsid	0.9999807592577004	1.1689554662760682e	7.46393587175640556	0.0	8.725176516096546e-08
rs17190183	31022315	-0.0967	0.1435	0.5	0.156665	0.0911516	0.0856643	rsid	0.9997575988382866	7.781730449814037e-	0.00015269839840193	0.0	1.18854588138458866-05
rs11756221	31022340	-0.0137	0.0294	0.641	0.00193021	0.0195063	0.921176	rsid	0.9999654092118474	2.13553397665458366	1.2958701342999042e	0.0	2.767470430112751e-07
rs2255314	31022354	-0.0314	0.02	0.116	-0.0624898	0.013417	3.20037e-06	rsid	0.6952371112556214	3.0845057737736966-	0.2917865701804061	0.0	0.012945473506234803
rs181574062	31022375	-0.0781	0.116	0.501	0.0795763	0.035863	0.0264942	rsid	0.9996740784298067	7.048067098148922e-	0.00023861749494950	0.0	1.68234042622962866-05
rs9262620	31022489	-0.0082	0.0155	0.597	-0.0358573	0.00950523	0.000161704	rsid	0.9923911651517933	1.1716352714508053e	0.00750847209227022	0.0	8.86464032220826e-05
rs2523854	31022500	-0.0107	0.0133	0.421	0.00416261	0.0085396	0.625942	rsid	0.9999813460142515	1.2177129760703527e	6.3989338858719986-	0.0	7.792210181620294e-08
rs190296067	31022551	-0.0857	0.1493	0.566	-0.0630036	0.0343818	0.0668821	rsid	0.9998048209511818	7.662434911701781e-	0.00011011552232487	0.0	8.43917737643736e-05
rs2523853	31022557	0.0301	0.02	0.132	0.0624773	0.013418	3.22033e-06	rsid	0.6973171313885452	2.80439245565666676	0.2909535774244004	0.0	0.011701247262497844
rs17190197	31022631	0.0036	0.0314	0.909	-0.00193896	0.0195086	0.920829	rsid	0.9999661519727713	2.06200017431796256	1.29607654334685256	0.0	2.672600520565824e-07
rs146711032	31022664	-0.0193	0.0367	0.599	0.00193422	0.0195063	0.921012	rsid	0.9999596108773563	2.707930515219795e-	1.2958886597368376-	0.0	3.50931794058761466-07
rs200424298	31022756	-0.0112	0.0196	0.568	0.0189703	0.0102686	0.0646874	rsid	0.9999468873437962	1.5213315517912046-	3.7331376510253846-	0.0	5.679641756544592e-07
rs2905758	31022799	-0.0376	0.0213	0.0775	-0.0608479	0.0146852	3.42042e-05	rsid	0.9512956685202909	6.160248883049369e-	0.04568437316053273	0.0	0.0029583558303459215
rs2904756	31022822	-0.0023	0.0373	0.951	-0.0557694	0.0327189	0.0882877	rsid	0.9998884851362195	2.4174727805396356e	8.527832574012986e-	0.0	2.06181023495505766-06
rs2905757	31022828	0.0085	0.0138	0.538	-0.00902048	0.00878591	0.304563	rsid	0.9999789446802827	1.10581582007145596	9.887818156876556-0	0.0	1.09343359698036146-07

(showing first 40 of 9300 rows; full data in the accompanying CSV)

Supplementary Table 49 - r5_pqtl_coloc_TRAIL_L12_ATOPIC_persnp

ST49_r5_pqtl_coloc_TRAIL_L12_ATOPIC_persnp.csv

snp	pos	beta_x	se_x	p_x	beta_y	se_y	p_y	match	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4
rs79653511	172024416	-0.0222	0.0622	0.721	-0.00110277	0.0684895	0.987154	rsid	0.9999163357021007	4.044409975214446e-	4.154000929565261e-	0.0	1.6801888514789797e-06
rs3733027	172024832	-0.0022	0.0194	0.91	-0.038259	0.0122933	0.00185708	rsid	0.9989720763778638	1.289589909935042e-	0.0010020915540736	0.0	1.2936168963302367e-05
rs116190318	172025052	0.1489	0.1874	0.427	-0.0245329	0.138355	0.859258	rsid	0.9998372805678826	8.830668534372476e-	6.837389233830572e-	0.0	6.038854435408847e-06
rs36032820	172025228	-0.0403	0.0612	0.51	-0.0326865	0.0332566	0.325677	rsid	0.9999186438676763	4.54940160705427e-0	3.4301477324890424	0.0	1.5606389282090745e-06
rs115069531	172025440	-0.0932	0.0983	0.343	0.00974076	0.080192	0.90332	rsid	0.9998739617731561	7.506186412314003e-	4.741672621059466e-	0.0	3.559636510257751e-06
rs74427876	172025631	-0.0341	0.0613	0.578	-0.0795683	0.0371938	0.0324123	rsid	0.9997400714305784	4.318593024556969e-	0.00020776766637670	0.0	8.974972799269394e-06
rs9846134	172025707	0.0196	0.0152	0.197	0.00548103	0.0109079	0.615327	rsid	0.9999686283244353	2.2959045156952083e	8.223813576896647e-	0.0	1.88816830774682e-07
rs55686393	172026017	-0.0185	0.0153	0.227	-0.00544286	0.0109001	0.61754	rsid	0.9999707016790251	2.0921520440831732e	8.205131674332357e-	0.0	1.7166885965361183e-07
rs4615104	172026019	0.0184	0.0153	0.229	0.0054402	0.0109001	0.617711	rsid	0.9999708665850051	2.0758961893825028e	8.204138735927963e-	0.0	1.7031436522986428e-07
rs145122076	172026256	0.0434	0.0652	0.506	-0.0664662	0.0559386	0.234755	rsid	0.9998839177586 21	4.802803857649641e-	6.493513363098807e-	0.0	3.11906917153949 35e-06
rs7613643	172026710	-0.0208	0.0151	0.168	-0.00517086	0.0108913	0.634951	rsid	0.9999660685234357	2.562239743015325e-	8.101492431638164e-	0.0	2.075867025841455e-07
rs79753549	172026814	0.0542	0.0619	0.381	0.0842557	0.0370462	0.0229456	rsid	0.9996581864568953	5.291817999679271e-	0.00027437117530328	0.0	1.4524187804725268e-05
rs76634719	172027068	0.0988	0.0731	0.177	-0.0675944	0.091207	0.458627	rsid	0.9998390592098669	9.163770083076589e-	6.348457259485986e-	0.0	5.818516707493114e-06
rs73167271	172027126	0.0199	0.0153	0.193	0.004935	0.0109141	0.65115	rsid	0.9999683380051477	2.34394667924273 68e	8.034204624511152e-	0.0	1.8832343519543123e-07
rs76714981	172027152	0.3682	0.2094	0.0787	-0.0974209	0.0549134	0.0760501	rsid	0.9997061190372413	0.00013729516759117	0.00013767774885066	0.0	1.890804631687391e-05
rs72352561	172027241	-0.0341	0.0217	0.116	-2.89881e-05	0.0145992	0.998416	rsid	0.999941861327584	4.7986328892246124e	9.687451272721864e-	0.0	4.6489225111873627e-07
rs149572671	172027788	-0.012	0.1303	0.927	0.321285	0.259297	0.215323	rsid	0.9998224451172956	6.573283096919703e-	0.00010492388306178	0.0	6.898168673461854 5e-06
rs114324091	172027797	-0.03	0.0489	0.54	0.0402953	0.0460698	0.381761	rsid	0.9999200818503704	3.674283475671831e-	4.164503590324388e-	0.0	1.5302789696941664e-06
rs73167272	172028351	-0.4069	0.1577	0.00987	-0.540124	0.372601	0.147169	rsid	0.9995027883744919	0.00035203579787116	0.00010736182155980	0.0	3.7814006077137695e-05
rs10936720	172028460	-0.0169	0.0157	0.282	-0.00472497	0.0109958	0.667409	rsid	0.999973371968521	1.846537269771664e-	8.014661135494297e-	0.0	1.4799764579877818e-07
rs12636516	172028521	0.017	0.0157	0.279	0.00472537	0.0109957	0.66738	rsid	0.999973244864848	1.8591400493097855e	8.014725696852703e-	0.0	1.4900896202742934e-07
rs572762130	172029160	0.1505	0.1731	0.385	0.00863795	0.104727	0.934265	rsid	0.99984865881969	8.88675545614507e-0	5.7374153290475694	0.0	5.099472458139159e-06
rs114511323	172029218	0.0635	0.0755	0.4	-0.0194049	0.0573914	0.735275	rsid	0.9999005807936396	5.961801712302148e-	3.7561617429808996	0.0	2.2395718075504287e-06
rs28757618	172029473	0.0171	0.0157	0.276	0.00509014	0.0109955	0.643416	rsid	0.9999729951884367	1.8719037324183913e	8.133518491261242e-	0.0	1.522557478526394e-07
rs12637253	172029560	0.0195	0.0168	0.246	0.00532019	0.0109959	0.628504	rsid	0.9999699572838671	2.16506237009314076	8.214243591898759e-	0.0	1.7784884005820363e-07
rs55825282	172030321	-0.0175	0.0157	0.265	-0.00521946	0.0109943	0.63497	rsid	0.9999724189152581	1.9246137801391817e	8.177556225546126e-	0.0	1.5739071500213834e-07
rs56189830	172030504	0.0182	0.0157	0.246	0.00558697	0.0109883	0.61114	rsid	0.9999712871631203	2.023576677754006e-	8.308927750161338e-	0.0	1.681423519676201e-07
rs190297450	172030635	-0.0685	0.1131	0.545	0.0100998	0.0500325	0.840022	rsid	0.9998979201885703	6.76720464425711e-0	3.222669588360412e-	0.0	2.181069103648689e-06
rs73167273	172030905	0.0159	0.0161	0.323	0.00521474	0.0109968	0.635354	rsid	0.9999743969495344	1.7284360110945274e	8.177346533505038e-	0.0	1.4134382107007497e-07
rs9877284	172031093	-0.0079	0.0133	0.553	0.00330324	0.00949129	0.727818	rsid	0.9999826989970787	1.0522324160452114e	6.7080928126124395e	0.0	7.058594828040445e-08
rs35394365	172031314	-0.0127	0.0191	0.506	-0.0312693	0.0119408	0.00882694	rsid	0.9997410865034715	1.5698350735099578e	0.00023945512178531	0.0	3.760024008064878e-06
rs6773689	172031363	-0.0051	0.0134	0.704	0.00338764	0.00949127	0.721151	rsid	0.999983645022176	9.561575208420238e-	6.729061139198798e-	0.0	6.434147646792807e-08
rs56966312	172031734	-0.0341	0.0191	0.0742	-0.00137062	0.0146059	0.925236	rsid	0.9999290632917405	6.061258246653443e-	9.734076426658657e-	0.0	5.900493663061531e-07
rs12631550	172032186	-0.0157	0.0161	0.329	-0.00479873	0.0109934	0.662468	rsid	0.9999747489982137	1.7077304539049495e	8.036452829305432e-	0.0	1.3724441793879733e-07
rs34818697	172033341	0.0351	0.0191	0.0661	0.000908775	0.0146074	0.950393	rsid	0.9999231009175212	6.654159155718434e-	9.711239868628867e-	0.0	6.462510529651706e-07
rs11311614	172033646	-0.0354	0.0218	0.104	-0.000915964	0.0146096	0.950009	rsid	0.9999374676195193	5.231112626226499e-	9.713118286368625e-	0.0	5.081359320279817e-07
rs55686146	172033943	-0.0351	0.0192	0.0675	-0.000910199	0.0146074	0.950315	rsid	0.999923928823379	6.572157937842053e-	9.711306299150525e-	0.0	6.382909433508638e-07
rs187759965	172034110	-0.028	0.0787	0.722	0.0118725	0.0704814	0.86623	rsid	0.9999060503877683	4.882447935697076e-	4.3024296619939806	0.0	2.10083625491995e-06
rs9837821	172034271	0.005	0.0131	0.703	-0.00308166	0.00945452	0.744466	rsid	0.9999839518623379	9.35321580679157e-0	6.632882082124133e-	0.0	6.20397731579282e-08
rs4894416	172034350	0.0111	0.0191	0.561	0.0316913	0.0119423	0.00796141	rsid	0.9997186942462011	1.4912121843073222e	0.00026247842039800	0.0	3.9152115576910185e-06

(showing first 40 of 1020 rows; full data in the accompanying CSV)

Supplementary Table 50 - r5_pqtl_drug_map

ST50_r5_pqtl_drug_map.csv

target	drugs	note	best_outcome	best_or	best_fdr
TSLP	tezepelumab	approved anti-TSLP mAb	ALLERG_RHINITIS	2.085956102765701	0.7966196045700854
IL5	mepolizumab; reslizumab	approved anti-IL5 mAbs	L12_ATOPIC	0.8084727846152194	0.7445028621239614
IL13	lebrikizumab; tralokinum	approved anti-IL13 mAbs	ALLERG_RHINITIS	1.7091160166349286	0.7966196045700854
IL4	dupilumab	acts via IL4RA blockade;	L12_ATOPIC	1.1411141117099532	0.7966196045700854
IL6	tocilizumab	acts via IL6R; direction	ALLERG_RHINITIS	0.5986944822686532	0.5489652801014233
IL1A	anakinra	IL1R1 antagonist; indire	L12_ATOPIC	1.1665913493208389	0.7445028621239614
TNF	etanercept; infliximab;	approved anti-TNF biolog	L12_ATOPIC	2.2611176967955213	0.00012716794945610843
IL33	itepekimab	investigational anti-IL3	L12_ATOPIC	0.7190469273730231	0.7966196045700854
IFNG	emapalumab	approved anti-IFN-gamma	ALLERG_RHINITIS	1.3321164130943206	0.7777233436234617
IL18	tadekinig alfa	investigational IL-18BP	L12_ATOPIC	1.236479549423418	0.0008820492580277062
OSM	vixarelimab	investigational anti-OSM	L12_ATOPIC	0.775827438004852	0.5919719627862193
IL10	none marketed		ALLERG_RHINITIS	1.1333703261393833	0.7966196045700854
IL17A	secukinumab; ixekizumab	approved anti-IL17A mAbs	ALLERG_RHINITIS	1.4325257166853218	0.7445028621239614
IL8	none marketed	anti-CXCL8 programmes di	ALLERG_RHINITIS	1.316946195408221	0.7445028621239614
CCL2	carlumab	anti-CCL2 mAb; developme	ALLERG_RHINITIS	1.36044127037089	0.7445028621239614
S100A12	none marketed		ALLERG_RHINITIS	1.1157032641545246	0.7445028621239614
IL2	aldesleukin	recombinant IL-2; approv	ALLERG_ASTHMA	0.7498241370261243	0.6653434499295083
IL7	none		ALLERG_RHINITIS	0.6129422830044939	0.5919719627862193
TNFB	none	lymphotoxin-alpha; no ap	ALLERG_ASTHMA	1.016237307989494	0.7988161269291195
TRAIL	none	TNFSF10 agonists trialle	L12_ATOPIC	1.162823161038227	0.027177368196705576
VEGFA	bevacizumab	approved anti-VEGFA mAb	ALLERG_RHINITIS	0.96300381590452	0.7445028621239614

Supplementary Table 51 - r5_pqtl_build_check

ST51_r5_pqtl_build_check.csv

protein	gene	ok	error	tss	mid	n_cached	n_in_5mb	build_flag	cache_path	global_min_snp	global_min_p	global_min_pos
ADA	ADA	True		44652252	44618574	9304	46705	grch38_ok	pqtl_cis_cache/ADA.raw.t	rs76418401	0.0	20:43233080
ARTN	ARTN	True		43933272	43935258	7696	40725	grch38_ok	pqtl_cis_cache/ARTN.raw	rs189746786	1.34e-07	1:21802667
CASP8	CASP8	True		201233443	201297639	7062	36945	grch38_ok	pqtl_cis_cache/CASP8.ra	rs56328050	1.56e-30	2:202164805
CCL2_MCP1	CCL2	True		34255274	34256241	8019	40769	grch38_ok	pqtl_cis_cache/CCL2_MC	rs35728689	5.44e-52	3:46390228
CCL3_MIP1A	CCL3	True		36090169	36089212	7500	40775	grch38_ok	pqtl_cis_cache/CCL3_MIP	rs8951	8.62e-203	17:34415720
CST5	CST5	True		23880086	23878010	11188	33601	grch38_ok	pqtl_cis_cache/CST5.raw	rs11696133	0.0	20:23810508
CX3CL1	CX3CL1	True		57372457	57378750	10508	49628	grch38_ok	pqtl_cis_cache/CX3CL1.r	rs671623	4.99e-67	16:57412802
GDNF	GDNF	True		37840063	37826370	8631	47369	grch38_ok	pqtl_cis_cache/GDNF.raw	rs11740708	7.56e-86	5:37868088
HGF	HGF	True		81770438	81734724	9363	46937	grch38_ok	pqtl_cis_cache/HGF.raw.t	rs59950280	1.34e-20	4:3452345
IFNG	IFNG	True		68159748	68157258	9114	44333	grch38_ok	pqtl_cis_cache/IFNG.raw.	rs62230781	5.98e-08	21:37335521
IL10	IL10	True		206774541	206771071	5920	43665	grch38_ok	pqtl_cis_cache/IL10.raw.	rs3135932	3.39e-25	11:117864063
IL13	IL13	True		132656263	132658686	8632	40972	grch38_ok	pqtl_cis_cache/IL13.raw.	rs13119159	2.16e-08	4:40938582
IL17A	IL17A	True		52186375	52188506	9880	46946	grch38_ok	pqtl_cis_cache/IL17A.raw	rs187663944	7.21e-08	4:117612924
IL17C	IL17C	True		88638571	88639519	12645	45427	grch38_ok	pqtl_cis_cache/IL17C.raw	rs17700884	1.16e-14	16:88684495
IL18	IL18	True		112164391	112153546	8472	44057	grch38_ok	pqtl_cis_cache/IL18.raw.	rs5744249	1.53e-51	11:112025306
IL1A	IL1A	True		112784500	112779208	7342	39232	grch38_ok	pqtl_cis_cache/IL1A.raw.	rs11759846	1.36e-34	6:32586222
IL20	IL20	True		206865621	206867422	5817	43031	grch38_ok	pqtl_cis_cache/IL20.raw.	rs114685711	9.88e-08	1:104071162
IL22RA1	IL22RA1	True		24143301	24131535	7361	40927	grch38_ok	pqtl_cis_cache/IL22RA1.r	rs148404159	2.87e-07	9:100911150
IL24	IL24	True		206897443	206900791	5823	43047	grch38_ok	pqtl_cis_cache/IL24.raw.	rs73169406	8.22e-08	22:48661109
IL2	IL2	True		122456725	122454097	10176	49593	grch38_ok	pqtl_cis_cache/IL2.raw.t	rs145516501	3.08e-07	13:95200973
IL33	IL33	True		6215136	6248366	10852	62085	grch38_ok	pqtl_cis_cache/IL33.raw.	rs1559941	3.04e-07	3:69700787
IL4	IL4	True		132673986	132678332	8583	40970	grch38_ok	pqtl_cis_cache/IL4.raw.t	rs75610895	1.68e-07	10:6800809
IL5	IL5	True		132556838	132549141	8794	40939	grch38_ok	pqtl_cis_cache/IL5.raw.t	rs34639000	1.95e-07	5:95530542
IL6	IL6	True		22725884	22728943	10500	52558	grch38_ok	pqtl_cis_cache/IL6.raw.t	rs2228145	1.83e-45	1:154426970
IL7	IL7	True		78805527	78740635	9175	46374	grch38_ok	pqtl_cis_cache/IL7.raw.t	rs144088095	5.12e-14	8:79713753
IL8	CXCL8	True		73740519	73743949	8576	48556	grch38_ok	pqtl_cis_cache/IL8.raw.t	rs6446951	3.87e-15	4:74574265
LAP_TGFB1	TGFB1	True		41353961	41321082	9488	45990	grch38_ok	pqtl_cis_cache/LAP_TGFB	rs1800472	2.55e-43	19:41847860
MMP10	MMP10	True		102780628	102775565	10539	52877	grch38_ok	pqtl_cis_cache/MMP10.ra	rs17860955	2.6e-129	11:102649482
MMP1	MMP1	True		102798167	102793784	10620	52877	grch38_ok	pqtl_cis_cache/MMP1.raw	rs484915	5.73e-192	11:102673248
NGF	NGF	True		115338770	115312337	9329	43921	grch38_ok	pqtl_cis_cache/NGF.raw.t	rs3128517	5.11e-11	9:90362040
NRTN	NRTN	True		5805053	5816688	11201	53876	grch38_ok	pqtl_cis_cache/NRTN.raw	rs527252348	2.72e-07	17:28260302
NTF3	NTF3	True		5432088	5476812	9795	50256	grch38_ok	pqtl_cis_cache/NTF3.raw	rs28735437	1.06e-12	15:88514855
OPG	TNFRSF11B	True		118952251	118937904	8654	44671	grch38_ok	pqtl_cis_cache/OPG.raw.t	rs2247769	1.42e-50	8:120081031
OSM	OSM	True		30266858	30264843	8805	51264	grch38_ok	pqtl_cis_cache/OSM.raw.	rs3859189	4.95e-21	17:38137033
PDL1	CD274	True		5450503	5460534	10259	64043	grch38_ok	pqtl_cis_cache/PDL1.raw	rs822335	3.77e-27	9:5448218
PLAU	PLAU	True		73909177	73913339	7934	43763	grch38_ok	pqtl_cis_cache/PLAU.raw	rs4251805	6.1e-42	19:44174441
S100A12	S100A12	True		153375660	153374681	7951	33040	grch38_ok	pqtl_cis_cache/S100A12.	rs3014874	2.15e-36	1:153337943
SCF	KITLG	True		88580851	88536820	7369	43094	grch38_ok	pqtl_cis_cache/SCF.raw.t	rs6073958	1.16e-87	20:44551855
SLAMF1	SLAMF1	True		160647056	160627578	9409	48166	grch38_ok	pqtl_cis_cache/SLAMF1.r	rs60094514	2.45e-42	1:160636559
TGFA	TGFA	True		70554193	70500738	8994	43879	grch38_ok	pqtl_cis_cache/TGFA.raw	rs72912115	9.65e-14	2:70774295

(showing first 40 of 47 rows; full data in the accompanying CSV)

Supplementary Table 52 - r5_pharmacovigilance_summary

ST52_r5_pharmacovigilance_summary.csv

source	drug	target	r5_09_axis	ae_domain	n_drug_reports	n_domain_reports	n_overlap_reports	db_total_reports	ror	ror_lower	ror_upper	signal_ror_gt2	note
FAERS	dupilumab	IL4RA	IL4/IL13 axis (R5-09 po	ageing_general	515	7	0	385288	49.7597	2.8381	872.4198	False	
FAERS	dupilumab	IL4RA	IL4/IL13 axis (R5-09 po	immunosenescence	515	1323	3	385288	1.9831	0.6921	5.6819	False	
FAERS	dupilumab	IL4RA	IL4/IL13 axis (R5-09 po	infection	515	32796	93	385288	2.3824	1.9038	2.9815	True	
FAERS	dupilumab	IL4RA	IL4/IL13 axis (R5-09 po	malignancy	515	16566	30	385288	1.3989	0.97	2.0176	False	
FAERS	mepolizumab	IL5	IL5 axis (R5-09 positive	ageing_general	471	7	0	385288	54.4094	3.1029	954.0582	False	
FAERS	mepolizumab	IL5	IL5 axis (R5-09 positive	immunosenescence	471	1323	2	385288	1.5452	0.4453	5.3616	False	
FAERS	mepolizumab	IL5	IL5 axis (R5-09 positive	infection	471	32796	144	385288	4.7587	3.9114	5.7895	True	
FAERS	mepolizumab	IL5	IL5 axis (R5-09 positive	malignancy	471	16566	42	385288	2.2054	1.6086	3.0237	True	
FAERS	reslizumab	IL5	IL5 axis (R5-09 positive	ageing_general	1	7	0	385288	17123.5778	644.518	454939.824	False	
FAERS	reslizumab	IL5	IL5 axis (R5-09 positive	immunosenescence	1	1323	0	385288	96.7043	3.9375	2375.0639	False	
FAERS	reslizumab	IL5	IL5 axis (R5-09 positive	infection	1	32796	0	385288	3.5826	0.1459	87.9509	False	
FAERS	reslizumab	IL5	IL5 axis (R5-09 positive	malignancy	1	16566	0	385288	7.419	0.3022	182.1356	False	
FAERS	benralizumab	IL5RA	IL5 axis (R5-09 positive	ageing_general	391	7	0	385288	65.5412	3.7367	1149.5855	False	
FAERS	benralizumab	IL5RA	IL5 axis (R5-09 positive	immunosenescence	391	1323	2	385288	1.863	0.5366	6.4686	False	
FAERS	benralizumab	IL5RA	IL5 axis (R5-09 positive	infection	391	32796	78	385288	2.6953	2.1041	3.4526	True	
FAERS	benralizumab	IL5RA	IL5 axis (R5-09 positive	malignancy	391	16566	20	385288	1.2284	0.7873	1.9169	False	
FAERS	omalizumab	IGHE	IgE (background; not in	ageing_general	631	7	0	385288	40.6071	2.3166	711.7805	False	
FAERS	omalizumab	IGHE	IgE (background; not in	immunosenescence	631	1323	6	385288	3.0236	1.3935	6.5606	True	
FAERS	omalizumab	IGHE	IgE (background; not in	infection	631	32796	343	385288	12.9215	11.0448	15.1172	True	
FAERS	omalizumab	IGHE	IgE (background; not in	malignancy	631	16566	29	385288	1.0899	0.7529	1.5778	False	
FAERS	tezepelumab	TSLP	TSLP axis (R5-09 positi	ageing_general	287	7	0	385288	89.2741	5.0866	1566.8289	False	
FAERS	tezepelumab	TSLP	TSLP axis (R5-09 positi	immunosenescence	287	1323	1	385288	1.5189	0.305	7.5643	False	
FAERS	tezepelumab	TSLP	TSLP axis (R5-09 positi	infection	287	32796	70	385288	3.4891	2.6666	4.5653	True	
FAERS	tezepelumab	TSLP	TSLP axis (R5-09 positi	malignancy	287	16566	15	385288	1.2662	0.7588	2.1129	False	
FAERS	lebrikizumab	IL13	IL13 axis (R5-09 positiv	ageing_general	41	7	0	385288	618.8602	34.7782	11012.285	False	
FAERS	lebrikizumab	IL13	IL13 axis (R5-09 positiv	immunosenescence	41	1323	0	385288	3.495	0.2149	56.8465	False	
FAERS	lebrikizumab	IL13	IL13 axis (R5-09 positiv	infection	41	32796	2	385288	0.6802	0.1894	2.4423	False	
FAERS	lebrikizumab	IL13	IL13 axis (R5-09 positiv	malignancy	41	16566	2	385288	1.4087	0.3923	5.0582	False	
FAERS	tralokinumab	IL13	IL13 axis (R5-09 positiv	ageing_general	6	7	0	385288	3951.5436	203.908	76577.1738	False	
FAERS	tralokinumab	IL13	IL13 axis (R5-09 positiv	immunosenescence	6	1323	0	385288	22.3161	1.2565	396.354	False	
FAERS	tralokinumab	IL13	IL13 axis (R5-09 positiv	infection	6	32796	1	385288	2.9313	0.4819	17.8302	False	
FAERS	tralokinumab	IL13	IL13 axis (R5-09 positiv	malignancy	6	16566	0	385288	1.7121	0.0964	30.3936	False	
CanadaVigilance	dupilumab	IL4RA	IL4/IL13 axis (R5-09 po	ageing_general	503	12	0	1154017	91.6387	5.4181	1549.916	False	
CanadaVigilance	dupilumab	IL4RA	IL4/IL13 axis (R5-09 po	immunosenescence	503	3095	3	1154017	2.6014	0.9086	7.4482	False	
CanadaVigilance	dupilumab	IL4RA	IL4/IL13 axis (R5-09 po	infection	503	126059	71	1154017	1.3483	1.0497	1.7317	False	
CanadaVigilance	dupilumab	IL4RA	IL4/IL13 axis (R5-09 po	malignancy	503	48588	29	1154017	1.4147	0.9752	2.0522	False	
CanadaVigilance	mepolizumab	IL5	IL5 axis (R5-09 positive	ageing_general	9248	12	0	1154017	4.9511	0.2931	83.6329	False	
CanadaVigilance	mepolizumab	IL5	IL5 axis (R5-09 positive	immunosenescence	9248	3095	44	1154017	1.8089	1.3445	2.4336	False	
CanadaVigilance	mepolizumab	IL5	IL5 axis (R5-09 positive	infection	9248	126059	3577	1154017	5.2648	5.0469	5.4921	True	
CanadaVigilance	mepolizumab	IL5	IL5 axis (R5-09 positive	malignancy	9248	48588	396	1154017	1.0191	0.9212	1.1275	False	

(showing first 40 of 87 rows; full data in the accompanying CSV)

Supplementary Table 53 - r5_canada_vigilance_ror

ST53_r5_canada_vigilance_ror.csv

drug	pt	a	b	c	d	n_drug_reports	n_pt_reports	ror	ror_lower	ror_upper	signal_ror_gt2
tocilizumab	hypersensitivity	514	826	24397	1128280	1340	24911	28.788184465212343	25.768940148058345	32.1611816412069	True
tocilizumab	anaphylaxis	21	1319	9675	1143002	1340	9696	1.9248763136809552	1.256409212430736	2.9489984523447945	False
tocilizumab	urticaria	173	1167	35444	1117233	1340	35617	4.684224324767711	3.9923309138552225	5.4960267568494485	True
tocilizumab	angioedema	2	1338	4671	1148006	1340	4673	0.45899676748958695	0.13268004105219194	1.5878652953010182	False
tocilizumab	pruritus	79	1261	50297	1102380	1340	50376	1.3812268016006308	1.1009269141484594	1.7328920321069052	False
tocilizumab	rash	512	828	68487	1084190	1340	68999	9.792546468366682	8.768826537065427	10.935780965649323	True
tocilizumab	eczema	0	1340	3161	1149516	1340	3161	0.13562045882237944	0.008476223594080347	2.1699414423232075	False
tocilizumab	allergic_rhinitis	0	1340	2766	1149911	1340	2766	0.15503756141392463	0.009689481460470393	2.4806947149068126	False
tocilizumab	asthma	77	1263	15422	1137255	1340	15499	4.523032654368001	3.5939988030570365	5.692217920350427	True
tocilizumab	bronchospasm	0	1340	1993	1150684	1340	1993	0.21529959312936758	0.013454409660599057	3.4452581697001285	False
tocilizumab	wheezing	70	1270	8941	1143736	1340	9011	7.0979027987309085	5.579373855077089	9.029727250556501	True
sarilumab	hypersensitivity	28	146	24883	1128960	174	24911	8.826215282671173	5.907734139067695	13.1864559884125	True
sarilumab	anaphylaxis	0	174	9696	1144147	174	9696	0.338097210215425	0.0210612446557811	5.427491367376355	False
sarilumab	urticaria	13	161	35604	1118239	174	35617	2.6253739971905836	1.506562317493863	4.575043823338253	True
sarilumab	angioedema	0	174	4673	1149170	174	4673	0.7045580718941125	0.04388599332957547	11.311173315441765	False
sarilumab	pruritus	15	159	50361	1103482	174	50376	2.129304657109661	1.2639598275556196	3.587090526094572	True
sarilumab	rash	25	149	68974	1084869	174	68999	2.6827977069030235	1.7627099808368532	4.0831467538109285	True
sarilumab	eczema	0	174	3161	1150682	174	3161	1.042886138611618	0.06495540309488677	16.74397273032806	False
sarilumab	allergic_rhinitis	0	174	2766	1151077	174	2766	1.1921982043658859	0.07425287599658234	19.141838473147672	False
sarilumab	asthma	2	172	15497	1138346	174	15499	1.0645443047148793	0.305404844628973	3.7106633919892196	False
sarilumab	bronchospasm	0	174	1993	1151850	174	1993	1.6555963040339556	0.10310438767811929	26.584699096298376	False
sarilumab	wheezing	10	164	9001	1144842	174	9011	8.118097338386864	4.348697202180053	15.154769654342847	True
anakinra	hypersensitivity	168	1027	24743	1128079	1195	24911	7.476470956982141	6.349467715845752	8.803512431616774	True
anakinra	anaphylaxis	7	1188	9689	1143133	1195	9696	0.7444879321180978	0.3630254020237422	1.526786494773256	False
anakinra	urticaria	66	1129	35551	1117271	1195	35617	1.8502746612463772	1.4445299317684313	2.3699864203293024	False
anakinra	angioedema	8	1187	4665	1148157	1195	4673	1.761524065226833	0.8966324616738205	3.4606900430313376	False
anakinra	pruritus	26	1169	50350	1102472	1195	50376	0.4961461497357564	0.3375566564154806	0.7292435128123471	False
anakinra	rash	146	1049	68853	1083969	1195	68999	2.197587736192087	1.8484028054466193	2.6127377885552177	True
anakinra	eczema	2	1193	3159	1149663	1195	3161	0.7622017982221976	0.2202602322413389	2.6375690940732484	False
anakinra	allergic_rhinitis	1	1194	2765	1150057	1195	2766	0.5222169893818918	0.10524707412243196	2.591146464383872	False
anakinra	asthma	3	1192	15496	1137326	1195	15499	0.21540768555231346	0.0754305457585437	0.6151416581756437	False
anakinra	bronchospasm	0	1195	1993	1150829	1195	1993	0.241443306387751	0.015087225674556781	3.863856182502682	False
anakinra	wheezing	2	1193	9009	1143813	1195	9011	0.2659329270576538	0.07687344631423469	0.9199577368285115	False
canakinumab	hypersensitivity	3	61	24908	1129045	64	24911	2.5796263103446417	0.8785418085137141	7.574451023884805	False
canakinumab	anaphylaxis	2	62	9694	1144259	64	9696	4.721272886688328	1.3334350793200633	16.716537622472423	True
canakinumab	urticaria	1	63	35616	1118337	64	35617	0.7417186208595039	0.14690866052902418	3.7448201525262292	False
canakinumab	angioedema	1	63	4672	1149281	64	4673	5.8102475954550625	1.1505541896355944	29.34149249518055	True
canakinumab	pruritus	2	62	50374	1103579	64	50376	0.8763001121599222	0.2475258626504491	3.102309707555317	False
canakinumab	rash	9	55	68990	1084963	64	68999	2.6918847228672496	1.3525811696640802	5.35734455256813	True
canakinumab	eczema	0	64	3161	1150792	64	3161	2.821721364233198	0.17456583980360768	45.61093663186163	False

(showing first 40 of 253 rows; full data in the accompanying CSV)

Supplementary Table 54 - r5_two_database_concordance

ST54_r5_two_database_concordance.csv

drug	faers_n_reports	faers_n_events	faers_ror	faers_ror_lower	faers_ror_upper	prr	chi2	p_chi2	faers_signal	cv_top_pt	cv_max_ror	cv_ror_lower	cv_ror_upper	cv_top_pt_n	cv_signal	faers_ror_gt2	concordance	concordant
golimumab	594	360	12.91032933766	10.94764323673	15.22488447994	5.691947920897	1546.024530922	0.0	True	hypersensitivity	37.75104699660	32.21609716387	44.23693975377	280	True	True	concordant_signa	True
tofacitinib	820	451	10.27533915123	8.951159614770	11.79540966946	5.173902618057	1682.462661811	0.0	True	hypersensitivity	18.47760454207	16.43867536749	20.76942709679	398	True	True	concordant_signa	True
sarilumab	84	42	8.331718307129	5.431638372534	12.78021937181	4.665859153564	135.3618446767	0.0	True	hypersensitivity	8.826215282671	5.907734139067	13.18645598841	28	True	True	concordant_signa	True
tocilizumab	1036	457	6.632570501350	5.864530874001	7.501195304538	4.147932741584	1209.411629440	0.0	True	hypersensitivity	28.78818446521	25.76894014805	32.16118164120	514	True	True	concordant_signa	True
certolizumab	1213	375	3.750016611901	3.318647894410	4.237456047454	2.899846595856	518.0869784860	0.0	True	hypersensitivity	2.960256741863	2.653542059634	3.302423621261	346	True	True	concordant_signa	True
abatacept	168	50	3.530293155328	2.535738403352	4.914927244104	2.777229716242	63.61920040302	1.554312234475	True	hypersensitivity	2.373683700257	1.918508192841	2.936851836176	89	True	True	concordant_signa	True
adalimumab	2854	751	3.009185956984	2.766871998942	3.272721010285	2.480489862486	729.8208143391	0.0	True	hypersensitivity	4.802355311139	4.440728658741	5.193430697240	710	True	True	concordant_signa	True
etanercept	619	150	2.668401729144	2.219636219943	3.207898539466	2.264104056492	118.1543792637	0.0	True	hypersensitivity	4.109157119407	3.474209196138	4.860148390242	149	True	True	concordant_signa	True
quercetin	87	21	2.649468412942	1.621243564600	4.329813869091	2.251320864991	16.35186163202	5.260446449473	True	angioedema	26.67299741222	13.13567337705	54.16157744872	8	True	True	concordant_signa	True
infliximab	4074	970	2.639827699437	2.454398058412	2.839266539847	2.249392532904	736.3028480517	0.0	True	hypersensitivity	7.296235947992	6.748770594714	7.888112103034	750	True	True	concordant_signa	True
azathioprine	2038	460	2.442652264878	2.200587606480	2.711344038084	2.117029084385	300.3079583457	0.0	True	hypersensitivity	3.376844491165	3.141009280145	3.630386828079	811	True	True	concordant_signa	True
methotrexate	7860	1535	2.059558042642	1.945977142402	2.179768322343	1.852634175536	649.6393756671	0.0	True	hypersensitivity	3.419958276047	3.319181223341	3.523795123827	5929	True	True	concordant_signa	True
anakinra	245	39	1.576497408541	1.119385511943	2.220275367709	1.484728433304	6.906741197064	0.008587135853	True	hypersensitivity	7.476470956982	6.349467715845	8.803512431616	168	True	False	cv_only	False
cyclosporine	2131	215	0.933745671738	0.810670714250	1.075505706772	0.940430176935	0.903998318505	0.341711872563	False	eczema	1.645938063685	0.761099265573	3.559472767913	6	False	False	neither	False
dasatinib	260	25	0.885490944463	0.586217982972	1.337547184666	0.896501430572	0.334397932711	0.563080829849	False	rash	1.109885606611	0.800376261125	1.539083702995	38	False	False	neither	False
canakinumab	39	3	0.693666602126	0.213606132574	2.252619571856	0.717230709655	0.374596775280	0.540509231819	False	angioedema	5.810247595455	1.150554189635	29.34149249518	1	True	False	cv_only	False
upadacitinib	409	29	0.635016786735	0.435248815645	0.926473099848	0.660895792077	5.648006252007	0.017475229865	False	bronchospasm	192.6299640498	7.844423671157	4730.277787808	0	False	False	neither	False
ruxolitinib	551	30	0.478946047041	0.331438578753	0.692102038452	0.507315590760	16.06696939303	6.114123776768	False	bronchospasm	1.286810754104	0.080208603130	20.64469211833	0	False	False	neither	False
sirolimus	438	22	0.439927419550	0.286491157792	0.675539643048	0.468058919024	14.88972358443	0.000113982220	False	anaphylaxis	1.057709513621	0.369194093998	3.030247323541	3	False	False	neither	False
everolimus	484	17	0.302737813655	0.186571523014	0.491233508396	0.327228427639	26.32832034300	2.880339610156	False	angioedema	5.999209176682	1.187251366863	30.31414555510	1	True	False	cv_only	False
baricitinib	98	3	0.262818040308	0.083269430363	0.829515970152	0.285384834993	6.012734009671	0.014203004743	False	bronchospasm	4.344791625414	0.268842740241	70.21656694648	0	False	False	neither	False

Supplementary Table 55 - r5_target_vs_ae_direction

ST55_r5_target_vs_ae_direction.csv

gene	MR_OR	MR_p	MR_significant	MR_direction	FAERS_signal	FAERS_max_ROR	FAERS_drugs	CV_max_ROR	CV_concordant_drugs	direction_verdict
BCL2	0.9130337238912126	0.7064114649417548	False	protective	True	2.6494684129429893	quercetin	26.672997412228398	quercetin	MR null: no directional
BCR-ABL			False		False	0.8854909444632214	dasatinib	1.1098856066111038		MR null: no directional
CDKN1A	1.0089503879722377	0.899604963180958	False	risk-enhancing	False					MR null: no directional
CDKN2A	0.9506275982714152	0.7530023649522026	False	protective	False					MR null: no directional
CTLA4			False		True	3.530293155328422	abatacept	2.373683700257228	abatacept	MR null: no directional
DHFR			False		True	2.059558042642763	methotrexate	3.4199582760478706	methotrexate	MR null: no directional
GATA3	0.5225341733156397	0.0084495188665798	True	protective	False					MR predicts target activ
IFNG	1.027818851019585	0.8522069643137928	False	risk-enhancing	False					MR null: no directional
IL1B	1.2555402939129297	0.240113680450916	False	risk-enhancing	False	0.6936666021265954	canakinumab	5.8102475954550625		MR null: no directional
IL1R1	0.9401537258380664	0.5480626384004903	False	protective	True	1.576497408541676	anakinra	7.476470956982141		MR null: no directional
IL4	0.5833726764248982	1.3300430534712861e-08	True	protective	False					MR predicts target activ
IL6	1.3996949790340494	0.0438453763397472	True	risk-enhancing	False					MR predicts target activ
IL6R	0.8278906334199958	0.1485721490591847	False	protective	True	8.331718307129533	sarilumab, tocilizumab	28.788184465212343	sarilumab; tocilizumab	MR null: no directional
IL7R	1.028611005454926	0.5357708814898983	False	risk-enhancing	False					MR null: no directional
JAK1	0.8568788594539481	0.4513733791625631	False	protective	True	10.275339151238343	baricitinib, ruxolitinib	192.6299640498286	tofacitinib	MR null: no directional
JAK2			False		False	0.4789460470418956	baricitinib, ruxolitinib	4.344791625414175		MR null: no directional
JAK3	0.8985863930971623	0.6355724254658544	False	protective	True	10.275339151238343	tofacitinib	18.477604542071326	tofacitinib	MR null: no directional
MTOR	0.9868793489872888	0.8199914473452496	False	protective	False	0.4399274195501101	everolimus, sirolimus	5.999209176682457		MR null: no directional
NLRP3	1.058753824824234	0.2294035583744755	False	risk-enhancing	False					MR null: no directional
PI3K			False		True	2.6494684129429893	quercetin	26.672997412228398	quercetin	MR null: no directional
PPP3CA			False		False	0.93374567173803	cyclosporine	1.6459380636850862		MR null: no directional
PPP3R1			False		False	0.93374567173803	cyclosporine	1.6459380636850862		MR null: no directional
RORC	0.6604790093983314	0.007004115449084	True	protective	False					MR predicts target activ
SRC	1.2109291503628594	0.2398008882434572	False	risk-enhancing	False	0.8854909444632214	dasatinib	1.1098856066111038		MR null: no directional
STAT5A	1.2206135931593989	0.0375905647982774	True	risk-enhancing	False					MR predicts target activ
STAT5B	1.0356176488907491	0.8179227175900448	False	risk-enhancing	False					MR null: no directional
TNF	0.981189296534815	0.6541829197537594	False	protective	True	12.910329337662656	adalimumab, certolizumab	37.75104699660663	adalimumab; certolizumab	MR null: no directional
TP53	1.0687433526792591	0.6811845529792437	False	risk-enhancing	False					MR null: no directional
TPMT			False		True	2.442652264878093	azathioprine	3.376844491165771	azathioprine	MR null: no directional

Supplementary Table 56 - r5_trial_ae_summary

ST56_r5_trial_ae_summary.csv

drug	n_trials	n_with_results	n_with_ae_module	n_allergy_events	n_allergy_affected	allergy_terms	note
abatacept	344	0	0	0	0		no resultsSection/advers
adalimumab	872	0	0	0	0		no resultsSection/advers
anakinra	583	0	0	0	0		no resultsSection/advers
azathioprine	473	0	0	0	0		no resultsSection/advers
baricitinib	202	0	0	0	0		no resultsSection/advers
canakinumab	125	0	0	0	0		no resultsSection/advers
certolizumab	201	0	0	0	0		no resultsSection/advers
cyclosporine	1000	0	0	0	0		no resultsSection/advers
dasatinib	448	0	0	0	0		no resultsSection/advers
etanercept	569	0	0	0	0		no resultsSection/advers
everolimus	1000	0	0	0	0		no resultsSection/advers
fisetin	34	0	0	0	0		no resultsSection/advers
golimumab	193	0	0	0	0		no resultsSection/advers
infliximab	746	0	0	0	0		no resultsSection/advers
methotrexate	1000	0	0	0	0		no resultsSection/advers
quercetin	154	0	0	0	0		no resultsSection/advers
ruxolitinib	491	0	0	0	0		no resultsSection/advers
sarilumab	83	0	0	0	0		no resultsSection/advers
sirolimus	1000	0	0	0	0		no resultsSection/advers
temsirolimus	1000	0	0	0	0		no resultsSection/advers
tocilizumab	693	0	0	0	0		no resultsSection/advers
tofacitinib	331	0	0	0	0		no resultsSection/advers
upadacitinib	171	0	0	0	0		no resultsSection/advers

Supplementary Table 57 - r6_replication_mr

ST57_r6_replication_mr.csv

trait	source	outcome	cohort	n_snps	n_steiger_remo	mean_F	ivw_beta	ivw_se	ivw_p	ivw_or	ivw_lo	ivw_hi	ivw_re_beta	ivw_re_se	ivw_re_p	ivw_re_or	ivw_re_lo	ivw_re_hi	cochran_q	cochran_q_p	egger_beta	egger_se	egger_p	egger_or	egger_intercep	egger_int_p	wmed_beta	wmed_se	wmed_p	wmed_or
GrimAge	EAGLE_AD	L12_ATOPI	EAGLE AD	3	0	36.550204	-0.061362	0.0540729	0.2564533	0.9404820	0.8459070	1.0456307	-0.061362	0.0676373	0.3642838	0.9404820	0.8237140	1.0738027	3.1292624	0.2091651	-0.068068	0.0940135	0.4690470	0.9341963	-0.005295	0.8103631	-0.092286	0.0869116	0.2883063	0.9118437	621671184
GrimAge	BuduAggr	L12_ATOPI	Budu-Aggr	5	0	35.839037	0.0236836	0.0167223	0.1566916	1.0239663	0.9909496	1.0580838	0.0236836	0.0345219	0.4926828	1.0239663	0.9569738	1.0956487	17.047202	0.0018925	0.0273228	0.0398368	0.4927956	1.0276995	0.0029956	0.7122847	-0.000129	0.0490178	0.9978893	0.9998703	419773595
GrimAge	HanY_asth	ALLERG_A	HanY UKB	5	0	35.839037	0.0738724	0.0150765	9.592124e-7	1.0766694	1.0453193	1.1089598	0.0738724	0.0767262	0.3356449	1.0766694	0.9263427	1.2513912	103.59622	0.0	0.0785578	0.0882215	0.3732184	1.0817258	0.0053632	0.7648326	-0.018117	0.1341462	0.8925652	0.9820454	881871897
GrimAge	TAGC_asth	ALLERG_A	TAGC EUR	4	0	37.082747	0.0621651	0.0333732	0.0625014	1.0641380	0.9967587	1.1360721	0.0621651	0.1306606	0.6342342	1.0641380	0.8237158	1.3747336	45.984630	5.7143234	0.0067046	0.1494149	0.9642088	1.0067271	-0.024575	0.4694311	-0.068142	0.1497004	0.6489692	0.9341270	311711106
GrimAge	Neale_rhin	ALLERG_R	UKBB Nea	5	0	35.839037	0.0022090	0.0024784	0.3727795	1.0022114	0.9973546	1.0070918	0.0022090	0.0071967	0.7588850	1.0022114	0.9881737	1.0164485	33.726058	8.4807539	0.0011957	0.0075485	0.8741325	1.0011964	-0.001195	0.4561919	-0.000422	0.0090500	0.9627719	0.9995776	747599762
GrimAge	Neale_asth	ALLERG_A	UKBB Nea	5	0	35.839037	0.0104560	0.0018925	3.2972436	1.0105108	1.0067694	1.0142662	0.0104560	0.0095423	0.2731851	1.0105108	0.9917870	1.0295882	101.68948	0.0	0.0111193	0.0108882	0.3071484	1.0111813	0.0007830	0.7242958	-0.000281	0.0158941	0.9858850	0.9997188	484194041
Hannum	EAGLE_AD	L12_ATOPI	EAGLE AD	56	0	54.663434	0.0120753	0.0113177	0.2859988	1.0121485	0.9899434	1.0348516	0.0120753	0.0113177	0.2859988	1.0121485	0.9899434	1.0348516	54.019516	0.5120994	0.0125051	0.0113916	0.2723164	1.0125836	0.0014014	0.7542965	0.038725	0.0087833	1.0387637	1.0394850	6975245281
Hannum	BuduAggr	L12_ATOPI	Budu-Aggr	57	0	54.363953	0.0140688	0.0039194	0.0003313	1.0141682	1.0064071	1.0219892	0.0140688	0.0039194	0.0003313	1.0141682	1.0064071	1.0219892	25.184221	0.9998770	0.0149373	0.0024921	2.0524386	1.0150492	0.0029804	0.0058518	0.0174527	0.0011724	0.0	1.0176058	91111642
Hannum	HanY_asth	ALLERG_A	HanY UKB	57	0	54.363953	0.0077455	0.0035635	0.0297387	1.0077755	1.0007612	1.0148390	0.0077455	0.0044595	0.0824149	1.0077755	0.9990053	1.0166228	87.701528	0.0043225	0.0085735	0.0044136	0.0520783	1.0086103	0.0030450	0.0954762	0.0013868	0.0125535	0.9120299	1.0013878	61438339
Hannum	TAGC_asth	ALLERG_A	TAGC EUR	10	0	55.795233	0.0385982	0.0173315	0.0259440	1.0393528	1.0046390	1.0752660	0.0385982	0.0173315	0.0259440	1.0393528	1.0046390	1.0752660	4.7001232	0.8596268	0.0378432	0.0126566	0.0027897	1.0385684	-0.003265	0.3833716	0.0449845	0.0202578	0.0263783	1.0460117	910270384
Hannum	Neale_rhin	ALLERG_R	UKBB Nea	57	0	54.289394	-0.004857	0.0005776	0.0	0.9951539	0.9940279	0.9962811	-0.004857	0.0008086	1.8836969	0.9951539	0.9935779	0.9967323	109.75619	2.3685586	-0.004796	0.0008097	3.1487958	0.9952150	0.0003454	0.2913455	-0.006148	0.0010399	3.3803360	0.9938704	2855942
Hannum	Neale_asth	ALLERG_A	UKBB Nea	57	0	54.289394	0.0006508	0.0004409	0.1399651	1.0006510	0.9997865	1.0015163	0.0006508	0.0005528	0.2390622	1.0006510	0.9995674	1.0017359	88.006502	0.0040578	0.0006906	0.0005541	0.2126572	1.0006908	0.0002237	0.3178270	0.0006535	0.0005283	0.2161392	1.0006537	140223817
IEAA	EAGLE_AD	L12_ATOPI	EAGLE AD	131	0	132.28035	-0.009695	0.0044781	0.0303775	0.9903510	0.9816966	0.9990818	-0.009695	0.0044781	0.0303775	0.9903510	0.9816966	0.9990818	121.03605	0.7011238	-0.010070	0.0042547	0.0179353	0.9899797	0.0059755	0.0232082	-0.017963	0.0006720	0.0	0.9821965	062519247
IEAA	BuduAggr	L12_ATOPI	Budu-Aggr	133	0	130.89309	-0.000309	0.0016750	0.8533114	0.9996902	0.9964124	1.0029788	-0.000309	0.0017502	0.8594985	0.9996902	0.9962667	1.0031255	144.01761	0.2239098	-0.000301	0.0017572	0.8639580	0.9996989	-0.000201	0.8516836	0.0022186	0.0001389	0.0	1.0022210	69972318
IEAA	HanY_asth	ALLERG_A	HanY UKB	133	0	130.89309	-0.007535	0.0014607	2.4623486	0.9924930	0.9896566	0.9953375	-0.007535	0.0019678	0.0001285	0.9924930	0.9886723	0.9963284	239.74738	2.9577241	-0.007631	0.0018968	5.7337318	0.9923971	0.0038997	0.0011193	-0.010797	0.0003529	0.0	0.9892605	807934333
IEAA	TAGC_asth	ALLERG_A	TAGC EUR	18	0	66.109459	-0.009489	0.0115640	0.4118594	0.9905550	0.9683560	1.0132630	-0.009489	0.0130546	0.4672689	0.9905550	0.9655312	1.0162275	21.664775	0.1979569	-0.009969	0.0130334	0.4443194	0.9900799	0.0038095	0.3148604	-0.010776	0.0158403	0.4962826	0.9892808	652288043
IEAA	Neale_rhin	ALLERG_R	UKBB Nea	133	0	130.89309	0.0005245	0.0002386	0.0279757	1.0005246	1.0000567	1.0009928	0.0005245	0.0002386	0.0279757	1.0005246	1.0000567	1.0009928	124.36384	0.6693458	0.0005199	0.0002311	0.0244616	1.0005201	0.0001836	0.2017890	0.0003609	5.4452027	3.3721692	1.0003610	45553419
IEAA	Neale_asth	ALLERG_A	UKBB Nea	133	0	130.89309	-0.000809	0.0001822	8.9188551	0.9991906	0.9988337	0.9995477	-0.000809	0.0002110	0.0001250	0.9991906	0.9987774	0.9996041	176.99100	0.0054616	-0.000816	0.0002080	8.6924306	0.9991837	0.0002834	0.0295854	-0.001167	2.4239307	0.0	0.9988327	786296952
PhenoAge	EAGLE_AD	L12_ATOPI	EAGLE AD	89	0	69.245802	-0.017819	0.0061252	0.0036235	0.9823381	0.9706150	0.9942027	-0.017819	0.0061252	0.0036235	0.9823381	0.9706150	0.9942027	39.699353	0.9999979	-0.018694	0.0040276	3.4588011	0.9814794	-0.005342	0.0201729	-0.024279	0.0009095	0.0	0.9760130	148365445
PhenoAge	BuduAggr	L12_ATOPI	Budu-Aggr	89	0	69.245802	0.0039299	0.0023181	0.0900275	1.0039376	0.9993864	1.0085095	0.0039299	0.0023181	0.0900275	1.0039376	0.9993864	1.0085095	27.188562	0.9999999	0.0039458	0.0013020	0.0024422	1.0039536	9.2247053	0.9010110	0.0030311	0.0001359	0.0	1.0030357	006804964
PhenoAge	HanY_asth	ALLERG_A	HanY UKB	89	0	69.245802	-0.013306	0.0020030	3.0650815	0.9867811	0.9829147	0.9906628	-0.013306	0.0020030	3.0650815	0.9867811	0.9829147	0.9906628	36.820795	0.9999997	-0.013512	0.0012884	0.0	0.9865780	-0.001243	0.0975155	-0.014729	0.0004418	0.0	0.9853781	380239894
PhenoAge	TAGC_asth	ALLERG_A	TAGC EUR	9	0	55.990022	-0.017815	0.0140469	0.2046920	0.9823420	0.9556649	1.0097637	-0.017815	0.0140469	0.2046920	0.9823420	0.9556649	1.0097637	4.9000080	0.7682080	-0.022717	0.0149787	0.1293643	0.9775390	-0.002911	0.6236766	-0.014205	0.0120784	0.2395451	0.9858947	856436279
PhenoAge	Neale_rhin	ALLERG_R	UKBB Nea	89	0	69.245802	0.0004115	0.0003277	0.2092210	1.0004116	0.9997692	1.0010544	0.0004115	0.0003277	0.2092210	1.0004116	0.9997692	1.0010544	32.673375	0.9999999	0.0004232	0.0002013	0.0355341	1.0004233	7.1226407	0.5413280	0.0004858	7.3928522	4.9602544	1.0004859	854698278
PhenoAge	Neale_asth	ALLERG_A	UKBB Nea	89	0	69.245802	-0.001459	0.0002502	5.5680715	0.9985420	0.9980523	0.9990320	-0.001459	0.0002502	5.5680715	0.9985420	0.9980523	0.9990320	33.182902	0.9999999	-0.001477	0.0001538	0.0	0.9985238	-0.000111	0.2138305	-0.001602	3.2407045	0.0	0.9983984	42881441
Frailty	EAGLE_AD	L12_ATOPI	EAGLE AD	17	22	57.461402	-0.289330	0.2612158	0.2680219	0.7487648	0.4487392	1.2493866	-0.289330	0.2612158	0.2680219	0.7487648	0.4487392	1.2493866	10.117164	0.8604384	-0.295104	0.2177408	0.1753222	0.7444541	-0.000816	0.8818645	-0.049976	0.2924933	0.8643318	0.9512521	209252524
Frailty	BuduAggr	L12_ATOPI	Budu-Aggr	98	0	48.919159	0.9777259	0.0427973	0.0	2.6584039	2.4445062	2.8910180	0.9777259	0.0912146	0.0	2.6584039	2.2231945	3.1788094	440.62361	0.0	0.9554283	0.0910503	0.0	2.5997839	0.0049685	0.0765139	1.1936732	0.0820307	0.0	3.2991776	80712377
Frailty	HanY_asth	ALLERG_A	HanY UKB	16	65	34.786098	0.8117883	0.0771887	0.0	2.2519316	1.9357568	2.6197484	0.8117883	0.1614316	4.9386207	2.2519316	1.6411256	3.0900718	65.608568	2.6722201	0.8133627	0.1620277	5.1699234	2.2554799	0.0031875	0.3611082	0.9921799	0.2135993	3.3999116	2.6971075	067295725
Frailty	TAGC_asth	ALLERG_A	TAGC EUR	9	8	37.078264	0.4575604	0.2085841	0.0282606	1.5802142	1.0499426	2.3782985	0.4575604	0.2910942	0.1159821	1.5802142	0.8931647	2.7957631	15.580973	0.0487857	0.3763086	0.2883776	0.1919213	1.4568966	0.0081574	0.2487962	0.4678223	0.2970849	0.1153235	1.5965138	175868931
Frailty	Neale_rhin	ALLERG_R	UKBB Nea	72	6	46.318983	0.1432214	0.0051323	0.0	1.1539853	1.1424350	1.1656523	0.1432214	0.0118274	0.0	1.1539853	1.1275415	1.1810493	377.05576	0.0	0.1417322	0.0117434	0.0	1.1522680	0.0006412	0.1192195	0.1753246	0.0142323	0.0	1.1916329	932066292
Frailty	Neale_asth	ALLERG_A	UKBB Nea	16	62	34.786098	0.0809220	0.0096711	0.0	1.0842864	1.0639267	1.1050356	0.0809220	0.0185412	1.2745183	1.0842864	1.0455899	1.1244150	55.132677	1.6964714	0.0811943	0.0180513	6.8603173	1.0845816	0.0005085	0.1978869	0.0927979	0.0186633	6.6195437	1.0972400	073589262
BloodInde	EAGLE_AD	L12_ATOPI	EAGLE AD	82	0	1142.1047	0.4754308	0.1026613	3.6379975	1.6087070	1.3154970	1.9672705	0.4754308	0.1457909	0.0011100	1.6087070	1.2088633	2.1408032	163.35494	1.6785417	0.4775289	0.1459317	0.0010668	1.6120858	0.0035968	0.3556627	0.4328557	0.1045067	3.4445369	1.5416537	603321956
BloodInde	BuduAggr	L12_ATOPI	Budu-Aggr	93	0	1184.5936	0.3702714	0.0400849	0.0	1.4481275	1.3387078	1.5664908	0.3702714	0.0812164	5.1381282	1.4481275	1.2350199	1.6980078	377.67010	0.0	0.3678870	0.0815745	6.4880306	1.4446788	-0.001285	0.5323430	0.2001537	0.1038533	0.0539456	1.2215906	153180298
BloodInde	HanY_asth	ALLERG_A	HanY UKB	93	0	1184.5936	0.9960806	0.0352504	0.0	2.7076488	2.5268917	2.9013367	0.9960806	0.1237065	8.8817841	2.7076488	2.1246685	3.4505909	1133.0352	0.0	0.9996412	0.1243150	8.8817841	2.7173067	0.0017858	0.5675688	0.7376301	0.5660504	0.1925347	2.0909742	87450236
BloodInde	TAGC_asth	ALLERG_A	TAGC EUR	22	0	988.68040	1.2473489	0.1017728	0.0	3.4811020	2.8515825	4.2495951	1.2473489	0.2518020	7.2819587	3.4811020	2.1250959	5.7023644	128.55051	0.0	1.3006048	0.2630310	7.6268615	3.6715166	0.0060361	0.4415946	1.8468693	0.7432789	0.0129639	6.3399404	44462622
BloodInde	Neale_rhin	ALLERG_R	UKBB Nea	92	0	1187.3035	0.1045167	0.0058086	0.0	1.1101732	1.0976057	1.1228846	0.1045167	0.0136290	1.7319479	1.1101732	1.0809099	1.1402288	500.98627	0.0	0.1047572	0.0137168	2.2204460	1.1104410	0.0001109	0.7472841	0.1146124	0.0381277	0.0026465	1.1214387	025861392
BloodInde	Neale_asth	ALLERG_A	UKBB Nea	92	0	1187.3035	0.1037281	0.0044353	0.0	1.1092988	1.0996972	1.1189842	0.1037281	0.0136474	2.9531932	1.1092988	1.0800195	1.1393719	861.58718	0.0	0.1041695	0.0137167	3.0864200	1.1097885	0.0002030	0.5556305	0.0795087	0.0557852	0.1540833	1.0827544	216594915

Supplementary Table 58 - r6_meta_analysis

ST58_r6_meta_analysis.csv

trait	outcome	outcome_pretty	scheme	k	fe_beta	fe_se	fe_p	re_beta	re_se	re_p	Q	Q_p	I2	tau2	re_or	re_or_lo	re_or_hi	fe_or	fe_or_lo	fe_or_hi	cohorts
GrimAge	ALLERG_ASTH	Asthma	primary	3	0.0779843961	0.0126210008	6.4533756116	0.0779843961	0.0126210008	6.4533756116	1.3625638060	0.5059679745	0.0	0.0	1.0811057890	1.0546903687	1.1081828011	1.0811057890	1.0546903687	1.1081828011	FinnGen R12 (discovery);
GrimAge	ALLERG_ASTH	Asthma	all_cohorts	4	0.0119410619	0.0018716151	1.7699064436	0.0596788466	0.0256885691	0.0201701607	29.360598732	1.8808335844	89.782224717	0.0021355518	1.0614955889	1.0093729042	1.1163098202	1.0120126410	1.0083070096	1.0157318910	FinnGen R12 (discovery);
GrimAge	L12_ATOPIC	Atopic dermati	primary	2	0.0045493668	0.0131708316	0.7297844543	0.0002431104	0.0251378419	0.9922837023	3.4485362918	0.0633078102	71.002190049	0.0009017337	1.0002431400	0.9521553642	1.0507595469	1.0045597309	0.9789590524	1.0308298907	FinnGen R12 (discovery);
GrimAge	L12_ATOPIC	Atopic dermati	all_cohorts	3	0.0008578878	0.0127966944	0.9465500180	-0.008421136	0.0231766601	0.7163467570	4.8511541233	0.0884270764	58.772697194	0.0008884868	0.9916142216	0.9475767045	1.0376983308	1.0008582559	0.9760674041	1.0262787633	FinnGen R12 (discovery);
GrimAge	ALLERG_RHINI	Allergic rhinitis	primary	2	0.0021923084	0.0024696839	0.3747086787	0.0021923084	0.0024696839	0.3747086787	0.0064139957	0.9361677233	0.0	0.0	1.0021947132	0.9973552315	1.0070576777	1.0021947132	0.9973552315	1.0070576777	FinnGen R12 (discovery);
GrimAge	ALLERG_RHINI	Allergic rhinitis	all_cohorts	2	0.0021923084	0.0024696839	0.3747086787	0.0021923084	0.0024696839	0.3747086787	0.0064139957	0.9361677233	0.0	0.0	1.0021947132	0.9973552315	1.0070576777	1.0021947132	0.9973552315	1.0070576777	FinnGen R12 (discovery);
Hannum	ALLERG_ASTH	Asthma	primary	3	0.0108851345	0.0031548334	0.0005599555	0.0156119749	0.0067674654	0.0210594986	4.6387257201	0.0983362195	56.884711003	7.4808134629	1.0157344785	1.0023505007	1.0292971671	1.0109445932	1.0047127311	1.0172151091	FinnGen R12 (discovery);
Hannum	ALLERG_ASTH	Asthma	all_cohorts	4	0.0008470247	0.0004367531	0.0524566748	0.0094094681	0.0049013144	0.0548851071	14.960521630	0.0018506986	79.947223272	6.0476991049	1.0094538763	0.9998029113	1.0191980009	1.0008473835	0.9999909884	1.0017045119	FinnGen R12 (discovery);
Hannum	L12_ATOPIC	Atopic dermati	primary	2	0.0122003384	0.0030633627	6.8147636148	0.0122003384	0.0030633627	6.8147636148	0.5840336256	0.4447355974	0.0	0.0	1.0122750661	1.0062153833	1.0183712419	1.0122750661	1.0062153833	1.0183712419	FinnGen R12 (discovery);
Hannum	L12_ATOPIC	Atopic dermati	all_cohorts	3	0.0121918067	0.0029569610	3.7382884278	0.0121918067	0.0029569610	3.7382884278	0.5841472583	0.7467135540	0.0	0.0	1.0122664297	1.0064166623	1.0181501988	1.0122664297	1.0064166623	1.0181501988	FinnGen R12 (discovery);
Hannum	ALLERG_RHINI	Allergic rhinitis	primary	2	-0.0050241145	0.0005755046	0.0	-0.015325328	0.0114135933	0.1793607191	11.416807077	0.0007278266	91.240983637	0.0002395040	0.9847915069	0.9630056779	1.0070701912	0.9949884848	0.9938667814	0.9961114543	FinnGen R12 (discovery);
Hannum	ALLERG_RHINI	Allergic rhinitis	all_cohorts	2	-0.0050241145	0.0005755046	0.0	-0.015325328	0.0114135933	0.1793607191	11.416807077	0.0007278266	91.240983637	0.0002395040	0.9847915069	0.9630056779	1.0070701912	0.9949884848	0.9938667814	0.9961114543	FinnGen R12 (discovery);
IEAA	ALLERG_ASTH	Asthma	primary	3	-0.0085703957	0.0013308761	1.1973799729	-0.009668213	0.0024058046	5.8519500151	3.1098588146	0.2112042915	35.688398760	6.8893334163	0.9903783737	0.9857193593	0.9950594089	0.9914662253	0.9888833387	0.9940558582	FinnGen R12 (discovery);
IEAA	ALLERG_ASTH	Asthma	all_cohorts	4	-0.0009525245	0.0001805880	1.3306511026	-0.006951408	0.0032555980	0.0327432798	36.488043227	5.9045270450	91.778128574	3.0133195563	0.9930726965	0.9867561017	0.9994297261	0.9990479286	0.9986943757	0.9994016068	FinnGen R12 (discovery);
IEAA	L12_ATOPIC	Atopic dermati	primary	2	-0.0016830416	0.0013470630	0.2115130894	-0.001951865	0.0019179645	0.3088321481	1.8987956051	0.1682131910	47.335037153	3.5672108954	0.9980500379	0.9943052010	1.0018089790	0.9983183738	0.9956860466	1.0009576602	FinnGen R12 (discovery);
IEAA	L12_ATOPIC	Atopic dermati	all_cohorts	3	-0.0023479174	0.0012899649	0.0687376204	-0.003445255	0.0022778018	0.1303977898	4.8347198075	0.0891566892	58.632556185	8.7895627925	0.9965606728	0.9921214528	1.0010197559	0.9976548367	0.9951356207	1.0001804302	FinnGen R12 (discovery);
IEAA	ALLERG_RHINI	Allergic rhinitis	primary	2	0.0005536611	0.0002379743	0.0199888182	0.0020513263	0.0022978897	0.3720178001	2.5507240546	0.1102437220	60.795445584	7.5544869952	1.0020534317	0.9975504675	1.0065767225	1.0005538144	1.0000872351	1.0010206113	FinnGen R12 (discovery);
IEAA	ALLERG_RHINI	Allergic rhinitis	all_cohorts	2	0.0005536611	0.0002379743	0.0199888182	0.0020513263	0.0022978897	0.3720178001	2.5507240546	0.1102437220	60.795445584	7.5544869952	1.0020534317	0.9975504675	1.0065767225	1.0005538144	1.0000872351	1.0010206113	FinnGen R12 (discovery);
PhenoAge	ALLERG_ASTH	Asthma	primary	3	-0.0190491709	0.0018499552	0.0	-0.029900175	0.0171527646	0.0813040208	62.764889944	2.3536728122	96.813505127	0.0008146057	0.9705424128	0.9384557311	1.0037261684	0.9811311188	0.9775800651	0.9846950716	FinnGen R12 (discovery);
PhenoAge	ALLERG_ASTH	Asthma	all_cohorts	4	-0.0017752251	0.0002480336	8.2334139506	-0.022080291	0.0086410072	0.0106098940	151.54968229	0.0	98.020451145	0.0002582430	0.9781616939	0.9617346811	0.9948692901	0.9982263493	0.9977411836	0.9987117510	FinnGen R12 (discovery);
PhenoAge	L12_ATOPIC	Atopic dermati	primary	2	0.0003989267	0.0019207064	0.8354648290	-0.001417377	0.0056239419	0.8010212559	7.3998298164	0.0065230047	86.486175698	5.4848267457	0.9985836265	0.9876367569	1.0096518301	1.0003990063	0.9966399996	1.0041721907	FinnGen R12 (discovery);
PhenoAge	L12_ATOPIC	Atopic dermati	all_cohorts	3	-0.0012320677	0.0018327168	0.5014162949	-0.006030931	0.0058721290	0.3044000549	15.454476486	0.0004406594	87.058765777	8.6931810827	0.9939872184	0.9826126315	1.0054934760	0.9987686909	0.9951874248	1.0023628445	FinnGen R12 (discovery);
PhenoAge	ALLERG_RHINI	Allergic rhinitis	primary	2	0.0003805778	0.0003269527	0.2444180111	-0.001089314	0.0027217245	0.6889875515	1.8558776485	0.1731013336	46.117137583	9.5529610404	0.9989112788	0.9935966946	1.0042542898	1.0003806502	0.9997397843	1.0010219270	FinnGen R12 (discovery);
PhenoAge	ALLERG_RHINI	Allergic rhinitis	all_cohorts	2	0.0003805778	0.0003269527	0.2444180111	-0.001089314	0.0027217245	0.6889875515	1.8558776485	0.1731013336	46.117137583	9.5529610404	0.9989112788	0.9935966946	1.0042542898	1.0003806502	0.9997397843	1.0010219270	FinnGen R12 (discovery);
Frailty	ALLERG_ASTH	Asthma	primary	3	0.9657509323	0.0550147330	0.0	0.8698270711	0.1949720123	8.1467010142	20.001009058	4.5377029951	90.000504503	0.0975250368	2.3864981234	1.6285367737	3.4972334580	2.6267594347	2.3582552755	2.9258347049	FinnGen R12 (discovery);
Frailty	ALLERG_ASTH	Asthma	all_cohorts	4	0.1074463857	0.0095251377	0.0	0.6476835256	0.3270817843	0.0476823729	270.92527434	0.0	98.892683597	0.4144067527	1.9111086633	1.0066250777	3.6282985630	1.1134311634	1.0928370531	1.1344133621	FinnGen R12 (discovery);
Frailty	L12_ATOPIC	Atopic dermati	primary	2	0.7456561274	0.0335391470	0.0	0.6779057833	0.3007196850	0.0241786817	76.204197087	0.0	98.687736321	0.1784928386	1.9697483299	1.0925290090	3.5513093485	2.1078239825	1.9737186425	2.2510411795	FinnGen R12 (discovery);
Frailty	L12_ATOPIC	Atopic dermati	all_cohorts	3	0.7288704631	0.0332660622	0.0	0.4123864289	0.2724349412	0.1301004004	91.648529615	0.0	97.817750040	0.2022156223	1.5104179928	0.8855148462	2.5763119871	2.0727380508	1.9419040895	2.2123868272	FinnGen R12 (discovery);
Frailty	ALLERG_RHINI	Allergic rhinitis	primary	2	0.1450114449	0.0051203455	0.0	0.3271554676	0.1910841076	0.0868778874	25.988634994	3.4343315480	96.152164204	0.0703185137	1.3870170967	0.9537350387	2.0171392981	1.1560528011	1.1445088209	1.1677132186	FinnGen R12 (discovery);
Frailty	ALLERG_RHINI	Allergic rhinitis	all_cohorts	2	0.1450114449	0.0051203455	0.0	0.3271554676	0.1910841076	0.0868778874	25.988634994	3.4343315480	96.152164204	0.0703185137	1.3870170967	0.9537350387	2.0171392981	1.1560528011	1.1445088209	1.1677132186	FinnGen R12 (discovery);
BloodIndex93	ALLERG_ASTH	Asthma	primary	3	0.9907189726	0.0302026904	0.0	1.0132397585	0.0892270962	0.0	10.723151544	0.0046935043	81.348766809	0.0188692758	2.7545105239	2.3125588759	3.2809232687	2.6931700921	2.5383688746	2.8574117883	FinnGen R12 (discovery);
BloodIndex93	ALLERG_ASTH	Asthma	all_cohorts	4	0.1224525269	0.0043882362	0.0	0.7939626848	0.3159608241	0.0119759676	854.99224643	0.0	99.649119625	0.3951779699	2.2121451145	1.1908644469	4.1092720672	1.1302654616	1.1205857994	1.1400287371	FinnGen R12 (discovery);
BloodIndex93	L12_ATOPIC	Atopic dermati	primary	2	0.4812536600	0.0307636073	0.0	0.5039959300	0.1350068403	0.0001891269	18.650718129	1.5698813108	94.638276161	0.0345022897	1.6553226260	1.2704644255	2.1567648343	1.6181016806	1.5234187542	1.7186693031	FinnGen R12 (discovery);
BloodIndex93	L12_ATOPIC	Atopic dermati	all_cohorts	3	0.4807738701	0.0294689395	0.0	0.4958102107	0.1003366749	7.7537564524	18.653670103	8.9003483200	89.278249326	0.0258775476	1.6418279270	1.3487123574	1.9986462844	1.6173255179	1.5265568104	1.7134913114	FinnGen R12 (discovery);
BloodIndex93	ALLERG_RHINI	Allergic rhinitis	primary	2	0.1074586952	0.0057861056	0.0	0.2891540456	0.1901985802	0.1284420078	33.190509013	8.3557374352	96.987090497	0.0702329279	1.3352974094	0.9197667519	1.9385558000	1.1134448692	1.1008888821	1.1261440614	FinnGen R12 (discovery);
BloodIndex93	ALLERG_RHINI	Allergic rhinitis	all_cohorts	2	0.1074586952	0.0057861056	0.0	0.2891540456	0.1901985802	0.1284420078	33.190509013	8.3557374352	96.987090497	0.0702329279	1.3352974094	0.9197667519	1.9385558000	1.1134448692	1.1008888821	1.1261440614	FinnGen R12 (discovery);

Supplementary Table 59 - r6_ldsc_h2

ST59_r6_ldsc_h2.csv

trait	n_snps	h2	h2_se	intercept	intercept_se	mean_chi2	lambda_gc	intercept_ratio	mean_N
GrimAge	1175366	0.10451458977535343	0.017717614389813847	1.0142382772633398	0.006221123835403891	1.0751721531148744	1.072745020366781	0.18940893234202813	29369.733933089778
Hannum	1175315	0.1189965445617679	0.024952146811927197	1.011000002380377	0.007555126479971174	1.0801995852976018	1.049011751552148	0.1371578461354691	29054.026428659552
IEAA	1175374	0.1639275075443605	0.025892039697555692	0.9996845143252803	0.007801800897899301	1.094745293994269	1.0637348001428908	-0.003329829497798719	29376.59296360137
PhenoAge	1175371	0.0965436809364225	0.019548370526330527	1.0188176077634274	0.00858449081697616	1.0741146793335423	1.057489895336982	0.25389852499720794	29384.259099467316
Frailty	1168664	0.11084777654924406	0.005337449159583173	1.017248302478022	0.009240558000876602	1.3997523109722234	1.329721223818186	0.043147474084822446	175226.0
ALLERG_ASTHMA	1175751	0.040992335550461986	0.004489646840371183	1.0742114562453442	0.012438508946307651	1.3043390448715404	1.2376963402647372	0.2438446774933804	283740.0
ALLERG_RHINITIS	1175751	0.014769505269137995	0.001746032298730723	1.0554598962965898	0.010114369795995238	1.1997334815163174	1.1673547402298554	0.27766950175581306	490219.0
L12_ATOPIC	1175751	0.025860955764112417	0.004371821932163779	1.1227826481517718	0.020217567198881417	1.3617640780343045	1.2410337176386	0.33939977904641155	464119.0

Supplementary Table 60 - r6_ldsc_rg

ST60_r6_ldsc_rg.csv

trait1	trait2	n_snps	gencov	gencov_se	rg	rg_se	z	p	rg_clip	cross_intercept	cross_intercept_se	mean_z1z2	q_bh
GrimAge	Hannum	1175301	0.02017511323461061	0.01313921180040091	0.1809090718698629	0.10103296947480978	1.7905944248720551	0.07335840235179872	0.1809090718698629	0.4149400983980739	0.00481328279602206	0.4291297993021229	0.12082560387355083
GrimAge	IEAA	1175357	0.03177193464506236	0.01284849040615945	0.2427332565922475	0.09183067580530138	2.6432698492480413	0.00821095663294286	0.24273325659222475	0.1331132366790157	0.00444686100892761	0.1523818053565054	0.016421913265885733
GrimAge	PhenoAge	1175357	0.06752391869476088	0.01296429985097195	0.6722135331411303	0.09923025704110183	6.774279873755592	1.25027435598878986	0.6722135331411303	0.3797311480762623	0.00477091775806597	0.4186843111815894	3.889742440854013e-11
GrimAge	Frailty	1165110	0.03149015619284336	0.00575933118494129	0.2925653314039992	0.0570318140130167	5.129861928242811	2.8995477529716546e-	0.2925653314039992	0.0005129612354405	0.00501565823963328	0.0433754184383972	7.380667007564209e-07
GrimAge	ALLERG_ASTHMA	1167344	0.01077222653676108	0.00470296953646318	0.16457569062724708	0.07046535227043399	2.3355547843659914	0.01951445748207015	0.16457569062724708	0.0054179223492271	0.00524434869886191	0.0260032750347839	0.034150300593622764
GrimAge	ALLERG_RHINITIS	1167344	-0.00014051568381413	0.00331495211350934	-0.0035764599577944	0.08500165613333324	-0.04207517971396213	0.9664387663632997	-0.0035764599577944	0.0072965306161656	0.00475667830160039	0.0070318118819354	0.9664387663632997
GrimAge	L12_ATOPIC	1167344	0.00472573512347617	0.00375897446294690	0.09089890048467074	0.07266278967203338	1.2509690433707106	0.21094577192644814	0.09089890048467074	0.0029229572658401	0.00539100971582472	0.0123805388775959	0.2684764369972976
Hannum	IEAA	1175303	0.0870512017436129	0.01438319805726418	0.6232778279515327	0.07598584991674645	8.202551246507399	2.35338364328537076	0.6232778279515327	0.3647065771439595	0.00494378774558525	0.4150033790121805	1.3178948402398075e-15
Hannum	PhenoAge	1175300	0.05581632790276415	0.01355511203008652	0.5207534543612996	0.08915583567944324	5.8409351490310835	5.1908609918180586-	0.5207534543612996	0.4704197332483621	0.00497342610944551	0.5022181997230669	1.45344107770905636-08
Hannum	Frailty	1165069	0.00663136440500868	0.00584883892797606	0.05773939453302669	0.05261563170544118	1.0973809999330606	0.2724748756090477	0.05773939453302669	0.0024380241383998	0.00456288870021758	0.0130245094705671	0.33170854421971024
Hannum	ALLERG_ASTHMA	1167293	0.00791503914075245	0.00444564741698315	0.1133273047363244	0.0648452058680512	1.7476589551882353	0.08052310010149774	0.1133273047363244	-0.0019773839258860	0.00466032918957808	0.0134370621166929	0.12525815571344093
Hannum	ALLERG_RHINITIS	1167293	0.00150422806606977	0.00355663446406389	0.03588091737857171	0.08406544466156352	0.4268212405587534	0.6695095344090213	0.03588091737857177	0.0030893762468486	0.00435225171773568	0.0078750305872266	0.6943061838315776
Hannum	L12_ATOPIC	1167293	0.00644817108895236	0.00383362956306634	0.11623772225383906	0.07066562136386823	1.6448977594821255	0.09999089705678482	0.11623772225383906	-0.0037759410322969	0.00521818664783905	0.0094735180179289	0.1473550061889460
IEAA	PhenoAge	1175361	0.07072926315623166	0.01562679957519246	0.5622264708427249	0.08208865922261262	6.84901515223982	7.4360121549941e-12	0.5622264708427249	0.3869134717949484	0.00640202372946479	0.4268975816802492	2.60260425424793536-11
IEAA	Frailty	1165120	0.00884734095676487	0.00604095004716362	0.06563316588426368	0.04544955111960833	1.444088319190188	0.14871413337716957	0.06563316588426368	-0.0011422697857780	0.00508997003561483	0.0116338317345265	0.2081997867280374
IEAA	ALLERG_ASTHMA	1167351	0.00677348656041383	0.00492149864256289	0.08262948083564081	0.05953112170222575	1.38800476915172	0.16513559288125146	0.08262948083564081	-0.0041462201141179	0.00603955093467547	0.0098912699065197	0.22018079050833528
IEAA	ALLERG_RHINITIS	1167351	-0.00155085035223955	0.00352253152414331	-0.03151817054432427	0.07262161943694755	-0.4340053387502518	0.6642845642099584	-0.0315181705443242	0.0092426115912152	0.00551603903276670	0.0060660399185709	0.6943061838315776
IEAA	L12_ATOPIC	1167351	0.01049431664345711	0.00430463660399022	0.16117807347139640	0.06648079918916923	2.424430443634842	0.01533242015429857	0.16117807347139640	-0.0112436453521983	0.00675898869378729	0.0106958165026353	0.02862051762135723
PhenoAge	Frailty	1165117	0.02502794138393086	0.00575016695196299	0.24193556181361292	0.05694322872188926	4.248715207127194	2.15000040218299356	0.24193556181361292	-0.0034355551108557	0.00523558051687408	0.0319729069697557	5.016667605093651e-05
PhenoAge	ALLERG_ASTHMA	1167349	0.00518913893360064	0.00515109770464893	0.08248634123700717	0.07804368067311366	1.0569253080528278	0.29054567715395960	0.08248634123700717	0.0010826449209755	0.00628037842544950	0.0117851043818251	0.33896995667961960
PhenoAge	ALLERG_RHINITIS	1167349	-0.00170869990989443	0.00343774172542311	-0.04525024092759870	0.09294240759160206	-0.4868632317599587	0.626355256701292	-0.0452502409275987	0.0085932108095878	0.00550458922229440	0.0045084389838171	0.6943061838315776
PhenoAge	L12_ATOPIC	1167349	0.01367146190355230	0.00473873074277066	0.2736092302557877	0.10220754422600188	2.676996422600483	0.00742854281299608	0.2736092302557877	-0.0129166746909303	0.00792996734569645	0.0163565431422233	0.01599993836645311
Frailty	ALLERG_ASTHMA	1160670	0.03088067488774526	0.00267858791864151	0.4581123063935907	0.03819606345757751	11.993704715209535	3.83367373751318646	0.4581123063935907	-0.0059425851849141	0.00586608180208312	0.1309303096600697	3.578095488345641e-32
Frailty	ALLERG_RHINITIS	1160670	0.01428770096198687	0.00182699745709096	0.3531147710661577	0.04576310979538401	7.716144568081239	1.19901307808438826	0.3531147710661577	-0.0006144413251261	0.00620319777507972	0.0836873818913005	5.595394364393812e-14
Frailty	L12_ATOPIC	1160670	0.01339960240933447	0.00193483839404464	0.25026846703164680	0.03319109891018862	7.540228412106664	4.69148929920137e-	0.2502684670316468	-0.0051102369025653	0.00574516139109118	0.0735957676835864	1.8765959719680547e-13
ALLERG_ASTHMA	ALLERG_RHINITIS	1175751	0.02226627119832697	0.00211691336395795	0.9049262550657243	0.04094659992567007	22.100156220746715	3.149754354570295e-	0.9049262550657243	0.3385945006313082	0.00877219389936634	0.5042738612828569	8.819312192796826e-107
ALLERG_ASTHMA	L12_ATOPIC	1175751	0.0205008035906413	0.00259582306611777	0.6296471872445731	0.05062592019906577	12.437249234556973	1.64056919602441 8e-	0.6296471872445731	0.1636443451777981	0.01160562841435628	0.3130603765639867	2.296796874434185e-34
ALLERG_RHINITIS	L12_ATOPIC	1175751	0.01404144044269981	0.00153182345415996	0.7184663005212137	0.06099423170665284	11.77924994574279	4.993543469211758e-	0.7184663005212137	0.1719646681284697	0.00907942996347728	0.3042177988192568	3.49548042844823046-31

Supplementary Table 61 - r6_mvmr_results

ST61_r6_mvmr_results.csv

outcome	model	exposure	exposure_label	n_snps	beta	se	ci_lo	ci_hi	p	or	resid_Q	resid_Q_df	resid_Q_p	marginal_mean_	cond_QA	cond_QA_over_d	cond_QA_chi2crit	cond_strong	univ_frailty_beta_same	univ_frailty_se_same	univ_frailty_p_same_snps
ALLERG_ASTH	MVMR_IVW	frailty	Frailty index (p	64	0.9857039432	0.1640921629	0.6640833039	1.3073245826	1.8897103881	2.6796975756	106.39678506	60	0.0002111916	45.542433753	631.76166431	9.8712760048	83.675260742	True	1.1001786311	0.0893491888	0.0
ALLERG_ASTH	MVMR_IVW_LD	frailty	Frailty index (p	64	0.8845248468	0.1774577384	0.5367076795	1.2323420140	6.2146238377	2.4218333762	109.69157074	60	9.5495160433	45.542433753	631.76166431	9.8712760048	83.675260742	True	1.1001786311	0.0893491888	0.0
ALLERG_ASTH	MVMR_IVW	bmi	BMI (Pan-UKBB	64	-0.1332096311	0.3168585941	-0.7542524761	0.4878332128	0.6741880152	0.8752815860	106.39678506	60	0.0002111916	14.626905801	299.63062186	4.6817284666	83.675260742	True			
ALLERG_ASTH	MVMR_IVW_LD	bmi	BMI (Pan-UKBB	64	-0.2855064284	0.3147918042	-0.902498364	0.3314855077	0.3644231560	0.7516335092	109.69157074	60	9.5495160433	14.626905801	299.63062186	4.6817284666	83.675260742	True			
ALLERG_ASTH	MVMR_IVW	smkinit	Smoking initia	64	-1.7479312709	0.5525266207	-2.8308834475	-0.6649790943	0.0015587145	0.1741338067	106.39678506	60	0.0002111916	3.3781606263	163.65937024	2.5571776600	83.675260742	True			
ALLERG_ASTH	MVMR_IVW_LD	smkinit	Smoking initia	64	-1.5381696944	0.5404723955	-2.5974955897	-0.4788437991	0.0044275987	0.2147738436	109.69157074	60	9.5495160433	3.3781606263	163.65937024	2.5571776600	83.675260742	True			
ALLERG_ASTH	MVMR_IVW	edu	Educational at	64	-1.1321251886	0.5735103617	-2.2562054976	-0.0080448796	0.0483786234	0.3223474787	106.39678506	60	0.0002111916	7.4725793925	91.620914074	1.4315767824	83.675260742	True			
ALLERG_ASTH	MVMR_IVW_LD	edu	Educational at	64	-1.5975445139	0.5793490922	-2.7330687347	-0.4620202931	0.0058248475	0.2023928812	109.69157074	60	9.5495160433	7.4725793925	91.620914074	1.4315767824	83.675260742	True			
ALLERG_RHINI	MVMR_IVW	frailty	Frailty index (p	67	0.4067403236	0.1492738251	0.1141636264	0.6993170209	0.0064341747	1.5019140435	86.185335378	63	0.0278472986	45.171647916	648.27954250	9.6758140672	87.108072195	True	0.4603943795	0.0797241067	7.702383841845517e-09
ALLERG_RHINI	MVMR_IVW_LD	frailty	Frailty index (p	67	0.3024393829	0.1610848426	-0.0132869085	0.6181656744	0.0604473949	1.3531556496	88.434459093	63	0.0189957427	45.171647916	648.27954250	9.6758140672	87.108072195	True	0.4603943795	0.0797241067	7.702383841845517e-09
ALLERG_RHINI	MVMR_IVW	bmi	BMI (Pan-UKBB	67	0.3858126870	0.2803643378	-0.1637014150	0.9353267891	0.1687870079	1.4708091433	86.185335378	63	0.0278472986	14.539886798	320.23556036	4.7796352293	87.108072195	True			
ALLERG_RHINI	MVMR_IVW_LD	bmi	BMI (Pan-UKBB	67	0.3800733240	0.2785912070	-0.1659654418	0.9261120899	0.1724829452	1.4623918139	88.434459093	63	0.0189957427	14.539886798	320.23556036	4.7796352293	87.108072195	True			
ALLERG_RHINI	MVMR_IVW	smkinit	Smoking initia	67	-0.8759749806	0.4990194075	-1.8540530194	0.1021030580	0.0791921750	0.4164557853	86.185335378	63	0.0278472986	3.2526242220	166.49562824	2.4850093767	87.108072195	True			
ALLERG_RHINI	MVMR_IVW_LD	smkinit	Smoking initia	67	-0.8653377395	0.4758723772	-1.7980475991	0.0673721194	0.0689993567	0.4209093707	88.434459093	63	0.0189957427	3.2526242220	166.49562824	2.4850093767	87.108072195	True			
ALLERG_RHINI	MVMR_IVW	edu	Educational at	67	0.0432411939	0.5024926802	-0.9416444591	1.0281268472	0.9314239827	1.0441897167	86.185335378	63	0.0278472986	7.5857474794	95.850195924	1.4305999391	87.108072195	True			
ALLERG_RHINI	MVMR_IVW_LD	edu	Educational at	67	-0.3416182761	0.4945359680	-1.3109087741	0.6276722206	0.4897003480	0.7106194128	88.434459093	63	0.0189957427	7.5857474794	95.850195924	1.4305999391	87.108072195	True			
L12_ATOPIC	MVMR_IVW	frailty	Frailty index (p	68	0.5812209882	0.1086819908	0.3682042861	0.7942376904	8.8978028056	1.7882204945	233.36806470	64	0.0	44.966847703	648.41833397	9.5355637349	88.250164421	True	0.4701792384	0.0574702271	2.220446049250313e-16
L12_ATOPIC	MVMR_IVW_LD	frailty	Frailty index (p	68	0.5839787689	0.1172909253	0.3540885551	0.8138689827	6.3949541573	1.7931588207	236.41169466	64	0.0	44.966847703	648.41833397	9.5355637349	88.250164421	True	0.4701792384	0.0574702271	2.220446049250313e-16
L12_ATOPIC	MVMR_IVW	bmi	BMI (Pan-UKBB	68	-0.2974073651	0.2008833947	-0.6911388196	0.0963240883	0.1387405252	0.7427413836	233.36806470	64	0.0	14.367060774	323.91086631	4.7633950928	88.250164421	True			
L12_ATOPIC	MVMR_IVW_LD	bmi	BMI (Pan-UKBB	68	-0.2294328741	0.1994308128	-0.6203172678	0.1614515184	0.2499633322	0.7949843304	236.41169466	64	0.0	14.367060774	323.91086631	4.7633950928	88.250164421	True			
L12_ATOPIC	MVMR_IVW	smkinit	Smoking initia	68	-1.4784710461	0.3593450930	-2.1827874285	-0.7741546636	3.8827398749	0.2279860020	233.36806470	64	0.0	3.2056334354	168.55214908	2.4787080747	88.250164421	True			
L12_ATOPIC	MVMR_IVW_LD	smkinit	Smoking initia	68	-1.1231425586	0.3441922759	-1.7975794195	-0.4485256977	0.0011019075	0.3252560506	236.41169466	64	0.0	3.2056334354	168.55214908	2.4787080747	88.250164421	True			
L12_ATOPIC	MVMR_IVW	edu	Educational at	68	-0.3746570614	0.3546399097	-1.0697512844	0.3204371616	0.2907655973	0.6875250172	233.36806470	64	0.0	7.6719047877	98.978179417	1.4555614620	88.250164421	True			
L12_ATOPIC	MVMR_IVW_LD	edu	Educational at	68	-0.3831987266	0.3476250080	-1.0645437424	0.2981462892	0.2703166669	0.6816774183	236.41169466	64	0.0	7.6719047877	98.978179417	1.4555614620	88.250164421	True			

Supplementary Table 62 - r6_frailty_immune_excluded

ST62_r6_frailty_immune_excluded.csv

outcome	instrument_set	steiger_remov	removed_immu	n_snps	mean_F	ivw_beta	ivw_se	ivw_p	ivw_or	ivw_ci_lo	ivw_ci_hi	ivw_re_beta	ivw_re_se	ivw_re_p	cochran_q	cochran_q_p	egger_beta	egger_p	egger_intercep	egger_int_p	wmed_beta	wmed_p
ALLERG_ASTH	full	5	0	66	46.45580341	1.2345712417	0.0846430400	0.0	3.4369046035	2.9115071945	4.0571128508	1.2345712417	0.1267354891	0.0	145.7227443	3.9630747356	1.2329836497	0.0	0.000991252	0.8029134826	1.4332237762	2.7277988823329324e-07
ALLERG_ASTH	immune_exclu	1	86	12	33.76795076	1.2329391255	0.1904110529	9.4721119830	3.4312997500	2.3625309051	4.9835614658	1.2329391255	0.2639964600	3.0077982242	21.144848742	0.0319010563	1.2255863087	6.1008048362	0.0037742200	0.5141336494	0.9122422680	0.11085661562464756
ALLERG_RHIN	full	1	0	70	45.83208167	0.5256676234	0.0748444946	2.1642687642	1.6915878151	1.4607827378	1.9588603166	0.5256676234	0.0907043185	6.8159355937	101.34102655	0.0068125666	0.5303999209	5.0765667225	-0.002848819	0.2976515221	0.6320320643	2.9001607182443934e-0
ALLERG_RHIN	immune_exclu	0	86	13	33.58794805	0.9823047529	0.1683217010	5.3508342201	2.6706042374	1.9201325477	3.7143930514	0.9823047529	0.2013850960	1.0730658244	17.177330671	0.1430476075	0.9800594947	3.0070219043	-0.001132526	0.7964827991	0.8542989690	0.005790542500843099
L12_ATOPIC	full	0	0	71	45.62663311	0.3762838630	0.0539932657	3.1898927943	1.4568606248	1.3105633237	1.6194889952	0.3762838630	0.1106447643	0.0006718197	293.95481933	0.0	0.3667750304	0.0009041815	0.0040311200	0.2234012897	0.3318492146	0.033297301626616704
L12_ATOPIC	immune_exclu	0	86	13	33.58794805	0.4704160556	0.1225597296	0.0001239094	1.6006600181	1.2588504780	2.0352794385	0.4704160556	0.1957689927	0.0162650115	30.617763926	0.0022522256	0.4673236293	0.0215641937	-0.001547750	0.7164816739	0.4171895163	0.10993615939007229

Supplementary Table 63 - r6_frailty_immune_excluded_snps

ST63_r6_frailty_immune_excluded_snps.csv

snp	chrom	pos	immune_locus	beta	se	p
rs9275160	6	32652620	HLA	0.0382	0.0035	7.176e-28
rs9273344	6	32623458	HLA	-0.0428	0.0039	7.751e-28
rs77062257	6	32620764	HLA	-0.0428	0.0043	1.0759999999999999e-23
rs182976814	6	32486900	HLA	0.0585	0.006	7.959e-23
rs28746860	6	32635033	HLA	-0.0429	0.0044	4.783e-22
rs1281935	6	32583820	HLA	0.0556	0.0059	3.395e-21
rs17612518	6	32614927	HLA	0.0395	0.0044	4.314e-19
rs575147125	6	32619144	HLA	-0.0452	0.0051	7.927e-19
rs6910879	6	32560739	HLA	-0.054	0.0061	1.457e-18
rs145244672	6	32556461	HLA	-0.0495	0.0057	1.99e-18
rs113315602	6	32574575	HLA	-0.0526	0.0061	3.835e-18
rs9274488	6	32633852	HLA	-0.0319	0.0037	1.186e-17
rs185320691	6	32490292	HLA	0.0519	0.0061	1.405e-17
rs181159261	6	32490653	HLA	0.0442	0.0054	1.929e-16
rs1281934	6	32584382	HLA	-0.0521	0.0063	1.943e-16
rs28417075	6	32519013	HLA	-0.0529	0.0064	2.267e-16
rs543993279	6	32489575	HLA	-0.0467	0.0058	1.238e-15
rs9268362	6	32333341	HLA	-0.0311	0.0039	1.515e-15
rs138748427	6	32484520	HLA	0.0419	0.0053	1.655e-15
rs141267330	6	32548054	HLA	-0.0513	0.0065	2.071e-15
rs115627609	6	32485579	HLA	0.0416	0.0053	3.296e-15
rs114281696	6	32525925	HLA	-0.0343	0.0045	1.806e-14
rs13195317	6	32656094	HLA	0.0402	0.0053	2.83e-14
rs28746878	6	32637515	HLA	0.0477	0.0064	8.913e-14
rs555911977	6	32535454	HLA	0.034	0.0046	2.103e-13
rs34422230	6	32776995	HLA	-0.0521	0.0071	2.176e-13
rs2523668	6	31438096	HLA	0.0326	0.0045	2.915e-13
rs35520662	6	32748029	HLA	-0.0499	0.0069	5.698e-13
rs34043227	6	32748590	HLA	0.0499	0.0069	5.712e-13
rs34153000	6	32747016	HLA	0.0499	0.0069	5.722e-13
rs34408145	6	32755177	HLA	-0.0498	0.0069	6.194e-13
rs41316544	6	32792886	HLA	0.0496	0.0069	7.88e-13
rs17213756	6	32787996	HLA	0.0495	0.0069	8.554e-13
rs41316548	6	32797167	HLA	0.0496	0.0069	8.932e-13
rs35921062	6	32774416	HLA	0.0494	0.0069	9.48e-13
rs35486885	6	32763506	HLA	-0.0493	0.0069	1.06e-12
rs34710970	6	32768115	HLA	-0.0493	0.0069	1.063e-12
rs41258084	6	32780957	HLA	0.0492	0.0069	1.105e-12
rs41316241	6	32782795	HLA	0.0492	0.0069	1.106e-12
rs72508422	6	32481498	HLA	0.0366	0.0052	1.674e-12

(showing first 40 of 86 rows; full data in the accompanying CSV)

Supplementary Table 64 - r6_mvmr_frailty_vs_index

ST64_r6_mvmr_frailty_vs_index.csv

outcome	model	exposure	n_snps	beta	se	ci_lo	ci_hi	p	or	resid_Q	resid_Q_df	resid_Q_p	marginal_mean_	cond_QA	cond_QA_over	cond_QA_chi2crit	cond_strong	univ_beta_same	univ_se_same_s	univ_p_same_sn	n_union	n_complete	n_overlap_instruments
ALLERG_ASTH	MVMR_IVW	frailty	157	1.346773130	0.084448626	1.181253822	1.512292437	0.0	3.844998182	826.1044363	155	0.0	20.68255013	3055.202842	19.45989071	187.2388699	True	1.526658942	0.081757036	0.0	200	157	0
ALLERG_ASTH	MVMR_IVW_L	frailty	157	1.410614871	0.084466571	1.245060392	1.576169351	0.0	4.098474665	828.1966743	155	0.0	20.68255013	3055.202842	19.45989071	187.2388699	True	1.526658942	0.081757036	0.0	200	157	0
ALLERG_ASTH	MVMR_IVW	index	157	0.627035348	0.073726297	0.482531804	0.771538891	0.0	1.872052360	826.1044363	155	0.0	680.9565038	101105.9414	643.9868882	187.2388699	True	0.921517517	0.071376455	0.0	200	157	0
ALLERG_ASTH	MVMR_IVW_L	index	157	0.572502546	0.065197117	0.444716197	0.700288896	0.0	1.772697764	828.1966743	155	0.0	680.9565038	101105.9414	643.9868882	187.2388699	True	0.921517517	0.071376455	0.0	200	157	0
ALLERG_RHIN	MVMR_IVW	frailty	157	0.525086758	0.077405749	0.373371488	0.676802028	1.172484331	1.690605516	510.6569729	155	0.0	20.68255013	3055.202842	19.45989071	187.2388699	True	0.634571648	0.074944962	0.0	200	157	0
ALLERG_RHIN	MVMR_IVW_L	frailty	157	0.568883743	0.077398647	0.417182394	0.720585091	1.980637875	1.766294312	524.0522736	155	0.0	20.68255013	3055.202842	19.45989071	187.2388699	True	0.634571648	0.074944962	0.0	200	157	0
ALLERG_RHIN	MVMR_IVW	index	157	0.382995046	0.067732676	0.250239001	0.515751092	1.562921614	1.466670765	510.6569729	155	0.0	680.9565038	101105.9414	643.9868882	187.2388699	True	0.497927353	0.065579403	3.130828929	200	157	0
ALLERG_RHIN	MVMR_IVW_L	index	157	0.425868649	0.059923744	0.308418111	0.543319188	1.187272502	1.530919675	524.0522736	155	0.0	680.9565038	101105.9414	643.9868882	187.2388699	True	0.497927353	0.065579403	3.130828929	200	157	0
L12_ATOPIC	MVMR_IVW	frailty	157	0.516399163	0.056353633	0.405946042	0.626852284	0.0	1.675981834	687.6224819	155	0.0	20.68255013	3055.202842	19.45989071	187.2388699	True	0.667696202	0.054541967	0.0	200	157	0
L12_ATOPIC	MVMR_IVW_L	frailty	157	0.587192862	0.056351810	0.476743314	0.697642410	0.0	1.798931473	705.6583781	155	0.0	20.68255013	3055.202842	19.45989071	187.2388699	True	0.667696202	0.054541967	0.0	200	157	0
L12_ATOPIC	MVMR_IVW	index	157	0.527392409	0.049408263	0.430552212	0.624232605	0.0	1.694507959	687.6224819	155	0.0	680.9565038	101105.9414	643.9868882	187.2388699	True	0.641269777	0.047819878	0.0	200	157	0
L12_ATOPIC	MVMR_IVW_L	index	157	0.461585530	0.043715573	0.375903005	0.547268054	0.0	1.586587574	705.6583781	155	0.0	680.9565038	101105.9414	643.9868882	187.2388699	True	0.641269777	0.047819878	0.0	200	157	0

Supplementary Table 65 - r6_joint_mvmr_snp_table

ST65_r6_joint_mvmr_snp_table.csv

snp	ea	oa	beta_frailty	se_frailty	beta_index	se_index	in_frailty_inst	in_index_inst
rs1002607	C	T	0.0001	0.0034	0.0050322932633617	0.0001227632222375	False	True
rs1008210	T	G	-0.0185	0.0033	0.0009067956136972807	0.0005692060172967167	True	False
rs10437340	C	G	-0.0011	0.0035	0.0121148478415067	0.0002473295597854	False	True
rs1081105	C	A	-0.0118	0.0101	-0.0300309143624057	0.000714355114038	False	True
rs10849448	G	A	0.0012	0.0039	0.0037913430625951	0.0001421694193212	False	True
rs10891490	T	C	0.0188	0.0034	-0.0003485091119153704	0.0005724794592185638	True	False
rs10947233	T	G	0.0433	0.0071	0.005124576769990907	0.0011927774084488814	True	False
rs11065394	G	A	0.0009	0.004	0.0097449493589159	0.0002878648890436	False	True
rs112019714	C	T	-0.0124	0.01	-0.029317199761471	0.0007134400953189	False	True
rs112849259	T	C	-0.0129	0.0104	-0.0294361521949601	0.000738054098864	False	True
rs113071828	A	G	0.0114	0.004	0.0447	0.0015655	False	True
rs113315602	A	C	-0.0526	0.0061	-0.0056682006844709	0.0010043486440186916	True	False
rs113542380	A	G	0.0054	0.0064	0.0240125627117426	0.0010963965111117	False	True
rs114403972	G	A	-0.0007	0.0045	0.0101018066593832	0.0003193415329823	False	True
rs114425738	C	G	0.0152	0.0142	-0.0164429468917217	0.0005251088622638	False	True
rs114536010	T	C	-0.0139	0.0104	-0.029628306125981	0.000737962596992	False	True
rs11466597	C	T	0.0193	0.0149	-0.0177462062056736	0.000548890966533	False	True
rs115881343	T	C	-0.013	0.01	-0.0292805990127051	0.0007136230990627	False	True
rs116359246	A	G	0.0005	0.009	-0.010254843360877	0.0003258148284879	False	True
rs116881820	C	T	-0.014	0.0104	-0.0294178518205772	0.0007377795932482	False	True
rs117310449	T	C	-0.028	0.0151	-0.0337916412981	0.0010797220885935	False	True
rs117520868	A	C	-0.0146	0.0076	-0.0235503382129602	0.0013318998932413	False	True
rs117866384	A	C	-0.0145	0.0076	-0.023504115763082	0.0013309754442437	False	True
rs118051449	G	A	-0.0148	0.0076	-0.0233423371885082	0.0013265843115053	False	True
rs118170342	C	T	-0.0106	0.0084	-0.0196180013385124	0.0005984222423221	False	True
rs11864884	T	C	0.0043	0.0037	0.0032557700744528	0.0001346067101636	False	True
rs12515180	T	C	-0.0104	0.0033	-0.0374	0.0013025	False	True
rs1260326	C	T	-0.0052	0.0034	-0.0077868092999412	0.0002401009119041	False	True
rs12619285	G	A	-0.0006	0.0037	-0.038745	0.00146	False	True
rs12739243	T	C	0.0242	0.004	0.0013328500525244219	0.0006827416357014788	True	False
rs1281934	A	G	-0.0521	0.0063	-0.0047194110478397475	0.0010600261019058127	True	False
rs1281935	T	G	0.0556	0.0059	0.0059885436354760475	0.0009868862162949119	True	False
rs12972156	G	C	0.008	0.0047	-0.018053319328771	0.0003320602931784	False	True
rs13195317	T	C	0.0402	0.0053	0.004821791103960102	0.0008842680366018732	True	False
rs13277237	A	G	0.0056	0.0033	0.0043911277322642	0.0001218595022753	False	True
rs1363103	T	C	0.0191	0.0034	0.0007188898056844803	0.000580342554693405	True	False
rs137893789	A	G	0.0464	0.0071	0.005127579937489338	0.001195618830116994	True	False
rs139795227	C	A	0.0092	0.0137	-0.0255181978808467	0.0005075101051046	False	True
rs1401884	C	G	-0.006	0.0041	-0.0061976163725523	0.0001496370000617	False	True
rs141094656	C	T	0.0194	0.0141	-0.0252470818921779	0.0005184498730684	False	True

(showing first 40 of 170 rows; full data in the accompanying CSV)

Supplementary Table 66 - r6_5q31_conditional_signals

ST66_r6_5q31_conditional_signals.csv

trait	signal_index	lead_rsid	lead_pos_b37	ea	oa	beta_marg	se_marg	p_marginal	z_conditional	p_conditional	ridge_lambda	candidates_tested
GWAS_ALLERG_ASTHMA	1	rs3749833	131799626	C	T	0.120072	0.0130867	4.511279999999999e-2	9.175116721557	4.5108550870858195e-	0.001	336
GWAS_ALLERG_ASTHMA	2	rs7713065	131788334	C	A	-0.120014	0.0130855	4.66444999999999997e-	-11.914117670274285	9.991830173312188e-3	0.001	333
GWAS_ALLERG_ASTHMA	3	rs847	131996669	C	T	-0.113991	0.0130813	2.9335900000000002e-	-8.14979282338907	3.645482731125592e-1	0.001	331
GWAS_ALLERG_ASTHMA	4	rs3756752	132073873	T	C	0.113895	0.0153556	1.19647e-13	7.639953332465897	2.1730048767897643e-	0.001	326
GWAS_ALLERG_ASTHMA	5	rs3815065	131785692	G	C	0.155924	0.0271642	9.46455e-09	6.928300285139503	4.259269170010685e-1	0.001	313
GWAS_ALLERG_RHINITIS	1	rs847	131996669	C	T	-0.0749036	0.0120551	5.18334e-10	-6.2134366367761364	5.183814132386067e-1	0.001	52
GWAS_ALLERG_RHINITIS	2	rs3756752	132073873	T	C	0.0818575	0.0142029	8.24271e-09	6.018999755907766	1.7549813457252541e-	0.001	47
GWAS_ALLERG_RHINITIS	3	rs185652639	132023258	G	C	0.16019	0.0289993	3.31535e-08	5.523926439603715	3.315063019095638e-0	0.001	34
GWAS_ALLERG_RHINITIS	4	rs17772853	132009599	T	C	0.159913	0.0290233	3.59228e-08	5.509814528327241	3.5921199112900176e-	0.001	33
GWAS_L12_ATOPIC	1	rs847	131996669	C	T	-0.116966	0.0087486	9.08657e-41	-13.369682006263858	9.092874555262906e-4	0.001	885
GWAS_L12_ATOPIC	2	rs2243266	132013789	A	G	0.095502	0.00886251	4.47095e-27	10.232644960344755	1.4159827928403234e-	0.001	880
GWAS_L12_ATOPIC	3	rs35081016	132020302	G	A	-0.10232	0.00851776	3.05492e-33	-10.779858753770808	4.285432961786565e-2	0.001	839
GWAS_L12_ATOPIC	4	rs60156285	131955818	T	C	0.0916206	0.00984284	1.2983699999999997e-	10.074102713068639	7.191417946177765e-2	0.001	806
GWAS_L12_ATOPIC	5	rs4705943	131755004	G	A	0.0753052	0.0123941	1.23308e-09	7.661839160867065	1.8328909659348865e-	0.001	753
IL4_eQTLGen	1	rs2070874	132009710	T	C	-0.13985918132477357	0.008845688528541748	2.6129e-56	-15.811	2.6128628418937777e-5	0.001	167
IL4_eQTLGen	2	rs2243293	132019799	G	A	-0.09609359088700163	0.008486509073222142	1.0073e-29	-14.661372129566443	1.139467725923611e-4	0.001	125
IL4_eQTLGen	3	rs2243300	132004086	T	G	0.13331232510385316	0.01862918700183803	8.3065e-13	9.754205705810099	1.7698064251668275e-	0.001	93
IL4_eQTLGen	4	rs75765453	131950225	T	C	0.17456077953595264	0.04073003395770979	1.8225e-05	6.034450029164792	1.5950497820138476e-0	0.001	91
IL4_eQTLGen	5	rs75639632	131915279	G	A	0.0996687097549279	0.02320953583935168	1.7532e-05	-19.733268591250607	1.1170868458525192e-	0.001	90
IL4_DICE_TREG_NAIVE	1	rs6879672	132025947	G	A	0.782607228262	0.16068626391390545	1.11369579381e-06	4.870405280449581	1.1136957938084860-0	0.001	25
IL4_DICE_TREG_NAIVE	2	rs187681	132101903	A	G	-0.718216176316	0.16531697280451704	1.3960636769e-05	-4.735054512052675	2.189962300546362e-0	0.001	1
IL13_DICE_TH1	1	rs78330488	131813917	C	T	1.35032520591	0.27559087562378143	9.59607815202e-07	4.89974569315341	9.596078152096472e-0	0.001	6
IL13_DICE_TH1	2	rs75893502	131813827	T	C	1.25718354702	0.2697483049839271	3.15319617726e-06	4.660579969519768	3.1531961772746137e-0	0.001	5
IL13_DICE_TH1	3	rs7725926	132938624	C	T	-0.814568045843	0.17822089348961234	4.86438790641e-06	-4.5705530361426305	4.864387906366994e-0	0.001	4

Supplementary Table 67 - r6_5q31_persignal_coloc

ST67_r6_5q31_persignal_coloc.csv

gwas_outcome	eqtl	gwas_signal	eqtl_signal	gwas_lead	eqtl_lead	n_snps_masked	n_snps	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4	note
ALLERG_ASTHMA	IL4_eQTLGen	FULL	FULL			0	175	5.111770592484172e-	1.20679716121318936	4.231539202521209e-	0.998989340107187	0.00101064782484136	full-region reference (r
ALLERG_ASTHMA	IL4_eQTLGen	1	1	rs3749833	rs2070874	94	81	6.3424000121298436-	1.4973266486509476-	4.1827531261120176	0.9874602954094599	0.01253955485787524	
ALLERG_ASTHMA	IL4_eQTLGen	1	2	rs3749833	rs2243293	103	72	6.8315326787343966-	1.5018316657169394	4.5022339471969216-	0.9897526400283871	0.01024720978844635	
ALLERG_ASTHMA	IL4_eQTLGen	1	3	rs3749833	rs2243300	131	44	6.8316301997664726-	1.5018531045625007	4.5022332824527796-	0.9897524937459689	0.01024735606872070	
ALLERG_ASTHMA	IL4_eQTLGen	1	4	rs3749833	rs75765453	136	39	6.8316302204339196-	1.5018531091059476	4.5022332823119016-	0.9897524937149675	0.01024735609972160	
ALLERG_ASTHMA	IL4_eQTLGen	1	5	rs3749833	rs75639632	134	41	6.83163021956477e-5	1.5018531089149223	4.5022332823178266-	0.9897524937162713	0.01024735609841789	
ALLERG_ASTHMA	IL4_eQTLGen	2	1	rs7713065	rs2070874	94	81	6.3424000121298436-	1.4973266486509476-	4.1827531261120176	0.9874602954094599	0.01253955485787524	
ALLERG_ASTHMA	IL4_eQTLGen	2	2	rs7713065	rs2243293	103	72	6.8315326787343966-	1.5018316657169394	4.5022339471969216-	0.9897526400283871	0.01024720978844635	
ALLERG_ASTHMA	IL4_eQTLGen	2	3	rs7713065	rs2243300	131	44	6.8316301997664726-	1.5018531045625007	4.5022332824527796-	0.9897524937459689	0.01024735606872070	
ALLERG_ASTHMA	IL4_eQTLGen	2	4	rs7713065	rs75765453	136	39	6.8316302204339196-	1.5018531091059476	4.5022332823119016-	0.9897524937149675	0.01024735609972160	
ALLERG_ASTHMA	IL4_eQTLGen	2	5	rs7713065	rs75639632	134	41	6.83163021956477e-5	1.5018531089149223	4.5022332823178266-	0.9897524937162713	0.01024735609841789	
ALLERG_ASTHMA	IL4_eQTLGen	3	1	rs847	rs2070874	94	81	6.3424000121298436-	1.4973266486509476-	4.1827531261120176	0.9874602954094599	0.01253955485787524	
ALLERG_ASTHMA	IL4_eQTLGen	3	2	rs847	rs2243293	103	72	6.8315326787343966-	1.5018316657169394	4.5022339471969216-	0.9897526400283871	0.01024720978844635	
ALLERG_ASTHMA	IL4_eQTLGen	3	3	rs847	rs2243300	131	44	6.8316301997664726-	1.5018531045625007	4.5022332824527796-	0.9897524937459689	0.01024735606872070	
ALLERG_ASTHMA	IL4_eQTLGen	3	4	rs847	rs75765453	136	39	6.8316302204339196-	1.5018531091059476	4.5022332823119016-	0.9897524937149675	0.01024735609972160	
ALLERG_ASTHMA	IL4_eQTLGen	3	5	rs847	rs75639632	134	41	6.83163021956477e-5	1.5018531089149223	4.5022332823178266-	0.9897524937162713	0.01024735609841789	
ALLERG_ASTHMA	IL4_eQTLGen	4	1	rs3756752	rs2070874	44	131	5.11177646109612056	1.20679854668699166	4.23153919761132e-5	0.9989893389468902	0.00101064898512424	
ALLERG_ASTHMA	IL4_eQTLGen	4	2	rs3756752	rs2243293	53	122	5.4908133566068426-	1.2070903788784002e	4.5450577631330566e	0.9991763730973983	0.00082361483169802	
ALLERG_ASTHMA	IL4_eQTLGen	4	3	rs3756752	rs2243300	81	94	5.4908196564487376-	1.2070917638241356-	4.5450577588388216-	0.9991763721524164	0.00082361577666607	
ALLERG_ASTHMA	IL4_eQTLGen	4	4	rs3756752	rs75765453	86	89	5.4908196577838326-	1.2070917641176399e	4.5450577588379116-	0.9991763721522161	0.00082361577686633	
ALLERG_ASTHMA	IL4_eQTLGen	4	5	rs3756752	rs75639632	84	91	5.4908196577276866-	1.2070917641052968e	4.5450577588379496-	0.9991763721522245	0.00082361577685791	
ALLERG_ASTHMA	IL4_eQTLGen	5	1	rs3815065	rs2070874	94	81	6.3424000121298436-	1.4973266486509476-	4.1827531261120176	0.9874602954094599	0.01253955485787524	
ALLERG_ASTHMA	IL4_eQTLGen	5	2	rs3815065	rs2243293	103	72	6.8315326787343966-	1.5018316657169394	4.5022339471969216-	0.9897526400283871	0.01024720978844635	
ALLERG_ASTHMA	IL4_eQTLGen	5	3	rs3815065	rs2243300	131	44	6.8316301997664726-	1.5018531045625007	4.5022332824527796-	0.9897524937459689	0.01024735606872070	
ALLERG_ASTHMA	IL4_eQTLGen	5	4	rs3815065	rs75765453	136	39	6.8316302204339196-	1.5018531091059476	4.5022332823119016-	0.9897524937149675	0.01024735609972160	
ALLERG_ASTHMA	IL4_eQTLGen	5	5	rs3815065	rs75639632	134	41	6.83163021956477e-5	1.5018531089149223	4.5022332823178266-	0.9897524937162713	0.01024735609841789	
ALLERG_RHINITIS	IL4_eQTLGen	FULL	FULL			0	175	1.47747523555662246	0.00034880534792662	4.1080279887658338e	0.9698017236273498	0.02984947102472352	full-region reference (r
ALLERG_RHINITIS	IL4_eQTLGen	1	1	rs847	rs2070874	94	81	7.8527269487933366-	0.00185388769275358	3.5566278556156264e	0.8394972633234834	0.15864884898376302	
ALLERG_RHINITIS	IL4_eQTLGen	1	2	rs847	rs2243293	103	72	9.5885225581234656-	0.00210792328492510	4.3285039801004986-	0.9515240405681262	0.04636803614694863	
ALLERG_RHINITIS	IL4_eQTLGen	1	3	rs847	rs2243300	131	44	9.5887479542594466-	0.00210797283559993	4.3284988016185086-	0.951522901050137	0.04636912611426296	
ALLERG_RHINITIS	IL4_eQTLGen	1	4	rs847	rs75765453	136	39	9.5887511887519366-	0.00210797354666485	4.3284987273059616-	0.951522884697761	0.04636914175557401	
ALLERG_RHINITIS	IL4_eQTLGen	1	5	rs847	rs75639632	134	41	9.5887504922061156-	0.00210797339353749	4.3284987433091196-	0.9515228882192344	0.04636913838722802	
ALLERG_RHINITIS	IL4_eQTLGen	2	1	rs3756752	rs2070874	44	131	1.4774814706242927e	0.00034880681991403	4.10802744949032e-5	0.9698015961881263	0.02984959699195958	
ALLERG_RHINITIS	IL4_eQTLGen	2	2	rs3756752	rs2243293	53	122	1.6234641911847606e	0.00035689940239450	4.5115013998418266-	0.9917923763756183	0.00785072422198731	
ALLERG_RHINITIS	IL4_eQTLGen	2	3	rs3756752	rs2243300	81	94	1.6234706524865358e	0.00035690082283528	4.5115012513932546-	0.9917923437097096	0.00785075546745526	
ALLERG_RHINITIS	IL4_eQTLGen	2	4	rs3756752	rs75765453	86	89	1.6234707452060404e	0.00035690084321856	4.5115012492630246-	0.9917923432409547	0.00785075591582688	
ALLERG_RHINITIS	IL4_eQTLGen	2	5	rs3756752	rs75639632	84	91	1.6234707025238961e	0.00035690083882904	4.5115012497217646-	0.9917923433419008	0.00785075581927037	
ALLERG_RHINITIS	IL4_eQTLGen	3	1	rs185652639	rs2070874	94	81	7.8527269487933366-	0.00185388769275358	3.5566278556156264e	0.8394972633234834	0.15864884898376302	
ALLERG_RHINITIS	IL4_eQTLGen	3	2	rs185652639	rs2243293	103	72	9.5885225581234656-	0.00210792328492510	4.3285039801004986-	0.9515240405681262	0.04636803614694863	
ALLERG_RHINITIS	IL4_eQTLGen	3	3	rs185652639	rs2243300	131	44	9.5887479542594466-	0.00210797283559993	4.3284988016185086-	0.951522901050137	0.04636912611426296	

(showing first 40 of 149 rows; full data in the accompanying CSV)

Supplementary Table 68 - r6_5q31_smr_heidi

ST68_r6_5q31_smr_heidi.csv

top_snp	b_smr	se_smr	z_smr	p_smr	n_heidi	p_heidi	chi2_heidi	df_heidi	heidi_pass	note	eqtl	outcome
rs2070874	-0.5389714088555713	0.09485326507949568	-5.682159790744832	1.3300430460458543e-	72	0.005753401255183792	99.81633579959416	67	False	ok	IL4_eQTLGen	ALLERG_ASTHMA
rs2070874	-0.3668532842392074	0.08740505903300798	-4.197163050947285	2.702793491600539e-0	72	0.9579430331081504	46.693110735760726	65	True	ok	IL4_eQTLGen	ALLERG_RHINITIS
rs2070874	-0.6866156307393606	0.06335779972444613	-10.837112932039448	2.2960275642388285e-	72	4.436611226972337e-0	87.81219252993051	42	False	ok	IL4_eQTLGen	L12_ATOPIC
rs6879672	-0.0791868484752265	0.01633744174379078	-4.846955215942686	1.2537071435120593e-	33	0.0	1556.1571707671903	11	False	ok	IL4_DICE_TREG_NAIVE	ALLERG_ASTHMA
rs6879672	-0.0438258155066741	0.015031167072305437	-2.9156628554426827	0.0035493381474052315	33	4.229071184342007e-16	784.4017955735419	11	False	ok	IL4_DICE_TREG_NAIVE	ALLERG_RHINITIS
rs6879672	-0.13018024408776963	0.010905074335887516	-11.937584291320176	7.538079073165796e-3	33	0.0	2400.4200010039235	12	False	ok	IL4_DICE_TREG_NAIVE	L12_ATOPIC

Supplementary Table 69 - r6_5q31_rs2070874_directions

ST69_r6_5q31_rs2070874_directions.csv

source	snp	ea	oa	beta	se	p
IL4_eQTLGen	rs2070874	T	C	-0.13985918132477357	0.008845688528541748	2.6129e-56
FinnGen_ALLERG_ASTHMA	rs2070874	T	C	0.0753801	0.0132661	1.32993e-08
FinnGen_ALLERG_RHINITIS	rs2070874	T	C	0.0513078	0.0122244	2.7029e-05
FinnGen_L12_ATOPIC	rs2070874	T	C	0.0960295	0.00886117	2.29615e-27

Supplementary Table 70 - r6_nhanes_index_age_correlation

ST70_r6_nhanes_index_age_correlation.csv

cycle	n	n_eff	weighted_pearson_r	fisher_p	spearman_rho_unweighted	spearman_p
D	8127	4422.5	0.20287474075913678	1.399219828407231e-42	0.28086657153022754	3.318585700247833e-147
J	7240	3081.0	0.2424096571198162	7.50254143314788e-43	0.28129914050073396	9.06450178671458e-132
pooled_FE	15367	7503.5	0.21919350274423316	6.199168131856196e-83		

Supplementary Table 71 - r6_nhanes_index_by_decade

ST71_r6_nhanes_index_by_decade.csv

cycle	decade	n	wmean	wsd	p25	median	p75
D	<20	3670	-0.13458653460296774	0.7043388750872378	-0.5735437267038822	-0.17659323911277472	0.23098173481391424
D	20-29	926	0.11711729245580514	0.5664556698641601	-0.2360855019103272	0.06218032118245047	0.40123754013212487
D	30-39	756	0.0848480943566814	0.47825604365609453	-0.20157679483097946	0.07134924149473205	0.33541989038712156
D	40-49	776	0.13186166843710198	0.48896650855418683	-0.1653883479483328	0.12595293112123512	0.3829000970016817
D	50-59	595	0.12103564613177749	0.5282382170787885	-0.20215320589492403	0.10167209782889387	0.385386963629515
D	60-69	628	0.17863647933252852	0.5691100661190402	-0.14971480192262915	0.17212303306846713	0.4741740513910504
D	70-79	449	0.25821248698644267	0.620483819383335	-0.05442058341811454	0.2502748641856887	0.5205723546360638
D	80+	327	0.26898255028213436	0.5452552001798865	-0.05499047577881262	0.22227887373169422	0.6049568083171538
J	<20	2337	-0.17504747171302773	0.6911539843601484	-0.591856419258641	-0.2102016694943162	0.14897198566311418
J	20-29	718	0.007089426939157077	0.5213983192637625	-0.306231665422427	-0.03189790590618319	0.3377286745837993
J	30-39	755	0.03279719650677347	0.48458850765590356	-0.2732102394196127	4.440245266223953e-05	0.3425723482815227
J	40-49	714	0.09889151864161173	0.5057050109083524	-0.18465865838470782	0.08165216598427547	0.3858748286097621
J	50-59	831	0.08458301764601914	0.564590357173032	-0.2861847025196412	0.02867720748783798	0.39148416117689994
J	60-69	994	0.18334859180717306	0.5622840353366453	-0.21720896349766686	0.15247540958071448	0.49559411963460537
J	70-79	537	0.26370558385531034	0.5503468837997094	-0.11884402987114456	0.21689228138068348	0.5539900248463175
J	80+	354	0.31712880148478756	0.6273810986573746	0.012924711865247532	0.32002176783718234	0.6111434810937394

Supplementary Table 72 - r6_nhanes_index_crp

ST72_r6_nhanes_index_crp.csv

cycle	dependent	n	beta_log1pcrp	se	p	weighted_r	weighted_r_p
D	I	8127	0.31747739457043495	0.024199632208671752	2.5595660480058443e-39	0.1696587945966524	4.749308813059427e-30
D	I_nocrp	8127	0.28905993361179133	0.02457331194844115	6.042557378897734e-32	0.15479244412787455	3.260583560575068e-25
J	I	7240	0.22994785074191387	0.013260779686429497	2.32917495600756e-67	0.29474438363801025	1.0109328847403396e-63
J	I_nocrp	7240	0.14050994559066987	0.012417303225764413	1.0977670141852363e-29	0.18584749316903248	1.7712340570729671e-25

Supplementary Table 73 - r6_nhanes_incremental_value

ST73_r6_nhanes_incremental_value.csv

cycle	outcome	n	n_events	l_or	l_or_lo	l_or_hi	l_wald_p	pseudo_lr_stat	pseudo_lr_p	aic_base	aic_full	delta_aic
D	allergy_composite	7857	2367	1.285000275256093	1.1449214682367774	1.4422174386782922	2.061292071957404e-0	63.10395143553251	1.960803541840307e-1	8294.625890878486	8233.521939442955	61.10395143553069
D	asthma_ever	7863	1119	1.450318432751778	1.3024995282765053	1.6149131041627844	1.2127603676278012e-	114.31247161000556	1.1131045555971793e-	6537.543249242583	6425.230777632578	112.31247161000556
D	hay_fever	5395	1521	1.1058363341354291	0.9696199044574454	1.2611890414712024	0.13361405460614945	5.746157798520471	0.01652476024069905	5374.06602193905	5370.319864140529	3.7461577985213808
J	allergy_composite	7021	1092	1.4348010732151717	1.2960547870630277	1.588400536959164	3.457776269560767e-1	98.5419301585489	3.1821500708736626e-	5968.665053434119	5872.123123275571	96.54193015854798
J	hay_fever	669	167	1.361698649664701	1.0719923942359209	1.7296981046402822	0.011419139671224373	14.270497018721358	0.00015832714099344	712.3549482438759	700.0844512251545	12.270497018721471

Supplementary Table 74 - r6_phewas_mr_summary

ST74_r6_phewas_mr_summary.csv

protein	endpoint	category	endpoint_label	n_cases	n_controls	n_snps	method	beta	se	p	or_	or_lo	or_hi	cochran_q	q_p	note	q_bh
IL18	L12_PSORIASIS	autoimmune	Psoriasis	12760	482181	3	ivw_fe	0.2111889520523	0.0479596987550	1.0653448264452	1.2351457163316	1.1243307193511	1.3568827341593	0.2341362002742	0.8895246143525		0.00021306896528905384
IL18	ST19_FRACT_FEM	skeletal	Fracture of femur	12229	476877	3	ivw_fe	0.1697655152973	0.0495465294643	0.0006116713527	1.1850269480480	1.0753587400096	1.3058794385093	0.2893825601299	0.8652893839628		0.006116713527706663
IL18	F5_DEPRESSIO	psychiatric	Depression	59333	434831	3	ivw_fe	-0.053177454248	0.0239623416623	0.0264724807741	0.9482117333512	0.9047074529888	0.9938079854373	16.897593708319	0.0002141579236		0.07563565935468983
IL18	K11_UC_STRICT2	autoimmune	Ulcerative colitis,	7220	492160	3	ivw_fe	0.1156199452878	0.0631611497695	0.0671670217090	1.1225691533536	0.9918574235762	1.2705067019788	0.6522235785653	0.7217245018575		0.14926004824243813
IL18	M13_RHEUMA	autoimmune	Rheumatoid arthr	16314	315115	3	ivw_fe	0.0771408347923	0.0437507593150	0.0778687061709	1.0801941946056	0.9914263120367	1.1769095567907	0.2846613558527	0.8673344007189		0.1557374123418156
IL18	AB1_OTHER_SEPS	infection	Other septicaemia	17133	439048	3	ivw_fe	0.0236437898464	0.0417672636211	0.5713365136170	1.0239255202553	0.9434423230807	1.1112745796763	2.1466234942514	0.3418744388742		0.8160780515816562
IL18	T1D	autoimmune	Type 1 diabetes	4721	403489	3	ivw_fe	0.0379343626722	0.0809825320068	0.6394796329747	1.0386630555736	0.8862188055014	1.2173302307697	0.5408492851817	0.7630553997076		0.8160780515816562
IL18	J10_PNEUMONIA	infection	Pneumonia	78791	421557	3	ivw_fe	0.0095162673330	0.0211571713940	0.6528624412653	1.0095616909785	0.9685532762637	1.0523063964258	1.8447369263539	0.3975762802983		0.8160780515816562
IL18	F5_DEMENTIA	psychiatric	Dementia	24864	469981	3	ivw_fe	-0.011796697200	0.0383260274227	0.7582355945563	0.9882726110276	0.9167543647314	1.0653701703329	2.7066008592608	0.2583860667869		0.875668250144805 4
IL18	I9_MI_STRICT	cardiovascular	Myocardial infarct	31666	416171	3	ivw_fe	-0.003195473482	0.0343665607673	0.9259178165927	0.9968096266093	0.9318774796080	1.0662661706547	1.5691818221063	0.4563063334859		0.9345647400041648
TNFSF14	ST19_FRACT_FEM	skeletal	Fracture of femur	12229	476877	3	ivw_fe	-0.110401807481	0.0398636159679	0.0056144466275	0.8954742547450	0.8281719353383	0.9682459724785	0.4600320866357	0.7945208556504		0.03369825986652053
TNFSF14	L12_PSORIASIS	autoimmune	Psoriasis	12760	482181	3	ivw_fe	0.1038259662825	0.0383687217661	0.0068097949351	1.1094073638262	1.0290368540568	1.1960550237434	0.2219619795548	0.8949557620755		0.03369825986652053
TNFSF14	AB1_OTHER_SEPS	infection	Other septicaemia	17133	439048	3	ivw_fe	-0.087332349830	0.0331487033537	0.0084245649666	0.9163724887765	0.8587273525206	0.9778872603994	3.3949343988396	0.1831468114480		0.03369825986652053
TNFSF14	M13_RHEUMA	autoimmune	Rheumatoid arthr	16314	315115	3	ivw_fe	-0.079650019637	0.0348346965670	0.0222241057618	0.9234394755212	0.8624949612131	0.9886903730446	5.2088082680436	0.0739471866851		0.07408035253915037
TNFSF14	K11_UC_STRICT2	autoimmune	Ulcerative colitis,	7220	492160	3	ivw_fe	0.1082619277412	0.0503453128483	0.0315245820602	1.1143395836146	1.0096311988925	1.2299072264926	0.2338773423758	0.8896397520395		0.07881145515066745
TNFSF14	J10_PNEUMONIA	infection	Pneumonia	78791	421557	3	ivw_fe	-0.021002520926	0.0169378219877	0.2149836561088	0.9792164960333	0.9472420214044	1.0122702798617	1.1572403534266	0.5606714606136		0.38177051563643 11
TNFSF14	F5_DEMENTIA	psychiatric	Dementia	24864	469981	3	ivw_fe	-0.036754306356	0.0305578517404	0.2290623093818	0.9639129335388	0.9078757982841	1.0234088684812	0.5828726883302	0.7471895750653		0.38177051563643 11
TNFSF14	T1D	autoimmune	Type 1 diabetes	4721	403489	3	ivw_fe	-0.048216331140	0.0642786255779	0.4531857279940	0.9529276168595	0.8401268074624	1.0808737858469	1.3013059599536	0.5217050025932		0.6972088122984877
TNFSF14	I9_MI_STRICT	cardiovascular	Myocardial infarct	31666	416171	3	ivw_fe	0.0073966266130	0.0275195843262	0.7881014251303	1.0074240492256	0.9545247000591	1.0632550576168	0.7990102393114	0.6706518563472		0.875668250144805 4
TNFSF14	F5_DEPRESSIO	psychiatric	Depression	59333	434831	3	ivw_fe	0.0015726902918	0.0191550694294	0.9345647400041	1.0015739276171	0.9646680313515	1.0398917553826	2.2349528398247	0.3271042276395		0.9345647400041648

Supplementary Table 75 - r6_phewas_mr_persnp

ST75_r6_phewas_mr_persnp.csv

protein	endpoint	category	rsid	bx	bx_se	by	by_se	orient	wald_beta	wald_se	wald_p
IL18	I9_MI_STRICT	cardiovascular	rs5744249	0.2039	0.0135	0.00679879	0.0107026	flipped	0.03334374693477195	0.05248945561549779	0.5252676109576255
IL18	I9_MI_STRICT	cardiovascular	rs10891343	0.1577	0.0115	0.000118964	0.00923825	flipped	0.0007543690551680406	0.05858116677235257	0.9897256606383629
IL18	I9_MI_STRICT	cardiovascular	rs7106524	0.1398	0.012	-0.0109226	0.0100802	aligned	-0.07813018597997137	0.07210443490701	0.2785556001710692
TNFSF14	I9_MI_STRICT	cardiovascular	rs344560	-0.4894	0.0273	-0.0111861	0.0196926	flipped	0.022856763383735185	0.04023825091949326	0.5700107211311092
TNFSF14	I9_MI_STRICT	cardiovascular	rs413141	0.2432	0.0189	0.00259117	0.0107854	flipped	0.010654481907894735	0.04434786184210526	0.8101380604907777
TNFSF14	I9_MI_STRICT	cardiovascular	rs62123250	-0.2703	0.0249	0.0135841	0.0193887	aligned	-0.05025564187939327	0.07173029966703662	0.48354054756456843
IL18	AB1_OTHER_SEPSIS	infection	rs5744249	0.2039	0.0135	0.0164649	0.0130036	flipped	0.08074987739087788	0.06377439921530162	0.2054485979596148
IL18	AB1_OTHER_SEPSIS	infection	rs10891343	0.1577	0.0115	0.00306217	0.0112266	flipped	0.019417691819911223	0.07118960050729232	0.7850375324258376
IL18	AB1_OTHER_SEPSIS	infection	rs7106524	0.1398	0.012	-0.0108927	0.0122594	aligned	-0.07791630901287554	0.08769241773962803	0.37426207122482724
TNFSF14	AB1_OTHER_SEPSIS	infection	rs344560	-0.4894	0.0273	0.063252	0.0235093	flipped	-0.12924397221087047	0.04803698406211688	0.007134295815459213
TNFSF14	AB1_OTHER_SEPSIS	infection	rs413141	0.2432	0.0189	-0.00237222	0.0130976	flipped	-0.009754194078947368	0.05385526315789473	0.8562744239464126
TNFSF14	AB1_OTHER_SEPSIS	infection	rs62123250	-0.2703	0.0249	0.0411967	0.0235344	aligned	-0.15241102478727342	0.08706770255271921	0.08003403156858369
IL18	J10_PNEUMONIA	infection	rs5744249	0.2039	0.0135	0.0037698	0.00658723	flipped	0.01848847474252084	0.032306179499754784	0.5671261255846456
IL18	J10_PNEUMONIA	infection	rs10891343	0.1577	0.0115	0.00512511	0.00568691	flipped	0.03249911223842739	0.03606157260621433	0.367475755333861
IL18	J10_PNEUMONIA	infection	rs7106524	0.1398	0.012	-0.00591473	0.00620937	aligned	-0.0423085121602289	0.044416094420600856	0.3408185612677912
TNFSF14	J10_PNEUMONIA	infection	rs344560	-0.4894	0.0273	0.00651219	0.0121273	flipped	-0.013306477319166326	0.024779934613812834	0.5912773027056901
TNFSF14	J10_PNEUMONIA	infection	rs413141	0.2432	0.0189	-0.0101898	0.00663139	flipped	-0.04189884868421053	0.02726722861842105	0.12439106621818152
TNFSF14	J10_PNEUMONIA	infection	rs62123250	-0.2703	0.0249	-0.00254186	0.0119442	aligned	0.009403847576766556	0.044188679245283025	0.831474197143979
IL18	K11_UC_STRICT2	autoimmune	rs5744249	0.2039	0.0135	0.0227702	0.019667	flipped	0.11167336929867583	0.09645414418832761	0.24695090606057035
IL18	K11_UC_STRICT2	autoimmune	rs10891343	0.1577	0.0115	0.0273187	0.016973	flipped	0.17323208623969563	0.10762840837032338	0.10749860445657977
IL18	K11_UC_STRICT2	autoimmune	rs7106524	0.1398	0.012	0.00497736	0.0185407	aligned	0.03560343347639485	0.13262303290414879	0.7883484286881406
TNFSF14	K11_UC_STRICT2	autoimmune	rs344560	-0.4894	0.0273	-0.0443769	0.0360108	flipped	0.09067613404168369	0.07358152840212506	0.21782883604409475
TNFSF14	K11_UC_STRICT2	autoimmune	rs413141	0.2432	0.0189	0.0337883	0.019743	flipped	0.13893215460526315	0.08118009868421053	0.08700608437104385
TNFSF14	K11_UC_STRICT2	autoimmune	rs62123250	-0.2703	0.0249	-0.022722	0.0354636	aligned	0.08406215316315205	0.13120088790233075	0.5217089748390409
IL18	L12_PSORIASIS	autoimmune	rs5744249	0.2039	0.0135	0.0482655	0.0149725	flipped	0.23671162334477686	0.07343060323688082	0.0012658569162011801
IL18	L12_PSORIASIS	autoimmune	rs10891343	0.1577	0.0115	0.0315527	0.0128672	flipped	0.20008053265694356	0.08159289790741915	0.01419933208839308
IL18	L12_PSORIASIS	autoimmune	rs7106524	0.1398	0.012	0.0252003	0.014044	aligned	0.1802596566523605	0.10045779685264662	0.07275222711067708
TNFSF14	L12_PSORIASIS	autoimmune	rs344560	-0.4894	0.0273	-0.0510693	0.0274654	flipped	0.10435083776052309	0.05612055578259093	0.06296976438606583
TNFSF14	L12_PSORIASIS	autoimmune	rs413141	0.2432	0.0189	0.0214221	0.0150254	flipped	0.08808429276315789	0.06178207236842105	0.1539475146019593 2
TNFSF14	L12_PSORIASIS	autoimmune	rs62123250	-0.2703	0.0249	-0.0387845	0.0270596	aligned	0.14348686644469108	0.10010950795412506	0.15177238253118608
IL18	M13_RHEUMA	autoimmune	rs5744249	0.2039	0.0135	0.0209597	0.0136273	flipped	0.10279401667484062	0.06683325159391859	0.12403224138928581
IL18	M13_RHEUMA	autoimmune	rs10891343	0.1577	0.0115	0.0103439	0.0117498	flipped	0.06559226379201014	0.07450729232720354	0.37867141307385777
IL18	M13_RHEUMA	autoimmune	rs7106524	0.1398	0.012	0.00645986	0.0128471	aligned	0.04620786838340486	0.09189628040057224	0.615086387909427
TNFSF14	M13_RHEUMA	autoimmune	rs344560	-0.4894	0.0273	0.0640771	0.024831	flipped	-0.13092991418062933	0.050737637923988554	0.00986492498300473
TNFSF14	M13_RHEUMA	autoimmune	rs413141	0.2432	0.0189	0.0046474	0.013658	flipped	0.019109375	0.05615953947368421	0.7336536295136535
TNFSF14	M13_RHEUMA	autoimmune	rs62123250	-0.2703	0.0249	0.0475027	0.0248231	aligned	-0.175740658527562	0.09183536810950796	0.05566502215407243
IL18	T1D	autoimmune	rs5744249	0.2039	0.0135	0.0170181	0.0251904	flipped	0.08346297204512017	0.12354291319274155	0.499308110387247
IL18	T1D	autoimmune	rs10891343	0.1577	0.0115	0.00813326	0.0217898	flipped	0.051574254914394414	0.13817247939124921	0.7089549695521404
IL18	T1D	autoimmune	rs7106524	0.1398	0.012	-0.00964293	0.0237722	aligned	-0.06897660944206008	0.17004434907010013	0.685007913125115
TNFSF14	T1D	autoimmune	rs344560	-0.4894	0.0273	-0.0142751	0.045858	flipped	0.0291685737637924	0.09370249284838579	0.7555807790873763

(showing first 40 of 60 rows; full data in the accompanying CSV)

Supplementary Table 76 - r6_ldsc_qc

ST76_r6_ldsc_qc.csv

trait	status	n_snps_ldmerged	mean_chi2
GrimAge	OK	1175366	1.0751721531148744
Hannum	OK	1175315	1.0801995852976018
IEAA	OK	1175374	1.094745293994269
PhenoAge	OK	1175371	1.0741146793335423
Frailty	OK	1168664	1.3997523109722234
ALLERG_ASTHMA	OK	1175751	1.3043390448715404
ALLERG_RHINITIS	OK	1175751	1.1997334815163174
L12_ATOPIC	OK	1175751	1.3617640780343045

Supplementary Table 77 - r6_hla_excluded_rg

ST77_r6_hla_excluded_rg.csv

instrument_set	outcome	n_snps	frailty_h2	frailty_h2_se	gencov	gencov_se	rg	rg_se	p	cross_intercept
full	ALLERG_ASTHMA	1160670	0.11084777654924406	0.005337449159583173	0.030883501398745492	0.002679686404487123	0.4600462761369134	0.038208966093440225	2.18239636095438e-33	-0.005948995204151473
full	ALLERG_RHINITIS	1160670	0.11084777654924406	0.005337449159583173	0.014288033950288378	0.001827107557809139	0.35358983320957416	0.0457645120812385	1.1072734101854537e-14	-0.0006154339208886736
full	L12_ATOPIC	1160670	0.11084777654924406	0.005337449159583173	0.01339958134382086	0.0019348295440988965	0.250252684095449	0.03319107944840421	4.7084686570506916e-14	-0.00511017580342552
hla_excluded	ALLERG_ASTHMA	1160670	0.11084777654924406	0.005337449159583173	0.030883501398745492	0.002679686404487123	0.4600462761369134	0.038208966093440225	2.18239636095438e-33	-0.005948995204151473
hla_excluded	ALLERG_RHINITIS	1160670	0.11084777654924406	0.005337449159583173	0.014288033950288378	0.001827107557809139	0.35358983320957416	0.0457645120812385	1.1072734101854537e-14	-0.0006154339208886736
hla_excluded	L12_ATOPIC	1160670	0.11084777654924406	0.005337449159583173	0.01339958134382086	0.0019348295440988965	0.250252684095449	0.03319107944840421	4.7084686570506916e-14	-0.00511017580342552

Supplementary Table 78 - r6_instrument_consistency

ST78_r6_instrument_consistency.csv

outcome	beta_full	se_full	beta_excluded	se_excluded	n_snps_full	n_snps_excluded	diff	se_diff	z	p	note
ALLERG_ASTHMA	1.2345712417627024	0.0846430400000468	1.2329391255200006	0.1904110529040856	66	12	-0.0016321162427017644	0.20837661406331567	-0.007832530776250363	0.9937506085201874	SEs treated as independe
ALLERG_RHINITIS	0.5256676234344337	0.0748444946867158	0.9823047529820976	0.1683217010362997	70	13	0.4566371295476639	0.18421154530773384	2.478873562375423 2	0.013179801710935453	SEs treated as independe
L12_ATOPIC	0.3762838636590032	0.0539932657056154	0.4704160556243372	0.1225597296358961	71	13	0.09413219196533401	0.13392594994989257	0.7028674577298343	0.48213835299968144	SEs treated as independe

Supplementary Table 79 - r6_ldsc_simulation_calibration

ST79_r6_ldsc_simulation_calibration.csv

part	scenario	truth	mean_est	bias	empirical_sd	mean_jk_se	se_ratio	coverage_95	type1	n_finite
univariate	h2=0.111	0.111	0.11019153508744886	-0.0008084649125511406	0.003921327833838469	0.004129574008416312	1.0531060353538453	0.96		
univariate	h2=0 (null)	0.0	9.58226292207337e-05	9.58226292207337e-05	0.0030987552883648332	0.0029037060604929733	0.9370556208151614	0.93	0.07	
bivariate	rg=0.5	0.5	0.4946090023736943	-0.0053909976263056825	0.04838318526529811	0.050574068684346704	1.0452819178199075	0.95		100.0
bivariate	rg=0 (null)	0.0	-0.0010543603714059748	-0.0010543603714059748	0.04580701270145672	0.04638599719499858	1.0126396474992887	0.94	0.06	100.0

Supplementary Table 80 - r7_gwas_phewas_mr

ST80_r7_gwas_phewas_mr.csv

exposure	endpoint	n_snp	b_ivw	se_ivw	se_re	Q	Q_p	or_ivw	or_lo	or_hi	p_ivw	p_ivw_re	b_egger	se_egger	egger_int	egger_int_p	b_wm	or_wm	q
frailty	AB1_OTHER_SEI	71	0.69036208244	0.07071785801	0.11918422088	198.827919469	2.79639568358	1.99443755333	1.57895148476	2.51925482990	1.63563027611	6.93897171687	0.69382996867	0.07092075461	-0.0013653856	0.51769717679	0.70253083700	2.01885564196	1.3106946576316419e-08
index	AB1_OTHER_SEI	88	0.05467280990	0.06358824364	0.08291772860	147.931406660	5.04402128056	1.05619498159	0.89776589724	1.24258210583	0.38990243781	0.50966325614	0.04672918961	0.06380129874	-0.0026063657	0.12731534595	-0.0894288590	0.91445331787	0.6826232270747397
frailty	ALLERG_ASTHM	71	1.34758663807	0.08095668624	0.16884205331	304.476545685	5.34733490885	3.84812739120	2.76393727910	5.35760508419	3.24744446666	1.44747541699	1.34578105042	0.08116314989	0.00075124102	0.75497275952	1.43322377622	4.19219212042	8.2023606963315527e-15
index	ALLERG_ASTHM	88	0.84944336898	0.07213222412	0.19520860403	637.174691473	1.33437489190	2.33834489431	1.59493742751	3.42825790555	5.17761077915	1.35231034814	0.85455253240	0.07237385344	0.00167876726	0.38724013281	0.84973199635	2.33901990205	7.6630919728202241e-05
frailty	ALLERG_RHINIT	71	0.57987962395	0.07421431940	0.10215444690	132.628596982	9.20989143034	1.78582344744	1.46178297737	2.18169552854	5.55854980579	1.37479293755	0.58151166791	0.07440386588	-0.0006793122	0.75846665145	0.63203206412	1.88142988359	2.3371479938458946e-08
index	ALLERG_RHINIT	88	0.49068169857	0.06626851121	0.14515091220	417.392233605	1.19915229795	1.63342934696	1.22898166355	2.17097741215	1.31697819641	0.00072355816	0.50143125283	0.06649955087	0.00345815195	0.05227191707	0.50132543299	1.65090798913	0.0030751222076243 63
frailty	ANAPH_SHOCK_	71	0.04233432965	0.44901148763	0.44901148763	51.6072715638	0.95139546560	1.04324320759	0.43269240429	2.51531198462	0.92488403612	0.92488403612	0.02402085693	0.45018701065	0.00752014224	0.57324156973	-0.0521710955	0.94916645477	0.9248840361246782
index	ANAPH_SHOCK_	88	0.29717892761	0.40123091556	0.43666234213	103.043821053	0.11536415562	1.34605612451	0.57196397922	3.16779929534	0.45889472435	0.49614414049	0.24707927263	0.40260944738	-0.0162283931	0.13231968134	0.04096554238	1.04181620642	0.6826232270747397
frailty	ASTHMA_NAS	71	0.74711328708	0.05493668216	0.11720754911	318.628306219	2.09198897979	2.11089765743	1.67763739448	2.65605007067	4.02991897289	1.83829076140	0.74309628129	0.05508385688	0.00163470081	0.31814739578	0.74839388489	2.11360259995	4.464420420544847e-10
index	ASTHMA_NAS	88	0.69133928377	0.04923432896	0.11826128337	501.959157457	1.26645720994	1.99638747294	1.58335682855	2.51716029531	8.64707240583	5.03948096048	0.69059792158	0.04939211241	-0.0002489356	0.85092917692	0.80495350957	2.23659251625	8.567117632816195e-08
frailty	CHRONNAS	71	-1.29224922362	0.17739574492	0.26932949823	161.353907454	3.57371572140	0.27465233331	0.16200387242	0.46563025357	3.22731596068	1.60239095813	-1.2749341750	0.17782779155	-0.0073914069	0.16220938730	-1.3543344947	0.25811901685	2.4764223898514404e-06
index	CHRONNAS	88	-0.11481135244	0.15895482077	0.27387210092	258.266116329	7.00079260233	0.89153431381	0.52121082203	1.52497492205	0.47011668334	0.67505885140	-0.1481228286	0.15950684385	-0.0107398060	0.01199172201	-0.4685214220	0.62592706637	0.7172500296149346
frailty	F5_DEMENTIA	71	0.04447345417	0.06523733479	0.08733249049	125.446142297	5.31844369720	1.04547722328	0.88099949416	1.24066203402	0.49541739221	0.61058194845	0.02521067833	0.06541102359	0.00784989516	5.26099541681	-0.0235873320	0.97668867471	0.6487433202308125
index	F5_DEMENTIA	88	-1.10643371001	0.05799570777	0.54590443258	7708.33868363	0.0	0.33073636212	0.11344848770	0.96419567544	3.85768098116	0.04268388138	-1.0104138649	0.05824010022	0.02810112675	1.47154303470	-0.0127715405	0.98730966946	0.1209376639180895
frailty	F5_DEPRESSIO	71	0.64539567692	0.04068953042	0.05176411224	113.289647293	0.00081532101	1.90674133357	1.72277784963	2.11034900056	1.17124178327	1.11572131140	0.64100754980	0.04079885084	0.00178224702	0.14150367755	0.63584051724	1.88860888285	1.8967262293941787e-34
index	F5_DEPRESSIO	88	0.05346347728	0.03643608822	0.03832048825	96.2316274638	0.23371791608	1.05491846258	0.97858788543	1.13720288106	0.14228845603	0.16296551214	0.04706196477	0.03656159128	-0.0020716571	0.03443387671	0.04101484062	1.04186756739	0.3957733866479835
frailty	FOOD_ALLERGY	71	-2.56588928969	0.89682204176	0.89682204176	58.3958760818	0.83735661017	0.07685080841	0.01325128842	0.44569603827	0.00422186122	0.00422186122	-2.50263430789	0.89908666748	-0.0264161398	0.32126135248	-1.84100968188	0.15865715195	0.0047847760525025002
index	FOOD_ALLERGY	88	-0.84467059779	0.79656033574	0.98828546517	133.920392856	0.00092416802	0.42969887865	0.06193250539	2.98132822390	0.28896379383	0.39272676989	-0.7570163012	0.79976786119	0.02624586768	0.22058481537	1.57680129240	4.83945103805	0.6676355088232454
frailty	I9_MI_STRICT	71	0.85921354227	0.05813928749	0.12791670220	338.854407303	6.76783718734	2.36130289834	1.83766695151	3.03414684209	2.01499076017	1.85533283541	0.85417342629	0.05830639860	0.00198369619	0.25322443518	0.80934816753	2.24644320473	5.256776367016840 3e-11
index	I9_MI_STRICT	88	-0.20592906659	0.05239607918	0.07653439264	185.624353140	4.08703640306	0.81389080627	0.70051711833	0.94561321515	8.48617849934	0.00713079327	-0.2098263347	0.05257519315	-0.0012648206	0.36876221741	-0.1719208400	0.84204582682	0.0242446971343 66774
frailty	J10_PNEUMONIA	71	0.20156539862	0.03598429044	0.04486062329	108.793452690	0.00205523118	1.22331623764	1.12034713244	1.33574904951	2.12525380637	7.01781320385	0.20196466459	0.03608055209	-0.0001624836	0.87950939098	0.18737132867	1.20607505215	9.941902038793735e-06
index	J10_PNEUMONIA	88	0.05197621870	0.03222513525	0.05548237314	257.893153507	7.93888415475	1.05335069217	0.94481207597	1.17435806433	0.10676422546	0.34885837098	0.05191359431	0.03233426934	-2.0433373280	0.98117724200	0.10311343399	1.10861715681	0.6589547007556799
frailty	K11_UC_STRICT	71	-1.76110600387	0.10825451232	0.28584639012	488.058013797	6.12667525514	0.17185468672	0.09813954068	0.30093918458	1.65966511880	7.22767794774	-1.7489412077	0.10849606107	-0.0053914841	0.09272187072	-1.3831846689	0.25077863305	1.5358815638958388e-09
index	K11_UC_STRICT	88	-0.23677125954	0.09623196521	0.20218429631	384.039048043	6.22079290284	0.78917178288	0.53096855373	1.17293594604	0.01387736723	0.24157204151	-0.2522260600	0.09655238061	-0.0050847284	0.04925296234	0.09293887240	1.09739465212	0.5133405882273062
frailty	L12_ATOPIC	71	0.37628386365	0.05399326570	0.11064476438	293.954819335	3.14779664199	1.45686062486	1.17283057917	1.80967559848	3.19000296931	0.00067181971	0.36677503042	0.05412616011	0.00403112002	0.01212292910	0.33184921465	1.39354270685	0.0008157810840553734
index	L12_ATOPIC	88	0.63027039962	0.04832791489	0.11753600778	514.594343056	6.54044327129	1.87811835311	1.49167526064	2.36467590592	7.10088182805	8.21415770889	0.62393351351	0.04848664049	-0.0020995631	0.10598113138	0.46021364985	1.58441245832	6.982034052553067e-07
frailty	L12_PSORIASIS	71	1.19129378453	0.08089978192	0.28056249938	841.904017086	9.39910044817	3.29133673448	1.89912126244	5.70416313798	4.41860008617	2.17532721994	1.20075197434	0.08112074542	-0.0038235003	0.11393559011	0.60011764705	1.82233317991	2.8446586722409154e-05
index	L12_PSORIASIS	88	-0.06768680193	0.07298768545	0.19839431659	642.803928982	1.16057824084	0.93455312797	0.63347188384	1.37873451260	0.35373295316	0.73297328069	-0.0642746814	0.07324222273	0.00109560235	0.57596953066	0.07245270029	1.07514195099	0.732973280695 1304
frailty	M13_RHEUMA	71	6.16716472474	0.07014189026	0.69543451351	6881.06387923	0.0	476.832237056	122.011228605	1863.50866960	0.0	7.44218723904	6.14341670095	0.07042325288	0.00813047058	0.00015923652	4.45740654205	86.2634979860	6.325859153187277e-18
index	M13_RHEUMA	88	0.13329597461	0.06663037856	0.28718142504	1616.17127497	4.78727019551	1.14258812537	0.65078261422	2.00605793042	0.04544322019	0.64253849446	0.15132588906	0.06685336449	0.00591704288	0.00095055654	0.00180909536	1.00181073276	0.7172500296149346
frailty	ST19_FRACT_FE	71	0.85031167361	0.08403981863	0.12374446925	151.767724133	5.61508707162	2.34037617177	1.83633642582	2.98276533013	4.59892632722	6.35247012410	0.83848539648	0.08427402005	0.00472496607	0.05961634647	0.62588995215	1.86990934700	2.1598398421957323e-11
index	ST19_FRACT_FE	88	-0.06264466400	0.07554629688	0.10096028198	155.379600441	9.28597360607	0.93927717338	0.77064548993	1.14480863115	0.40697870383	0.53493641796	-0.0423171054	0.07579609985	0.00671073874	0.00094629668	0.20546418892	1.22809500077	0.6826232270747397
frailty	T1D	71	13.3991058148	0.12237740350	1.78580568802	14906.0980930	0.0	659413.323444	19908.9899761	21840682.6090	0.0	6.23187340613	13.5444392997	0.12289051182	-0.0493171536	2.19847667020	19.9465198237	459900087.968	2.6485461976058634e-13
index	T1D	88	0.31695419823	0.12317810904	0.54681620097	1714.48991035	2.62489375825	1.37293968743	0.47010201451	4.00969008243	0.01007811912	0.56216030464	0.36121872745	0.12360521475	0.01420996064	1.62137190421	0.06473469010	1.06687593408	0.6826232270747397

Supplementary Table 81 - r7_gwas_sensitivity

ST81_r7_gwas_sensitivity.csv

exposure	endpoint	n_snp	ivw	ivw_se	n_outliers_presso	ivw_outlier_corrected	or_outlier_corrected	p_outlier_corrected	loo_max_abs_delta
frailty	AB1_OTHER_SEPSIS	71	0.6903620824418696	0.0707178580129378	0	0.6903620824418696	1.9944375533392638	1.6356302761106476e-22	0.0419738607175596
frailty	ALLERG_ASTHMA	71	1.3475866380783847	0.08095668624488415	2	1.338747929001435	3.8142647840245028	4.642103396784575e-59	0.10928398613193835
frailty	ALLERG_RHINITIS	71	0.5798796239531949	0.07421431940275652	1	0.5256676234344336	1.6915878151416148	2.16433740281717e-12	0.056033092499809145
frailty	ANAPH_SHOCK_ADVER_EFFECT	71	0.04233432965155498	0.4490114876328819	0	0.04233432965155498	1.0432432075904	0.9248840361246782	0.14575127018589118
frailty	ASTHMA_NAS	71	0.7471132870830333	0.05493668216780518	2	0.6416636363341127	1.8996385596903345	1.5317590827304037e-30	0.05901145847636058
frailty	CHRONNAS	71	-1.29224922362275	0.17739574492736404	1	-1.4407395516029828	0.23675260315465607	9.890397586116893e-16	0.1714803319088114
frailty	F5_DEMENTIA	71	0.0444734541728052	0.0652373347924307	1	-0.003417192830660229	0.9965886391278939	0.9587037917313274	0.0626781066005346
frailty	F5_DEPRESSIO	71	0.645395676922893	0.04068953042168465	1	0.6650949386715662	1.9446751374472653	3.1065211393168996e-59	0.028227948129535663
frailty	FOOD_ALLERGY	71	-2.565889289695969	0.8968220417682092	0	-2.565889289695969	0.0768508084125771	0.00422186122502206	0.5105079454263777
frailty	I9_MI_STRICT	71	0.8592135422715816	0.058139287493237816	0	0.8592135422715816	2.36130289834925	2.0149907601706836e-49	0.0393671399431027
frailty	J10_PNEUMONIA	71	0.20156539862455572	0.035984290443692495	1	0.17981741011086988	1.196998783292054	7.137418401750463e-07	0.02174798851368584
frailty	K11_UC_STRICT2	71	-1.7611060038785338	0.10825451232867037	2	-1.440938232482204	0.23670556961179096	5.297032902642976e-38	0.19244806486590083
frailty	L12_ATOPIC	71	0.37628386365900324	0.053993265705615495	0	0.37628386365900324	1.4568606248683944	3.1900029693178512e-12	0.06777245625328648
frailty	L12_PSORIASIS	71	1.191293784534204	0.08089978192798304	1	1.0664636720235705	2.9050879697276475	4.0593455430984066e-39	0.20959382982759145
frailty	M13_RHEUMA	71	6.167164724742683	0.07014189026579595	0	6.167164724742683	476.832237056913	0.0	0.21867367329658371
frailty	ST19_FRACT_FEMUR	71	0.8503116736188023	0.08403981863558904	0	0.8503116736188023	2.340376171776025	4.5989263272235715e-24	0.04332615425376052
frailty	T1D	71	13.399105814820679	0.12237740350457257	1	14.241428577413794	1530995.2581542963	0.0	0.8423227625931151
index	AB1_OTHER_SEPSIS	88	0.054672809904817536	0.063588243648253	2	0.03753411110941174	1.038247412248872	0.5561988283589374	0.03447838107899191
index	ALLERG_ASTHMA	88	0.8494433689845724	0.07213222412156442	0	0.8494433689845724	2.338344894311672	5.177610779150447e-32	0.0965150988200909
index	ALLERG_RHINITIS	88	0.4906816985776417	0.06626851121378051	0	0.4906816985776417	1.633429346966227	1.3169781964180848e-13	0.058605446323664045
index	ANAPH_SHOCK_ADVER_EFFECT	88	0.29717892761688275	0.40123091556687834	0	0.29717892761688275	1.3460561245168774	0.4588947243508317	0.26925919313445973
index	ASTHMA_NAS	88	0.6913392837731115	0.04923432896908098	0	0.6913392837731115	1.9963874729483162	8.647072405838578e-45	0.06723650742733056
index	CHRONNAS	88	-0.11481135244469073	0.1589548207798089	2	0.07594615206798594	1.0789044758181967	0.6399486553513742	0.17619360130746445
index	F5_DEMENTIA	88	-1.106433710019649	0.05799570777684635	1	-0.8983806151725391	0.40722858586237326	9.159950871361499e-54	0.20805309484711
index	F5_DEPRESSIO	88	0.053463477287852945	0.03643608822126006	0	0.053463477287852945	1.054918462580653	0.14228845603283433	0.026922759590414236
index	FOOD_ALLERGY	88	-0.8446705977964357	0.7965603357447947	0	-0.8446705977964357	0.429698878658385	0.2889637938304993	0.45493145993415673
index	I9_MI_STRICT	88	-0.20592906659865434	0.05239607918608174	0	-0.20592906659865434	0.8138908062786173	8.486178499340245e-05	0.05774676282273386
index	J10_PNEUMONIA	88	0.051976218704062147	0.0322251352534578	1	0.03590928976060939	1.0365618154557763	0.26625338994789816	0.01934613475357283
index	K11_UC_STRICT2	88	-0.2367712595489313	0.09623196521863718	1	-0.2232775707191898	0.7998927916601754	0.02034410495924617	0.09525876904682609
index	L12_ATOPIC	88	0.6302703996273565	0.04832791489100902	1	0.6359434418485163	1.8888032771829435	1.5420669840507427e-39	0.07785772680096759
index	L12_PSORIASIS	88	-0.06768680193388536	0.0729876854540804	3	-0.1788571416433746	0.8362253526440323	0.015198999145870348	0.11930826761523354
index	M13_RHEUMA	88	0.1332959746192688	0.0666303785637419	1	0.09472515281395068	1.0993566585150039	0.1551768871766939	0.05410911203348301
index	ST19_FRACT_FEMUR	88	-0.06264466400797736	0.07554629688592254	0	-0.06264466400797736	0.9392771733840547	0.4069787038315448	0.044299346328488566
index	T1D	88	0.3169541982393621	0.12317810904041178	3	0.35247208404081215	1.4225799425106984	0.004230840192326531	0.13488741878050736

Supplementary Table 82 - r7_gwas_phewas_snp

ST82_r7_gwas_phewas_snp.csv

snp	exposure	ea	oa	bx	se_x	by	se_out	pval_out	endpoint
rs9275160	frailty	A	G	0.0382	0.0035	0.0329669	0.0121912	0.00684778	AB1_OTHER_SEPSIS
rs9273344	frailty	T	C	-0.0428	0.0039	-0.0351046	0.0111872	0.00170153	AB1_OTHER_SEPSIS
rs77062257	frailty	A	T	-0.0428	0.0043	0.048886	0.0213091	0.0217831	AB1_OTHER_SEPSIS
rs28746860	frailty	T	G	-0.0429	0.0044	-0.024037	0.0125986	0.0564027	AB1_OTHER_SEPSIS
rs1281935	frailty	T	G	0.0556	0.0059	0.0409582	0.0295411	0.1656	AB1_OTHER_SEPSIS
rs6910879	frailty	A	G	-0.054	0.0061	-0.0753278	0.0807341	0.350801	AB1_OTHER_SEPSIS
rs1281934	frailty	A	G	-0.0521	0.0063	-0.0115069	0.0494709	0.816072	AB1_OTHER_SEPSIS
rs9268362	frailty	A	G	-0.0311	0.0039	-0.0564657	0.0143622	8.44034e-05	AB1_OTHER_SEPSIS
rs28746878	frailty	C	G	0.0477	0.0064	-0.0100812	0.0240637	0.675263	AB1_OTHER_SEPSIS
rs34422230	frailty	T	C	-0.0521	0.0071	0.0311082	0.0375761	0.407743	AB1_OTHER_SEPSIS
rs2523668	frailty	A	C	0.0326	0.0045	0.00748631	0.0146967	0.610481	AB1_OTHER_SEPSIS
rs35520662	frailty	T	G	-0.0499	0.0069	0.0400708	0.0377073	0.287927	AB1_OTHER_SEPSIS
rs34043227	frailty	T	C	0.0499	0.0069	-0.0383838	0.0377002	0.308615	AB1_OTHER_SEPSIS
rs34153000	frailty	A	C	0.0499	0.0069	-0.0400708	0.0377073	0.287927	AB1_OTHER_SEPSIS
rs34408145	frailty	T	C	-0.0498	0.0069	0.0400708	0.0377073	0.287927	AB1_OTHER_SEPSIS
rs17213756	frailty	T	C	0.0495	0.0069	-0.028482	0.0370717	0.442312	AB1_OTHER_SEPSIS
rs41316548	frailty	T	C	0.0496	0.0069	-0.0296689	0.0370797	0.423631	AB1_OTHER_SEPSIS
rs35921062	frailty	T	C	0.0494	0.0069	-0.028696	0.0370672	0.438836	AB1_OTHER_SEPSIS
rs35486885	frailty	A	G	-0.0493	0.0069	0.0287485	0.0370614	0.437927	AB1_OTHER_SEPSIS
rs34710970	frailty	A	G	-0.0493	0.0069	0.0286932	0.0370672	0.438881	AB1_OTHER_SEPSIS
rs41258084	frailty	T	C	0.0492	0.0069	-0.0281093	0.0370981	0.44863	AB1_OTHER_SEPSIS
rs41316241	frailty	A	C	0.0492	0.0069	-0.0279143	0.0371019	0.451829	AB1_OTHER_SEPSIS
rs41317090	frailty	A	C	0.0502	0.0071	-0.013467	0.0366705	0.713438	AB1_OTHER_SEPSIS
rs2857702	frailty	C	G	0.0287	0.0041	0.0267175	0.0137246	0.0515727	AB1_OTHER_SEPSIS
rs2228396	frailty	T	C	0.0372	0.0055	0.0057668	0.0203991	0.777408	AB1_OTHER_SEPSIS
rs8192574	frailty	A	G	0.0474	0.007	0.0998472	0.0321262	0.00188378	AB1_OTHER_SEPSIS
rs41268932	frailty	A	G	-0.0481	0.0072	-0.100031	0.0321436	0.00185815	AB1_OTHER_SEPSIS
rs2022059	frailty	C	G	0.0455	0.0068	0.101104	0.0316196	0.00138612	AB1_OTHER_SEPSIS
rs9268835	frailty	A	G	0.0234	0.0035	0.0601067	0.013172	5.03721e-06	AB1_OTHER_SEPSIS
rs137893789	frailty	A	G	0.0464	0.0071	0.0865515	0.0331334	0.0089958	AB1_OTHER_SEPSIS
rs2844456	frailty	T	C	-0.0454	0.0069	-0.0318949	0.0301234	0.289688	AB1_OTHER_SEPSIS
rs2856437	frailty	A	G	0.0443	0.0068	0.101909	0.0316288	0.00127286	AB1_OTHER_SEPSIS
rs34562262	frailty	C	G	0.0449	0.007	0.0858299	0.0331441	0.00960882	AB1_OTHER_SEPSIS
rs644045	frailty	A	G	-0.0216	0.0034	0.000418999	0.0116464	0.971301	AB1_OTHER_SEPSIS
rs35029304	frailty	T	C	-0.0479	0.0075	-0.0150096	0.0475114	0.752068	AB1_OTHER_SEPSIS
rs82334	frailty	A	C	0.0223	0.0035	0.015136	0.0112974	0.180318	AB1_OTHER_SEPSIS
rs41268928	frailty	C	G	0.0437	0.007	0.0853008	0.0256347	0.000876113	AB1_OTHER_SEPSIS
rs805284	frailty	A	G	0.0427	0.0069	0.0301305	0.0301638	0.317845	AB1_OTHER_SEPSIS
rs41268924	frailty	A	G	0.0432	0.007	0.086839	0.0326568	0.00783394	AB1_OTHER_SEPSIS
rs10947233	frailty	T	G	0.0433	0.0071	0.0844621	0.0256292	0.000982313	AB1_OTHER_SEPSIS

(showing first 40 of 2703 rows; full data in the accompanying CSV)

Supplementary Table 83 - r7_hrs_cox_perSD

ST83_r7_hrs_cox_perSD.csv

outcome	n	events	HR	lower	upper	p	term	HR_perSD	lower_perSD	upper_perSD
mortality	36528	18276	37.98596226572751	33.99725883410355	42.44263739892552	0.0		1.6113110651328255	1.5880359251506524	1.6349273385444194
mortality	36528	18276	0.9333038604427636	0.9242965892066104	0.9423989075466078	3.1425051816625135e-44	fi_x_age	0.9909877827477753	0.9897281047973375	0.9922490639552399
diabe	34023	6565	19.549993431563685	16.064282424863915	23.792052023601112	1.7549996344973594e-193		1.4768743114433103	1.4393212371130413	1.5154071763549255
cancre	36385	5422	3.0549637390808204	2.424982874524491	3.848605920125813	2.5980306178489654e-21		1.1577430234058206	1.1232008423423907	1.193347492020718
hearte	33715	8459	34.14882464729435	28.358725715768443	41.12110806668728	1.2203408734890524e-303		1.5889626934000745	1.5507105032676438	1.628158470389526
stroke	37266	4341	30.933435580606687	24.316738376310923	39.35056675823442	7.006730425219645e-172		1.5684865584828258	1.51974894175114	1.618787167113652
lunge	36858	3936	77.2959366052818	60.87132906281361	98.15231419577584	1.0867013219129057e-278		1.7686661740761902	1.7141115612679023	1.8249570833111008

Supplementary Table 84 - r7_hrs_timevarying

ST84_r7_hrs_timevarying.csv

model	n_intervals	events	HR	lower	upper	p
time_varying_mortality	267563	17376	109.0666228691886	99.34572682407924	119.7387004390671	0.0

Supplementary Table 85 - r7_hrs_age_stratified

ST85_r7_hrs_age_stratified.csv

age_band	n	events	HR	lower	upper	p
50-65	25178	8148	62.44607331743946	53.548025356697465	72.82270535265027	0.0
65-80	8765	7608	27.939712613830675	23.24504675439466	33.58253262260551	1.003661322491663e-275
80-200	2585	2520	15.283840535034804	11.015539758146616	21.20602227663112	7.185156657878207e-60

Supplementary Table 86 - r7_hrs_ews

ST86_r7_hrs_ews.csv

metric	n_dead	n_alive	median_dead	median_alive	p
var	11927	14482	0.005875345358090247	0.0025837133201414833	0.0
ar1	11927	14482	0.5544150055877016	0.449149048808123	2.410239523395484e-66

Supplementary Table 87 - r7_onek1k_coverage

ST87_r7_onek1k_coverage.csv

cell	onek1k_cell	gene	dice_snp	dice_ea	wald_beta	exact_snp_present	n_gene_eqtl_records	onek1k_lead_snp	onek1k_lead_p	onek1k_lead_rho	gene_level_same_direction	note
B_CELL_NAIVE	bin	TSLP	rs6898145	T	-0.0763679027344687	False	0					no cis-eQTL record for g
B_CELL_NAIVE	bin	TSLP	rs6898145	T	-0.105489452755952	False	0					no cis-eQTL record for g
B_CELL_NAIVE	bin	GATA3	rs3781093	C	0.0501581330860805	True	5514	rs11255648	0.0006179141959396	-0.1090651756645		exact SNP present
B_CELL_NAIVE	bin	GATA3	rs3781093	C	0.0427909534359813	True	5514	rs11255648	0.0006179141959396	-0.1090651756645		exact SNP present
B_CELL_NAIVE	bin	GATA3	rs3781093	C	-0.0555473329676327	True	5514	rs11255648	0.0006179141959396	-0.1090651756645		exact SNP present
CD4_NAIVE	cd4nc	IL5	rs13165038	C	0.1073027038805357	False	0					no cis-eQTL record for g
CD4_NAIVE	cd4nc	IL5	rs13165038	C	0.0509208713077404	False	0					no cis-eQTL record for g
CD4_NAIVE	cd4nc	IL5	rs13165038	C	0.0401485998562045	False	0					no cis-eQTL record for g
CD4_STIM	cd4et	TSLP	rs2289277	G	0.1176032768055994	False	0					no cis-eQTL record for g
CD4_STIM	cd4et	TSLP	rs2289277	G	0.1137032934440771	False	0					no cis-eQTL record for g
CD8_NAIVE	cd8et	TSLP	rs3806933	T	0.1686332959611813	False	0					no cis-eQTL record for g
CD8_NAIVE	cd8et	TSLP	rs3806933	T	0.1621588076441139	False	0					no cis-eQTL record for g
CD8_NAIVE	cd8et	STAT6	rs3024949	TGGGGAGGCAA	-0.0458360387742567	False	2244	rs111566142	0.0052915130432809	-0.0889288537306751		gene present, exact SNP
CD8_STIM	cd8et	TSLP	rs2289277	G	0.1190536826189359	False	0					no cis-eQTL record for g
CD8_STIM	cd8et	TSLP	rs2289277	G	0.1151056006100535	False	0					no cis-eQTL record for g
MONOCYTES	monoc	IL1R1	rs549445698	A	-0.0583830499469875	False	4157	rs62151004	0.0031810375480559	0.0946666737609232	0.0	gene present, exact SNP
MONOCYTES	monoc	IL1R1	rs549445698	A	-0.0425930394639849	False	4157	rs62151004	0.0031810375480559	0.0946666737609232	0.0	gene present, exact SNP
NK	nk	IL5	rs368020909	C	0.1114778858645588	False	0					no cis-eQTL record for g
NK	nk	IL5	rs368020909	C	0.0517441203418477	False	0					no cis-eQTL record for g
NK	nk	IL5	rs368020909	C	0.0425355037862469	False	0					no cis-eQTL record for g
TFH	cd4et	STAT6	rs3024949	TGGGGAGGCAA	-0.0477445391000122	False	2244	rs10747763	0.0003426391357988	-0.114037898244386		gene present, exact SNP
TFH	cd4et	IL1R1	rs4851546	G	0.0619544234118056	True	4157	rs11680437	0.0023064706732075	-0.0971476741361739		exact SNP present
TFH	cd4et	IL1R1	rs4851546	G	0.0603872784541407	True	4157	rs11680437	0.0023064706732075	-0.0971476741361739		exact SNP present
TH1	cd4et	IL13	rs78330488	C	0.1206027994495233	False	0					no cis-eQTL record for g
TH1	cd4et	IL13	rs78330488	C	-0.0962797698146984	False	0					no cis-eQTL record for g
TH2	cd4et	COL1A1	rs2214615	T	0.0615083557939695	False	0					no cis-eQTL record for g
THSTAR	cd4et	IL5	rs7734334	A	0.0838668221446272	False	0					no cis-eQTL record for g
THSTAR	cd4et	IL5	rs7734334	A	0.0410254128717665	False	0					no cis-eQTL record for g
TREG_NAIVE	cd4nc	IL4	rs370314650	G	-0.0833233331930977	False	3354	rs56132027	0.0243577529171238	0.071844948091905	1.0	gene present, exact SNP
TREG_NAIVE	cd4nc	IL4	rs370314650	G	-0.0421207052037857	False	3354	rs56132027	0.0243577529171238	0.071844948091905	1.0	gene present, exact SNP
TREG_NAIVE	cd4nc	IL4	rs370314650	G	-0.1310569336083457	False	3354	rs56132027	0.0243577529171238	0.071844948091905	1.0	gene present, exact SNP
TREG_NAIVE	cd4nc	TSLP	rs3806932	G	0.1789715836088386	True	3415	rs247534	0.0068570482177934	-0.0862278806396051	0.0	exact SNP present
TREG_NAIVE	cd4nc	TSLP	rs3806932	G	0.1725373597092305	True	3415	rs247534	0.0068570482177934	-0.0862278806396051	0.0	exact SNP present

Supplementary Table 88 - r7_onek1k_celltype_mr

ST88_r7_onek1k_celltype_mr.csv

outcome	onek1k_cell	gene	snp	exposure_rho	exposure_aligned	beta_outcome	wald_beta	wald_se	wald_p	fstat	or	or_lower	or_upper	onek1k_p	outcome_p	wald_q
ALLERG_ASTHMA	monoc	TNF	rs3130932	0.1520998850526	0.1520998850526	0.0802165	0.5273935609632	0.0848396433405	5.0882774293190	23.208604083210	1.6945099109871	1.4349182228343	2.0010644458623	1.9720375816191	5.08827e-10	1.882662648848039e-08
ALLERG_ASTHMA	nk	GATA3	rs369421	0.1502071347307	0.1502071347307	-0.0605073	-0.4028257386605	0.1121964015238	0.0003302185512	22.621326165522	0.6684285704330	0.5364774842456	0.8328341205212	2.2688130641381	0.000330225	0.003054521598672722708
ALLERG_ASTHMA	bin	IL6	rs62449498	0.1384708882486	0.1384708882486	0.0556964	0.4022246170617	0.0926151349370	1.4057027220641	19.158043052491	1.4951471305170	1.2469479835862	1.7927491533881	1.3324134193867	1.40579e-05	0.000260055003581872 5
ALLERG_ASTHMA	cd4et	GATA3	rs61834016	-0.1244002254471	-0.1244002254471	-0.0144563	0.1162079887559	0.1160753523403	0.3167578359588	15.404295654327	1.1232294669393	0.8946705743354	1.4101776358722	9.2855032192426	0.316758	0.67616235140032 59
ALLERG_ASTHMA	cd8et	GATA3	rs1572599	0.1221707246904	0.1221707246904	-0.0157592	-0.1289932595548	0.1068545679259	0.2273608050389	14.848800789609	0.8789798902413	0.7128899540044	1.0837656543045	0.0001240615843	0.227362	0.6213078581416658
ALLERG_ASTHMA	nk	TNF	rs184003	0.1204555317796	0.1204555317796	-0.0150661	-0.1250760324362	0.2383609916106	0.5997683635551	14.428698136178	0.8824298067458	0.5530739418539	1.4079172871954	0.0001545281328	0.599768	0.9246428938142034
ALLERG_ASTHMA	cd4nc	RORC	rs4845604	0.1154876382113	0.1154876382113	-0.0528713	-0.4578091717766	0.1697696853416	0.0070041154490	13.247331576415	0.6326681956364	0.4535917316378	0.8824434350346	0.0002872294667	0.0070039	0.037021753088011566
ALLERG_ASTHMA	bin	TNF	rs2844480	-0.1141432124954	-0.1141432124954	0.0664965	-0.5825707770637	0.1433585023783	4.8293922125735	12.936646840462	0.5584608412568	0.4216610689789	0.7396426517925	0.0003382996740	4.82959e-05	0.000595625039550 7322
ALLERG_ASTHMA	cd4et	STAT6	rs10747763	-0.1140378982443	-0.1140378982443	-0.013456	0.1179958610878	0.1903805693917	0.5353964406199	12.912471463036	1.1252394540905	0.7748002266518	1.6341810256217	0.0003426391357	0.535397	0.861289262146518
ALLERG_ASTHMA	monoc	STAT6	rs76167224	0.1138735130089	0.1138735130089	0.024712	0.2170127130271	0.2797112265914	0.4378401800394	12.874783015442	1.2423598961924	0.7180462542182	2.1495246338240	0.0003827143021	0.43784	0.7363675755209799
ALLERG_ASTHMA	bin	STAT6	rs1048691	-0.1130758464985	-0.1130758464985	-0.0171794	0.1519281131379	0.1323695595786	0.2510688165431	12.692715041259	1.1640765516137	0.8980619525021	1.5088872368343	0.0003847582836	0.25107	0.6213078581416658
ALLERG_ASTHMA	monoc	IL6	rs4722074	0.1126561055046	0.1126561055046	-0.0206046	-0.1828982096238	0.1133919900992	0.1067503907507	12.597449390684	0.8328529278353	0.6668790990983	1.0401345616344	0.0004422896818	0.10675	0.438862717530941 5
ALLERG_ASTHMA	cd4nc	STAT5A	rs799923	0.1112598424784	0.1112598424784	-0.00477958	-0.0429587162225	0.1467016277967	0.7696516868833	12.283228544006	0.9579509370839	0.7185687398542	1.2770803222614	0.0004777242156	0.769651	0.9646041975631998
ALLERG_ASTHMA	nk	STAT5A	rs150495898	0.1102279161233	0.1102279161233	0.0107965	0.0979470571494	0.2191042056258	0.6548504199864	12.053643726606	1.1029043926603	0.7178481763267	1.6945060800654	0.0005394761169	0.65485	0.9646041975631998
ALLERG_ASTHMA	bin	GATA3	rs11255648	-0.1090651756645	-0.1090651756645	0.0301715	-0.2766373392439	0.2546954133687	0.2774127801009	11.797643772045	0.7583294635843	0.4603167914575	1.2492778582315	0.0006179141959	0.277414	0.6415170539835091
ALLERG_ASTHMA	cd4nc	IL6	rs9655226	0.1086268496267	0.1086268496267	-0.00214407	-0.0197379377876	0.1202695286192	0.8696412413394	11.701876322622	0.9804555800009	0.7745553053358	1.2410903879073	0.0006501404142	0.869641	0.9646041975631998
ALLERG_ASTHMA	bin	STAT5A	rs28625298	0.1072519915558	0.1072519915558	-0.00736386	-0.0686594243442	0.2134039626488	0.7476538138587	11.404111273271	0.9336446025595	0.6145093151777	1.4185175429543	0.0007616271397	0.747655	0.9646041975631998
ALLERG_ASTHMA	cd4nc	IL13	rs1100636	-0.1060876900920	-0.1060876900920	-0.0122572	0.1155383814028	0.1229134123731	0.3472185047731	11.155051651169	1.1224775959862	0.8821687569662	1.4282482161621	0.0008696363188	0.347217	0.676162351400325 9
ALLERG_ASTHMA	cd4nc	GATA3	rs434645	0.1055186521257	0.1055186521257	-0.0293203	-0.2778684091327	0.2082020529775	0.1820042322503	11.034360853037	0.7573964814169	0.5036143748845	1.1390648453872	0.0009274325445	0.182003	0.6213078581416658
ALLERG_ASTHMA	nk	STAT6	rs66491720	-0.1053852475665	-0.1053852475665	0.0150098	-0.1424279047265	0.1709594124037	0.4047829955809	11.006164413310	0.8672500766698	0.6203271039587	1.2124614428162	0.0009414868347	0.404782	0.7131890874522269
ALLERG_ASTHMA	cd8et	IL6	rs61480128	-0.1004469186448	-0.1004469186448	0.0542981	-0.5405651137191	0.1867354444820	0.0037937850360	9.9885723301688	0.5824190263756	0.4039086320446	0.8398234040385	0.0016234641286	0.00379376	0.023395007722591 4
ALLERG_ASTHMA	monoc	STAT5A	rs33986943	-0.1005918171003	-0.1005918171003	-0.0302457	0.3006775389079	0.2485868206860	0.2264526180144	10.017705689669	1.3507736993407	0.8298148879459	2.1987910958638	0.0017172962093	0.226452	0.6213078581416658
ALLERG_ASTHMA	cd4nc	COL1A1	rs60124448	0.0997964670649	0.0997964670649	0.0030011	0.0300722068452	0.2849509690706	0.9159515599040	9.8583305568472	1.0305289425109	0.5895288074887	1.8014215554223	0.0017412546620	0.915952	0.9646041975631998
ALLERG_ASTHMA	cd8et	IL4	rs29552	-0.0991559792020	-0.0991559792020	0.0406779	-0.4102415237825	0.3104805201637	0.1863974760062	9.7309437934749	0.6634899821726	0.3610339274159	1.2193284980008	0.0018648605457	0.186397	0.6213078581416658
ALLERG_ASTHMA	cd8et	TNF	rs2229094	-0.0977282205833	-0.0977282205833	-0.0449951	0.4604105112260	0.1389833961871	0.0009239952449	9.4500445299005	1.5847243986400	1.2068372018687	2.0809363647031	0.0021698791465	0.000923975	0.006837564812714266
ALLERG_ASTHMA	cd4et	IL1R1	rs11680437	-0.0971476741361	-0.0971476741361	0.00267519	-0.0275373551017	0.1629580959170	0.8658090131080	9.3370370583104	0.9728383413994	0.7068511143993	1.3389162430635	0.0023064706732	0.865809	0.9646041975631998
ALLERG_ASTHMA	cd8et	STAT5A	rs4408592	-0.0964733410669	-0.0964733410669	-0.0241434	0.2502598099431	0.1341075146451	0.0620255226385	9.2066506955497	1.2843590625987	0.9874880152252	1.6704792121483	0.0024749737347	0.0620255	0.2868680422031 6536
ALLERG_ASTHMA	cd4et	RORC	rs6701221	0.0963544870031	0.0963544870031	0.000968876	0.0100553283000	0.1845933754950	0.9565584345120	9.1837672362287	1.0101060529889	0.7034575906102	1.4504275053735	0.0025058134334	0.956558	0.9646041975631998
ALLERG_ASTHMA	cd4nc	STAT6	rs73125460	0.0960758531451	0.0960758531451	0.00482143	0.0501835772690	0.2022880813834	0.8040727007881	9.1302354397120	1.0514641035682	0.7072995138712	1.5630956043522	0.0025794927608	0.804072	0.9646041975631998
ALLERG_ASTHMA	cd8et	RORC	rs6684085	-0.0958820542179	-0.0958820542179	-0.0142239	0.1483478854934	0.1764511632337	0.4004988955837	9.0930971458749	1.1599163442518	0.8207831000012	1.6391735230178	0.0026318996796	0.4005	0.7131890874522269
ALLERG_ASTHMA	cd4et	STAT5A	rs62079659	-0.0945966846321	-0.0945966846321	0.00107667	-0.0113816885252	0.2564793903121	0.9646041975631	8.8487453762625	0.9886828378532	0.5980497745051	1.6344688946240	0.0030048103434	0.964604	0.9646041975631998
ALLERG_ASTHMA	monoc	IL1R1	rs62151004	0.0946666737609	0.0946666737609	0.0246918	0.2608288536930	0.2276408280111	0.2518815641114	8.8619624889420	1.2980054975753	0.8308158001147	2.0279083179488	0.0031810375480	0.251881	0.6213078581416658
ALLERG_ASTHMA	cd8et	STAT6	rs111566142	-0.0889288537306	-0.0889288537306	-0.00387284	0.0435498697838	0.2463182542118	0.8596629300681	7.8119538000638	1.0445120826050	0.6445300009690	1.6927148295154	0.0052951304328	0.859663	0.9646041975631998
ALLERG_ASTHMA	cd4nc	TSLP	rs247535	-0.0862278806396	-0.0862278806396	0.00212301	-0.0246209228877	0.1499804924355	0.8696045318347	7.3411255361588	0.9756796997779	0.7271789464521	1.3091012620803	0.0068570482177	0.869605	0.9646041975631998
ALLERG_ASTHMA	cd4nc	IL1R1	rs11680510	-0.0817310102014	-0.0817310102014	0.00077344	-0.0094632379814	0.1673905652979	0.9549165070512	6.5903823454425	0.9905813975452	0.7135171927743	1.3752317604953	0.0104005317199	0.954916	0.9646041975631998
ALLERG_ASTHMA	monoc	GATA3	rs1931901	-0.0820791879868	-0.0820791879868	-0.01299	0.1582617995939	0.1635615596265	0.3332452937351	6.6470342625285	1.1714728456806	0.8501700784035	1.6142048079886	0.0105871040565	0.333244	0.676162351400325 9
ALLERG_ASTHMA	cd4nc	IL4	rs56132027	0.0718449480919	0.0718449480919	0.00337568	0.0469856279342	0.2225152975202	0.8327648850639	5.0847083566673	1.0481069455094	0.6776364426052	1.6211173132922	0.0243577529171	0.832765	0.9646041975631998
ALLERG_RHINITIS	monoc	TNF	rs3130932	0.1520998850526	0.1520998850526	0.0513735	0.3377615964813	0.0780072910370	1.4919594022847	23.208604083210	1.4018062680186	1.2030587909111	1.6333871859813	1.9720375816191	1.49204e-05	0.000552024978845361
ALLERG_RHINITIS	nk	GATA3	rs369421	0.1502071347307	0.1502071347307	-0.00625818	-0.0416636667174	0.1042067677281	0.6892911003099	22.621326165522	0.9591923346331	0.7819934366798	1.1765443182302	2.2688130641381	0.68929	0.9445841004247998
ALLERG_RHINITIS	bin	IL6	rs62449498	0.1384708882486	0.1384708882486	0.0348541	0.2517070587243	0.0851225853252	0.0031065316089	19.158043052491	1.2862191953993	1.0885720499756	1.5197522466709	1.3324134193867	0.00310649	0.038313889844 14543

(showing first 40 of 111 rows; full data in the accompanying CSV)

Supplementary Table 89 - r7_bloodtrait_mr

ST89_r7_bloodtrait_mr.csv

exposure	outcome	n_snp	f_stat	b_ivw	se_ivw	se_re	p_ivw	p_ivw_re	or	lo	hi	Q	Q_p	b_egger	egger_int	egger_int_p	b_wm	or_wm
WBC	ALLERG_ASTHMA	175	67.42417082580	0.094690595876	0.030784983249	0.058613210875	0.002098895759	0.106198939598	1.099318668771	0.980011181743	1.233150761973	630.7575114309	4.848563134017	0.094690394145	0.000212697011	0.856762420654	0.055232286024	1.056786062800965
WBC	ALLERG_RHINITIS	175	67.42417082580	0.012360777032	0.028251281911	0.043562432627	0.661727187059	0.776603086701	1.012437487176	0.929580759307	1.102679519963	413.7111045806	1.438039378549	0.012358048023	0.001721755741	0.111336769524	0.003594632290	1.003601100728886
WBC	L12_ATOPIC	175	67.42417082580	-0.000450767784	0.020600683625	0.038348877110	0.982542717300	0.990621567016	0.999549333795	0.927173501035	1.077574876305	602.9638664996	1.109907286288	-0.000446361997	-0.002523587431	0.001346875759	-0.022695000521	0.977560593778707
Eosinophil	ALLERG_ASTHMA	167	85.50005467411	0.477431291090	0.028043455746	0.072767394360	5.387210220529	5.342636719665	1.611928504559	1.397670885701	1.859030999638	1117.683079688	9.281260793918	0.477076363048	-0.000175186491	0.885660607101	0.224255524517	1.251390739593429
Eosinophil	ALLERG_RHINITIS	167	85.50005467411	0.265371645857	0.025743392500	0.057375681859	6.460751688542	3.743113219681	1.303915480526	1.165226291064	1.459111928210	824.5789636433	6.463906569463	0.271018062256	0.002765774921	0.013398163841	0.171207173437	1.186736583852599
Eosinophil	L12_ATOPIC	167	85.50005467411	0.238522307467	0.018745576821	0.054169234659	4.339437580268	1.066264064872	1.269372022196	1.141508470585	1.411557927299	1386.168003465	2.086640362775	0.238702974499	8.869682812237	0.913245916372	0.071962380368	1.074614916686079

Supplementary Table 90 - r7_sc_age_module_celltype

ST90_r7_sc_age_module_celltype.csv

tissue	cell_type	score	n_cells	n_donors	age_range	slope_per_year	se	p	q
Blood	B cell	aging_sasp_score	1815	9	36.0	0.0006028319994327002	0.0003236605127939251	0.06252642625858362	0.2000845640274676
Blood	B cell	th2_score	1815	9	36.0	0.0003404088098201744	0.0008998102800835951	0.705199011020582	0.9232326575758685
Blood	B cell	alarmin_score	1815	9	36.0	0.000485151010394323	0.0001606649418813454 4	0.002530715277189556	0.012653576385947779
Blood	B cell	treg_score	1815	9	36.0	-0.0003255620580966932	0.0016996499345238558	0.8480974246335551	0.9232326575758685
Blood	B cell	targets_score	1815	9	36.0	-0.00012169878456456723	0.0005322450847800232	0.8191395780080861	0.9232326575758685
Blood	B cell	t2_receptor_score	1815	9	36.0	-0.0002835080326216633	0.0005021863326723395	0.5723815912662051	0.7894918500223519
Blood	CD4-positive, alpha-beta	aging_sasp_score	2343	8	36.0	-0.0003626949633292 3436	0.0011502453318071489	0.75251896168543	0.9232326575758685
Blood	CD4-positive, alpha-beta	th2_score	2343	8	36.0	-6.126550852831217e-05	0.0006575298114184958	0.9257643316828169	0.9382054883629489
Blood	CD4-positive, alpha-beta	alarmin_score	2343	8	36.0	0.0009802157526096265	0.0007364696068935481	0.1832003121671476	0.40711180481588355
Blood	CD4-positive, alpha-beta	treg_score	2343	8	36.0	0.013375053939780746	0.009289243379192 23	0.1499117650964786	0.3548552215780809
Blood	CD4-positive, alpha-beta	targets_score	2343	8	36.0	-0.001033234683661877	0.001699508105064876	0.5432133261499199	0.7894918500223519
Blood	CD4-positive, alpha-beta	t2_receptor_score	2343	8	36.0	0.0017760833762429735	0.0013663434665516937	0.1936418923733525	0.4186851726991405
Blood	CD8-positive, alpha-beta	aging_sasp_score	2147	9	36.0	0.000398444871851794	0.0012898540476441104	0.7573923145560738	0.9232326575758685
Blood	CD8-positive, alpha-beta	th2_score	2147	9	36.0	0.0002509158310888653	0.0012998142300869261	0.8469278272406531	0.9232326575758685
Blood	CD8-positive, alpha-beta	alarmin_score	2147	9	36.0	0.0008430516289682422	8.58606754114305e-05	9.341874841164478e-23	7.473499872931583e-21
Blood	CD8-positive, alpha-beta	treg_score	2147	9	36.0	-0.002243239632039262	0.002753868225265987	0.4153140821358582	0.7222853602362751
Blood	CD8-positive, alpha-beta	targets_score	2147	9	36.0	0.0005785571568851057	0.0008578737181439151	0.5000517665967458	0.7693104101488396
Blood	CD8-positive, alpha-beta	t2_receptor_score	2147	9	36.0	-0.0012368323047852671	0.0017316218884946936	0.4750649739095303	0.7451999590737731
Blood	classical monocyte	aging_sasp_score	4939	9	36.0	0.0029276584912434024	0.0012861196255315557	0.022825065936259856	0.09610554078425203
Blood	classical monocyte	th2_score	4939	9	36.0	-0.000939723926454808	0.0005757748070677896	0.10265773995400909	0.29330782844002595
Blood	classical monocyte	alarmin_score	4939	9	36.0	0.00042074299965919537	0.00013517815273223806	0.001855053680227071	0.009893619627877713
Blood	classical monocyte	treg_score	4939	9	36.0	3.8745061492992836e-05	0.0001170293823493168 06	0.7405906832303091	0.9232326575758685
Blood	classical monocyte	targets_score	4939	9	36.0	-0.002256566208991059	0.0009025215945320586	0.012409148848323044	0.05515177265921353
Blood	classical monocyte	t2_receptor_score	4939	9	36.0	0.00016339314101140282	0.0007929600140039607	0.8367481803838017	0.9232326575758685
Blood	erythrocyte	aging_sasp_score	5484	8	36.0	0.001217665117493974	0.0001721991881917223	1.5353389345785853e-12	3.07067786915717e-11
Blood	erythrocyte	th2_score	5484	8	36.0	0.0003103907595941517	4.739512247420167e-05	5.792279280315609e-11	9.267646848504975e-10
Blood	erythrocyte	alarmin_score	5484	8	36.0	7.550113744368464e-05	1.1980462204046762e-05	2.937872851600394e-10	3.3575689732575934e-09
Blood	erythrocyte	treg_score	5484	8	36.0	-1.1285468614593728e-05	6.664454576561839e-05	0.8655306164773767	0.9232326575758685
Blood	erythrocyte	targets_score	5484	8	36.0	0.0006096399417202698	9.655264243351423e-05	2.71794567677719e-10	3.3575689732575934e-09
Blood	erythrocyte	t2_receptor_score	5484	8	36.0	0.00022939150543237544	4.967507819617042e-05	3.877572043445643e-06	2.8200523952331948e-05
Blood	intermediate monocyte	aging_sasp_score	464	5	24.0	0.005215032345246359	0.004992231937442695	0.29619427758467143	0.5641795763517552
Blood	intermediate monocyte	th2_score	464	5	24.0	-0.0015083371415575103	0.0013998790511337248	0.2812673286949863	0.5641795763517552
Blood	intermediate monocyte	alarmin_score	464	5	24.0	-0.0001930766510832918	2.2053721515835448e-05	2.0440574139511404e-18	8.176229655804561e-17
Blood	intermediate monocyte	treg_score	464	5	24.0	-8.665147915593133e-05	0.0003984711481559872	0.8278499873987418	0.9232326575758685
Blood	intermediate monocyte	targets_score	464	5	24.0	0.003960574343211608	0.0010922086856228867	0.00028761527221099394	0.0016435158412056797
Blood	intermediate monocyte	t2_receptor_score	464	5	24.0	-0.0030386974095632196	0.0016187492933705 49	0.06049227567806508	0.2000845640274676
Blood	macrophage	aging_sasp_score	574	8	36.0	-0.0014180597003088894	0.0019163270948464998	0.4593070711408539	0.7348913138253663
Blood	macrophage	th2_score	574	8	36.0	-0.0024356985817339943	0.00253541996125859	0.3367187770718726	0.6264535387383675
Blood	macrophage	alarmin_score	574	8	36.0	-0.00028512970110869195	0.0001943525548490819 2	0.1423557566562051	0.3548552215780809
Blood	macrophage	treg_score	574	8	36.0	0.001194317815410294	0.0006387480458816528	0.06151448481333987	0.2000845640274676

(showing first 40 of 463 rows; full data in the accompanying CSV)

Supplementary Table 91 - r7_sc_age_gene_celltype

ST91_r7_sc_age_gene_celltype.csv

tissue	cell_type	gene	n_cells	n_donors	age_range	frac_detected	slope_per_year	se	p	q
Blood	B cell	STAT6	1815	9	36.0	0.3867768595041322	-0.0009228103308527249	0.0008920109577762831	0.30088938411734023	0.5621442236513435
Blood	B cell	IL4R	1815	9	36.0	0.6044077134986225	-0.0012827364312584754	0.002454157340539785	0.6011976752323663	0.7651606775684663
Blood	B cell	IL2RA	1815	9	36.0	0.09696969696969697	0.0005627160601411775	0.0008329642245016662	0.4993209990537931	0.7133157129339902
Blood	CD4-positive, alpha-beta	GATA3	2343	8	36.0	0.3542466922748613	-0.003259770196376538	0.002378396453047785	0.17050754917408206	0.3729852638183045
Blood	CD4-positive, alpha-beta	STAT6	2343	8	36.0	0.2987622705932565	-0.0030727464821593206	0.001666850799159503	0.06526417783155795	0.16920342400774283
Blood	CD4-positive, alpha-beta	FOXP3	2343	8	36.0	0.06103286384976526	0.0004638624630033837	0.00043904293550642965	0.29072569466382314	0.5621442236513435
Blood	CD4-positive, alpha-beta	CDKN2A	2343	8	36.0	0.05761843790012804	6.736787223907462e-05	0.0007102933146355932	0.9244378281487129	0.95024239044205
Blood	CD4-positive, alpha-beta	IL4R	2343	8	36.0	0.39223218096457535	0.006888609825804854	0.001087129947520037	2.350278753155298e-10	8.225975636043544e-09
Blood	CD4-positive, alpha-beta	IL2RA	2343	8	36.0	0.30985915492957744	0.006243184684612117	0.001593555799554305	8.937201512607068e-05	0.0008937201512607068
Blood	CD8-positive, alpha-beta	GATA3	2147	9	36.0	0.2966930600838379	0.0011237047112367068	0.0010572750758984666	0.28785860492315896	0.5621442236513435
Blood	CD8-positive, alpha-beta	STAT6	2147	9	36.0	0.3022822543083372	0.0029511597463423534	0.0011994853326429778	0.013879964273606483	0.05831888998638127
Blood	CD8-positive, alpha-beta	IL18	2147	9	36.0	0.09874243129948766	0.003232156554957043	0.0009498337869736688	0.0006668311606781281	0.004243471022497179
Blood	CD8-positive, alpha-beta	TNFSF14	2147	9	36.0	0.11318118304611086	-0.0005764586937580183	0.0014533707282183587	0.6916361339839898	0.8024449041035943
Blood	CD8-positive, alpha-beta	CDKN2A	2147	9	36.0	0.05775500698649278	-0.00026305552163221235	0.000740216748699 3906	0.7223076688179264	0.8155086583428202
Blood	CD8-positive, alpha-beta	IL4R	2147	9	36.0	0.239869585468095	0.0036397662398318124	0.0013527693388225167	0.007132285642325821	0.038404614997139035
Blood	classical monocyte	STAT6	4939	9	36.0	0.7363838833772018	-0.004746562235471208	0.0012524575183518714	0.00015076924069757738	0.0011104284531143498
Blood	classical monocyte	IL18	4939	9	36.0	0.49605183235472766	-0.0007194188700262805	0.0007457130674818443	0.33467529324821055	0.5621442236513435
Blood	classical monocyte	TNFSF14	4939	9	36.0	0.07511642032800162	-0.0006209039668406658	0.00027963250017986875	0.026389695687613273	0.10262659434071829
Blood	classical monocyte	CDKN2A	4939	9	36.0	0.06539785381656206	-0.00012423492929519654	0.0002287639190194498	0.5870812827714702	0.7651606775684663
Blood	classical monocyte	IL4R	4939	9	36.0	0.4909900789633529	-0.0019449015628831092	0.000974030639560706	0.04585169111386263	0.1374837166645399
Blood	classical monocyte	CXCL8	4939	9	36.0	0.28487548086657216	0.010179447443137863	0.005029597620010229	0.04297952487734437	0.1374837166645399
Blood	classical monocyte	IL1B	4939	9	36.0	0.10042518728487548	0.00413766806600431	0.0016302517853343263	0.011147058748103589	0.05201960749115008
Blood	intermediate monocyte	STAT6	464	5	24.0	0.9051724137931034	-0.010065053596486465	0.0026647210819329154	0.00015863263615919284	0.0011104284531143498
Blood	intermediate monocyte	IL18	464	5	24.0	0.6271551724137931	0.012829689267825056	0.004893746615241291	0.008750532636411566	0.04375266318205783
Blood	intermediate monocyte	TNFSF14	464	5	24.0	0.07543103448275862	-0.0021704830712338677	0.0002479182694502952	2.0440574139551706e-18	1.4308401897686194e-16
Blood	intermediate monocyte	CDKN2A	464	5	24.0	0.07112068965517242	-0.0006404590522986767	0.0008221041618343812	0.4359511180859533	0.6534527559939537
Blood	intermediate monocyte	TNF	464	5	24.0	0.06465517241379311	0.002307268091009451	0.0021652577152122495	0.2866108758820387	0.5621442236513435
Blood	intermediate monocyte	IL4R	464	5	24.0	0.5754310344827587	4.596192604826993e-07	0.005090272263421602	0.999927956091087	0.999927956091087
Blood	intermediate monocyte	CXCL8	464	5	24.0	0.21551724137931033	-0.0026988530793538943	0.008417760246030895	0.7485028101460172	0.8316697890511302
Blood	intermediate monocyte	IL1B	464	5	24.0	0.12931034482758622	0.013297274380725655	0.009174272134237155	0.14722334259604136	0.33243980586202887
Blood	macrophage	STAT6	574	8	36.0	0.5853658536585366	-0.007794687158290256	0.0015669257039110285	6.541297929599083e-07	9.157817101438716e-06
Blood	macrophage	IL18	574	8	36.0	0.2804878048780488	-0.003140852451830072	0.002137059018601717	0.14164089114896075	0.3304954126809084
Blood	macrophage	CDKN2A	574	8	36.0	0.05749128919860627	-0.0012977560110801838	0.0003401195688161 2945	0.00013585924359460483	0.0011104284531143498
Blood	macrophage	IL4R	574	8	36.0	0.3588850174216028	0.0034797006916472287	0.0077069297079233575	0.6516271748192161	0.7902039298829434
Blood	macrophage	CXCL8	574	8	36.0	0.15853658536585366	-0.0025320067936139487	0.006554484075247047	0.6992734164331321	0.8024449041035943
Blood	macrophage	IL1B	574	8	36.0	0.08013937282229965	-0.0002373410533273224	0.0029869604885479663	0.9366674991500206	0.95024239044205
Blood	monocyte	STAT6	4499	8	36.0	0.6110246721493665	-0.00421855635175183	0.008966556675855903	0.6380144487013053	0.7902039298829434
Blood	monocyte	IL18	4499	8	36.0	0.3656368081795955	0.005028383425378068	0.001182689104646075	2.1219867047312526e-05	0.0002475651155519 7944
Blood	monocyte	IL4R	4499	8	36.0	0.39497666148032895	-0.007831455373900925	0.008156980218088136	0.33700867243024535	0.5621442236513435
Blood	monocyte	CXCL8	4499	8	36.0	0.13358524116470327	-0.0036141864145900404	0.008082055955574 15	0.6547403990458674	0.7902039298829434

(showing first 40 of 428 rows; full data in the accompanying CSV)

Supplementary Table 92 - r7_index_celltype_anchoring

ST92_r7_index_celltype_anchoring.csv

compartment	n_donors	mean_fraction	frac_slope_per_year	se	p
lymphocyte_T	9	0.17746799604543392	0.004127949408946983	0.004361171941647421	0.37540147717237143
lymphocyte_B	9	0.10442684912751009	-0.0030027736948551828	0.0031830569531368824	0.37690738574379085
monocyte	9	0.2925017923316086	-0.0031820749492204145	0.006950998852887003	0.6609767038703505
NK	9	0.07779183178722565	0.0010361041572128093	0.003404996227405 7496	0.7697541294784567
neutrophil	9	0.34696918987142983	0.0009808872969219457	0.012589426141764795	0.9400771516023919

Supplementary Table 93 - r7_target_cell_localization

ST93_r7_target_cell_localization.csv

tissue	cell_type	gene	n_cells	frac_detected	mean_expr
Blood	B cell	IL18	1815	0.022589531680440773	0.018999752952376017
Blood	CD4-positive, alpha-beta	IL18	2343	0.020486555697823303	0.014346073145393443
Blood	CD8-positive, alpha-beta	IL18	2147	0.09874243129948766	0.09421330070080566
Blood	classical monocyte	IL18	4939	0.49605183235472766	0.36253031898809607
Blood	erythrocyte	IL18	5484	0.0036469730123997084	0.0025338414570087067
Blood	hematopoietic precursor	IL18	62	0.45161290322580644	0.42145383574308887
Blood	intermediate monocyte	IL18	464	0.6271551724137931	0.39512913648424475
Blood	macrophage	IL18	574	0.2804878048780488	0.21789854453415822
Blood	mature NK T cell	IL18	99	0.5555555555555556	0.6252139451527836
Blood	monocyte	IL18	4499	0.3656368081795955	0.27057369154072497
Blood	naive thymus-derived CD4	IL18	230	0.004347826086956522	0.0028628193813821545
Blood	natural killer cell	IL18	1408	0.13636363636363635	0.13109337065411222
Blood	neutrophil	IL18	15117	0.05543427928821856	0.06864320000477976
Blood	non-classical monocyte	IL18	158	0.36075949367088606	0.19037186599607708
Blood	plasma cell	IL18	374	0.07219251336898395	0.03478696265003898
Blood	platelet	IL18	206	0.024271844660194174	0.02599223927386756
Blood	B cell	TNFSF14	1815	0.0011019283746556473	0.0007282892534555482
Blood	CD4-positive, alpha-beta	TNFSF14	2343	0.04822876653862569	0.035460028131987315
Blood	CD8-positive, alpha-beta	TNFSF14	2147	0.11318118304611086	0.10604401578945397
Blood	classical monocyte	TNFSF14	4939	0.07511642032800162	0.039973388876864774
Blood	erythrocyte	TNFSF14	5484	0.004558716265499635	0.0033789205342465993
Blood	hematopoietic precursor	TNFSF14	62	0.0	0.0
Blood	intermediate monocyte	TNFSF14	464	0.07543103448275862	0.033945827873359466
Blood	macrophage	TNFSF14	574	0.017421602787456445	0.005906561292214676
Blood	mature NK T cell	TNFSF14	99	0.10101010101010101	0.08070725172457069
Blood	monocyte	TNFSF14	4499	0.0484552122693932	0.025400624807864414
Blood	naive thymus-derived CD4	TNFSF14	230	0.013043478260869565	0.008744869802309119
Blood	natural killer cell	TNFSF14	1408	0.08522727272727272	0.08356195533732799
Blood	neutrophil	TNFSF14	15117	0.170536482106238	0.21944574246119827
Blood	non-classical monocyte	TNFSF14	158	0.012658227848101266	0.00904029758670662
Blood	plasma cell	TNFSF14	374	0.016042780748663103	0.015130820918210688
Blood	platelet	TNFSF14	206	0.0	0.0
Blood	B cell	IL4	1815	0.0	0.0
Blood	CD4-positive, alpha-beta	IL4	2343	0.002134016218523261	0.0008536721744689482
Blood	CD8-positive, alpha-beta	IL4	2147	0.0	0.0
Blood	classical monocyte	IL4	4939	0.00020247013565499088	4.306131995359059e-06
Blood	erythrocyte	IL4	5484	0.0	0.0
Blood	hematopoietic precursor	IL4	62	0.0	0.0
Blood	intermediate monocyte	IL4	464	0.0	0.0
Blood	macrophage	IL4	574	0.0	0.0

(showing first 40 of 651 rows; full data in the accompanying CSV)

Supplementary Table 94 - r8_onset_mr

ST94_r8_onset_mr.csv

n_snp	b_ivw	se_ivw	p_ivw	q	q_p	b_wm	se_wm	p_wm	b_egger	se_egger	p_egger	egger_int	egger_int_p	exposure	stratum	or_ivw	or_lo	or_hi	q_fdr
95	2.30197444844	0.11095490649	1.30654694516	424.430365938	1.70867018554	6.60216370672	0.05117210593	0.0	2.68144386610	0.11421895457	7.11599366563	0.00162129051	0.67248777679	Frailty	adult_onset	9.99389541853	8.04059295998	12.4217139374	5.226187780670739e-95
95	3.20414765022	0.19073683397	2.49081454154	618.908174820	3.17877726565	11.0142564286	0.07284696497	0.0	3.89491507760	0.17816368658	6.05026569262	0.01207294872	0.04351335789	Frailty	childhood_onset	24.6344938600	16.9506038005	35.8015734945	4.981629083080923e-63
92	0.80235084490	0.11143254355	6.00679598336	454.644901504	9.76654803133	21.1011547930	0.04885653391	0.0	0.82934386515	0.11207422472	1.36233899907	0.00185681068	0.50786097344	index	adult_onset	2.23077898456	1.79309478929	2.77529939169	8.009061311145749e-13
92	1.47713408101	0.21805296718	1.25097827589	879.719018518	2.48378655604	49.2511466194	0.06872848679	0.0	1.57779310361	0.22149833509	1.05393790716	0.00107704933	0.84590824924	index	childhood_onset	4.38037387796	2.85693755259	6.71616895976	1.250978275894967e-11

Supplementary Table 95 - r8_onset_mr_snp

ST95_r8_onset_mr_snp.csv

rsid	bx	se_x	by	se_y	exposure	stratum
rs9275160	0.0382	0.0035	0.1112522825240873	0.00875794	Frailty	adult_onset
rs9273344	-0.0428	0.0039	-0.1434720663814634	0.00979539	Frailty	adult_onset
rs77062257	-0.0428	0.0043	-0.14141265405032855	0.0106235	Frailty	adult_onset
rs182976814	0.0585	0.006	0.1563217014568496	0.0149707	Frailty	adult_onset
rs28746860	-0.0429	0.0044	-0.12796065233701848	0.0111235	Frailty	adult_onset
rs1281935	0.0556	0.0059	0.16519191049531523	0.0148073	Frailty	adult_onset
rs17612518	0.0395	0.0044	0.14230001054695884	0.011164	Frailty	adult_onset
rs575147125	-0.0452	0.0051	-0.1360087444423612	0.0129946	Frailty	adult_onset
rs6910879	-0.054	0.0061	-0.14884925723378178	0.0153125	Frailty	adult_onset
rs145244672	-0.0495	0.0057	-0.14548252374427706	0.0142093	Frailty	adult_onset
rs113315602	-0.0526	0.0061	-0.13388116807453815	0.014969	Frailty	adult_onset
rs9274488	-0.0319	0.0037	-0.12807089971747626	0.00935918	Frailty	adult_onset
rs185320691	0.0519	0.0061	0.16605812624898805	0.0151227	Frailty	adult_onset
rs181159261	0.0442	0.0054	0.12219023309962243	0.0135168	Frailty	adult_onset
rs1281934	-0.0521	0.0063	-0.1544422223216414	0.0159199	Frailty	adult_onset
rs28417075	-0.0529	0.0064	-0.14380339806655712	0.0159591	Frailty	adult_onset
rs543993279	-0.0467	0.0058	-0.15450524267723723	0.0146734	Frailty	adult_onset
rs9268362	-0.0311	0.0039	-0.10159428348894753	0.00988543	Frailty	adult_onset
rs138748427	0.0419	0.0053	0.12978755240547293	0.0131883	Frailty	adult_onset
rs141267330	-0.0513	0.0065	-0.16846599002351373	0.0161363	Frailty	adult_onset
rs115627609	0.0416	0.0053	0.14103144785635188	0.0132212	Frailty	adult_onset
rs114281696	-0.0343	0.0045	-0.08950466446739819	0.0112097	Frailty	adult_onset
rs13195317	0.0402	0.0053	0.06991615472463183	0.0130967	Frailty	adult_onset
rs28746878	0.0477	0.0064	0.1355482823226974	0.0159679	Frailty	adult_onset
rs555911977	0.034	0.0046	0.13373598498068312	0.0116125	Frailty	adult_onset
rs34422230	-0.0521	0.0071	-0.11999372743578322	0.0179733	Frailty	adult_onset
rs2523668	0.0326	0.0045	0.07418059017496216	0.0114024	Frailty	adult_onset
rs35520662	-0.0499	0.0069	-0.1239613320847482	0.0174874	Frailty	adult_onset
rs34043227	0.0499	0.0069	0.12386285536111372	0.0174866	Frailty	adult_onset
rs34153000	0.0499	0.0069	0.12387191028716331	0.0174864	Frailty	adult_onset
rs34408145	-0.0498	0.0069	-0.12386738281388947	0.0174889	Frailty	adult_onset
rs41316544	0.0496	0.0069	0.12342039575379514	0.0174868	Frailty	adult_onset
rs17213756	0.0495	0.0069	0.12361953493010769	0.0174854	Frailty	adult_onset
rs41316548	0.0496	0.0069	0.12379155068355313	0.0175469	Frailty	adult_onset
rs35921062	0.0494	0.0069	0.12329256025189197	0.0174813	Frailty	adult_onset
rs35486885	-0.0493	0.0069	-0.12321564056844667	0.0174844	Frailty	adult_onset
rs34710970	-0.0493	0.0069	-0.12310818915193553	0.0174805	Frailty	adult_onset
rs41258084	0.0492	0.0069	0.12329029782380906	0.0174759	Frailty	adult_onset
rs41316241	0.0492	0.0069	0.12326767382449573	0.0174764	Frailty	adult_onset
rs72508422	0.0366	0.0052	0.13930919488077467	0.0130502	Frailty	adult_onset

(showing first 40 of 374 rows; full data in the accompanying CSV)

Supplementary Table 96 - r8_onset_ldsc_h2

ST96_r8_onset_ldsc_h2.csv

trait	n_snps	h2	h2_se	intercept	mean_chi2	lambda_gc
ADULT_ONSET_ASTHMA	1165799	0.039239331453765805	0.0034390243048111604	1.031652399397501	1.2848036605770248	1.2014908222172767
CHILD_ONSET_ASTHMA	1165832	0.06693134594358828	0.009729386703957964	1.0282200684029748	1.4457251891528278	1.241087262367308
Frailty	1168664	0.11084777654924406	0.005337449159583173	1.017248302478022	1.3997523109722234	1.329721223818186
GrimAge	1175366	0.10451458977535343	0.017717614389813847	1.0142382772633398	1.0751721531148744	1.072745020366781
Hannum	1175315	0.1189965445617679	0.024952146811927197	1.011000002380377	1.0801995852976018	1.049011751552148
IEAA	1175374	0.1639275075443605	0.025892039697555692	0.9996845143252803	1.094745293994269	1.0637348001428908
PhenoAge	1175371	0.0965436809364225	0.019548370526330527	1.0188176077634274	1.0741146793335423	1.057489895336982

Supplementary Table 97 - r8_onset_ldsc_rg

ST97_r8_onset_ldsc_rg.csv

trait1	trait2	n_snps	rg	rg_se	p	gencov	intercept	q
Frailty	ADULT_ONSET_ASTHMA	1156835	0.4866291429022408	0.036888819364955736	9.784890195836285e-40	0.03209389348351904	0.10852175951732783	1.0763379215419913e-38
Frailty	CHILD_ONSET_ASTHMA	1156855	0.20881566203367607	0.03021897810874219	4.843685130709093e-12	0.01798627965947269	0.048726039665128264	1.7760178812600008e-11
GrimAge	ADULT_ONSET_ASTHMA	1163142	0.08281200490994103	0.06408947007643022	0.1963117070997511	0.005303255597714295	-0.0002984482137520751	0.2699285972621578
GrimAge	CHILD_ONSET_ASTHMA	1163158	0.05681648288535482	0.07260570945354616	0.43390043454018645	0.004752012620917739	-0.002350963006465393	0.5303227533268946
Hannum	ADULT_ONSET_ASTHMA	1163101	0.10318776404918573	0.05395949429687218	0.05583529530900063	0.007051088682212726	-9.50843196472442e-05	0.08774117834271529
Hannum	CHILD_ONSET_ASTHMA	1163117	0.008496921195926899	0.05315088595297104	0.8729880895425015	0.0007583043457866156	0.006591751933532888	0.8729880895425015
IEAA	ADULT_ONSET_ASTHMA	1163152	0.15341834133250531	0.05742898953295241	0.007552562702937398	0.012304500850675998	-0.011215007503869402	0.018258613772677124
IEAA	CHILD_ONSET_ASTHMA	1163168	0.13013579496959823	0.049854686090101435	0.009046227931247678	0.013631305980742695	-0.012978590172586275	0.018258613772677124
PhenoAge	ADULT_ONSET_ASTHMA	1163151	0.19163825312798943	0.07435790479038473	0.009959243876005703	0.011795189693451285	-0.00727102940037263	0.018258613772677124
PhenoAge	CHILD_ONSET_ASTHMA	1163167	0.013051335557124448	0.06962205129166585	0.8513001427308982	0.0010491356906870465	0.006470647730931659	0.8729880895425015
ADULT_ONSET_ASTHMA	CHILD_ONSET_ASTHMA	1165774	0.7227418132589214	0.08156603831084182	7.94295116614723e-19	0.037038957959841676	0.10150068614421578	4.368623141380977e-18

Supplementary Table 98 - r8_onset_rg_contrast

ST98_r8_onset_rg_contrast.csv

contrast	rg_adult	se_adult	rg_child	se_child	z	p_conservative	note
Frailty rg: adult vs chi	0.4866291429022408	0.0368888193649557	0.208815662033676	0.0302189781087421	5.825870071068855	5.681580716328051e-09	independence assumption;

Supplementary Table 99 - r8_mediation_mr

ST99_r8_mediation_mr.csv

exposure	outcome	group	n_snp	b_ivw	se_ivw	p_ivw	q
IL18	VEGFA	other	3	0.11936590481727724	0.0429106611623249	0.005406988093972243	0.24331446422875094
IL18	IL33	readout	3	0.11273138370814796	0.04703169718861428	0.01653337180526472	0.3021421305723109
IL18	MMP10	other	3	-0.10013278265319499	0.04312755787103612	0.020244683096881006	0.3021421305723109
IL18	OSM	alarmin	3	0.1008340507357349	0.04325428721367482	0.01974323485292579	0.3021421305723109
IL18	SCF	other	3	-0.09470550144403193	0.04278369194803443	0.0268570782730943	0.3021421305723109
IL18	SLAMF1	other	3	0.09547555488798835	0.0429106611623249	0.026082280395489883	0.3021421305723109
IL18	IL17C	other	3	-0.09560360555971589	0.047507752318609354	0.04417999500772293	0.31502297661088735
IL18	IL2	other	3	0.08640232562570051	0.047164336137743415	0.06696007061684815	0.3544944915009608
IL18	IL22RA1	other	3	0.08248386464020904	0.048537800555023075	0.08924878278811556	0.4462439139405778
IL18	CASP8	other	3	-0.07094655097865908	0.04325428721367482	0.10095941307245716	0.4543173588260572
IL18	CST5	other	3	0.06536346987651137	0.04222328389067081	0.12161188240811178	0.4955439127177796
IL18	CX3CL1	other	3	-0.06268448473411693	0.04222328389067081	0.13765108686604988	0.4955439127177796
IL18	IL20	other	3	0.0702865323318604	0.04668846895723496	0.13221185226945684	0.4955439127177796
IL18	CCL2_MCP1	inflammation	3	0.06143742204505539	0.0429106611623249	0.15221482807462364	0.5073827602487454
IL18	IL13	type2	3	0.06600700071406565	0.04759411695521679	0.16548046052545518	0.513560049906585
IL18	IL10	inflammation	3	0.05752699634834563	0.04312755787103612	0.18224311525143644	0.5467293457543093
IL18	IL17A	inflammation	3	-0.0602215759525988	0.04759411695521679	0.20575823691044637	0.5776315023711452
IL18	IL6	inflammation	3	0.054681196592104235	0.04312755787103612	0.20483560224919717	0.5776315023711452
IL18	LAP_TGFB1	other	3	-0.05245000538593371	0.04256699371877977	0.21788354531253562	0.5776315023711452
IL18	NRTN	negative_control	3	0.05548202551059384	0.04703169718861428	0.23813027793071906	0.5953256948267976
IL18	IL24	other	3	0.04763324669217005	0.04841321595414441	0.32516992966386105	0.6651203106760794
IL18	HGF	other	3	0.03561934024709127	0.042092355097690666	0.3974307222000317	0.7451610748982938
IL18	IFNG	inflammation	3	0.03790604990075132	0.04668846895723496	0.41685292359868154	0.7451610748982938
IL18	IL8	other	3	0.03147631359370524	0.043040411977723306	0.4645837240987618	0.7451610748982938
IL18	NGF	negative_control	3	-0.029686452222528397	0.04092985340323277	0.46826756294414473	0.7451610748982938
IL18	NTF3	negative_control	3	-0.03247430369437324	0.04222328389067081	0.4418286614235092	0.7451610748982938
IL18	OPG	other	3	0.029537104309714807	0.04106175622824265	0.4719353474355861	0.7451610748982938
IL18	PDL1	other	3	-0.03249256533166629	0.04222328389067081	0.44157197273000903	0.7451610748982938
IL18	PLAU	other	3	0.03137632501752156	0.04222328389067081	0.4574182325656403	0.7451610748982938
IL18	TSLP	readout	3	-0.03902004866798539	0.048280738915575556	0.4189807571271298	0.7451610748982938
IL18	IL1A	alarmin	3	-0.031800219777833955	0.047507752318609354	0.5032600796432692	0.7676848672524446
IL18	IL5	type2	3	0.028429231970057978	0.048280738915575556	0.5559741451920424	0.8008032362523241
IL18	IL4	type2	3	-0.02487734245385943	0.048280738915575556	0.6063683231627077	0.8385380990874768
IL18	TNF	inflammation	3	0.023526466260324327	0.04759411695521679	0.6210840249612275	0.8385380990874768
IL18	TRAIL	other	3	-0.02034888719672883	0.04256699371877977	0.6326190347967802	0.8385380990874768
IL18	TNFB	other	3	0.021715427728686083	0.04819448382577272	0.6522929239760309	0.8386623308263255
IL18	ADA	other	3	-0.018302059470007254	0.04222328389067081	0.6646813964352072	0.8425538828051922
IL18	ARTN	other	3	-0.013771767883002248	0.047507752318609354	0.7719044895401722	0.8906590263925065
IL18	TGFA	other	3	0.009331536140826996	0.0429106611623249	0.8278465393872979	0.9431163106943901
IL18	GDNF	negative_control	3	0.004947367627523974	0.042092355097690666	0.9064353069666644	0.9906205595863785

(showing first 40 of 90 rows; full data in the accompanying CSV)

Supplementary Table 100 - r8_gtex_context

ST100_r8_gtex_context.csv

tissue	gene	n_pairs	frac_pos_slope	top_variant	top_p	top_slope
Whole_Blood	IL4	0				
Whole_Blood	IL13	0				
Whole_Blood	IL5	0				
Whole_Blood	RAD50	0				
Whole_Blood	IL4R	108	0.23148148148148148	chr16_27314947_G_T_b38	3.6191e-20	0.156016
Whole_Blood	GATA3	2	1.0	chr10_8487232_G_A_b38	1.68505e-05	0.361834
Whole_Blood	STAT6	0				
Whole_Blood	IL18	0				
Whole_Blood	TNFSF14	0				
Whole_Blood	IL33	1	0.0	chr9_6878721_A_AT_b38	1.84454e-05	-0.552588
Whole_Blood	TSLP	0				
Lung	IL4	71	0.5211267605633803	chr5_133626561_C_G_b38	2.83351e-05	-0.872499
Lung	IL13	0				
Lung	IL5	0				
Lung	RAD50	269	0.8550185873605948	chr5_132496995_C_T_b38	8.25281e-18	-0.190673
Lung	IL4R	103	0.1553398058252427	chr16_27252452_T_C_b38	1.04376e-07	-0.12132
Lung	GATA3	0				
Lung	STAT6	0				
Lung	IL18	134	0.48507462686567165	chr11_112155192_T_C_b38	1.20035e-22	-0.487168
Lung	TNFSF14	0				
Lung	IL33	0				
Lung	TSLP	29	0.0	chr5_111071044_C_T_b38	5.05852e-06	-0.160549
Skin_Sun_Exposed_Lower_l	IL4	0				
Skin_Sun_Exposed_Lower_l	IL13	0				
Skin_Sun_Exposed_Lower_l	IL5	0				
Skin_Sun_Exposed_Lower_l	RAD50	6	0.0	chr5_132433145_C_T_b38	9.79571e-09	-0.600655
Skin_Sun_Exposed_Lower_l	IL4R	0				
Skin_Sun_Exposed_Lower_l	GATA3	0				
Skin_Sun_Exposed_Lower_l	STAT6	0				
Skin_Sun_Exposed_Lower_l	IL18	229	0.2096069868995633	chr11_112155192_T_C_b38	3.10783e-14	-0.210603
Skin_Sun_Exposed_Lower_l	TNFSF14	0				
Skin_Sun_Exposed_Lower_l	IL33	118	0.6949152542372882	chr9_6535956_G_C_b38	4.24088e-08	-0.170871
Skin_Sun_Exposed_Lower_l	TSLP	267	0.352059925093633	chr5_111086186_A_G_b38	8.02611e-40	-0.591373
Skin_Not_Sun_Exposed_Sup	IL4	0				
Skin_Not_Sun_Exposed_Sup	IL13	2	1.0	chr5_133473554_C_T_b38	3.78453e-05	0.556192
Skin_Not_Sun_Exposed_Sup	IL5	0				
Skin_Not_Sun_Exposed_Sup	RAD50	41	0.14634146341463414	chr5_132728941_G_A_b38	5.01348e-07	-0.627339
Skin_Not_Sun_Exposed_Sup	IL4R	0				
Skin_Not_Sun_Exposed_Sup	GATA3	5	0.4	chr10_8477232_G_T_b38	2.05747e-05	-0.192278
Skin_Not_Sun_Exposed_Sup	STAT6	12	0.0	chr12_57098080_G_A_b38	5.38637e-06	-0.17371

(showing first 40 of 110 rows; full data in the accompanying CSV)

Supplementary Table 101 - r8_gene_burden

ST101_r8_gene_burden.csv

target	ensembl	disease	efo	n_evidence	score	datasource	study	cohort	tail	odds_ratio	p_value	method	study_cases
IL18	ENSG00000150782	asthma	MONDO_0004979	0									
IL18	ENSG00000150782	atopic dermatitis	MONDO_0004980	0									
IL18	ENSG00000150782	allergic rhinitis	MONDO_0011786	0									
TNFSF14	ENSG00000125735	asthma	MONDO_0004979	0									
TNFSF14	ENSG00000125735	atopic dermatitis	MONDO_0004980	0									
TNFSF14	ENSG00000125735	allergic rhinitis	MONDO_0011786	0									
IL4	ENSG00000113520	asthma	MONDO_0004979	0									
IL4	ENSG00000113520	atopic dermatitis	MONDO_0004980	0									
IL4	ENSG00000113520	allergic rhinitis	MONDO_0011786	0									
IL13	ENSG00000169194	asthma	MONDO_0004979	0									
IL13	ENSG00000169194	atopic dermatitis	MONDO_0004980	0									
IL13	ENSG00000169194	allergic rhinitis	MONDO_0011786	0									
IL5	ENSG00000113525	asthma	MONDO_0004979	0									
IL5	ENSG00000113525	atopic dermatitis	MONDO_0004980	0									
IL5	ENSG00000113525	allergic rhinitis	MONDO_0011786	0									
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.8374511998938867	gene_burden		UK Biobank 450k		0.674	1.47e-15	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.8252951590479218	gene_burden		UK Biobank 470k		0.6929000020027161	2.1349999999999984	Burden test carried out	63896.0
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.8191285220527578	gene_burden		UK Biobank 450k		0.676	2.58e-15	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.782041040684083	gene_burden		UK Biobank 470k		0.6363999843597412	8.056e-15	Burden test carried out	38346.0
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.6766184604208185	gene_burden		UK Biobank 450k		0.725	2.0499999999999984	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.6685033188632682	gene_burden		UK Biobank 450k		0.539	2.63e-13	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.6612103076537519	gene_burden		UK Biobank 450k		0.727	3.29e-13	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.6531788428123247	gene_burden		UK Biobank 450k		0.542	4.21e-13	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.6167057717275563	gene_burden		UK Biobank 450k		0.707	1.29e-12	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.6029150330378805	gene_burden		UK Biobank 450k		0.709	1.97e-12	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.5502288063628464	gene_burden		UK Biobank 450k		0.689	9.93e-12	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.5487224995525914	gene_burden		UK Biobank 450k		0.706	1.04e-11	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.5463086482997364	gene_burden		UK Biobank 450k		0.694	1.12e-11	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.5457321361497646	gene_burden		UK Biobank 450k		0.689	1.1399999999999996	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.5414542485769872	gene_burden		UK Biobank 470k		0.7125999927520752	1.3e-11	Burden test carried out	48317.0
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.5409569551595612	gene_burden		UK Biobank 450k		0.695	1.32e-11	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.5388085665508464	gene_burden		UK Biobank 450k		0.707	1.4099999999999984	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.5332962646639312	gene_burden		UK Biobank 450k		0.696	1.6699999999999984	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.5324491886809615	gene_burden		UK Biobank 470k		0.7397000193595886	1.714e-11	Burden test carried out	60383.0
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.5275860192692721	gene_burden		UK Biobank 450k		0.696	1.9899999999999974	Burden test carried out	
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.5105051718172411	gene_burden		UK Biobank 470k		0.7127000093460083	3.3619999999999964	Burden test carried out	48613.0
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.4872661964709884	gene_burden		UK Biobank 470k		0.7444000244140625	6.8619999999999994	Burden test carried out	59566.0
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.4081849150890206	gene_burden		UK Biobank 450k			7.7779999999999994	Burden test carried out	53242.0
IL33	ENSG00000137033	asthma	MONDO_0004979	24	0.2843832721781117	gene_burden		UK Biobank 450k			3.48e-08	Burden test carried out	53242.0
IL33	ENSG00000137033	atopic dermatitis	MONDO_0004980	1	0.4592060481321133	gene_burden		UK Biobank 470k		0.7784000039100647	1.6240000000000003	Burden test carried out	96419.0

(showing first 40 of 47 rows; full data in the accompanying CSV)

Supplementary Table 102 - r9_coloc_prior_sensitivity

ST102_r9_coloc_prior_sensitivity.csv

pair	w2_prior	p12	n_snps	pp_h0	pp_h1	pp_h2	pp_h3	pp_h4	verdict
IL4_eQTLGen_vs_ASTHMA	0.15^2	1e-06	175	5.116424405354996e-58	1.2078958427887236e-08	4.235391642932569e-50	0.9998988311280539	0.00010115679298767751	H3_dominant
IL4_eQTLGen_vs_ASTHMA	0.15^2	5e-06	175	5.1143549983556585e-58	1.2074072929903682e-08	4.2336785815411713e-50	0.9994944085321489	0.0005055793937780867	H3_dominant
IL4_eQTLGen_vs_ASTHMA	0.15^2	1e-05	175	5.111770592478838e-58	1.2067971612131 89e-08	4.2315392025164143e-50	0.9989893401071868	0.0010106478248415667	H3_dominant
IL4_eQTLGen_vs_ASTHMA	0.15^2	5e-05	175	5.091188996139159e-58	1.20193821623785e-08	4.214501694634521e-50	0.9949670947866466	0.005032893193971246	H3_dominant
IL4_eQTLGen_vs_ASTHMA	0.15^2	0.0001	175	5.0656938997881735e-58	1.1959192822218169e-08	4.1933967765382604e-50	0.9899846079909526	0.010015380049854543	H3_dominant
IL4_eQTLGen_vs_ASTHMA	0.20^2	1e-06	175	6.0173511899083286e-58	1.420588463944235e-08	4.235419745521413e-50	0.9999055326809807	9.445311313454875e-05	H3_dominant
IL4_eQTLGen_vs_ASTHMA	0.20^2	5e-06	175	6.0150786183033374e-58	1.4200519506340344e-08	4.233820155544841e-50	0.9995278985942325	0.0004720872052479172	H3_dominant
IL4_eQTLGen_vs_ASTHMA	0.20^2	1e-05	175	6.0122403165750075e-58	1.4193818786097822e-08	4.2318223663508056e-50	0.9990562569180187	0.0009437288881625101	H3_dominant
IL4_eQTLGen_vs_ASTHMA	0.20^2	5e-05	175	5.989629969249469e-58	1.4140439819900368e-08	4.2159076709153306e-50	0.9952990869152838	0.004700898944276365	H3_dominant
IL4_eQTLGen_vs_ASTHMA	0.20^2	0.0001	175	5.961605066287167e-58	1.4074278061021858e-08	4.196181844114342e-50	0.9906421781458822	0.00935780777983974	H3_dominant
IL4_eQTLGen_vs_RHINITIS	0.15^2	1e-06	175	1.5182626385279445e-53	0.00035843452068204656	4.2214348256202855e-50	0.9965742154793961	0.003067349999921904	H3_dominant
IL4_eQTLGen_vs_RHINITIS	0.15^2	5e-06	175	1.499860253138555e-53	0.0003540900482442294	4.1702681377324 93e-50	0.9844950518900438	0.015150858061711887	H3_dominant
IL4_eQTLGen_vs_RHINITIS	0.15^2	1e-05	175	1.4774752355549672e-53	0.00034880534792662706	4.1080279887612317e-50	0.9698017236273622	0.029849471024711194	H3_dominant
IL4_eQTLGen_vs_RHINITIS	0.15^2	5e-05	175	1.319883891637322e-53	0.00031160086407294556	3.6698550596853598e-50	0.8663601543325116	0.13332824480341537	H3_dominant
IL4_eQTLGen_vs_RHINITIS	0.15^2	0.0001	175	1.1646086627500104e-53	0.00027494317334956673	3.238121944381546e-50	0.7644388625316477	0.2352861942950027	H3_dominant
IL4_eQTLGen_vs_RHINITIS	0.20^2	1e-06	175	1.9069729284915803e-53	0.0004502020337141672	4.2214247572869143e-50	0.996572741461803	0.002977056504482823	H3_dominant
IL4_eQTLGen_vs_RHINITIS	0.20^2	5e-06	175	1.8845315014345857e-53	0.00044490401613376103	4.171746655226923e-50	0.984844984790294	0.014710111193572158	H3_dominant
IL4_eQTLGen_vs_RHINITIS	0.20^2	1e-05	175	1.8572117106607616e-53	0.00043845430456037753	4.1112694248407817e-50	0.9705678241757651	0.02899372151967444	H3_dominant
IL4_eQTLGen_vs_RHINITIS	0.20^2	5e-05	175	1.6642056499914753e-53	0.00039288904260293366	3.6840160796869923e-50	0.869704001660949	0.12990310929644802	H3_dominant
IL4_eQTLGen_vs_RHINITIS	0.20^2	0.0001	175	1.4728746529671196e-53	0.00034771923306554334	3.260470786721618e-50	0.7697155574715463	0.2299367232953881	H3_dominant
IL4_eQTLGen_vs_ATOPIC	0.15^2	1e-06	175	1.1076164078833264e-76	2.614883263176097e-27	4.235725309400334e-50	0.9999783993453704	2.160065462961895e-05	H3_dominant
IL4_eQTLGen_vs_ATOPIC	0.15^2	5e-06	175	1.1075207151934703e-76	2.6146573499345343e-27	4.2353593632609817e-50	0.9998920060578114	0.00010799394218870551	H3_dominant
IL4_eQTLGen_vs_ATOPIC	0.15^2	1e-05	175	1.1074011225806585e-76	2.614375013270491e-27	4.234902019497113e-50	0.9997840354384869	0.00021596456151304013	H3_dominant
IL4_eQTLGen_vs_ATOPIC	0.15^2	5e-05	175	1.106445310675226e-76	2.612118513153238e-27	4.2312468220388124e-50	0.998921109201148	0.0010788907988520226	H3_dominant
IL4_eQTLGen_vs_ATOPIC	0.15^2	0.0001	175	1.1052528635303582e-76	2.609303359766972e-27	4.226686688660786e-50	0.9978445439040452	0.00215545609595479	H3_dominant
IL4_eQTLGen_vs_ATOPIC	0.20^2	1e-06	175	1.3344583603126033e-76	3.150416341750703e-27	4.2357263760726613e-50	0.9999786537111938	2.1346288806324835e-05	H3_dominant
IL4_eQTLGen_vs_ATOPIC	0.20^2	5e-06	175	1.3343444271065705e-76	3.1501473659288803e-27	4.235364738797003e-50	0.9998932776684711	0.0001067223315287806	H3_dominant
IL4_eQTLGen_vs_ATOPIC	0.20^2	1e-05	175	1.3342020379543496e-76	3.1498112107325954e-27	4.2349127790314036e-50	0.9997865781138237	0.0002134218861762691 8	H3_dominant
IL4_eQTLGen_vs_ATOPIC	0.20^2	5e-05	175	1.3330640178060541e-76	3.1471245497028783e-27	4.231300570436518e-50	0.9989338007701213	0.001066199229878624	H3_dominant
IL4_eQTLGen_vs_ATOPIC	0.20^2	0.0001	175	1.331644219764469e-76	3.1437726617120477e-27	4.226793965965151e-50	0.9978698726803505	0.0021301273196494947	H3_dominant
IL4_TREG_NAIVE_vs_ASTHMA	0.15^2	1e-06	42	2.2541762805613613e-05	8.198748337001241e-06	0.5304208237429587	0.1901271161640318	0.27942131958186683	inconclusive
IL4_TREG_NAIVE_vs_ASTHMA	0.15^2	5e-06	42	1.0644529211355209e-05	3.8715612848178045e-06	0.2504719795577373	0.08978062892999512	0.6597328754217714	inconclusive
IL4_TREG_NAIVE_vs_ASTHMA	0.15^2	1e-05	42	6.413399028834818 5e-06	2.3326411991652353e-06	0.15091101903617316	0.054093420850737844	0.7949868140728611	inconclusive
IL4_TREG_NAIVE_vs_ASTHMA	0.15^2	5e-05	42	1.5343253480487575e-06	5.580551753743454e-07	0.03610357016085061	0.012941173066076176	0.9509531643925497	H4_pass
IL4_TREG_NAIVE_vs_ASTHMA	0.15^2	0.0001	42	7.864490937313128e-07	2.8604232308575276e-07	0.018505605782746633	0.006633256657448028	0.9748600650683886	H4_pass
IL4_TREG_NAIVE_vs_ASTHMA	0.20^2	1e-06	42	2.757185826688944e-05	1.0028263053916924e-05	0.5304380217223328	0.1901336774942318	0.2793907006621146	inconclusive
IL4_TREG_NAIVE_vs_ASTHMA	0.20^2	5e-06	42	1.302056223900657e-05	4.735757088939449e-06	0.2504945879569231	0.08978892019468304	0.659698735529066	inconclusive
IL4_TREG_NAIVE_vs_ASTHMA	0.20^2	1e-05	42	7.845135963700048e-06	2.8533835614629307e-06	0.15092774525557032	0.05409952919320675	0.7949620270316977	inconclusive
IL4_TREG_NAIVE_vs_ASTHMA	0.20^2	5e-05	42	1.8768949877501327e-06	6.826524523499209e-07	0.03610842818956141	0.012942941416464914	0.9509460708465337	H4_pass
IL4_TREG_NAIVE_vs_ASTHMA	0.20^2	0.0001	42	9.620435007389675e-07	3.4990841753699095e-07	0.018508163156910652	0.006634187182246842	0.9748563377089243	H4_pass

(showing first 40 of 100 rows; full data in the accompanying CSV)

Supplementary Table 103 - r9_ldsc_external_benchmark

ST103_r9_ldsc_external_benchmark.csv

comparison_type	quantity	our_traits	our_estimate	our_se	published_estimate	published_se	published_source	published_method	note	abs_delta
same_GWAS	h2 Frailty index	Frailty	0.110847776549244	0.0053374491595831	0.11	0.005	Atkins et al. 2021 Aging	official LDSR	identical input GWAS; de	0.0008
same_GWAS	h2 IEAA	IEAA	0.1639275075443605	0.0258920396975556	0.17	0.02	McCartney et al. 2021 Ge	official LDSC	identical input GWAS; de	0.0061
same_GWAS	rg(GrimAge, PhenoAge)	GrimAge x PhenoAge	0.6722135331411303	0.0992302570411018	0.66		McCartney et al. 2021 Ge	official LDSC	identical input GWAS; de	0.0122
same_GWAS	rg(GrimAge, IEAA)	GrimAge x IEAA	0.2427332565922247	0.0918306758053013	0.28		McCartney et al. 2021 Ge	official LDSC	identical input GWAS; de	0.0373
different_cohort	rg(asthma, hay fever/rhi	ALLERG_ASTHMA x ALLERG_F	0.9049262550657244	0.04094659992567	0.665	0.0457	Guo et al. 2020 Front Ge	official LDSC	FinnGen (doctor-diagnose	0.2399
different_cohort	rg(asthma, allergic dise	ALLERG_ASTHMA x ALLERG_F	0.9049262550657244	0.04094659992567	0.7455		Zhu et al. 2018 Nat Gene	official LDSC	Zhu allergic-disease com	0.1594
different_cohort	rg(asthma, eczema/AD)	ALLERG_ASTHMA x L12_ATOP	0.6296471872445731	0.0506259201990657	0.4519	0.0577	Guo et al. 2020 Front Ge	official LDSC	concordant direction; Zh	0.1777
different_cohort	rg(hay fever/rhinitis, e	ALLERG_RHINITIS x L12_AT	0.7184663005212137	0.0609942317066528	0.3297	0.0714	Guo et al. 2020 Front Ge	official LDSC	concordant direction; la	0.3888

Supplementary Table 104 - r9_hrs_component_exclusion

ST104_r9_hrs_component_exclusion.csv

outcome	excluded_deficit	n	events	HR	lower	upper	p	HR_perSD	lower_perSD	upper_perSD
all_mortality	(none)	40651	19117	41.351134936823925	37.13774952503934	46.042541145648734	0.0	1.6293501252947864	1.6065455961090078	1.6524783592995531
incident_diabe	full	34023	6565	19.549993431563685	16.064282424863915	23.792052023601112	1.7549996344971599e-193	1.4768743114433105	1.4393212371130413	1.5154071763549255
incident_diabe	excl_diabe	34023	6565	15.724115877145117	13.087315309457896	18.89217263216749	3.056920093538552e-190	1.435286540261961	1.401145724143818	1.4702592436742865
incident_cancre	full	36385	5422	3.05496373908082	2.4249828745244906	3.848605920125812	2.5980306178490038e-21	1.1577430234058204	1.1232008423423907	1.193347492020718
incident_cancre	excl_cancre	36385	5422	2.836993567443008	2.286517805603048	3.519995550434948	2.6861073906300707e-21	1.1465572904091967	1.1145727832530117	1.1794596458328033
incident_hearte	full	33715	8459	34.14882464729431	28.35872571576842	41.12110806668724	1.2203408734890524e-303	1.5889626934000742	1.5507105032676436	1.6281584703895258
incident_hearte	excl_hearte	33715	8459	26.96576030847319	22.664917907037545	32.08272061679273	2.60376988138306e-302	1.5405002892397144	1.5057914779980115	1.5760091459029877
incident_stroke	full	37266	4341	30.9334355806067	24.316738376310933	39.35056675823444	7.006730425218448e-172	1.5684865584828258	1.5197489417511405	1.618787167113652
incident_stroke	excl_stroke	37266	4341	24.618260043613777	19.6626272952868	30.82287623486943	1.0272145727685345e-171	1.5222073294807987	1.4779875148546224	1.567750153933459
incident_lunge	full	36858	3936	77.2959366052818	60.87132906281361	98.15231419577584	1.0867013219129057e-278	1.7686661740761902	1.7141115612679023	1.8249570833111008
incident_lunge	excl_lunge	36858	3936	57.49641691858096	45.98541910552926	71.88882960681315	9.109409285311096e-277	1.701335839051419	1.6522093251226495	1.7519230724750463

Supplementary Table 105 - r9_hrs_ews_mean_matched

ST105_r9_hrs_ews_mean_matched.csv

metric	stratum	n_dead	n_alive	median_dead	median_alive	p
fi_var	unmatched	13734	17808	0.0065443538950586	0.0025185648669046	0.0
fi_var	mean_decile_0	562	2593	0.0017223919285851	0.0008182417733441	1.825917485133749e-37
fi_var	mean_decile_1	824	2330	0.0031185060926213	0.0015256513389002	2.955276297269084e-49
fi_var	mean_decile_2	1040	2114	0.0042567768087894	0.0022757177859057	1.22307703862262e-50
fi_var	mean_decile_3	1174	1980	0.0051186107561689	0.0026883484407528	4.080981499316105e-53
fi_var	mean_decile_4	1324	1830	0.0063540779937731	0.002932812239165	5.753156168613877e-71
fi_var	mean_decile_5	1510	1644	0.0068134573057032	0.0033885419562845	4.454039236875327e-72
fi_var	mean_decile_6	1615	1539	0.0075113465437367	0.0035773507232994	1.8198403966341586e-74
fi_var	mean_decile_7	1777	1377	0.0079869677893871	0.004195201262746	4.9897565204726825e-56
fi_var	mean_decile_8	1880	1274	0.0091313756915694	0.0052130444135436	1.3174804145411363e-49
fi_var	mean_decile_9	2028	1127	0.0089338858947436	0.0054201292705447	7.124258271962729e-42
fi_var	regression_adj_meanFI	13734	17808			0.0
fi_ar1	unmatched	13734	17808	0.5323702986838525	0.3768468860906452	1.379460603436829e-118
fi_ar1	mean_decile_0	562	2593	0.3985339058857834	0.2050976076538099	7.02937096399887e-08
fi_ar1	mean_decile_1	824	2330	0.4717558043629986	0.3537724873347113	9.235077188187375e-08
fi_ar1	mean_decile_2	1040	2114	0.5414093781980283	0.4131671823760567	1.2146840771594248e-07
fi_ar1	mean_decile_3	1174	1980	0.5549850215799961	0.4838587428292435	3.792575229700356e-05
fi_ar1	mean_decile_4	1324	1830	0.5829568443667565	0.4603519339475448	6.876804107520362e-14
fi_ar1	mean_decile_5	1510	1644	0.5883709426633305	0.4726757925611489	5.6559548865760847e-11
fi_ar1	mean_decile_6	1615	1539	0.5728330262340106	0.4277423042762996	3.412911490798623e-13
fi_ar1	mean_decile_7	1777	1377	0.5412036232303972	0.3848142699986188	7.509787436971796e-14
fi_ar1	mean_decile_8	1880	1274	0.5271249357117576	0.3516987249632558	1.865102832568233e-13
fi_ar1	mean_decile_9	2028	1127	0.4659009835511694	0.2052182453638274	5.0094942003116203e-23
fi_ar1	regression_adj_meanFI	13734	17808			3.084062754528584e-116

Supplementary Table 106 - r9_pqtl_replication

ST106_r9_pqtl_replication.csv

protein	source	layer	snp	status	rep_beta	rep_se	rep_f	outcome_beta	outcome_se	wald_logor	wald_se	wald_p	wald_or	n_snps	discovery_or	direction_concordan	replicates_p005
IL18	SCALLOP_Olink_Fo	cross-cohort (Olin	rs5744249	ok	0.2183	0.012	330.936736111	0.0432815	0.00994051	0.198266147503	0.0455360054970	1.3364382697292	1.2192868607956				
IL18	SCALLOP_Olink_Fo	cross-cohort (Olin	rs10891343	ok	0.1683	0.0106	252.09051263794	0.0228999	0.00852292	0.136065953654	0.0506412358882	0.0072126995230	1.145757457994				
IL18	SCALLOP_Olink_Fo	cross-cohort (Olin	rs7106524	ok	0.1404	0.0107	172.17363961918	0.0267177	0.00929466	0.190297008547	0.0662012820512	0.0040464022015	1.209608808464				
IL18	SCALLOP_Olink_Fo	cross-cohort (Olin	(IVW)	summary						0.1745721755963	0.0301459456916	7.0009994559739	1.190736681413	3.0	1.24	True	True
IL18	HELIC_Olink_Gilly2	cross-cohort (Olin	rs5744249	not_assayed													
IL18	HELIC_Olink_Gilly2	cross-cohort (Olin	rs10891343	not_assayed													
IL18	HELIC_Olink_Gilly2	cross-cohort (Olin	rs7106524	not_assayed													
TNFSF14	INTERVAL_SomaSc	cross-platform (So	rs344560	ok	0.0244	0.0531	0.2111497689396	-0.0782515	0.0185387	-3.207028688524	0.7597827868852	2.4324146758131	0.040476703664				
TNFSF14	INTERVAL_SomaSc	cross-platform (So	rs413141	ok	0.1597	0.0367	18.935540393053	0.0263193	0.0098578	0.164804633688	0.0617269881026	0.0075874762051	1.179162727533				
TNFSF14	INTERVAL_SomaSc	cross-platform (So	rs62123250	ok	0.1022	0.0467	4.7892557625556	-0.0773179	0.018236	-0.756535225048	0.1784344422700	2.2366944919407	0.469289596265				
TNFSF14	INTERVAL_SomaSc	cross-platform (So	(IVW)	summary						0.0471473621656	0.0581638773004	0.4175981504495	1.048276473989	3.0	1.17	True	False
TNFSF14	AGES_SomaScan_	cross-platform (So	rs344560	ok	-0.0745547	0.0346352	4.6335585416215	-0.0782515	0.0185387	1.049585069754	0.2486590382631	2.4324146758131	2.856465638144				
TNFSF14	AGES_SomaScan_	cross-platform (So	rs413141	ok	0.194464	0.0263077	54.640248757142	0.0263193	0.0098578	0.135342788382	0.0506921589600	0.0075874762051	1.144929185515				
TNFSF14	AGES_SomaScan_	cross-platform (So	rs62123250	ok	-0.019509	0.0303267	0.4138278428693	-0.0773179	0.018236	3.963191347583	0.9347480649956	2.2366944919407	52.62500292493				
TNFSF14	AGES_SomaScan_	cross-platform (So	(IVW)	summary						0.182497689169	0.0496005406817	0.0002338302869	1.200211377465	3.0	1.17	True	True
IL18	AholaOlli_Luminex	cross-platform (Lu	rs5744249	ok	0.2649	0.0275	92.789434710743	0.0432815	0.00994051	0.163388070970	0.0375255190631	1.3364382697292	1.177493552100				
IL18	AholaOlli_Luminex	cross-platform (Lu	rs10891343	ok	0.194	0.0237	67.004931545870	0.0228999	0.00852292	0.118040721649	0.0439325773195	0.0072126995230	1.125289934096				
IL18	AholaOlli_Luminex	cross-platform (Lu	rs7106524	ok	0.1362	0.0261	27.231602589509	0.0267177	0.00929466	0.196165198237	0.0682427312775	0.0040464022015	1.216727890036				
IL18	AholaOlli_Luminex	cross-platform (Lu	(IVW)	summary						0.151983239702	0.0263250240546	7.7716710098817	1.164140724923	3.0	1.24	True	True

Supplementary Table 107 - p2_bbj_index_mr

ST107_p2_bbj_index_mr.csv

analysis	n_snp	ivw_b	ivw_se	ivw_p	ivw_rand_b	ivw_rand_se	ivw_rand_p	cochran_Q	cochran_Q_p	wm_b	wm_se	wm_p	egger_b	egger_se	egger_p	egger_intercept	egger_intercept_p	outcome	n_steiger_removed	ivw_q
BBJ index -> as	28	0.38000411359	0.19217738082	0.04800080631	0.38000411359	0.30192923511	0.20817966067	66.645304649	3.33127872340	-0.34159650731	0.74787681523	0.64784712231	0.29069616735	0.28089802772	0.30072426351	0.01055420714	0.02393477274	asthma	19.0	0.0480008063165959
EUR index -> F	93	0.84148100435	0.07162353416	0.0	0.84148100435	0.18989386188	9.36567501508	646.69304055	0.0	0.84973199635	0.47356746638	0.07276196615	0.84381306054	0.19121810511	1.02027937707	0.00101776882	0.83571871100	asthma		
BBJ index -> al	28	0.52092418480	0.23550244376	0.02696878357	0.52092418480	0.23550244376	0.02696878357	21.533420549	0.76069210706	0.09963296287	0.46536558055	0.83047210529	0.51037917015	0.21327319085	0.01670773379	0.00227006506	0.52625384395	allergic rhinitis	19.0	0.040453175355554904
EUR index -> F	93	0.48508073578	0.06580151918	1.68309810533	0.48508073578	0.14256532748	0.00066768569	431.86083337	0.0	0.50132543299	0.16144836656	0.00190166471	0.49176104769	0.14313586809	0.00059119048	0.00283511218	0.44044549871	allergic rhinitis		
BBJ index -> at	27	-1.19816119900	0.34097191492	0.00044148158	-1.19816119900	0.54589238052	0.02817313601	66.642333527	2.01842588836	-0.95139145285	0.79062518656	0.22884451142	-1.24673020235	0.54734855820	0.02274071187	0.00929451812	0.31394600309	atopic dermatit	20.0	0.001324444750862019
EUR index -> F	93	0.64029738001	0.04798585709	0.0	0.64029738001	0.11423819803	2.08336232709	521.41557871	0.0	0.46021364985	0.27383832064	0.09283994862	0.63667711765	0.11486997831	2.98031250789	-0.00159851311	0.58737594260	atopic dermatit		

Supplementary Table 108 - p2_bbj_index_mr_sensitivity

ST108_p2_bbj_index_mr_sensitivity.csv

analysis	n_snp	ivw_b	ivw_se	ivw_p	ivw_rand_b	ivw_rand_se	ivw_rand_p	cochran_Q	cochran_Q_p	wm_b	wm_se	wm_p	egger_b	egger_se	egger_p	egger_intercept	egger_intercept_p	outcome
BBJ full-LC index	47	0.263888742597	0.159601200119	0.098244234889	0.263888742597	0.316743931069	0.404771644925	181.176610736	0.0	-0.41425600800	0.560488458709	0.459847372981	0.268134652834	0.306133173409	0.381097530721	0.009963809454	0.045145235328	asthma
BBJ full-LC index	47	0.573547662206	0.200923444140	0.004309673640	0.573547662206	0.210336176964	0.006394904770	50.4109121301	0.303283270250	0.125749149519	0.304989240926	0.680114537363	0.572888517093	0.212126016676	0.006919393273	-0.00160575630	0.634641457258	allergic rhinitis
BBJ full-LC index	47	-0.44215598095	0.289182637620	0.126268106143	-0.44215598095	0.487463273588	0.364377733360	130.706505547	4.743053727551	-1.15216674048	0.934219152676	0.217466149591	-0.44279604746	0.492670031284	0.368776260330	-0.00145881839	0.852412281068	atopic dermatitis

Supplementary Table 109 - p2_bbj_component_mr

ST109_p2_bbj_component_mr.csv

analysis	n_snp	ivw_b	ivw_se	ivw_p	ivw_rand_b	ivw_rand_se	ivw_rand_p	cochran_Q	cochran_Q_p	wm_b	wm_se	wm_p	egger_b	egger_se	egger_p	egger_intercept	egger_intercept_	component	outcome	ivw_q
BBJ wbc -> asth	4	0.66515231489	0.16273580833	4.36396675851	0.66515231489	0.68340166625	0.33040605959	52.9062610479	1.91969773410	0.31757519161	1.29626776929	0.80646265063	-1.2456168758	0.52330226465	0.01729850405	0.11064378790	0.05148688092	wbc	asthma	0.000785514016532484848
BBJ wbc -> alle	4	-0.0603367484	0.20418775383	0.76761469368	-0.0603367484	0.25116470358	0.81015365025	4.53919757439	0.20882048827	0.04738204705	0.53662865833	0.92964159129	0.58901282424	0.29652788510	0.04699273075	-0.0375760396	0.12652153425	wbc	allergic rhinitis	0.793499997537761
BBJ wbc -> ato	4	0.55710485534	0.29379701717	0.05793047573	0.55710485534	0.61124231446	0.36206861776	12.985339642	0.00466841679	0.14895702922	1.23974132453	0.90436297734	1.68762694957	1.16364888900	0.14697800129	-0.0653478326	0.37762545334	wbc	atopic dermatit	0.173791427196441 8
BBJ eosinophil_	13	0.25585480246	0.10001078695	0.01051937565	0.25585480246	0.21673545026	0.23780397178	56.356947428	1.03233405046	-0.1707982536	0.46705838436	0.71459693125	0.24449252975	0.22466259589	0.27647796049	0.00637563093	0.61134696074	eosinophil_pct	asthma	0.063116253950938 9
BBJ eosinophil_	13	0.29354237859	0.12595625404	0.01977918320	0.29354237859	0.14628550825	0.04478798979	16.186180666	0.18285670282	0.04981648143	0.26577751503	0.85131818783	0.28747367599	0.15237446834	0.05921060982	0.00337663577	0.69107506403	eosinophil_pct	allergic rhinitis	0.089006324422473 25
BBJ eosinophil_	13	-0.7081477480	0.18126633971	9.35768642142	-0.7081477480	0.32070982849	0.02723982919	37.563967383	0.00018089959	-0.5203706129	0.46672256444	0.26487341904	-0.6849169259	0.32896841148	0.03734116662	-0.0128476485	0.48741087109	eosinophil_pct	atopic dermatit	0.000842191779286581
BBJ lymphocyte	3	-0.4771796897	0.32715910429	0.14468750691	-0.4771796897	0.63516829975	0.45249321028	7.5385809847	0.02306842470	-0.2013547986	1.27880596923	0.87488602076	0.13632889385	1.03418354267	0.89512451060	0.02619786880	0.56832980529	lymphocyte_pc	asthma	0.325546890550963 7
BBJ lymphocyte	3	-0.3619093003	0.41175137508	0.37942782136	-0.3619093003	0.41175137508	0.37942782136	0.0048494460	0.99757821423	-0.3720634939	0.03718457648	0.0	-0.3505617063	0.03892276402	0.0	0.00048862137	0.75793391600	lymphocyte_pc	allergic rhinitis	0.569141732051 8791
BBJ lymphocyte	3	1.20605622159	0.59298762374	0.04196530253	1.20605622159	0.60803510898	0.04730819674	2.1027907158	0.34944980120	0.87466048181	1.22776110911	0.47621536706	1.10036768249	1.24741680699	0.37771349185	-0.0045509979	0.92632117030	lymphocyte_pc	atopic dermatit	0.151075089127566 65
BBJ neutrophil_	6	0.20679471945	0.14125262167	0.14319224357	0.20679471945	0.22266381194	0.35302866296	12.424423593	0.02941297806	0.12558472809	0.16567448204	0.44843844166	0.20286492997	0.24875590177	0.41477583212	-0.0020287317	0.86083302319	neutrophil_pct	asthma	0.325546890550963 7
BBJ neutrophil_	6	0.04944964886	0.17847470360	0.78172780787	0.04944964886	0.21118807220	0.81486867087	7.00092519869	0.22057150883	0.03140436910	0.12083302646	0.79494176975	0.03582683467	0.22480221933	0.87337700542	-0.0065569447	0.53946036550	neutrophil_pct	allergic rhinitis	0.793499997537761
BBJ neutrophil_	6	0.27885738531	0.25726455271	0.27839477052	0.27885738533	0.55047584242	0.61245294442	22.892152585	0.00035397283	-0.0599256479	0.38049332804	0.87485505619	0.25443389265	0.60301646484	0.67307229213	-0.0117640002	0.67716147076	neutrophil_pct	atopic dermatit	0.487605558415033 56
BBJ monocyte_	12	0.03118400038	0.11347559236	0.78346375290	0.03118400038	0.11347559236	0.78346375290	4.67962473339	0.94568438372	-0.0149272775	0.07425975673	0.84068734466	0.04701299289	0.07961479539	0.55485164514	0.00228892818	0.52141976560	monocyte_pct	asthma	0.793499997537761
BBJ monocyte_	12	-0.1490101779	0.14317267873	0.29798117458	-0.1490101779	0.14317267873	0.29798117458	7.7429542038	0.73612782156	-0.1756658297	0.12169083604	0.14886785726	-0.1412459399	0.13183859690	0.28400983869	0.00111836202	0.84858615240	monocyte_pct	allergic rhinitis	0.487605558415033 56
BBJ monocyte_	12	-0.2718811800	0.20627752844	0.18749164311	-0.2718811800	0.20627752844	0.18749164311	9.0322367991	0.61891661092	0.00314595462	0.33264564646	0.99245421896	-0.2159677083	0.19682240356	0.27252251604	0.00808944977	0.36536581285	monocyte_pct	atopic dermatit	0.374983286269753 97
BBJ crp -> asth	9	-0.0417465021	0.11607794095	0.71911487524	-0.0417465021	0.19370648675	0.82936642757	22.278139766	0.00442585221	-0.0881094896	0.24190918966	0.71568964494	-0.0448744181	0.19927960549	0.82183676269	0.00730969369	0.47784687189	crp	asthma	0.793499997537761
BBJ crp -> aller	9	0.04334933220	0.14693069105	0.76796903659	0.04334933220	0.20482960984	0.83239097236	15.547136805	0.04934012641	-0.1874514530	0.35481967030	0.59729069895	0.04716665478	0.20825090473	0.82082068190	-0.0087716436	0.41730869274	crp	allergic rhinitis	0.793499997537761
BBJ crp -> atop	9	-0.0553107610	0.21129654905	0.79349999753	-0.0553107610	0.35429859539	0.87594363018	22.492826845	0.00408023538	0.00366610121	0.59162305165	0.99505579298	-0.0605843025	0.36576560772	0.86844266075	0.01272269657	0.49853214386	crp	atopic dermatit	0.793499997537761

Supplementary Table 110 - p2_bbj_sign_concordance

ST110_p2_bbj_sign_concordance.csv

outcome	n_snp	concordant	concordance	binomial_p
asthma	45	27	0.6	0.23269319190831084
allergic rhinitis	45	36	0.8	6.57466324582856e-05
atopic dermatitis	45	20	0.444	0.5514843298025198

Supplementary Table 111 - p2_bbj_alignment

ST111_p2_bbj_alignment.tsv

snp[A]component[A]beta_x[A]se_x[A]As_beta[A]AR_beta[A]AD_beta
rs3828037[A]crp[A]-0.0023081
rs6698842[A]crp[A]0.00187573
rs4845620[A]crp[A]-0.0033731
rs1811472[A]crp[A][A][A][A]
rs10437340[A]crp[A][A][A][A]
rs2701185[A]crp[A]-0.0017172
rs147636938[A]crp[A][A][A][A]
rs12972156[A]crp[A]-0.003960
rs77301115[A]crp[A][A][A][A]
rs112849259[A]crp[A][A][A][A]
rs116881820[A]crp[A][A][A][A]
rs79398853[A]crp[A][A][A][A]
rs75687619[A]crp[A][A][A][A]
rs76366838[A]crp[A][A][A][A]
rs114536010[A]crp[A][A][A][A]
rs115881343[A]crp[A][A][A][A]
rs112019714[A]crp[A][A][A][A]
rs1081105[A]crp[A][A][A][A]
rs150966173[A]crp[A][A][A][A]
rs140480140[A]crp[A][A][A][A]
rs4239225[A]wbc[A]0.00241711
rs2095044[A]eosinophil_pct
rs56177354[A]monocyte_pct[A]
rs1401884[A]monocyte_pct[A]-
rs76428106[A]monocyte_pct[A]
rs391855[A]monocyte_pct[A]0.
rs139795227[A]monocyte_pct
rs141094656[A]monocyte_pct
rs1002607[A]monocyte_pct[A]0
rs1610009[A]monocyte_pct[A]-
rs6740847[A]monocyte_pct[A]-
rs150649461[A]monocyte_pct
rs4857909[A]eosinophil_pct
rs17575497[A]monocyte_pct[A]
rs9807989[A]eosinophil_pct
rs13277237[A]monocyte_pct[A]
rs34856868[A]monocyte_pct[A]
rs876036[A]monocyte_pct[A]0.
rs3093068[A]crp[A]0.01199918
rs2524079[A]wbc[A]0.00206614

(showing first 40 of 101 rows; full data in the accompanying CSV)

Supplementary Table 112 - r11_incident_allergy_synthetic_validation

ST112_incident_allergy_synthetic_validation.csv

analysis	term	hr	lo	hi	p	n	events	cohort
main	index	1.3180518148258693	1.2912971749663926	1.3453607893248754	1.6284859654423998e-153	30000	9324	synthetic_hr1.30_age_inv
interaction	index:age	1.001402385467957	0.9997596492775083	1.0030478208893592	0.0943215905458839	30000	9324	synthetic_hr1.30_age_inv
age_stratum_<50	index	1.2953537989627109	1.2513087477724998	1.3409492001666992	1.1284038251737697e-48	10138	3205	synthetic_hr1.30_age_inv
age_stratum_50-65	index	1.3255789313192878	1.2853359436399698	1.367081899368229	8.398577373827047e-72	13798	4259	synthetic_hr1.30_age_inv
age_stratum_>65	index	1.3476768018748513	1.2880415846000712	1.410073078405738	3.39366125127469e-38	6064	1860	synthetic_hr1.30_age_inv

Supplementary Table 113 - r11_tnfsf14_multiplatform_sensitivity

ST113_tnfsf14_multiplatform_sensitivity.csv

protein	source	n_snp	ivw_or	ivw_p	direction_stable_under_LOO	single_snp_dominance	loo_min_or	cochran_q	q_p	i2	per_snp_f_min	per_snp_f_max
IL18	AholaOlli_Luminex_2017	3	1.1641407249232432	7.771671009881564e-0	True	0.5290646984889268	1.1513459551335445	1.1084440836101002	1.1490380783625826	0.0	27.23160258950984	92.7894347107438
IL18	SCALLOP_Olink_Folkerse	3	1.1907366814136648	7.000999455973844e-0	True	0.49776173413655334	1.1689259307305377	0.9053360950864937	1.2718583937044283	0.0	172.17363961918073	330.9367361111111
TNFSF14	AGES_SomaScan_Gudjon	3	1.2002113774650207	0.0002338302869714211	True	0.7100174054895236	1.157852981638114	29.383743004763225	8.32590159029115e-07	0.9319351520439111	0.4138278428693804	54.6402487571427
TNFSF14	INTERVAL_SomaScan_Su	3	1.0482764739895991	0.4175981504495637	False	0.5960201557776057	0.41286877105014247	42.26428585229281	1.3287917774681869e-(	0.9526787224800226	0.211149768939676	18.935540393053625

Supplementary Table 114 - r12_il4_independent_onek1k_check

ST114_il4_independent_onek1k_check.csv

outcome	onek1k_cell	gene	snp	exposure_rho	fstat	onek1k_p	outcome_p	or	or_lower	or_upper	wald_p	wald_q	dice_treg_or	dice_direction	onek1k_direction	direction_concordant_with_dice
ALLERG_ASTHMA	cd8et	IL4	rs29552	-0.0991559792020	9.7309437934749	0.00186486054579	0.186397	0.6634899821726	0.3610339274159	1.2193284980008	0.1863974760062	0.6213078581416	0.92	protective	protective	True
ALLERG_ASTHMA	cd4nc	IL4	rs56132027	0.0718449480919	5.0847083566673	0.02435775291712	0.832765	1.0481069455094	0.6776364426052	1.6211173132922	0.8327648850639	0.9646041975631	0.92	protective	risk	False
ALLERG_RHINITIS	cd8et	IL4	rs29552	-0.0991559792020	9.7309437934749	0.00186486054579	0.675403	0.8868588769973	0.5055672727195	1.5557151543416	0.6754033326397	0.9445841004247		risk	protective	True
ALLERG_RHINITIS	cd4nc	IL4	rs56132027	0.0718449480919	5.0847083566673	0.02435775291712	0.906113	1.0244424686797	0.6858081939016	1.5302855535509	0.9061130009892	0.9774601786529		risk	risk	False
L12_ATOPIC	cd8et	IL4	rs29552	-0.0991559792020	9.7309437934749	0.00186486054579	0.711851	0.9255336921824	0.6138460499160	1.3954844467633	0.7118500794707	0.8835715256094	0.88	protective	protective	True
L12_ATOPIC	cd4nc	IL4	rs56132027	0.0718449480919	5.0847083566673	0.02435775291712	0.323349	1.1586231290833	0.8650517374210	1.5518234311035	0.3233480062699	0.5836028869001	0.88	protective	risk	False

Supplementary Table 115 - r12_il4_independent_validation_search

ST115_il4_independent_validation_search.csv

resource	study	dataset	cell_type	n	il4_cis_snps	status
eQTL Catalogue	Nathan_2022 (ref 27)	QTD000687	CD4+ Treg	195	0	IL4 not a detectable cis
eQTL Catalogue	Nathan_2022	QTD000682	CD4+ RORC+ Treg	62	0	IL4 not a detectable cis
eQTL Catalogue	Cytoimmgen	QTD000747	nTreg unstimulated	93	0	IL4 not a detectable cis
eQTL Catalogue	Cytoimmgen	QTD000703	nTreg stimulated 40h	90	0	IL4 not a detectable cis
eQTL Catalogue	Perez_2022	QTD000600	CD4+ T	165	0	IL4 not a detectable cis
eQTL Catalogue	Schmiedel_2018 (DICE, re	QTD000469	Treg naive	89	0	IL4 not recovered by eQT
OneK1K	Yazar_2022 (ref 46)	ST114	CD4 naive / CD8 effector	982	2	IL4 cis-eQTL present but

Supplementary Table 116 - r13_immune_treg_il4_age

ST116_immune_treg_il4_age.csv

gene	n_donors	mean_detect_frac	raw_rho	raw_p	partial_r_adj_nGenes	partial_p_adj_nGenes	robust
IL4	12	0.00076	0.116	0.7196	0.608	0.0358	False
STAT6	12	0.18686	0.726	0.0075	0.598	0.04	False
IL5	12	0.00264	-0.137	0.6713	0.562	0.0574	False
GATA3	12	0.36995	0.298	0.3464	-0.479	0.1155	False
IL18	12	0.01208	0.364	0.2449	0.467	0.126	False
TSLP	12	0.00227	0.445	0.1471	0.374	0.2314	False
CD3D	12	0.79539	0.617	0.0326	0.221	0.4899	False
IL13	12	0.00038	0.088	0.7865	0.134	0.6786	False
FOXP3	12	0.71687	0.293	0.3549	-0.132	0.6818	False
IL6	12	0.00076	0.642	0.0244	0.123	0.7042	False
IL2RA	12	0.51793	0.487	0.1085	0.1	0.756	False
TNFSF14	12	0.02869	0.668	0.0176	0.097	0.7642	False
IL1RL1	12	0.01246	-0.323	0.3052	-0.035	0.9133	False
IL33	12	0.00076	-0.011	0.9735	-0.016	0.9595	False

Supplementary Table 116b - r13_immune_treg_il4_per_donor

ST116b_immune_treg_il4_per_donor.csv

donor	age	n_treg	FOXP3_detect_f	FOXP3_mean	GATA3_detect_f	GATA3_mean	IL4_detect_fr	IL4_mean	IL5_detect_fr	IL5_mean	IL1RL1_detect_f	IL1RL1_mean	TNFSF14_detect_f	TNFSF14_mean	IL2RA_detect_f	IL2RA_mean	IL6_detect_fr	IL6_mean	IL33_detect_f	IL33_mean	TSLP_detect_f	TSLP_mean	IL18_detect_f	IL18_mean	STAT6_detect_	STAT6_mean	CD3D_detect_	CD3D_mean	IL13_detect_f	IL13_mean
TSP10	33	74	1.0	1.6422981	0.7297297	1.2604182	0.0	0.0	0.0	0.0	0.0405405	0.0399091	0.0270270	0.0242374	0.7162162	1.4341872	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0675675	0.0637323	0.8378378	1.3290273	0.0	0.0
TSP26	37	36	0.2777777	0.3778400	0.25	0.4545885	0.0	0.0	0.0277777	0.0302347	0.0277777	0.0536895	0.0	0.0	0.3888888	0.7489577	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0833333	0.1067525	0.4722222	0.7095059	0.0	0.0
TSP4	38	3	0.6666666	0.8136872	0.3333333	0.3245660	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3333333	0.7755126	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6666666	1.2125090	0.0	0.0
TSP21	45	217	0.6036866	0.9427820	0.4470046	0.6371025	0.0	0.0	0.0	0.0	0.0276497	0.0334079	0.0552995	0.0638076	0.5299539	0.8658704	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0599078	0.0625013	0.7465437	1.3957095	0.0	0.0
TSP27	56	563	0.3143872	0.4059304	0.3090586	0.4023670	0.0	0.0	0.0053285	0.0070809	0.0035523	0.0042743	0.0159857	0.0149465	0.3516873	0.4805876	0.0	0.0	0.0017761	0.0019972	0.0	0.0	0.0355239	0.0340956	0.2060390	0.2220705	0.5541740	0.8979213	0.0	0.0
TSP20	56	224	0.9642857	1.2646297	0.1875	0.2338359	0.0044642	0.0062799	0.0044642	0.0024236	0.0044642	0.0099186	0.0401785	0.0526701	0.5580357	1.0097084	0.0	0.0	0.0	0.0	0.0	0.0	0.0089285	0.0108991	0.0892857	0.1042178	0.96875	2.2065612	0.0	0.0
TSP1	59	1	1.0	2.8968243	1.0	1.5993809	0.0	0.0	0.0	0.0	1.0	1.2901444	1.0	0.8401246	1.0	2.6508841	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	1.8352375	0.0	0.0
TSP14	59	1042	0.8291746	1.0714368	0.3618042	0.4316976	0.0	0.0	0.0009596	0.0006623	0.0134357	0.0185972	0.0163147	0.0131719	0.5047984	0.8030461	0.0	0.0	0.0009596	0.0006818	0.0028790	0.0020752	0.0038387	0.0026756	0.1967370	0.1808744	0.8522072	1.4365852	0.0009596	0.0008399548
TSP25	60	96	0.8541666	1.2120761	0.4375	0.4420326	0.0104166	0.0098789	0.0	0.0	0.0208333	0.0120223	0.0416666	0.0209943	0.75	1.2960528	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2708333	0.2876639	0.9895833	2.0841953	0.0	0.0
TSP2	61	87	0.4482758	0.5516849	0.4022988	0.4214247	0.0	0.0	0.0	0.0	0.0114942	0.0001552	0.0459770	0.0356084	0.5402298	1.1511664	0.0	0.0	0.0	0.0	0.0	0.0	0.0459770	0.0277148	0.2988505	0.2010753	0.8045977	1.5062030	0.0	0.0
TSP6	67	11	1.0	1.2340940	0.7272727	0.5784471	0.0	0.0	0.0	0.0	0.0	0.0	0.0909090	0.0587213	1.0	2.0791583	0.0909090	0.0125664	0.0	0.0	0.0	0.0	0.0	0.0	0.2727272	0.1400995	1.0	2.0576369	0.0	0.0
TSP7	69	295	0.9898305	0.9943990	0.4745762	0.4653603	0.0	0.0	0.0033898	0.0023842	0.0067796	0.0042243	0.0576271	0.0497483	0.7084745	1.0773121	0.0033898	0.0001094	0.0	0.0	0.0101694	0.0047277	0.0067796	0.0025566	0.2644067	0.1794175	0.9152542	1.5146273	0.0	0.0

Supplementary Table 117 - r13_epigenetic_clock_validity_gse40279

ST117_epigenetic_clock_validity_gse40279.csv

clock	n	pearson_r	spearman_rho	p
horvath_age	656	0.918	0.915	1.8562612709738036e-265
hannum_age	656	0.946	0.94	2.37e-322
pheno_age	656	0.852	0.847	5.102183396591418e-186

Supplementary Table 118 - onset_h2_power_report

ONSET01_onset_h2_power_report.csv

trait	n_snps_h2	h2	h2_se	h2_z	mean_chi2	rg_frailty	rg_se	rg_z	rg_p	verdict
adult-onset asthma	1165799	0.0392	0.00344	11.41	1.285	0.4866	0.0369	13.19	9.784890195836285e-40	PASS: childhood-onset GW
childhood-onset asthma	1165832	0.0669	0.00973	6.88	1.446	0.2088	0.0302	6.91	4.843685130709093e-12	PASS: childhood-onset GW

Supplementary Table 119 - dimension_disease_coupling_rg

COUP01_dimension_disease_coupling_rg.csv

exposure	endpoint	rg	se	z	p	q
GrimAge	ALLERG_ASTHMA	0.1976	0.0764	2.5856	0.0097216	0.051646000000000004
GrimAge	ALLERG_RHINITIS	-0.022	0.0932	-0.2364	0.81312	0.899896103896104
GrimAge	L12_ATOPIC	0.0508	0.074	0.6868	0.49222	0.6537296874999999
GrimAge	ASTHMA_NAS	0.2085	0.0668	3.1233	0.0017883	0.011692730769230771
GrimAge	FOOD_ALLERGY	-0.427	0.2564	-1.6652	0.095872	0.26849999999999996
GrimAge	CHRONNAS	-0.078	0.1089	-0.7167	0.47355	0.6389166666666667
GrimAge	L12_PSORIASIS	0.1039	0.0829	1.253	0.21021	0.4187666666666667
GrimAge	AB1_OTHER_SEPSIS	0.3533	0.1682	2.1003	0.035702	0.12644458333333333
GrimAge	J10_PNEUMONIA	0.4128	0.106	3.894	9.8617e-05	0.0007620404545454544
GrimAge	I9_MI_STRICT	0.1192	0.0676	1.7635	0.077818	0.22808724137931036
GrimAge	ST19_FRACT_FEMUR	0.0086	0.1152	0.0748	0.94035	0.9736801204819276
GrimAge	M13_RHEUMA	0.0777	0.0783	0.9918	0.32131	0.5223490566037736
GrimAge	T1D	-0.0067	0.108	-0.0617	0.95077	0.9736801204819276
GrimAge	K11_UC_STRICT2	-0.0213	0.0685	-0.3115	0.75538	0.8805068493150685
GrimAge	F5_DEPRESSIO	0.1034	0.064	1.6162	0.10605	0.26849999999999996
GrimAge	F5_DEMENTIA	0.1628	0.1176	1.3835	0.16652	0.38254594594594593
GrimAge	ANAPH_SHOCK_ADVER_EFFECT	0.4931	0.3667	1.3448	0.17868	0.38943076923076925
Hannum	ALLERG_ASTHMA	0.1318	0.0716	1.8419	0.0655	0.21250000000000002
Hannum	ALLERG_RHINITIS	0.0356	0.0895	0.3977	0.6909	0.8511086956521738
Hannum	L12_ATOPIC	0.0844	0.0704	1.1985	0.2307	0.42629347826086955
Hannum	ASTHMA_NAS	0.0678	0.0687	0.986	0.3241	0.5223490566037736
Hannum	FOOD_ALLERGY	-0.3173	0.2502	-1.2684	0.2047	0.4187666666666667
Hannum	CHRONNAS	0.009	0.0966	0.0934	0.9256	0.9736801204819276
Hannum	L12_PSORIASIS	0.0639	0.0723	0.8841	0.3766	0.5615964912280701
Hannum	AB1_OTHER_SEPSIS	0.2492	0.1592	1.5659	0.1174	0.28511428571428576
Hannum	J10_PNEUMONIA	0.1513	0.0787	1.9223	0.0546	0.18564
Hannum	I9_MI_STRICT	0.0044	0.0582	0.0755	0.9398	0.9736801204819276
Hannum	ST19_FRACT_FEMUR	0.1337	0.1085	1.233	0.2176	0.4187666666666667
Hannum	M13_RHEUMA	0.0833	0.0762	1.0934	0.2742	0.48556249999999995
Hannum	T1D	-0.0291	0.0938	-0.3104	0.7562	0.8805068493150685
Hannum	K11_UC_STRICT2	0.0352	0.066	0.5335	0.5937	0.7646136363636363
Hannum	F5_DEPRESSIO	0.0669	0.058	1.1532	0.2488	0.44995744680851063
Hannum	F5_DEMENTIA	0.1759	0.0962	1.8284	0.0675	0.21250000000000002
Hannum	ANAPH_SHOCK_ADVER_EFFECT	-0.0378	0.2599	-0.1454	0.8844	0.9637692307692307
IEAA	ALLERG_ASTHMA	0.1032	0.0641	1.6108	0.1072	0.26849999999999996
IEAA	ALLERG_RHINITIS	-0.0003	0.0741	-0.0036	0.9971	0.9971
IEAA	L12_ATOPIC	0.1215	0.0547	2.2205	0.0264	0.10685714285714284
IEAA	ASTHMA_NAS	0.0216	0.0564	0.383	0.7017	0.8520642857142857
IEAA	FOOD_ALLERGY	-0.0785	0.1887	-0.4159	0.6775	0.849
IEAA	CHRONNAS	-0.1078	0.0882	-1.2222	0.2217	0.4187666666666667

(showing first 40 of 85 rows; full data in the accompanying CSV)

Supplementary Table 120 - cohort_period_decomposition

DRIFT01_cohort_period_decomposition.csv

block	term	beta	se	lo	hi	p	n
B1_period_level_shift	cycleJ	0.016684351813217303	0.0321861286772081	-0.04640046039411058	0.07976916402054518	0.604199528640091	9360
B2_period_x_age_interact	a:cycleJ	0.03343804036278357	0.0133484429819971	0.007275092118069253	0.059600988607497886	0.01224464476391143	9360
B2_drift_rate_cycleD	a (D)	0.055789172935006875	0.009206761567462798	0.03774392026277979	0.07383442560723397	1.3647147692664007e-09	9360
B2_drift_rate_cycleJ	a + a:cycleJ (J)	0.08922721329779044	0.009647228375598706				9360
B3_cohort_main_effect	bc10	-0.014470571820803177	0.009523873063781937	-0.03313736302581577	0.004196219384209419	0.12866186607476154	9360
B3_age_x_cohort_interact	a:bc10	-0.008370121214198696	0.003874376924843776	-0.015963899986892495	-0.0007763424415048962	0.03074336837688643	9360
B3_drift_beta_cohort_adj	a	0.07006877071741635	0.010277101138678872	0.049925652485605765	0.09021188894922694	9.234820635723545e-12	9360
B4_same_age_level_diff_J	age 25-35 (birth 1976 vs	-0.016940432620772994					1603
B4_same_age_level_diff_J	age 35-45 (birth 1966 vs	-0.014228794520148011					1482
B4_same_age_level_diff_J	age 45-55 (birth 1956 vs	0.04240815528890714					1478
B4_same_age_level_diff_J	age 55-65 (birth 1946 vs	-0.017936244021434237					1627
B4_same_age_level_diff_J	age 65-75 (birth 1936 vs	0.18340657682480976					1264

Supplementary Table 121 - noccrp_crp_sensitivity

DRIFT02_noccrp_crp_sensitivity.csv

block	term	beta	se	lo	hi	p	n
S1_noccrp_drift	drift_beta_D	0.04651673573045731	0.009623521741226087	0.02765463311765418	0.06537883834326044	1.3405219494543123e-06	4486
S1_noccrp_drift	drift_beta_J	0.08649176370162082	0.00914601547058165	0.06856557337928078	0.10441795402396085	3.1760271441715015e-21	5024
S1_noccrp_drift	drift_beta_pooled	0.06848966416401753	0.007117144513065179	0.05454006091840978	0.08243926740962527	6.381794128488754e-22	9510
S1_noccrp_period_level_s	cycleJ	0.008352620089958556	0.030112540518228252	-0.05066795932576881	0.06737319950568593	0.781488245228	9510
S1_noccrp_period_x_age	a:cycleJ	0.04068220185043859	0.013217016168314309	0.014776850160542546	0.06658755354033463	0.002083829533355475	9510
S2_component_period_x_ag	a:cycleJ [eos]	0.026867411259162546	0.013278193006943275	0.0008421529655537295	0.05289266955277136	0.043029476280263636	9510
S2_component_drift_poole	a [eos]	0.024900853832242352	0.00878856210039229	0.0076752721154734645	0.04212643554901124	0.004606650531158798	9510
S2_component_period_x_ag	a:cycleJ [neu]	0.01916386063848515	0.01336207600266089	-0.007025808326730193	0.045353529603700496	0.15151581121190869	9510
S2_component_drift_poole	a [neu]	0.032214993218988124	0.007063455875966786	0.01837061970209322	0.046059366735883026	5.095966182475813e-06	9510
S2_component_period_x_ag	a:cycleJ [lymp]	-0.026811077068121723	0.013588241823709764	-0.05344403104259286	-0.0001781230936505858607	0.04848318844020939	9510
S2_component_drift_poole	a [lymp]	-0.0638522603808043	0.008160788101505915	-0.07984740505975589	-0.04785711570185271	5.105858398546971e-15	9510
S2_component_period_x_ag	a:cycleJ [mono]	-0.001977110649060052	0.01616225487017545	-0.033655130194603936	0.02970090889648383	0.9026385553071692	9510
S2_component_drift_poole	a [mono]	0.08041661152439519	0.012464973856644839	0.05598526276537131	0.10484796028341908	1.1081691959729414e-10	9510
S2_component_period_x_ag	a:cycleJ [wbc]	0.036914907567585246	0.010349569145014137	0.016629752043357537	0.057200063091812955	0.0003613584954923644	9510
S2_component_drift_poole	a [wbc]	-0.0414891000433222	0.007966269128402127	-0.05710298753499037	-0.02587521255165403	1.9078748483798244e-07	9510
S2_component_period_x_ag	a:cycleJ [crp]	-0.051507416806943415	0.014664919462093763	-0.08025065895264719	-0.02276417466123964	0.00044426672471776005	9360
S2_component_drift_poole	a [crp]	0.08498677156949093	0.011076220654810915	0.06327737908606154	0.10669616405292032	1.68146069994164e-14	9360

Supplementary Table 122 - geroprotector_triple_filter

GER01_geroprotector_triple_filter.csv

compound	mean_z|ageing_v2_AA_cura	mean_z|alarmin_type2	z_score|ageing_v2_AA_cura	z_score|alarmin_type2	p|ageing_v2_AA_curated	p|alarmin_type2	q|ageing_v2_AA_curated	q|alarmin_type2	n_sig|ageing_v2_AA_curat	n_sig|alarmin_type2	pass_ageing_suppress	pass_alarmin_safe	triple_pass	geroprotective_phenotypes
metoprolol	-0.047509294003248	0.005479359999299	-2.781792987576026	0.075639250895828	0.005405951041177	0.939706108196678	0.021623804164711	0.977294352524545	42.0	42.0	True	True	True	Extended Lifespan
ly-294002	-0.052198916673660	0.020380059257149	-2.222319793012262	0.144915119749768	0.026261701452877	0.884777888617503	0.068280423777482	0.958509379335629	14.0	14.0	True	True	True	Extended Lifespan
rutin	-0.040572028607130	-0.019203677773475	-1.717515295595416	-0.299921195908562	0.085885062163065	0.764237266173138	0.165408267869607	0.883118618688960	42.0	42.0	False	True	False	Extended Lifespan; Exten
phenformin	-0.030188543722033	-0.041772004216909	-1.460863630438581	-0.645910984710770	0.144052869326229	0.518337005255137	0.249691640165463	0.691116007006850	42.0	42.0	False	True	False	Extended Lifespan
cinnarizine	-0.032261949032545	0.038616426289081	-1.412054247846762	0.581398129017359	0.157933988791727	0.560972158156074	0.256642731786557	0.711476883515021	42.0	42.0	False	True	False	Extended Lifespan
acarbose	-0.037730325013399	0.017253203317523	-1.361279883236641	0.191971567816036	0.173425257782492	0.847764476775452	0.273276163778473	0.937952187070712	42.0	42.0	False	True	False	Extended Lifespan
inositol	-0.021208865568041	-0.019659351557493	-1.060795595031290	-0.246252413686906	0.288782804525070	0.805486844184911	0.405856914467666	0.910550345600334	42.0	42.0	False	True	False	Extended Lifespan
mianserin	-0.026760129258036	-0.158092752099037	-1.040651438669282	-2.359169787177533	0.298037348680553	0.018315873539999	0.407840582404967	0.045353591622854	42.0	42.0	False	True	False	Extended Lifespan
oxatomide	-0.025773266330361	0.071175843477249	-1.02253470168949	0.786788219789911	0.306527897147751	0.431405850648680	0.408703862863669	0.628129332230324	42.0	42.0	False	True	False	Extended Lifespan
lonidamine	-0.014640237204730	-0.068066507577896	-0.724545826532979	-0.826418356588580	0.468730703680363	0.408566813915739	0.594487721740948	0.624866891871131	42.0	42.0	False	True	False	Extended Lifespan
chloramphenicol	-0.015316191129386	0.093623831868171	-0.581280680369251	1.205184248790356	0.561051299237131	0.228132181646736	0.663060626371155	0.409064601573458	42.0	42.0	False	True	False	Extended Lifespan
quercetin	-0.009513750672340	-0.047975320369005	-0.388576260580651	-0.550449733439108	0.697589630668625	0.582010945095400	0.762815481822677	0.720584979641924	42.0	42.0	False	True	False	Extended Lifespan; Exten
trehalose	-0.007687923964112	-0.061328962445259	-0.320113172811746	-0.835965896074455	0.748882540206248	0.403174089181912	0.789298891473733	0.624866891871131	42.0	42.0	False	True	False	Extended Lifespan
doxycycline	0.000668968830723	0.002061111619696	0.029655184949809	0.025002274875681	0.976342053416245	0.980053149088517	0.976342053416245	0.980053149088517	42.0	42.0	False	True	False	Extended Lifespan
celecoxib	0.006364138331264	0.087933741509144	0.286989883142396	1.085982858136714	0.774120066637700	0.277486576266730	0.789298891473733	0.467274197385773	42.0	42.0	False	True	False	Extended Lifespan
hesperidin	0.006509329658001	0.075094714760780	0.298287779082071	0.668251098020081	0.765483530866537	0.503973323090410	0.789298891473733	0.689647705281614	42.0	42.0	False	True	False	Extended Lifespan
genistein	0.016299607232213	0.142458528280258	0.379741423615555	1.254298963040658	0.704137367836317	0.209733360709693	0.762815481822677	0.393936344070802	42.0	42.0	False	True	False	Extended Lifespan
aspirin	0.010286511853337	-0.060617912560701	0.499004481248928	-1.083543373038071	0.617776227092393	0.278567309979980	0.698355734974010	0.467274197385773	42.0	42.0	False	True	False	Extended Lifespan
simvastatin	0.013417690992355	0.093367233872413	0.560876656294584	1.521663770062719	0.574881624653392	0.128093350005078	0.664307655155031	0.277535591677670	42.0	42.0	False	True	False	Extended Lifespan
caffeine	0.018824474886059	0.136621370911598	0.639748769133237	1.524127226950726	0.522335943864264	0.127476932493320	0.631662071649807	0.277535591677670	42.0	42.0	False	True	False	Extended Lifespan
nebivolol	0.021625978872179	0.036253105849027	0.695469105383621	0.411347528988249	0.486761366004647	0.680817723014460	0.602656929339086	0.804602763562544	42.0	42.0	False	True	False	Extended Lifespan
vinpocetine	0.017391990870237	0.097237929701805	0.852961675019547	1.247758338377426	0.393680557721036	0.212119569884278	0.511784725037347	0.393936344070802	42.0	42.0	False	True	False	Extended Lifespan
nitrendipine	0.027895303443074	-0.107801303267478	1.134950739069675	-1.359881261929683	0.256395958881854	0.173867501540659	0.370349718384900	0.347735003081319	42.0	42.0	False	True	False	Extended Lifespan
celastrol	0.018119033426046	-0.021620895713567	1.22113534471202	-0.472699535443038	0.222034779279524	0.636427558074909	0.329880243501007	0.769633326044076	136.0	136.0	False	True	False	Extended Lifespan
staurosporine	0.090228535234928	0.253303885459899	1.298604939570232	1.449219062512289	0.194079541932878	0.147276415654945	0.296827534720872	0.306334944562287	33.0	33.0	False	True	False	Extended Lifespan
indirubin	0.027970930561423	0.176574036478996	1.415425585678996	2.564606745677514	0.156943752323299	0.010329280129615	0.256642731786557	0.028269608775789	42.0	42.0	False	False	False	Extended Lifespan
amoxapine	0.049710229039192	0.196500226855278	1.582563095068243	2.268724416228417	0.113521080665971	0.023285089847785	0.203555041194156	0.055037485094765	42.0	42.0	False	False	False	Extended Lifespan
melatonin	0.058830641210079	0.226990252733230	1.641141851767086	2.583800521845645	0.100767972837493	0.009771829364887	0.187140520983916	0.028229729276341	42.0	42.0	False	False	False	Extended Lifespan
navitoclax	0.056607689708471	-0.008670110255479	1.740941008346141	-0.097750844035902	0.081693919510812	0.922130141140991	0.163387839021625	0.977294352524545	42.0	42.0	False	True	False	Senescence Retardated
sulforaphane	0.056878384202718	0.061488237231969	1.765122271980719	0.593343677522509	0.077543218277297	0.552951173712539	0.161289894016778	0.711476883515021	42.0	42.0	False	True	False	Extended Lifespan; Senes
clofibrate	0.057263068854808	0.242546990513801	1.932921173652255	2.362794977714085	0.053245909076745	0.018137700649338	0.115366136332949	0.045353591622854	42.0	42.0	False	False	False	Extended Lifespan
metformin	0.046507488936185	-0.002037251368165	2.026549209244801	-0.026573230087120	0.042708535332072	0.978800125013305	0.096558427707295	0.980053149088517	42.0	42.0	False	True	False	Extended Lifespan; Exten
lamotrigine	0.055925630033016	0.056470006704330	2.119698243925343	0.780904324445579	0.034031501194212	0.434858768467147	0.080438093731774	0.628129332230324	42.0	42.0	False	True	False	Extended Lifespan
arbutin	0.087524138391017	0.271441608667373	2.183983443163135	2.778327703075319	0.028963464706342	0.005463948214145	0.071719055463325	0.016713253360917	42.0	42.0	False	False	False	Extended Lifespan
danazol	0.159644797444343	0.415425062179565	2.305393454299605	3.109159503152789	0.021144547766255	0.001876204013368	0.057869288623437	0.007504816053475	42.0	42.0	False	False	False	Extended Lifespan
metergoline	0.114906415343284	0.432486861944198	2.31296408610396	2.982692806233531	0.020724613789183	0.002857246183312	0.057869288623437	0.009286050095765	42.0	42.0	False	False	False	Extended Lifespan
reserpine	0.051170252263545	0.063796602189540	2.352618574501929	0.737982747375014	0.018641745173484	0.460524933203878	0.057021808765952	0.647224230448693	42.0	42.0	False	True	False	Extended Lifespan
tanespimycin	0.063285723328590	0.079665981233119	2.385673900702446	0.931989392498891	0.017047860227919	0.351342007393405	0.055405545740737	0.570930762014284	59.0	59.0	False	True	False	Extended Lifespan
ivabradine	0.110491670668125	0.348480612039566	2.560647892304265	4.690857252220105	0.010447718483502	2.720627403134710	0.036218757409474	3.536815624075124	42.0	42.0	False	False	False	Extended Lifespan
bezafibrate	0.143250331282615	0.699803650379180	2.689763742825313	3.664427443580074	0.007150262061016	0.000247892498426	0.026558116226632	0.001611301239770	42.0	42.0	False	False	False	Extended Lifespan

(showing first 40 of 52 rows; full data in the accompanying CSV)

Supplementary Table 123 - dual_signature_pass_list

GER02_dual_signature_pass_list.csv

compound	z_curated_ageing	q_curated_ageing	z_curated_alarmin	q_curated_alarmin	triple_pass	geroprotective_phenotype	z_internal_ageing	p_internal_ageing	z_internal_alarmin	q_internal_alarmin	pass_internal_definition	dual_direction_suppressio	editorial_note
metoprolol	-2.7817929875760266	0.0216238041647112	0.0756392508958288	0.9772943525245456	True	Extended Lifespan					False	False	curated triple-pass (z=-
ly-294002	-2.2223197930122622	0.0682804237774821	0.1449151197497682	0.958509379335629	True	Extended Lifespan					False	False	curated triple-pass (z=-
rutin	-1.7175152955954165	0.1654082678696076	-0.2999211959085623	0.8831186186889606	False	Extended Lifespan; Ext					False	False	
phenformin	-1.4608636304385818	0.2496916401654635	-0.6459109847107708	0.6911160070068503	False	Extended Lifespan					False	False	
cinnarizine	-1.412054247846762	0.2566427317865575	0.5813981290173597	0.7114768835150213	False	Extended Lifespan					False	False	
acarbose	-1.3612798832366415	0.2732761637784731	0.1919715678160363	0.9379521870707128	False	Extended Lifespan					False	False	
inositol	-1.0607955950312908	0.4058569144676664	-0.2462524136869068	0.9105503456003344	False	Extended Lifespan					False	False	
mianserin	-1.040651438669282	0.4078405824049675	-2.359169787177533	0.0453535916228548	False	Extended Lifespan					False	False	
oxatomide	-1.02253470168949	0.4087038628636691	0.7867882197899114	0.6281293322303247	False	Extended Lifespan					False	False	
lonidamine	-0.7245458265329793	0.5944877217409485	-0.8264183565885803	0.6248668918711313	False	Extended Lifespan					False	False	
chloramphenicol	-0.5812806803692517	0.6630606263711557	1.205184248790356	0.4090646015734588	False	Extended Lifespan					False	False	
quercetin	-0.3885762605806515	0.7628154818226776	-0.5504497334391082	0.7205849796419247	False	Extended Lifespan; Ext	-2.9693518	0.0029842875419288	0.8328635416306158	0.6264432250607387	True	False	suppresses internal agei
trehalose	-0.3201131728117469	0.7892988914737334	-0.8359658960744552	0.6248668918711313	False	Extended Lifespan					False	False	
doxycycline	0.0296551849498092	0.9763420534162456	0.0250022748756819	0.9800531490885176	False	Extended Lifespan					False	False	
celecoxib	0.2869898831423966	0.7892988914737334	1.0859828581367146	0.4672741973857733	False	Extended Lifespan					False	False	
hesperidin	0.2982877790820719	0.7892988914737334	0.6682510980200815	0.6896477052816149	False	Extended Lifespan					False	False	
genistein	0.3797414236155552	0.7628154818226776	1.254298963040659	0.3939363440708029	False	Extended Lifespan					False	False	
aspirin	0.4990044812489287	0.6983557349740102	-1.0835433730380717	0.4672741973857733	False	Extended Lifespan					False	False	
simvastatin	0.5608766562945849	0.6643076551550317	1.521663770062719	0.2775355916776707	False	Extended Lifespan					False	False	
caffeine	0.6397487691332379	0.6316620716498077	1.5241272269507269	0.2775355916776707	False	Extended Lifespan					False	False	
nebivolol	0.6954691053836217	0.6026569293390869	0.411347528988249	0.8046027635625442	False	Extended Lifespan					False	False	
vinpocetine	0.8529616750195477	0.5117847250373474	1.2477583383774264	0.3939363440708029	False	Extended Lifespan					False	False	
nitrendipine	1.1349507390696754	0.3703497183849005	-1.3598812619296836	0.3477350030813198	False	Extended Lifespan					False	False	
celastrol	1.22113534471202	0.3298802435010077	-0.4726995354430385	0.7696333260440769	False	Extended Lifespan					False	False	
staurosporine	1.2986049395702328	0.2968275347208729	1.4492190625122892	0.3063349445622875	False	Extended Lifespan					False	False	
indirubin	1.4154255856789966	0.2566427317865575	2.5646067456775143	0.0282696087757892	False	Extended Lifespan					False	False	
amoxapine	1.5825630950682437	0.2035550411941565	2.268724416228417	0.0550374850947653	False	Extended Lifespan					False	False	
melatonin	1.6411418517670868	0.187140520983916	2.583800521845645	0.0282297292763419	False	Extended Lifespan					False	False	
navitoclax	1.7409410083461418	0.1633878390216255	-0.0977508440359023	0.9772943525245456	False	Senescence Retardate	-3.4141395	0.0006398380381069	-0.1219314646242758	0.902953296373278	True	True	
sulforaphane	1.7651222719807198	0.1612898940167781	0.5933436775225096	0.7114768835150213	False	Extended Lifespan; Ser					False	False	
clofibrate	1.932921173652255	0.1153661363329495	2.362794977714085	0.0453535916228548	False	Extended Lifespan					False	False	
metformin	2.026549209244801	0.0965584277072953	-0.0265732300871206	0.9800531490885176	False	Extended Lifespan; Ext					False	False	
lamotrigine	2.119698243925344	0.0804380937317745	0.7809043244455794	0.6281293322303247	False	Extended Lifespan					False	False	
arbutin	2.1839834431631355	0.071719055463325	2.77832770307532	0.016713253360917	False	Extended Lifespan					False	False	
danazol	2.305393454299605	0.0578692886234374	3.1091595031527897	0.0075048160534752	False	Extended Lifespan					False	False	
metergoline	2.31296408610396	0.0578692886234374	2.982692806233531	0.0092860500957655	False	Extended Lifespan					False	False	
reserpine	2.3526185745019297	0.0570218087659522	0.737982747375014	0.6472242304486938	False	Extended Lifespan					False	False	
tanespimycin	2.385673900702446	0.0554055457407376	0.9319893924988912	0.570930762014284	False	Extended Lifespan					False	False	
ivabradine	2.560647892304265	0.0362187574094745	4.690857252220105	3.536815624075124e-	False	Extended Lifespan					False	False	
bezafibrate	2.689763742825313	0.0265581162266324	3.6644274435800743	0.0016113012397703	False	Extended Lifespan					False	False	

(showing first 40 of 60 rows; full data in the accompanying CSV)

Supplementary Table 124 - r14_index_weight_sensitivity

ST124_index_weight_sensitivity.csv

cycle	outcome	index	or_	lo	hi	p	n	beta
J	self-reported allergy	idx_pub	1.312214004140804	1.2015042653782828	1.4331248271691446	1.5208465095636342e-09	4903	
J	self-reported allergy	idx_equal	1.5682057497655606	1.4396184754571741	1.7082784887272189	6.502332634731923e-25	4903	
J	self-reported allergy	idx_noeos	1.6277613922311394	1.4532679000292386	1.8232062718683524	3.716580888123635e-17	4903	
J	self-reported allergy	idx_eosonly	1.0687173944010628	0.981897268951398	1.1632142233322897	0.12419671179931335	5024	
D	log total IgE	idx_pub		0.15154589471580632	0.26759955328571744	1.4536792902202248e-12	4451	0.20957272400076188
D	high IgE	idx_pub	1.3157445557882486	1.20263179169681	1.439496068571299	2.1883903389835427e-09	4457	
D	log total IgE	idx_equal		0.12754159383721234	0.25312377632722516	2.830082799019103e-09	4451	0.19033268508221876
D	high IgE	idx_equal	1.303721261870385	1.1948821181528606	1.4224743201282628	2.4745642555289024e-09	4457	
D	log total IgE	idx_noeos		-0.05641264846311882	0.06600380577501966	0.8779541169248554	4451	0.004795578655950417
D	high IgE	idx_noeos	1.036043485513352	0.945955078376333	1.1347114978409427	0.44551398720689106	4457	
D	log total IgE	idx_eosonly		0.28438102119742215	0.41790305244412684	6.417466545190506e-25	4458	0.3511420368207745
D	high IgE	idx_eosonly	1.5345333290604874	1.401578455623652	1.6801004100406671	2.020108462404321e-20	4486	

Supplementary Table 125 - r14_independent_ageing_mr

ST125_independent_ageing_mr.csv

exposure	outcome	n_snp	ivw_beta	ivw_se	ivw_or	ivw_p	wmedian_or
Telomere length (Codd 20	Allergic rhinitis	90	-0.01184178667068369	0.05143500804317983	0.9882280513443155	0.817914489928592	0.9772833963175629
Telomere length (Codd 20	Asthma	90	-0.09118521295683266	0.0558584117021118	0.9128486238484292	0.1025879313321125	1.0201256098809128
Telomere length (Codd 20	Atopic dermatitis	90	0.07748012615933343	0.037498502420010446	1.080560757352842	0.038807785497755265	1.0280367252443408

Supplementary Table 126 - r14_independent_ldsc_rg

ST126_independent_ldsc_rg.csv

exposure	outcome	rg_independent	se_independent	z	p
Frailty	ALLERG_ASTHMA	0.458481	0.038021	12.058568	1.747958e-33
GrimAge	ALLERG_ASTHMA	0.166996	0.071582	2.332922	0.019652
Hannum	ALLERG_ASTHMA	0.117546	0.066063	1.779293	0.075192
PhenoAge	ALLERG_ASTHMA	0.119459	0.067532	1.768915	0.076908
Frailty	ALLERG_RHINITIS	0.354849	0.046426	7.643366	2.116156e-14
GrimAge	ALLERG_RHINITIS	-0.004528	0.086854	-0.052131	0.958424
Hannum	ALLERG_RHINITIS	0.038893	0.08597	0.452407	0.650976
PhenoAge	ALLERG_RHINITIS	-0.014554	0.082792	-0.175795	0.860455
Frailty	L12_ATOPIC	0.250524	0.032992	7.593393	3.116354e-14
GrimAge	L12_ATOPIC	0.090611	0.073199	1.23787	0.215764
Hannum	L12_ATOPIC	0.110542	0.070774	1.561894	0.118313
PhenoAge	L12_ATOPIC	0.202576	0.071685	2.825903	0.004715

Supplementary Table 127 - r15_pqtl_helic_il18

ST127_pqtl_helic_il18.csv

protein	source	layer	snp	status	rep_beta	rep_se	wald_logor	wald_se	wald_p	wald_or	n_snps	discovery_or	direction_concordant	replicates_p005
IL18	HELIC_Olink_Gilly202	cross-cohort (Olink)	rs5744249	ok	0.1703283	0.05942742	0.254106334649027	0.058360883071104	1.336438269729244	1.289308895187776				
IL18	HELIC_Olink_Gilly202	cross-cohort (Olink)	rs7106524	ok	0.1284985	0.04093954	0.207922271466203	0.072332828787884	0.004046402201550	1.231117472873637				
IL18	HELIC_Olink_Gilly202	cross-cohort (Olink)	rs10891343	ok	0.154944	0.04081114	0.147794687112763	0.055006453944650	0.007212699523021	1.159274857859455				
IL18	HELIC_Olink_Gilly202	cross-cohort (Olink)	(IVW)	summary			0.200178977517524	0.035023504576817	1.093426752233328	1.221621381357458	3.0	1.24	True	True

Supplementary Table 128 - r15_pqtl_crossplatform_meta

ST128_pqtl_crossplatform_meta.csv

protein	k_platforms	platforms	meta_logor	meta_se	meta_or	or_lo	or_hi	meta_p	cochran_q	q_p	i2_pct	discovery_or
IL18	3	SCALLOP_Olink_Folkerse	0.17108259979968582	0.017255241894796373	1.1865887569675726	1.1471290304638566	1.227405846047248	3.589064647669066e-2	1.229954401755562	0.5406532215876789	0.0	1.24
TNFSF14	2	INTERVAL_SomaScan_Su	0.1255104465581919	0.0377409229617408	1.1337270124438739	1.0528894454294366	1.2207710356719057	0.000882348760542237	3.1351919239136556	0.0766193681670743	68.10402602875739	1.17

Supplementary Table 129 - r15_open_targets_constraint

ST129_open_targets_constraint.csv

| gene | ensembl | approvedSym | biotype | pLI | oe_lof | oe_lof_upper_bin | oe_lof_upper_bin6 | Approved_SM | Advanced_SM | Phase3_SM | Phase2_SM | Phase1_SM | Entered_Clin_SM | High_Quality_SM | Medium_Quality_SM | Low_Quality_SM | Druggable_SM | Approved_AB | Advanced_AB | Phase3_AB | Phase2_AB | Phase1_AB | Entered_Clin_AB | High_Quality_AB | Medium_Quality_AB | Low_Quality_AB | Druggable_AB | Approved_OC | Advanced_OC | Phase3_OC | Phase2_OC | Phase1_OC | Entered_Clin_OC | High_Quality_OC | Medium_Quality_OC | Low_Quality_OC | Druggable_OC | Approved_DR | Advanced_DR | Phase3_DR | Phase2_DR | Phase1_DR | Entered_Clin_DR | High_Quality_DR | Medium_Quality_DR | Low_Quality_DR | Druggable_DR | Approved_PR | Advanced_PR | Phase3_PR | Phase2_PR | Phase1_PR | Entered_Clin_PR | High_Quality_PR | Medium_Quality_PR | Low_Quality_PR | Druggable_PR | Approved_UD | Advanced_UD | Phase3_UD | Phase2_UD | Phase1_UD | Entered_Clin_UD | High_Quality_UD | Medium_Quality_UD | Low_Quality_UD | Druggable_UD | Approved_GC | Advanced_GC | Phase3_GC | Phase2_GC | Phase1_GC | Entered_Clin_GC | High_Quality_GC | Medium_Quality_GC | Low_Quality_GC | Druggable_GC | Approved_OC | Advanced_OC | Phase3_OC | Phase2_OC | Phase1_OC | Clinical |
|---|---|---

Supplementary Table 130 - r15_type2_spectrum_mr_panukbb

ST130_type2_spectrum_mr_panukbb.csv

outcome	n_snp	ivw_logor	ivw_se	ivw_or	or_lo	or_hi	ivw_p	wm_or	egger_intercept	Q	Q_p	I2_pct
UKB_icd10_J45_asthma	101	1.0236204251071586	0.04822323707527603	2.7832531046627746	2.5322364394699983	3.0591526619987865	5.412706088393219e-10	4.226034011861847	0.004343477610980059	776.7099860133632	3.0624679256165515e-	87.12518162496245
UKB_icd10_L50_urticaria	101	0.036936204767321464	0.3750495616501007	1.037626823082017	0.49749939619288175	2.164162272795708	0.921548349511409	1.0482523974714943	-0.004094289248706173	71.51366805475493	0.9859900740843464	0.0
UKB_phecode_476_allerg	101	0.4587163933838193	0.20224645475328487	1.582041961585174	1.0642957811258433	2.351655256557223	0.023322748007071504	1.9546562338386215	0.008142361734414842	98.50194896800396	0.523619485741547	0.0
UKB_phecode_250.1_T1D	101	-0.15831781577545143	0.1353427440584908	0.8535784581556841	0.654692516432223	1.1128830190361019	0.24209968408944282	1.0166245957429836	0.016922345150617385	476.15886590967193	2.3887667139842395e-	78.99860589407356
UKB_phecode_714.1_RA	101	0.09222503570395864	0.10616527540867561	1.0966115710729747	0.8906008745673936	1.3502759453220567	0.3850150499918503	1.0223347894514132	0.008967344241621	702.7963742773375	2.5987665588696454e-	85.77112750434624

Supplementary Table 131 - r15_nonHLA_frailty_readjudication

ST131_nonHLA_frailty_readjudication.csv

cohort	outcome	n_snp	ivw_or	or_lo	or_hi	ivw_p	wm_or	Q	I2_pct
FinnGen_R12	asthma	13	3.85266599457749	2.688865876102671	5.520184326667767	1.9745399563189 21e-13	2.489899300372189	26.3311445112461	54.426591693062306
FinnGen_R12	allergic_rhinitis	13	2.6706042374827605	1.9201325477646451	3.7143930514400494	5.350834228932366e-09	2.349726572061049	17.17733067303289	30.140484406933542
FinnGen_R12	atopic_dermatitis	13	1.6006600183008857	1.2588504780484964	2.035279438554805	0.00012390945771094546	1.5176901124370314	30.617763926468673	60.807066026052446
PanUKB	asthma	13	2.3810261266285746	1.862040764962396	3.044662352385376	4.645124131701532e-12	2.3574387577790166	17.813101734604214	32.633854682991704
PanUKB	allergic_rhinitis	13	0.8027873199570034	0.286107609818345 27	2.252535266338885	0.6764541834537071	1.6061917097955212	22.017332188674413	45.49748399502856
PanUKB	urticaria	13	2.015058184303514	0.297630218122236 45	13.64263182598263	0.47273851343282336	1.5176528267769676	2.5676991123008333	0.0

Supplementary Table 132 - r15_independent_ageing_readjudication

ST132_independent_ageing_readjudication.csv

exposure	outcome	n_snp	ivw_or	or_lo	or_hi	ivw_p	Q	I2_pct	direction_positive
healthspan	asthma	12	0.6734512096487068	0.5301820315849008	0.8554355009382832	0.0011975142080785767	5.699655249059651	0.0	False
healthspan	allergic_rhinitis	12	0.8573664250032155	0.6886353619049639	1.067440371765631	0.16872422066876158	16.769443336975044	34.404501217127255	False
healthspan	atopic_dermatitis	12	0.9292134917254253	0.7922025489287173	1.0899203926725178	0.3670284706993079	8.902615097808187	0.0	False

Supplementary Table 133 - r16_eosinophil_free_instrument_mr

ST133_eosinophil_free_instrument_mr.csv

instrument	outcome	n_snp	mean_F	ivw_beta	ivw_se	ivw_or	or_lo	or_hi	ivw_p	ivw_re_p	wm_or	egger_or	egger_int	egger_int_p	Q	Q_p	I2_pct
eosinophil_free_in	ALLERG_ASTHMA	73	1228.916074418	0.3501117951657	0.1755980406098	1.4192262023532	0.5709100368085	3.5280567577786	0.0461711626501	0.4511130198040	1.7414380009547	1.4687909426591	0.0020387180634	0.3709996941476	504.04592720220	4.4525022783218	85.715587387116
eosinophil_only	ALLERG_ASTHMA	15	954.9607193707	0.9508045632994	0.0791153968650	2.5877908635287	1.6330156438815	4.1007944892954	2.8599252459988	5.1669508472389	2.3390199020547	2.5857099732216	-0.003307738171	0.3979981255565	123.40124498397	1.3545240533382	88.654895660236250
full_index	ALLERG_ASTHMA	88	1182.219138898	0.8494433689845	0.0721322241215	2.3383448943116	1.5949374275175	3.4282579055555	5.1776107791504	1.3523103481447	2.3390199020547	2.3503224521001	0.0016787672619	0.3872401328184	637.17469147347	1.3343748919096	86.345973692267 7
eosinophil_free_in	ALLERG_RHINITIS	73	1228.916074418	-0.222153016254	0.1607972164553	0.8007928206413	0.3926106845011	1.6333461286349	0.1671029951255	0.5412840749898	1.0716318994380	0.8343724616253	0.0024277309916	0.2455404301908	368.28065018483	2.4307143932469	80.449692384361 86
eosinophil_only	ALLERG_RHINITIS	15	954.9607193707	0.6365254985054	0.0727324289355	1.8899029877199	1.5595007986617	2.2903055298578	2.1033926011180	8.4444132865184	1.6509079891328	1.8904224055554	0.0009722822213	0.7868891159195	25.438088092312	0.0304809019920	44.964417336730 89
full_index	ALLERG_RHINITIS	88	1182.219138898	0.4906816985776	0.0662685112137	1.6334293469662	1.2289816635541	2.1709774121564	1.3169781964180	0.0007235581664	1.6509079891328	1.6510826971823	0.0034581519564	0.0522719170786	417.39223360544	1.1991522979585	79.156296405304 08
eosinophil_free_in	L12_ATOPIC	73	1228.916074418	0.7041382978135	0.1174394434553	2.0221034829773	1.2253541095676	3.3369149896692	2.0254041353542	0.0058653345229	3.3748350957940	1.9109948182330	-0.003370760843	0.0269576154550	340.96441836273	1.4257266293495	78.883427090211 36
eosinophil_only	L12_ATOPIC	15	954.9607193707	0.6152111843223	0.0530258067560	1.8500472592830	1.2836462227414	2.6663692853556	4.0204685915735	0.0009702250588	1.5844124583224	1.8508571589576	0.0020313559014	0.4382573380523	173.15364510626	1.5777626930203	91.914695187959 67
full_index	L12_ATOPIC	88	1182.219138898	0.6302703996273	0.0483279148910	1.8781183531131	1.4916752606416	2.3646759059228	7.1008818280590	8.2141577088918	1.5844124583224	1.8662545604586	-0.002099563199	0.1059811313874	514.59434305631	6.5404432712965	83.093479130904 7

Supplementary Table 134 - r16_frailty_vs_clock_rg_contrast

ST134_frailty_vs_clock_rg_contrast.csv

outcome	clock	source	rg_frailty	se_frailty	rg_clock	se_clock	clock_rg_hi95	frailty_rg_lo95	ci_nonoverlap	diff	se_diff_rho0	z_rho0	p_rho0	se_diff_rho0p5	p_rho0p5	q_bh_rho0
ALLERG_ASTHMA	GrimAge	in_house_ST60	0.4581123063935	0.0381960634575	0.1645756906272	0.0704653522704	0.30268778107729	0.3832480220167	True	0.2935366157663	0.08015176313875	3.6622602457063	0.0002499969262	0.0610966944062	1.55169826700721	0.000428570901738896515
ALLERG_ASTHMA	Hannum	in_house_ST60	0.4581123063935	0.0381960634575	0.1133273047363	0.0648452058680	0.24042390823770	0.3832480220167	True	0.3447850016572	0.0752584878118	4.5813437351980	4.6199775279182	0.0564535950092	1.01263675578584	2.771986516750919 8e-05
ALLERG_ASTHMA	IEAA	in_house_ST60	0.4581123063935	0.0381960634575	0.0826294808356	0.0595311217022	0.19931047937200	0.3832480220167	True	0.3754828255579	0.0707311368124	5.3085931101830	1.1047464601964	0.0522402068577	6.59298032867210	1.325695752235780 2e-06
ALLERG_ASTHMA	PhenoAge	in_house_ST60	0.4581123063935	0.0381960634575	0.0824863412370	0.0780436806731	0.23545195535630	0.3832480220167	True	0.3756259651565	0.0868893282092	4.3230391222706	1.5389443076752	0.0675928544833	2.74160879999261	4.616832923025867e-05
ALLERG_RHINITIS	GrimAge	in_house_ST60	0.3531147710661	0.0457631097953	-0.0035764599577	0.0850016561333	0.16302678606351	0.2634190758672	True	0.3566912310239	0.0965377841239	3.6948354912118	0.0002200291986	0.0736858442384	1.29381111831386	0.000428570901738896515
ALLERG_RHINITIS	Hannum	in_house_ST60	0.3531147710661	0.0457631097953	0.0358809173785	0.0840654446615	0.20064918891521	0.2634190758672	True	0.3172338536875	0.0957144775062	3.3143769046530	0.0009184761047	0.0728983198039	1.35067121536173	0.001377714157143002
ALLERG_RHINITIS	IEAA	in_house_ST60	0.3531147710661	0.0457631097953	-0.0315181705443	0.0726216194369	0.11082020355209	0.2634190758672	True	0.3846329416104	0.0858379975756	4.4809169886763	7.4323016919677	0.0635985116490	1.46810389702147	2.972920676787097 2e-05
ALLERG_RHINITIS	PhenoAge	in_house_ST60	0.3531147710661	0.0457631097953	-0.0452502409275	0.0929424075916	0.13691687795194	0.2634190758672	True	0.3983650119937	0.1035980373707	3.8452949699054	0.0001204075075	0.0804936006388	7.45853576572941	0.000288978018065317
L12_ATOPIC	GrimAge	in_house_ST60	0.2502684670316	0.0331910989101	0.0908989004846	0.0726627896720	0.23331796824185	0.1852139131676	False	0.1593695665469	0.0798844794048	1.9950003772228	0.0460428387222	0.0630061283582	0.01142480887185	0.061390451629602154
L12_ATOPIC	Hannum	in_house_ST60	0.2502684670316	0.0331910989101	0.1162377222538	0.0706656213638	0.25474234012701	0.1852139131676	False	0.1340307447778	0.0780722683774	1.7167522804607	0.0860244433230	0.0612356878084	0.02861367005856	0.10322933198766107
L12_ATOPIC	IEAA	in_house_ST60	0.2502684670316	0.0331910989101	0.1611780734713	0.0664807991891	0.29148043988216	0.1852139131676	False	0.0890903935602	0.0743057582405	1.1989702503524	0.2305395136120	0.0575740820698	0.12176573730626	0.2514976512131437
L12_ATOPIC	PhenoAge	in_house_ST60	0.2502684670316	0.0331910989101	0.2736092302557	0.1022075442260	0.47393601693875	0.1852139131676	False	-0.0233407632241	0.1074617659615	-0.2172006295940	0.8280519940510	0.0903086398624	0.79605540064693	0.828051994051049
ALLERG_ASTHMA	GrimAge	independent_ST12	0.458481	0.038021	0.166996	0.071582	0.30729672	0.38395984000000	True	0.291485	0.0810529405080	3.5962297959725	0.0003228626093			0.000581152696914 4772
ALLERG_ASTHMA	Hannum	independent_ST12	0.458481	0.038021	0.117546	0.066063	0.24702947999999	0.38395984000000	True	0.3409350000000	0.0762228076759	4.4728738076610	7.7175325798128			5.476812867943 73e-05
ALLERG_ASTHMA	PhenoAge	independent_ST12	0.458481	0.038021	0.119459	0.067532	0.25182171999999	0.38395984000000	True	0.3390220000000	0.0774994675142	4.3745074756474	1.2170695262097			5.476812867943 73e-05
ALLERG_RHINITIS	GrimAge	independent_ST12	0.354849	0.046426	-0.004528	0.086854	0.16570584	0.26385404	True	0.359377	0.0984834544073	3.6491104232962	0.0002631499802			0.000581152696914 4772
ALLERG_RHINITIS	Hannum	independent_ST12	0.354849	0.046426	0.038893	0.08597	0.20739420000000	0.26385404	True	0.315956	0.0977047305712	3.2337840568487	0.0012216178524			0.001832426778615 8267
ALLERG_RHINITIS	PhenoAge	independent_ST12	0.354849	0.046426	-0.014554	0.082792	0.14771832	0.26385404	True	0.3694030000000	0.0949204337326	3.8917120947880	9.9539320416442			0.000298617961249 3266
L12_ATOPIC	GrimAge	independent_ST12	0.250524	0.032992	0.090611	0.073199	0.23408104	0.18585968000000	False	0.1599130000000	0.0802905079383	1.9916800143145	0.0464061779140			0.059665085889526105
L12_ATOPIC	Hannum	independent_ST12	0.250524	0.032992	0.110542	0.070774	0.24925904000000	0.18585968000000	False	0.1399820000000	0.0780860495863	1.7926633597365	0.0730267686031			0.082155114678544
L12_ATOPIC	PhenoAge	independent_ST12	0.250524	0.032992	0.202576	0.071685	0.3430786	0.18585968000000	False	0.0479480000000	0.0789126814206	0.6076082973837	0.5434472963734			0.543447296373478 9

Supplementary Table 135 - r16_steiger_directionality

ST135_steiger_directionality.csv

exposure	outcome	n_snp	median_r2_exposure	median_r2_outcome	sumR2_ratio	n_correct_direction	prop_correct	binom_p	n_steiger_sig_correct	n_steiger_sig_wrong
Frailty	ALLERG_ASTHMA	71	0.00021559183443180003	0.0002825655082128717	0.39106068399829913	34	0.4788732394366197	0.8125886494061619	2	10
GrimAge	ALLERG_ASTHMA	5	0.0144367247448398	0.0002425855114906745	13.614298896894832	5	1.0	0.0625	5	0
Hannum	ALLERG_ASTHMA	56	0.029256985804817587	4.724185027852495e-05	223.97670192653524	56	1.0	2.7755575615628914e-17	56	0
IEAA	ALLERG_ASTHMA	133	0.06930070473387501	0.00021141916874701245	294.9047287830103	133	1.0	1.8367099231598242e-40	133	0
PhenoAge	ALLERG_ASTHMA	89	0.06360905938334215	0.00021141916874701245	282.1106168730458	89	1.0	3.2311742677852644e-27	89	0
composite_index	ALLERG_ASTHMA	94	4.3984395344583945e-05	0.00017558330977699883	0.2374422865651545	35	0.3723404255319149	0.017209595585930455	3	19
Frailty	ALLERG_RHINITIS	71	0.00021559183443180003	4.983177615670846e-05	1.389408862178681	57	0.8028169014084507	2.6659171387086086e-07	0	1
GrimAge	ALLERG_RHINITIS	5	0.0144367247448398	0.0002372308522332248	38.558478739248116	5	1.0	0.0625	5	0
Hannum	ALLERG_RHINITIS	56	0.029256985804817587	5.6751209869380066e-05	202.82678419119642	56	1.0	2.7755575615628914e-17	56	0
IEAA	ALLERG_RHINITIS	133	0.06930070473387501	2.1998978482305316e-06	1225.0808440136477	133	1.0	1.8367099231598242e-40	133	0
PhenoAge	ALLERG_RHINITIS	89	0.06360905938334215	1.3421851650869673e-06	3953.379649210177	89	1.0	3.2311742677852644e-27	89	0
composite_index	ALLERG_RHINITIS	94	4.3984395344583945e-05	9.602596658690233e-05	0.4579083577097344	29	0.30851063829787234	0.000261683232376927	5	17
Frailty	L12_ATOPIC	71	0.00021559183443180003	6.941278048903184e-05	1.4989237827572912	51	0.7183098591549296	0.000303163413558493	19	3
GrimAge	L12_ATOPIC	5	0.0144367247448398	0.0003474033862652553	70.14449861856647	5	1.0	0.0625	5	0
Hannum	L12_ATOPIC	56	0.029256985804817587	4.274986618343803e-06	1390.8424058550636	56	1.0	2.7755575615628914e-17	56	0
IEAA	L12_ATOPIC	133	0.06930070473387501	3.9556592196646e-06	2572.359480639485	133	1.0	1.8367099231598242e-40	133	0
PhenoAge	L12_ATOPIC	89	0.06360905938334215	3.9504746635591085e-06	3673.1225741843823	89	1.0	3.2311742677852644e-27	89	0
composite_index	L12_ATOPIC	94	4.3984395344583945e-05	6.83661409210324e-05	0.5828131800253651	32	0.3404255319148936	0.002587308956921308	9	15

Supplementary Table 135b - r16_steiger_directionality_persnp

ST135b_steiger_directionality_persnp.csv

exposure	outcome	snp	chrom	pos	r2_exposure	r2_outcome	steiger_z	steiger_p	direction	steiger_sig	in_HLA	in_5q31	in_immune_locus_r6
Frailty	ALLERG_ASTHMA	rs9275160	6	32684843	0.00065462647525679	0.00195914373123456	-2.0291435944529175	0.04244366673075552	wrong	True	True	False	True
Frailty	ALLERG_ASTHMA	rs9273344	6	32655681	0.0008471563900799	0.0067139308865286	-5.702250768361949	1.182357131829151e-	wrong	True	True	False	True
Frailty	ALLERG_ASTHMA	rs77062257	6	32652987	0.00066901669125119	1.9565852743308336	2.294237476740179	0.02177686172355622	correct	True	True	False	True
Frailty	ALLERG_ASTHMA	rs28746860	6	32667256	0.00059091586133579	0.0014932253808608	-1.5601901138186454	0.11871496118218977	wrong	False	True	False	True
Frailty	ALLERG_ASTHMA	rs1281935	6	32616043	0.00053855843834879	0.0004986955844965	0.09254850693865185	0.9262622535934678	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs6910879	6	32592962	0.00047955276288000	2.07126545309843758	2.331568433660818	0.01972340605944908	correct	True	True	False	True
Frailty	ALLERG_ASTHMA	rs1281934	6	32616605	0.00039956115199999	0.00061520137673229	-0.520545315936362	0.6026835519539894	wrong	False	True	False	True
Frailty	ALLERG_ASTHMA	rs9268362	6	32365564	0.00034373248683179	0.00124133728751426	-1.8150382393160247	0.06951803319611729	wrong	False	True	False	True
Frailty	ALLERG_ASTHMA	rs28746878	6	32669738	0.0003599528802552	0.00017406549146009	0.6253399893864799	0.5317479393154416	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs34422230	6	32809218	0.00029468851015679	5.581853014461968e-	1.0308389815772188	0.30261633469526805	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs2523668	6	31470319	0.00029184560761519	0.00162462031807495	-2.550990287486771	0.01074173192461985	wrong	True	True	False	True
Frailty	ALLERG_ASTHMA	rs35520662	6	32780252	0.00028393838011019	7.87549519343177e-0	0.8472160838131249	0.39687469020648536	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs34043227	6	32780813	0.0002839383801102	7.189538167189274e-	0.8890124122128028	0.37399640911271903	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs34153000	6	32779239	0.0002839383801102	7.87549519343177e-0	0.8472160838131252	0.39687469020648525	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs34408145	6	32787400	0.00028280149084079	7.87549519343177e-0	0.8436288017648258	0.3988768645607995	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs17213756	6	32820219	0.00027983501568000	7.254803464636909e-	0.8740488097749854	0.3820916731740389	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs41316548	6	32829390	0.00028010220564479	7.20970715024187e-0	0.877137536175975	0.38041193876627366	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs35921062	6	32806639	0.00027827673996719	7.254859929273695e-	0.8691907257446561	0.3847428192527469	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs35486885	6	32795729	0.00027715125379979	7.215358626783635e-	0.8679869445309685	0.3854014797633962	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs34710970	6	32800338	0.00027715125379979	7.255004350939932e-	0.8655861245643287	0.3867171674324179	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs41258084	6	32813180	0.0002764533547008	6.874648362558348e-	0.8872064407750931	0.37496776543047494	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs41316241	6	32815018	0.0002764533547008	6.911550344193748e-	0.8848350968316739	0.37624557719342766	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs41317090	6	32834577	0.00027314083550000	7.014217576495511e-	0.8698118700563744	0.38440322353968526	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs2857702	6	31609506	0.0002734913721848	0.00020853967739017	0.22679578566788475	0.8205825408758651	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs2228396	6	32830032	0.00025174418734079	0.0004319959371394	-0.526137206001214	0.5987928795041488	wrong	False	True	False	True
Frailty	ALLERG_ASTHMA	rs8192574	6	32201368	0.0002463022333152	0.00030125743471868	-0.1787935652973846	0.8580997980203036	wrong	False	True	False	True
Frailty	ALLERG_ASTHMA	rs41268932	6	32186934	0.00024584563303820	0.00028650212313460	-0.1340262570311122	0.8933818116398868	wrong	False	True	False	True
Frailty	ALLERG_ASTHMA	rs2022059	6	32188712	0.0002444178555	0.00031029171089512	-0.2127830281039387	0.8314962022179593	wrong	False	True	False	True
Frailty	ALLERG_ASTHMA	rs9268835	6	32460338	0.00023710552632000	0.00322385154829446	-4.543725249263305	5.526868660918331e-	wrong	True	True	False	True
Frailty	ALLERG_ASTHMA	rs137893789	6	32087709	0.00023601935953919	0.00028034846806135	-0.1483717310887143	0.8820494084459272	wrong	False	True	False	True
Frailty	ALLERG_ASTHMA	rs2844456	6	31896897	0.00023322421142720	4.719137349741666e-	0.900921347867919	0.3676301398387656	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs2856437	6	32189587	0.0002320384983822	0.00030806926343148	-0.2491212568258773	0.803266987134	wrong	False	True	False	True
Frailty	ALLERG_ASTHMA	rs34562262	6	32053184	0.00022492022799020	0.00028256550821287	-0.1947268455566630	0.8456067944765947	wrong	False	True	False	True
Frailty	ALLERG_ASTHMA	rs644045	6	31916180	0.0002214071304192	0.00077286192414030	-1.3949954036046548	0.1630172256319664	wrong	False	True	False	True
Frailty	ALLERG_ASTHMA	rs35029304	6	32941756	0.00022662158310380	0.00037335073379360	-0.458716849592055	0.6464375125638111	wrong	False	True	False	True
Frailty	ALLERG_ASTHMA	rs82334	4	3223644	0.00021559183443180	0.00140275762275369	-2.4500409648979122	0.01428399628170486	wrong	True	False	False	False
Frailty	ALLERG_ASTHMA	rs41268928	6	32179380	0.00021272174407680	4.498645363373112e-	1.3413742836560207	0.17979896269981743	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs805284	6	31714252	0.00020823197050880	4.36286740798269466	0.8398407751091829	0.4009976679163779	correct	False	True	False	True
Frailty	ALLERG_ASTHMA	rs41268924	6	32167360	0.00020656499834880	0.00023605276108354	-0.1065057495489925	0.9151810938772382	wrong	False	True	False	True
Frailty	ALLERG_ASTHMA	rs10947233	6	32156647	0.00020487298007999	7.736724536876913e-	1.2402859711966367	0.21486963943118187	correct	False	True	False	True

(showing first 40 of 1344 rows; full data in the accompanying CSV)

Supplementary Table 136 - r16_eosinophil_free_panukbb_mr

ST136_eosinophil_free_panukbb_mr.csv

instrument	outcome	n_snp	mean_F	ivw_beta	ivw_se	ivw_or	or_lo	or_hi	ivw_p	ivw_re_p	wm_or	egger_or	egger_int_p	Q	Q_p	I2_pct
eosinophil_free_inc	UKB_asthma	81	1212.76742329220	0.6183298384892	0.1183215316879	1.85582592302030	1.04849699019870	3.28478754707860	1.73362702232960	0.0337901724027(	2.73895923174171	1.9320006212991(	3.1431098619833(	484.93118131720	2.9245309237144	83.50281378427806
eosinophil_only	UKB_asthma	17	946.97422531732	1.10717619001630	0.0529600091909	3.02580202936751	1.97457732064779	4.63667734111381	4.73278194705361	3.69097733166110	4.36928303522434	3.02287134670460	0.0654120014783(	270.52349038656(	3.1279576812583(	94.08554134166444
full_index	UKB_asthma	98	1166.66043997001	1.02558627286331	0.0483387972233(	2.78872993808009	2.13550676241410	3.64176541344800	6.70647941039860	4.99809366246849	4.22603401186184	2.80326908494960	0.0001341195178(	769.67510294398(	2.9698123536953(	87.39727975752656
eosinophil_free_inc	UKB_allergic_rhinit	81	1212.76742329220	0.42547786879342	0.4949307033285(	1.53032153786701	0.57160405557290	4.09703882683700	0.38997009505090	0.39709566895422	1.47433865854151	1.65614420018924	0.0217059233058(	82.446707800251	0.4035786148179(	2.967623408540377
eosinophil_only	UKB_allergic_rhinit	17	946.97422531732	0.45838793619220	0.2222300403542	1.58152241385472	1.02307935731925	2.44478898692545	0.03914369308494	0.03914369308494	1.95465623383861	1.57970779881900	0.7063581586427(	15.216996799287(	0.5088064375427(	0.0
full_index	UKB_allergic_rhinit	98	1166.66043997001	0.45286612958480	0.2027312072397(	1.57281361920785	1.05564120762814	2.34335554816380	0.02549456051410	0.02600211122201	1.95465623383861	1.59112042766695	0.0653066323955(	97.667384231932(	0.4618929123242(	0.683323544682428
eosinophil_free_inc	UKB_urticaria_cont	81	1212.76742329220	-0.9467918733774	0.9191223557002(	0.38798372994191	0.06403844386639	2.35064073408459	0.30296105582960	0.30296105582960	0.24513474780800	0.39384320214725	0.8114106911773(	54.184822714206(	0.9880815281048(	0.0
eosinophil_only	UKB_urticaria_cont	17	946.97422531732	0.27508830245514	0.4119901593801(	1.31664693289134	0.58718672458682	2.95231324773564	0.50432127758889	0.50432127758889	1.04825239747149	1.30563578319918	0.1559211996703(	13.831718060853(	0.6112489480738(	0.0
full_index	UKB_urticaria_cont	98	1166.66043997001	0.07066004705150	0.3759494738379(	1.07321632079401	0.51365629618004	2.24234235963681	0.85091513063891	0.85091513063891	1.04825239747149	1.06827190456991	0.6899327891246(	69.488161364901(	0.9842832515888(	0.0

Supplementary Table 137 - r17_canonical_ldsc_rg

ST137_canonical_ldsc_rg.csv

trait1	trait2	ld_reference	n_valid	rg	rg_se	z	p	q_bh_mhc_free_family	h2_trait2	h2_trait2_se	h2_int	gcov_int
GrimAge	ALLERG_ASTHMA	canonical_bulik_ldsc_mh	1137100	0.1976	0.0764	2.5856	0.0097	0.0342	0.0438	0.0048	1.0835	
GrimAge	ALLERG_RHINITIS	canonical_bulik_ldsc_mh	1137124	-0.022	0.0932	-0.2364	0.8131	0.8734285714285714	0.0159	0.002	1.0606	
GrimAge	L12_ATOPIC	canonical_bulik_ldsc_mh	1137128	0.0508	0.074	0.6868	0.4922	0.6711818181818182	0.0293	0.0047	1.1187	0.0025
Hannum	ALLERG_ASTHMA	canonical_bulik_ldsc_mh	1137360	0.1318	0.0716	1.8419	0.0655	0.14035714285714287	0.0439	0.0048	1.0833	
Hannum	ALLERG_RHINITIS	canonical_bulik_ldsc_mh	1137383	0.0356	0.0895	0.3977	0.6909	0.863625	0.0158	0.002	1.0615	
Hannum	L12_ATOPIC	canonical_bulik_ldsc_mh	1137386	0.0844	0.0704	1.1985	0.2307	0.34604999999999997	0.0293	0.0047	1.1186	-0.004
IEAA	ALLERG_ASTHMA	canonical_bulik_ldsc_mh	1137104	0.1032	0.0641	1.6108	0.1072	0.17866666666666667	0.0438	0.0048	1.0835	
IEAA	ALLERG_RHINITIS	canonical_bulik_ldsc_mh	1137128	-0.0003	0.0741	-0.0036	0.9971	0.9971	0.0159	0.002	1.0605	
IEAA	L12_ATOPIC	canonical_bulik_ldsc_mh	1137132	0.1215	0.0547	2.2205	0.0264	0.066	0.0292	0.0047	1.1187	-0.007
PhenoAge	ALLERG_ASTHMA	canonical_bulik_ldsc_mh	1137104	0.1193	0.0727	1.6407	0.1009	0.17866666666666667	0.0438	0.0048	1.0835	
PhenoAge	ALLERG_RHINITIS	canonical_bulik_ldsc_mh	1137128	-0.0205	0.0877	-0.2337	0.8152	0.8734285714285714	0.0159	0.002	1.0606	
PhenoAge	L12_ATOPIC	canonical_bulik_ldsc_mh	1137132	0.188	0.0743	2.5291	0.0114	0.0342	0.0293	0.0047	1.1183	-0.0026
Frailty	ALLERG_ASTHMA	canonical_bulik_ldsc_mh	1140501	0.4672	0.0393	11.8941	1.2705e-32	1.9057499999999998e-	0.0445	0.0048	1.0803	
Frailty	ALLERG_RHINITIS	canonical_bulik_ldsc_mh	1140538	0.3613	0.0463	7.8116	5.6461e-15	4.234575e-14	0.016	0.0019	1.0598	
Frailty	L12_ATOPIC	canonical_bulik_ldsc_mh	1140540	0.2464	0.0337	7.3203	2.4749e-13	1.23745e-12	0.0289	0.0041	1.1218	-0.0061
Frailty	ALLERG_ASTHMA	canonical_bulik_ldsc_mh	1143177	0.4918	0.0434	11.3373	8.5701e-30		0.0371	0.0044	1.105	
Frailty	ALLERG_RHINITIS	canonical_bulik_ldsc_mh	1143214	0.3798	0.0527	7.202	5.9328e-13		0.0132	0.0017	1.0732	
Frailty	L12_ATOPIC	canonical_bulik_ldsc_mh	1143216	0.2792	0.0389	7.1754	7.2083e-13		0.0243	0.0041	1.1448	0.0033

Supplementary Table 138 - r17_canonical_rg_contrast

ST138_canonical_rg_contrast.csv

outcome	clock	source	rg_frailty	se_frailty	rg_clock	se_clock	clock_rg_hi95	frailty_rg_lo95	ci_nonoverlap	diff	se_diff_rho0	z_rho0	p_rho0	se_diff_rho0p5	p_rho0p5	q_bh_rho0	q_bh_rho0p5
ALLERG_ASTHMA	GrimAge	canonical_bulik_lo	0.4672	0.0393	0.1976	0.0764	0.347344	0.390172	True	0.2696	0.0859153653312	3.1379718745370	0.0017012119998	0.0661734841156	4.6184809542063	0.0025518179998	6.92772143130958e-05
ALLERG_ASTHMA	Hannum	canonical_bulik_lo	0.4672	0.0393	0.1318	0.0716	0.272136	0.390172	True	0.3354000000000	0.0816764960071	4.1064445268414	4.0179589905788	0.0621061188611	6.6479842306823	0.0001205387697	1.99439526920471146e-07
ALLERG_ASTHMA	IEAA	canonical_bulik_lo	0.4672	0.0393	0.1032	0.0641	0.2288359999999	0.390172	True	0.364	0.0751884299609	4.8411703793896	1.2907664105075	0.0559836583299	7.9313214220499	1.5489196926090	9.51758570645992e-10
ALLERG_ASTHMA	PhenoAge	canonical_bulik_lo	0.4672	0.0393	0.1193	0.0727	0.261792	0.390172	True	0.3479	0.0826424830217	4.2096992645831	2.5571083020845	0.0630291202540	3.3963516694947	0.0001205387697	1.35854066779790 97e-07
ALLERG_RHINITIS	GrimAge	canonical_bulik_lo	0.3613	0.0463	-0.022	0.0932	0.160672	0.270552	True	0.3833000000000	0.1040669496045	3.6832058733003	0.0002303190004	0.0807141251578	2.0456749369669	0.0004606380009	4.09134987393386 8e-06
ALLERG_RHINITIS	Hannum	canonical_bulik_lo	0.3613	0.0463	0.0356	0.0895	0.2110199999999	0.270552	True	0.3257	0.1007667603925	3.2322166429817	0.0012283389152	0.0775247702350	2.6545949914078	0.0021057238547	4.55073427098485 9e-05
ALLERG_RHINITIS	IEAA	canonical_bulik_lo	0.3613	0.0463	-0.0003	0.0741	0.144936	0.270552	True	0.3616000000000	0.0873756258918	4.1384539030071	3.4965410449902	0.0648357154660	2.4445876405583	0.0001205387697	1.35854066779790 97e-07
ALLERG_RHINITIS	PhenoAge	canonical_bulik_lo	0.3613	0.0463	-0.0205	0.0877	0.151392	0.270552	True	0.3818000000000	0.0991714676708	3.8498976466397	0.0001181671947	0.0759899335438	5.0514120674453	0.0002836012675	1.21233889618699 381e-06
L12_ATOPIC	GrimAge	canonical_bulik_lo	0.2464	0.0337	0.0508	0.074	0.1958400000000	0.180348	False	0.1956	0.0813122991926	2.4055401451199	0.0161485775871	0.0641707877464	0.0023028107407	0.0215314367829	0.00307041432105 4171
L12_ATOPIC	Hannum	canonical_bulik_lo	0.2464	0.0337	0.0844	0.0704	0.222384	0.180348	False	0.162	0.0780503042915	2.0755844768383	0.0379323983648	0.0609866378807	0.0078998711492	0.0455188780371	0.00947984537909291
L12_ATOPIC	IEAA	canonical_bulik_lo	0.2464	0.0337	0.1215	0.0547	0.228712	0.180348	False	0.1249000000000	0.0642478015188	1.9440353918320	0.0518911833470	0.0477952926552	0.0089691429032	0.0566085636513	0.00978451953086 7675
L12_ATOPIC	PhenoAge	canonical_bulik_lo	0.2464	0.0337	0.188	0.0743	0.3336280000000	0.180348	False	0.0584000000000	0.0815854153632	0.7158142143415	0.4741060742386	0.0644381098419	0.3647792119613	0.4741060742386	0.36477921196134 74

Supplementary Table 139 - r17_treg_donor_pseudobulk

ST139_treg_donor_pseudobulk.csv

cell_type	gene	donor	gene_counts	total_counts	age	n_cells
regulatory T cell	IL4	TSP1	0.0	2921.0	59.0	23709
regulatory T cell	IL4	TSP10	2.0	149486.0	33.0	12459
regulatory T cell	IL4	TSP14	189.0	2633424.0	59.0	132053
regulatory T cell	IL4	TSP2	1783.0	266647.0	61.0	71621
regulatory T cell	IL4	TSP20	18.0	444729.0	56.0	9546
regulatory T cell	IL4	TSP21	9.0	442463.0	45.0	95511
regulatory T cell	IL4	TSP25	7.0	264821.0	60.0	98467
regulatory T cell	IL4	TSP26	0.0	70294.0	37.0	3895
regulatory T cell	IL4	TSP27	11.0	1294468.0	56.0	57162
regulatory T cell	IL4	TSP4	0.0	6056.0	38.0	6340
regulatory T cell	IL4	TSP6	0.0	50034.0	67.0	7414
regulatory T cell	IL4	TSP7	637.0	1024081.0	69.0	51945
regulatory T cell	IL13	TSP1	0.0	2921.0	59.0	23709
regulatory T cell	IL13	TSP10	0.0	149486.0	33.0	12459
regulatory T cell	IL13	TSP14	0.0	2633424.0	59.0	132053
regulatory T cell	IL13	TSP2	123.0	266647.0	61.0	71621
regulatory T cell	IL13	TSP20	0.0	444729.0	56.0	9546
regulatory T cell	IL13	TSP21	0.0	442463.0	45.0	95511
regulatory T cell	IL13	TSP25	0.0	264821.0	60.0	98467
regulatory T cell	IL13	TSP26	0.0	70294.0	37.0	3895
regulatory T cell	IL13	TSP27	0.0	1294468.0	56.0	57162
regulatory T cell	IL13	TSP4	0.0	6056.0	38.0	6340
regulatory T cell	IL13	TSP6	0.0	50034.0	67.0	7414
regulatory T cell	IL13	TSP7	19.0	1024081.0	69.0	51945
regulatory T cell	GATA3	TSP1	0.0	2921.0	59.0	23709
regulatory T cell	GATA3	TSP10	0.0	149486.0	33.0	12459
regulatory T cell	GATA3	TSP14	6.0	2633424.0	59.0	132053
regulatory T cell	GATA3	TSP2	0.0	266647.0	61.0	71621
regulatory T cell	GATA3	TSP20	6.0	444729.0	56.0	9546
regulatory T cell	GATA3	TSP21	1.0	442463.0	45.0	95511
regulatory T cell	GATA3	TSP25	0.0	264821.0	60.0	98467
regulatory T cell	GATA3	TSP26	0.0	70294.0	37.0	3895
regulatory T cell	GATA3	TSP27	3.0	1294468.0	56.0	57162
regulatory T cell	GATA3	TSP4	0.0	6056.0	38.0	6340
regulatory T cell	GATA3	TSP6	0.0	50034.0	67.0	7414
regulatory T cell	GATA3	TSP7	0.0	1024081.0	69.0	51945
regulatory T cell	FOXP3	TSP1	1.0	2921.0	59.0	23709
regulatory T cell	FOXP3	TSP10	132.0	149486.0	33.0	12459
regulatory T cell	FOXP3	TSP14	1848.0	2633424.0	59.0	132053
regulatory T cell	FOXP3	TSP2	10.0	266647.0	61.0	71621

(showing first 40 of 60 rows; full data in the accompanying CSV)

Supplementary Table 139b - r17_treg_detection_by_donor

ST139b_treg_detection_by_donor.csv

gene	donor	age	n_cells	n_positive	detection
IL4	TSP1	59.0	1	0	0.0
IL4	TSP10	33.0	74	2	0.02702702702702703
IL4	TSP14	59.0	1042	91	0.0873320537428023
IL4	TSP2	61.0	87	8	0.09195402298850575
IL4	TSP20	56.0	224	17	0.07589285714285714
IL4	TSP21	45.0	217	9	0.041474654377880185
IL4	TSP25	60.0	96	7	0.07291666666666667
IL4	TSP26	37.0	36	0	0.0
IL4	TSP27	56.0	563	11	0.019538188277087035
IL4	TSP4	38.0	3	0	0.0
IL4	TSP6	67.0	11	0	0.0
IL4	TSP7	69.0	295	28	0.09491525423728814
IL13	TSP1	59.0	1	0	0.0
IL13	TSP10	33.0	74	0	0.0
IL13	TSP14	59.0	1042	0	0.0
IL13	TSP2	61.0	87	13	0.14942528735632185
IL13	TSP20	56.0	224	0	0.0
IL13	TSP21	45.0	217	0	0.0
IL13	TSP25	60.0	96	0	0.0
IL13	TSP26	37.0	36	0	0.0
IL13	TSP27	56.0	563	0	0.0
IL13	TSP4	38.0	3	0	0.0
IL13	TSP6	67.0	11	0	0.0
IL13	TSP7	69.0	295	4	0.013559322033898305
GATA3	TSP1	59.0	1	0	0.0
GATA3	TSP10	33.0	74	0	0.0
GATA3	TSP14	59.0	1042	4	0.003838771593090211
GATA3	TSP2	61.0	87	0	0.0
GATA3	TSP20	56.0	224	2	0.008928571428571428
GATA3	TSP21	45.0	217	1	0.004608294930875576
GATA3	TSP25	60.0	96	0	0.0
GATA3	TSP26	37.0	36	0	0.0
GATA3	TSP27	56.0	563	3	0.005328596802841918 5
GATA3	TSP4	38.0	3	0	0.0
GATA3	TSP6	67.0	11	0	0.0
GATA3	TSP7	69.0	295	0	0.0
FOXP3	TSP1	59.0	1	1	1.0
FOXP3	TSP10	33.0	74	31	0.4189189189189189
FOXP3	TSP14	59.0	1042	190	0.18234165067178504
FOXP3	TSP2	61.0	87	7	0.08045977011494253

(showing first 40 of 60 rows; full data in the accompanying CSV)

Supplementary Table 140 - r17_treg_age_analysis

ST140_treg_age_analysis.csv

gene	n_donors	total_gene_counts	primary_rr_per_decade	primary_ci_lo	primary_ci_hi	primary_p	q_primary_family	spearman_rho	spearman_p	pooled_poisson_rr_per_decade	pooled_poisson_p
IL4	12	2656.0	1.9733093478272752	0.7754998924592041	5.021212536695936	0.1842072853927855	0.46051821348196376	0.4411164427501008	0.15115751331223262	3.1537446101976787	2.9462324825365933e-254
IL13	12	142.0	1.1413254852774315	0.327093942216834	3.9824151267229557	0.8399174992610294	0.8399174992610294	0.5718129807661293	0.05207589417623057	2.658715742835054	1.0394564386795623e-11
GATA3	12	16.0	0.682392663151488	0.30080454838632353	1.5480475585260531	0.382032892874173	0.624166180522244	-0.16491579051513755	0.6085141755527179	0.7450355899054658	0.33435834216766236
FOXP3	12	2713.0	0.8173225807199835	0.4649645901292211	1.4367033858839047	0.49933294441779524	0.624166180522244	-0.0912286317569255	0.7779647906315891	1.2877871540845236	5.094091644942401e-18
IL5	12	5034.0	1.721795700411643	1.1492358045074993	2.5796102264900114	0.024973862083109283	0.12486931041554641	0.5676634425762845	0.05420083742111326	1.8385195821880007	4.674842780038236e-152

Supplementary Table 141 - r17_immune_cell_source_rate

ST141_cell_source_expression_rate.csv

gene	cell_type	n_cells	rate_per_1e4
IL4	B cell	114495	2.9551576745169608
IL4	CD4-positive, alpha-beta	97130	3.0455174275063124
IL4	CD4-positive, alpha-beta	4685	4.097405822699664
IL4	CD8-positive, alpha-beta	76277	3.592952506442057
IL4	CD8-positive, alpha-beta	731	66.21665899237034
IL4	T cell	6290	0.8226198121212356
IL4	activated CD4-positive,	572	2.6052835455014294
IL4	activated CD8-positive,	1147	0.13744004178177271
IL4	basophil	1669	1.851486531508992
IL4	classical monocyte	18200	0.2637686724247896
IL4	erythrocyte	14611	1.436722377099139
IL4	erythroid progenitor cel	971	13.919773318717894
IL4	gamma-delta T cell	1977	0.07308421222318832
IL4	granulocyte	769	6.08756923839647
IL4	hematopoietic precursor	2140	0.26780900418863995
IL4	hematopoietic stem cell	858	18.971826501503763
IL4	innate lymphoid cell	971	87.00894458139172
IL4	intermediate monocyte	5478	0.14611786170728916
IL4	macrophage	69072	0.8783093674172948
IL4	mast cell	6222	6.935732272497646
IL4	mature NK T cell	5394	2.6236554962504846
IL4	monocyte	25753	0.8466130953032079
IL4	mononuclear phagocyte	5336	0.15254874675205435
IL4	myeloid dendritic cell	571	17.29132270751622
IL4	myeloid leukocyte	1002	0.12900590994502875
IL4	naive thymus-derived CD4	9827	4.872236518681197
IL4	naive thymus-derived CD8	1056	0.2764196396487006
IL4	natural killer cell	14231	5.676544535406991
IL4	neutrophil	69539	0.04810985792710168
IL4	non-classical monocyte	1284	1.367834493360776
IL4	plasma cell	27348	3.8874355397995792
IL4	regulatory T cell	2649	3.994330937536845
IL4	tissue-resident macropha	1355	0.12327894191701444
IL5	B cell	114495	9.980129855579987
IL5	CD4-positive, alpha-beta	97130	9.929988526891998
IL5	CD4-positive, alpha-beta	4685	4.009203349021197
IL5	CD8-positive, alpha-beta	76277	16.959666899344374
IL5	CD8-positive, alpha-beta	731	147.76558235935084
IL5	T cell	6290	7.1335943284045555
IL5	activated CD4-positive,	572	62.88484113483714

(showing first 40 of 132 rows; full data in the accompanying CSV)

Supplementary Table 141b - r17_immune_cell_source_fraction

ST141_cell_source_expression.csv

gene	cell_type	n_cells	n_expressing	pct_expressing	total_counts
IL18	common myeloid progenito	456	6	1.3157894736842106	9.0
IL18	myeloid cell	81	1	1.2345679012345678	199.0
IL18	plasmacytoid dendritic c	99	1	1.0101010101010102	1.0
IL18	microglial cell	410	3	0.7317073170731707	3.0
IL18	naive thymus-derived CD8	1056	4	0.3787878787878788	4.0
IL18	classical monocyte	18200	62	0.3406593406593406	67.0
IL18	monocyte	25753	80	0.3106434201840562	218.0
IL18	innate lymphoid cell	971	3	0.3089598352214212	6.0
IL18	colon macrophage	328	1	0.3048780487804878	1.0
IL18	intermediate monocyte	5478	16	0.2920774005111354	16.0
IL18	plasma cell	27348	75	0.2742430890741553	973.0
IL18	CD8-positive, alpha-beta	731	2	0.2735978112175102	8073.0
IL18	non-classical monocyte	1284	3	0.2336448598130841	4.0
IL18	macrophage	69072	156	0.2258512856150104	825.0
IL18	mast cell	6222	12	0.1928640308582449	1228.0
IL18	hematopoietic precursor	2140	4	0.1869158878504672	4.0
IL18	tissue-resident macropha	1355	2	0.1476014760147601	2.0
IL18	CD4-positive, alpha-beta	4685	6	0.128068303094984	78.0
IL18	B cell	114495	143	0.12489628368051	1746.0
IL18	natural killer cell	14231	16	0.1124306092333637	16.0
IL18	CD8-positive, alpha-beta	76277	81	0.1061919058169566	1561.0
IL18	erythroid progenitor cel	971	1	0.1029866117404737	1.0
IL18	mature NK T cell	5394	5	0.0926955876900259	5.0
IL18	CD4-positive, alpha-beta	97130	82	0.0844229383300731	123.0
IL18	mononuclear phagocyte	5336	4	0.0749625187406296	4.0
IL18	neutrophil	69539	39	0.0560836365205136	42.0
IL18	naive thymus-derived CD4	9827	5	0.0508802279434211	7.0
IL18	gamma-delta T cell	1977	1	0.0505816894284269	1.0
IL18	regulatory T cell	2649	1	0.0377500943752359	1.0
IL18	T cell	6290	1	0.0158982511923688	1.0
IL18	erythrocyte	14611	2	0.0136883170214222	2.0
IL18	Langerhans cell	76	0	0.0	0.0
IL18	T follicular helper cell	6	0	0.0	0.0
IL18	activated CD4-positive,	572	0	0.0	0.0
IL18	activated CD8-positive,	1147	0	0.0	0.0
IL18	basophil	1669	0	0.0	0.0
IL18	erythroid lineage cell	2	0	0.0	0.0
IL18	granulocyte	769	0	0.0	0.0
IL18	hematopoietic cell	100	0	0.0	0.0
IL18	hematopoietic stem cell	858	0	0.0	0.0

(showing first 40 of 180 rows; full data in the accompanying CSV)

Supplementary Table 142 - r17_translational_pseudobulk

ST142_translational_pseudobulk.csv

cell_type	gene	donor	gene_counts	total_counts	age	n_cells
classical monocyte	IL18	TSP1	6.0	1800192.0	59.0	23709
classical monocyte	IL18	TSP10	5.0	7269456.0	33.0	12459
classical monocyte	IL18	TSP11	0.0	31607.0	42.0	2552
classical monocyte	IL18	TSP14	15.0	14011192.0	59.0	132053
classical monocyte	IL18	TSP15	0.0	5640.0	46.0	561
classical monocyte	IL18	TSP17	0.0	117140.0	60.0	1846
classical monocyte	IL18	TSP19	1.0	377702.0	36.0	5410
classical monocyte	IL18	TSP2	9.0	9024728.0	61.0	71621
classical monocyte	IL18	TSP21	15.0	16182866.0	45.0	95511
classical monocyte	IL18	TSP25	2.0	1385438.0	60.0	98467
classical monocyte	IL18	TSP27	0.0	1866320.0	56.0	57162
classical monocyte	IL18	TSP30	0.0	7008.0	26.0	365
classical monocyte	IL18	TSP5	0.0	18488.0	40.0	430
classical monocyte	IL18	TSP6	0.0	275258.0	67.0	7414
classical monocyte	IL18	TSP7	14.0	12331789.0	69.0	51945
classical monocyte	IL18	TSP8	0.0	885923.0	56.0	7390
classical monocyte	IL18	TSP9	0.0	1361859.0	37.0	2724
naive thymus-derived CD8	IL18	TSP14	0.0	531605.0	59.0	132053
naive thymus-derived CD8	IL18	TSP2	3.0	1679080.0	61.0	71621
naive thymus-derived CD8	IL18	TSP21	0.0	143257.0	45.0	95511
naive thymus-derived CD8	IL18	TSP25	0.0	30024.0	60.0	98467
naive thymus-derived CD8	IL18	TSP27	1.0	364911.0	56.0	57162
naive thymus-derived CD8	IL18	TSP7	0.0	253804.0	69.0	51945
mononuclear phagocyte	TNFSF14	TSP20	194874.0	104086.0	56.0	9546
mononuclear phagocyte	TNFSF14	TSP25	62174.0	11102166.0	60.0	98467
mononuclear phagocyte	TNFSF14	TSP27	40335.0	6296350.0	56.0	57162
tissue-resident macropha	TNFSF14	TSP14	22533.0	2377627.0	59.0	132053
tissue-resident macropha	TNFSF14	TSP20	62449.0	9103.0	56.0	9546
tissue-resident macropha	TNFSF14	TSP25	1077.0	216123.0	60.0	98467
tissue-resident macropha	TNFSF14	TSP27	8780.0	1858574.0	56.0	57162

Supplementary Table 143 - r17_translational_age_glm

ST143_translational_age_glm_results.csv

rr_per_decade	ci_lo	ci_hi	p	n_donors	total_gene_counts	gene	cell_type	q_translational_family
0.9120418941302656	0.10901287579164609	7.6304786073028925	0.9322998499040822	6	4.0	IL18	naive thymus-derived CD8	1.0
1.1474626426485302	0.9221782165308677	1.4277831471959044	0.2173924612262229	17	67.0	IL18	classical monocyte	0.8695698449048916
				3	297383.0	TNFSF14	mononuclear phagocyte	1.0
				4	94839.0	TNFSF14	tissue-resident macropha	1.0

Supplementary Table 144 - r17_treg_age_sensitivity

ST144_treg_age_sensitivity.csv

gene	test	rr_per_decade	ci_lo	ci_hi	p	note
IL4	equal_weight_WLS_lograte	1.9733093478272752	0.7754998924592041	5.021212536695936	0.1842072853927855	
IL4	LOO_equal_weight_minmax_	1.5655333029676455		2.624127623786048		exp(min/max LOO beta)
IL4	large_donors_only_WLS	6.881449833092256	0.6193008885070669	76.46420776098253	0.19150647528698153	donors above median Treg
IL13	equal_weight_WLS_lograte	1.1413254852774315	0.327093942216834	3.9824151267229557	0.8399174992610294	
IL13	LOO_equal_weight_minmax_	0.9091875438079953		1.6082224517261696		exp(min/max LOO beta)
IL13	large_donors_only_WLS	5.339373766012374	0.20583132277537913	138.5061895768533	0.3703023126884115	donors above median Treg
GATA3	equal_weight_WLS_lograte	0.682392663151488	0.30080454838632353	1.5480475585260531	0.382032892874173	
GATA3	LOO_equal_weight_minmax_	0.5255726683135287		0.8806204329499677		exp(min/max LOO beta)
GATA3	large_donors_only_WLS	0.4233468345327097	0.14468104568777865	1.2387423760789462	0.19169216766150107	donors above median Treg
FOXP3	equal_weight_WLS_lograte	0.8173225807199835	0.4649645901292211	1.4367033858839047	0.49933294441779524	
FOXP3	LOO_equal_weight_minmax_	0.6882254531867117		1.0177534878838286		exp(min/max LOO beta)
FOXP3	large_donors_only_WLS	1.9593572166611377	0.48068996898885474	7.9866045687570715	0.4012537984233774	donors above median Treg
IL5	equal_weight_WLS_lograte	1.721795700411643	1.1492358045074993	2.5796102264900114	0.024973862083109283	
IL5	LOO_equal_weight_minmax_	1.5721214184862031		1.9785548240207396		exp(min/max LOO beta)
IL5	large_donors_only_WLS	2.2368738911487553	0.8250931120756003	6.064291086270165	0.18877480177251998	donors above median Treg
IL4	ambient_proxy_spearman_t				0.044668642496620906	rho=0.587 n=12 donors
IL5	ambient_proxy_spearman_t				0.11138981647538268	rho=0.483 n=12 donors

Supplementary Table 145 - r17_lung_cell_source

ST145_lung_cell_source_expression.csv

gene	cell_type	n_cells	n_expressing	pct_expressing	total_counts
IL18	mesothelial cell	18	14	77.77777777777777	17.273688316345215
IL18	macrophage	16486	12407	75.2577944922965	10467.114068798488
IL18	pulmonary alveolar type	3116	2223	71.34146341463415	2536.0261881276965
IL18	basophil	1322	910	68.83509833585477	1303.4137329161167
IL18	pulmonary ionocyte	25	17	68.0	18.408747375011444
IL18	basal cell	4015	2375	59.15317559153176	2302.407297298196
IL18	pulmonary alveolar type	11594	6362	54.87321028117992	3444.8953789226944
IL18	intermediate monocyte	2785	1416	50.843806104129264	1504.8313273601234
IL18	myeloid dendritic cell	33	16	48.48484848484848	17.17259225063026
IL18	lung multiciliated epith	1209	578	47.808105872622	433.2228840419557
IL18	respiratory tract goblet	1040	460	44.23076923076923	314.4268364548916
IL18	monocyte	506	217	42.88537549407115	253.57270285487172
IL18	non-classical monocyte	527	183	34.72485768500949	220.0488695120439
IL18	mast cell	3	1	33.333333333333336	0.8103750944137573
IL18	classical monocyte	1569	504	32.122370936902485	534.504541970673
IL18	serous cell of epitheliu	15	3	20.0	1.017066270112991
IL18	club cell	1747	347	19.86262163709216	236.6316543892026
IL18	adventitial cell	581	93	16.006884681583475	85.65518675744534
IL18	plasmacytoid dendritic c	19	3	15.789473684210526	2.1747710704803467
IL18	mature NK T cell	163	23	14.11042944785276	31.41723281145096
IL18	natural killer cell	1019	113	11.089303238469087	137.65304574370384
IL18	alveolar adventitial fib	1113	80	7.187780772686433	67.75916688889265
IL18	CD4-positive, alpha-beta	2124	64	3.0131826741996237	71.49218386411667
IL18	bronchial smooth muscle	220	6	2.727272727272727	5.877671130001545
IL18	neutrophil	371	10	2.6954177897574123	14.852520167827606
IL18	B cell	663	16	2.4132730015082955	19.039239153265957
IL18	vascular associated smoo	131	3	2.2900763358778624	3.0797133445739746
IL18	CD8-positive, alpha-beta	1898	40	2.107481559536354	48.79350411891937
IL18	pericyte	739	15	2.029769959404601	15.764164159074426
IL18	endothelial cell of arte	1754	18	1.026225769669327	12.8213629424572
IL18	capillary endothelial ce	7243	51	0.7041281237056468	46.94478058768436
IL18	vein endothelial cell	1336	5	0.374251497005988	3.5615711596328765
IL18	endothelial cell of lymp	315	0	0.0	0.0
IL18	plasma cell	148	0	0.0	0.0
IL33	vein endothelial cell	1336	1025	76.72155688622755	1501.4399301223457
IL33	endothelial cell of arte	1754	1311	74.74344355758267	2287.280550841242
IL33	basal cell	4015	2505	62.39103362391034	3070.645125316689
IL33	endothelial cell of lymp	315	120	38.095238095238095	170.32482779026031
IL33	adventitial cell	581	213	36.660929432013766	174.03452306741383
IL33	respiratory tract goblet	1040	312	30.0	198.45908404514196

(showing first 40 of 170 rows; full data in the accompanying CSV)

Supplementary Table 146 - r18_nhanes_target_trial

ST146_nhanes_target_trial.csv

outcome	n	cases	estimand	OR	ci_lo	ci_hi	p	prevalence	RR	E_value_point	E_value_ci_limit
allergy_composite	4403	689	high_index_vs_low_IPTW	1.855672508784867	1.3845330203359594	2.487134946788301	3.5145753737831064e-05	0.15648421530774473	1.6365409213860875	2.6571903548279217	1.9380531692711442
allergy_composite	4403	689	per_SD_IPTW_sensitivity	1.4464197631912512	1.1699269007056834	1.7882571381923877	0.0006499214360283483				

Supplementary Table 147 - r18_gene_triangulation

ST147_gene_level_triangulation.csv

gene	genetic_MR_significant	genetic_OR	genetic_p	coloc_pp4	PRISM_candidate	PRISM_score	FAERS_allergy_signal	ChEMBL_n_mechanisms	ChEMBL_top_action	lines_of_evidence
RORC	True	0.66	0.007004115449084	0.96	H-151	0.18	False	1	ANTAGONIST	3
IL6	True	1.4	0.0438453763397472	0.9	PF-06873600	1.74	False	9	INHIBITOR	3
TNF	False				ABBV-744	1.37	False	15	INHIBITOR	2
SRC	False				PD-166285	4.22	False	13	INHIBITOR	2
MTOR	False				PF-06873600	0.8	False	27	INHIBITOR	2
JAK1	False				INCB-057643	1.42	False	20	INHIBITOR	2
NLRP3	False				dapansutrile	0.02	False	1	INHIBITOR	2
JAK3	False				AT-406	0.62	False	15	INHIBITOR	2
IL1R1	False				AMG-176	0.68	False	3	ANTAGONIST	2
TSLP	False				AMG-176	0.66	False	2	INHIBITOR	2
IL4	True	0.58	1.3300430534712861e-08	1.0			False	2	INHIBITOR	2
IFNG	False				AMG-176	0.63	False	3	INHIBITOR	2
TP53	False				siremadlin	0.11	False	6	INHIBITOR	2
STAT5A	True	1.22	0.0375905647982774	0.9			False	0		1
GATA3	True	0.52	0.0084495188665798	0.95			False	0		1
IL33	False						False	3	INHIBITOR	1
IL13	False						False	8	INHIBITOR	1
IL5	False						False	3	INHIBITOR	1
BCL2	False						False	4	INHIBITOR	1
IL1B	False						False	5	INHIBITOR	1
IL6R	False						False	5	INHIBITOR	1
CDKN2A	False						False	0		0
CDKN1A	False						False	0		0
TBX21	False						False	0		0
IL7R	False						False	0		0
FOXP3	False						False	0		0
IL2	False						False	0		0
STAT5B	False						False	0		0

Supplementary Table 148 - r18_gene_mr_detail

ST148_gene_level_mr_detail.csv

gene	snp	outcome	exposure_source	ea	oa	eaf_exposure	beta_exposure	se_exposure	fstat	p_exposure	beta_outcome	se_outcome	p_outcome	beta_iv	se_iv	or	or_lower	or_upper	p_iv	approx_pp_h4
CDKN2A	rs2188127	ALLERG_ASTHM	eqtlgen	G	C	0.10337	0.12980260343	0.01368792612	89.9272890000	2.4755e-21	-0.00657228	0.0208854	0.753004	-0.0506328827	0.16090124117	0.95062759827	0.69350327594	1.30308372279	0.75300236495	0.566120984580277
CDKN1A	rs12199346	ALLERG_ASTHM	eqtlgen	A	C	0.19203	-0.2292658997	0.01010351360	514.913248890	5.39830000000	-0.00204289	0.0161928	0.899605	0.00891057066	0.07062890737	1.00895038795	0.87851518375	1.15875161207	0.89960496318	0.525762454018063
TNF	rs1121800	ALLERG_ASTHM	eqtlgen	A	T	0.660413	0.31387989316	0.01282011048	599.43687556	2.23059999999	-0.00596054	0.013306	0.654183	-0.0189898752	0.04239201137	0.98118929653	0.90295892172	1.06619737893	0.65418291975	0.595592099483702
BCL2	rs11152375	ALLERG_ASTHM	eqtlgen	C	T	0.820121	0.06825886978	0.01040499828	43.0362240399	5.3736e-11	-0.00621036	0.0164871	0.706412	-0.0909824616	0.24153784047	0.91303372389	0.56870318229	1.46584476211	0.70641146494	0.579754737578325
IL1B	rs6542082	ALLERG_ASTHM	eqtlgen	G	A	0.489359	-0.0653709273	0.00804674200	65.9977512100	4.5107e-16	-0.0148762	0.0126638	0.240113	0.22756599307	0.19372220206	1.25554029395	0.85887702917	1.83539828879	0.24011368045	0.754993356817511
IFNG	rs2870960	ALLERG_ASTHM	eqtlgen	C	T	0.337652	-0.0908602997	0.00844410882	115.781904039	5.294e-27	-0.00249311	0.013382	0.852208	0.02743893654	0.14728104609	1.02781885103	0.77010230155	1.37178085091	0.85220696433	0.538424437385967
IL4	rs2070874	ALLERG_ASTHM	eqtlgen	T	C	0.338504	-0.1398701720	0.00884638365	249.987721000	2.6129e-56	0.0753801	0.0132661	1.32993e-08	-0.5389290577	0.09484581172	0.58337267642	0.48440843765	0.70255522643	1.33004305347	0.999942338704489
SRC	rs6017938	ALLERG_ASTHM	eqtlgen	C	T	0.02508	-0.2456168117	0.02759460413	79.22602081	5.53460000000	-0.0470081	0.0399904	0.2398	0.19138795779	0.16281621650	1.21092915036	0.88008936408	1.66613694817	0.23980088824	0.755153108694197
IL7R	rs10058453	ALLERG_ASTHM	eqtlgen	C	T	0.725104	0.31220991722	0.00895847203	1214.57826064	4.1695e-266	0.00880724	0.0142231	0.535772	0.02820935375	0.04555620822	1.02861100545	0.94074721080	1.12468109220	0.53577088148	0.634017759993739 5
IL6	rs10265117	ALLERG_ASTHM	eqtlgen	A	G	0.11398	-0.1208382913	0.01270965242	90.39445776	1.9474e-21	-0.0406324	0.0201593	0.043845	0.33625434076	0.16682874090	1.39969497903	1.00931310820	1.94106865194	0.04384537633	0.895304011516103 6
GATA3	rs10905355	ALLERG_ASTHM	eqtlgen	A	G	0.326243	0.05493241175	0.00851744530	41.5947603599	1.1244e-10	-0.0356547	0.0135386	0.00844967	-0.6490648938	0.24645923177	0.52253417331	0.32234754342	0.84704216878	0.00844951886	0.954033654826864 1
IL1R1	rs12624116	ALLERG_ASTHM	eqtlgen	T	C	0.239414	0.14444836471	0.00930927939	240.76487556	2.6787e-54	-0.00891418	0.0148405	0.548061	-0.0617118789	0.10273913469	0.94015372583	0.76867995229	1.14987912142	0.54806263840	0.629844289521288 1
MTOR	rs4845986	ALLERG_ASTHM	eqtl_catalogue	G	C	0.743689	0.25077442522	0.00911539070	756.86062321	1.2967999999	-0.0033121	0.0145551	0.819992	-0.0132074871	0.05804060751	0.98687934898	0.88076266861	1.10578125545	0.81999144734	0.547232951729037 5
JAK3	rs3212797	ALLERG_ASTHM	eqtlgen	T	C	0.119414	-0.0870910766	0.01468676980	35.16371401	3.0293e-09	0.00931286	0.0196516	0.635572	-0.1069324247	0.22564424238	0.89858639309	0.57741436120	1.39840218759	0.63557242546	0.601358632224014
JAK1	rs310233	ALLERG_ASTHM	eqtlgen	G	A	0.239893	-0.0720308938	0.01017356768	50.1292320400	1.4374e-12	0.0111258	0.0147728	0.451372	-0.1544587245	0.20508977748	0.85687885945	0.57324923688	1.28084144300	0.45137337916	0.664078278848142 3
STAT5A	rs9909659	ALLERG_ASTHM	eqtlgen	A	G	0.197769	-0.1665321674	0.01002686363	275.84559396	6.01850000000	-0.0331988	0.0159664	0.0375907	0.19935367752	0.09587577131	1.22061359315	1.01150282018	1.47295441404	0.03759056479	0.903058393865172 6
IL6R	rs10908839	ALLERG_ASTHM	eqtlgen	C	G	0.673053	0.10331743600	0.00848421988	148.29394176	4.09150000000	-0.019514	0.0135083	0.148573	-0.1888742185	0.13074559837	0.82789063341	0.64073739166	1.06970954060	0.14857214905	0.807274158452998 3
STAT5B	rs185478092	ALLERG_ASTHM	eqtlgen	C	T	0.144334	0.11800899220	0.01146441851	105.95614225	7.54640000000	0.00413008	0.0179399	0.817923	0.03499801093	0.15202146603	1.03561764889	0.76876952447	1.39509161139	0.81792271759	0.54780452235 7429
NLRP3	rs10925027	ALLERG_ASTHM	eqtl_catalogue	T	C	0.612212	0.27459435394	0.00816912014	1129.88082769	1.0617e-247	0.0156773	0.0130438	0.229402	0.05709257956	0.04750206916	1.05875382482	0.96462922328	1.16206272266	0.22940355837	0.760520355771011
RORC	rs4845604	ALLERG_ASTHM	eqtlgen	A	G	0.127227	0.12746524322	0.01192132986	114.323140839	1.1095e-26	-0.0528713	0.0196063	0.0070039	-0.4147899353	0.15381683276	0.66047900939	0.48857077895	0.89287476994	0.00700411544	0.958155346817013 6
TP53	rs35850753	ALLERG_ASTHM	eqtlgen	T	C	0.0264954	-0.2458488904	0.02606678583	88.95319225	4.0388e-21	-0.0163449	0.0397834	0.681185	0.06648352153	0.16182053916	1.06874335267	0.77826769327	1.46763429057	0.68118455297	0.587330337897041

Supplementary Table 149 - r18_lincs_aging_connectivity

ST149_lincs_aging_signature_connectivity.csv

pert_iname	signature	n_sig	mean_z	se_z	z_score	p	connectivity_score
quercetin	aging_signature_genes	42	-0.22428343	0.075532794	-2.9693518	0.0029842875419288095	0.22428343
tofacitinib	aging_signature_genes	48	-0.20753719	0.060636967	-3.4226182	0.0006202113700430889	0.20753719
navitoclax	aging_signature_genes	42	-0.18924694	0.055430345	-3.4141395	0.0006398380381069657	0.18924694
ruxolitinib	aging_signature_genes	129	-0.11929675	0.02649978	-4.5018015	6.7379899555408684e-06	0.11929675
azithromycin	aging_signature_genes	42	-0.09635806	0.044371862	-2.1716027	0.029885641233910576	0.09635806
dasatinib	aging_signature_genes	654	-0.0652636	0.013039884	-5.004922	5.588468054984475e-07	0.0652636
azathioprine	aging_signature_genes	42	-0.030878777	0.034256164	-0.9014079	0.3673714839236828	0.030878777
everolimus	aging_signature_genes	180	-0.0014664924	0.02511882	-0.058382217	0.9534441795138897	0.0014664924
sirolimus	aging_signature_genes	255	0.019250615	0.019165954	1.0044173	0.31517751463993493	-0.019250615
temsirolimus	aging_signature_genes	84	0.042106055	0.036316674	1.1594138	0.2462875484231093	-0.042106055
baricitinib	aging_signature_genes	42	0.063097216	0.06008981	1.0500485	0.2936958283155966	-0.063097216

Supplementary Table 150 - r18_prism_scores

ST150_prism_target_drug_scores.csv

prism_name	target_gene	chembl_id	chembl_name	matrix_row_id	prism_id	name	dose	mean_lfc	median_lfc	std_lfc	n_cell_lines	target_median_gene_effe	abs_essentiality	senolytic_proxy
PD-166285	SRC	CHEMBL3545196	PD-0166285	BRD:BRD-K93217218	BRD-K93217218-300	PD-166285	2.5	-4.840203469037358	-4.89479060731119	1.6307053439534826	893	-0.147115517344791	0.147115517344791	4.219456014544111
PF-06873600	IL6	CHEMBL2109613	PF-04236921	BRD:BRD-K00004750	BRD-K00004750-001	PF-06873600	2.5	-1.754816852367211	-1.67174939415494	1.1214837414175911	890	0.006465410950045	0.006465410950045	1.7435441231018223
INCB-057643	JAK1	CHEMBL3545225	INCB-047986	BRD:BRD-K00003102	BRD-K00003102-001	INCB-057643	2.5	-1.485222732462227	-1.27477384679264	1.4958705227429671	895	-0.047436173462594	0.047436173462594	1.4179601297828073
JNJ-64619178	SRC	CHEMBL3545133	JNJ-26483327	BRD:BRD-K30421675	BRD-K30421675-001	JNJ-64619178	2.5	-1.604863508741922	-1.521045175987385	1.0074273989727236	890	-0.147115517344791	0.147115517344791	1.3990426286418591
ABBV-744	TNF	CHEMBL4297572	ABBV-3373	BRD:BRD-K00003108	BRD-K00003108-001	ABBV-744	2.5	-1.439102220677563	-1.25355838464331	1.5257742382783714	895	-0.049811348988813	0.049811348988130	1.3708198354539174
PD-161570	SRC	CHEMBL3545196	PD-0166285	BRD:BRD-K97152684	BRD-K97152684-001	PD-161570	2.5	-1.351840872907508	-1.14669781462647	1.2650648788084296	893	-0.147115517344791	0.147115517344791	1.1784696941739503
PF-06459988	IL6	CHEMBL2109613	PF-04236921	BRD:BRD-K00003311	BRD-K00003311-001	PF-06459988	2.5	-0.825217301010410	-0.690002405291834	0.9160307928777606	895	0.006465410950045	0.006465410950045	0.8199162057953423
PF-06873600	MTOR	CHEMBL1234354	PF-04691502	BRD:BRD-K00004750	BRD-K00004750-001	PF-06873600	2.5	-1.754816852367211	-1.67174939415494	1.1214837414175911	890	-1.197786192457064	1.197786192457064	0.7984474824666064
CC-223	MTOR	CHEMBL3586573	CC-115	BRD:BRD-K88737571	BRD-K88737571-001	CC-223	2.5	-1.728943774306761	-1.63692474945249	0.908897401653491	895	-1.197786192457064	1.197786192457064	0.7866751462178628
CC-90003	MTOR	CHEMBL3586573	CC-115	BRD:BRD-K00003116	BRD-K00003116-001	CC-90003	2.5	-1.680192210891092	-1.39950143153849	1.4608362881727908	895	-1.197786192457064	1.197786192457064	0.7644930233239313
SBI-115	MTOR	CHEMBL3586573	CC-115	BRD:BRD-K00005340	BRD-K00005340-001	SBI-115	2.5	-1.640108257755764	-1.53908329531297	0.9517896885339275	895	-1.197786192457064	1.197786192457064	0.7462546918279472
AMG-176	IL1R1	CHEMBL2109458	AMG-108	BRD:BRD-K43900408	BRD-K43900408-001	AMG-176	2.5	-0.684387223940705	-0.383383822297619	1.1737700723378715	890	-0.011760825204883	0.011760825204883	0.6764318274549869
triciribine-phosphate	JAK1	CHEMBL1795071	RUXOLITINIB PHOSPH	BRD:BRD-K00004666	BRD-K00004666-001	triciribine-phosphate	2.5	-0.706297190503097	-0.560788037809482	0.9772458966465009	890	-0.047436173462594	0.047436173462594	0.6743104815334322
AMG-176	TSLP	CHEMBL2109595	AMG-157	BRD:BRD-K43900408	BRD-K43900408-001	AMG-176	2.5	-0.684387223940705	-0.383383822297619	1.1737700723378715	890	0.030325546033514	0.030325546033514	0.6642436718912944
AMG-176	IFNG	CHEMBL2109450	AMG-811	BRD:BRD-K43900408	BRD-K43900408-001	AMG-176	2.5	-0.684387223940705	-0.383383822297619	1.1737700723378715	890	0.080331813279410	0.080331813279410	0.63349724179945
AT-406	JAK3	CHEMBL495727	AT-9283	BRD:BRD-K83030136	BRD-K83030136-001	AT-406	2.5	-0.692329237301555	-0.439335243234272	1.0556578209627585	895	-0.118825286634706	0.118825286634706	0.6188001340084116
AZM-475271	SRC	CHEMBL3545150	AZM-475271	BRD:BRD-K51329597	BRD-K51329597-001	AZM-475271	2.5	-0.564601024111108	-0.421310963177947	0.8380572725122019	893	-0.147115517344791	0.147115517344791	0.4921919506572281
PF-06747775	IL6	CHEMBL2109613	PF-04236921	BRD:BRD-K00004648	BRD-K00004648-001	PF-06747775	2.5	-0.488621017161835	-0.246715987763041	0.9842775956143629	895	0.006465410950045	0.006465410950045	0.4854821753890233
PF-429242	IL6	CHEMBL2109613	PF-04236921	BRD:BRD-K11821338	BRD-K11821338-300	PF-429242	2.5	-0.452211895143372	-0.296094884257479	0.946218541146213	893	0.006465410950045	0.006465410950045	0.4493069411262833
PF-06459988	MTOR	CHEMBL1234354	PF-04691502	BRD:BRD-K00003311	BRD-K00003311-001	PF-06459988	2.5	-0.825217301010410	-0.690002405291834	0.9160307928777606	895	-1.197786192457064	1.197786192457064	0.37547660634260366
PF-06409577	IL6	CHEMBL2109613	PF-04236921	BRD:BRD-K00003731	BRD-K00003731-001	PF-06409577	2.5	-0.337795469790134	-0.320791654165143	0.4878061875322386	890	0.006465410950045	0.006465410950045	0.3356255129237625
PF-4191834	IL6	CHEMBL2109613	PF-04236921	BRD:BRD-K00004633	BRD-K00004633-001	PF-4191834	2.5	-0.279972949512833	-0.193899818846308	0.5239998790702735	895	0.006465410950045	0.006465410950045	0.2781744374588646
PF-06260414	IL6	CHEMBL2109613	PF-04236921	BRD:BRD-K00004644	BRD-K00004644-001	PF-06260414	2.5	-0.261543729706188	-0.205112272325678	0.5381039881771751	895	0.006465410950045	0.006465410950045	0.2598636047107736
TP-003	IL6	CHEMBL2109615	MRA 003 US	BRD:BRD-K87941707	BRD-K87941707-001	TP-003	2.5	-0.232263585271772	-0.217062738066831	0.4339869097795129	895	0.006465410950045	0.006465410950045	0.2307715523502483
PF-06747775	MTOR	CHEMBL1234354	PF-04691502	BRD:BRD-K00004648	BRD-K00004648-001	PF-06747775	2.5	-0.488621017161835	-0.246715987763041	0.9842775956143629	895	-1.197786192457064	1.197786192457064	0.22232418186937938
PF-429242	MTOR	CHEMBL1234354	PF-04691502	BRD:BRD-K11821338	BRD-K11821338-300	PF-429242	2.5	-0.452211895143372	-0.296094884257479	0.946218541146213	893	-1.197786192457064	1.197786192457064	0.205757910708235
PF-06305591	IL6	CHEMBL2109613	PF-04236921	BRD:BRD-K00004712	BRD-K00004712-001	PF-06305591	2.5	-0.184104475008390	-0.183340093266558	0.4253301276993474	890	0.006465410950045	0.006465410950045	0.1829218103328625
H-151	RORC	CHEMBL3545357	JTE-151	BRD:BRD-K00005333	BRD-K00005333-001	H-151	2.5	-0.215626857933865	-0.172259458846688	0.5175506432350907	895	-0.181153695338268	0.181153695338268	0.18255613878608143
AMG-510	IL1R1	CHEMBL2109458	AMG-108	BRD:BRD-K00010385	BRD-K00010385-001	AMG-510	2.5	-0.178210968684693	-0.140702027364949	0.5504878424610691	895	-0.011760825204883	0.011760825204883	0.17613942371074298
AMG-510	TSLP	CHEMBL2109595	AMG-157	BRD:BRD-K00010385	BRD-K00010385-001	AMG-510	2.5	-0.178210968684693	-0.140702027364949	0.5504878424610691	895	0.030325546033514	0.030325546033514	0.17296568969949327
betamethasone-phos	JAK1	CHEMBL1795071	RUXOLITINIB PHOSPH	BRD:BRD-K00005349	BRD-K00005349-304	betamethasone-phos	2.5	-0.181073422765429	-0.086467399321798	0.7247663498086904	893	-0.047436173462594	0.047436173462594	0.17287298964178405
PF-06650833	IL6	CHEMBL2109613	PF-04236921	BRD:BRD-K00003296	BRD-K00003296-001	PF-06650833	2.5	-0.173904530184660	-0.158010346964836	0.4854703812082989	895	0.006465410950045	0.006465410950045	0.1727873887096676
AMG-510	IFNG	CHEMBL2109450	AMG-811	BRD:BRD-K00010385	BRD-K00010385-001	AMG-510	2.5	-0.178210968684693	-0.140702027364949	0.5504878424610691	895	0.080331813279410	0.080331813279410	0.16495947494476176
PD-81723	SRC	CHEMBL3545196	PD-0166285	BRD:BRD-K77951111	BRD-K77951111-001	PD-81723	2.5	-0.183657757584126	-0.165531076151371	0.4041405096209133	895	-0.147115517344791	0.147115517344791	0.16010397802763168
PF-06409577	MTOR	CHEMBL1234354	PF-04691502	BRD:BRD-K00003731	BRD-K00003731-001	PF-06409577	2.5	-0.337795469790134	-0.320791654165143	0.4878061875322386	890	-1.197786192457064	1.197786192457064	0.1536980580501729
delgocitinib	JAK1	CHEMBL4297507	DELGOCITINIB	BRD:BRD-K00003500	BRD-K00003500-001	delgocitinib	2.5	-0.157668707945692	-0.134349910322879	0.4766522501342686	895	-0.047436173462594	0.047436173462594	0.15052822495567802
JNJ-38877618	SRC	CHEMBL3545133	JNJ-26483327	BRD:BRD-K00003094	BRD-K00003094-001	JNJ-38877618	2.5	-0.167771210376543	-0.097530695295422	0.6176416214859807	895	-0.147115517344791	0.147115517344791	0.1462548521398097
PF-06412562	IL6	CHEMBL2109613	PF-04236921	BRD:BRD-K00004652	BRD-K00004652-001	PF-06412562	2.5	-0.146106910371015	-0.109453136490365	0.4420926042515951	895	0.006465410950045	0.006465410950045	0.14516833741270793
JNJ-678	SRC	CHEMBL3545133	JNJ-26483327	BRD:BRD-K00004600	BRD-K00004600-001	JNJ-678	2.5	-0.166295076275539	-0.151869726946275	0.4899756296999944	895	-0.147115517344791	0.147115517344791	0.14496802960216273
PF-06840003	IL6	CHEMBL2109613	PF-04236921	BRD:BRD-K00003400	BRD-K00003400-001	PF-06840003	2.5	-0.142362118441806	-0.131335391127593	0.4105596763643008	895	0.006465410950045	0.006465410950045	0.14144760156975952

(showing first 40 of 99 rows; full data in the accompanying CSV)

Supplementary Table 151 - r18_lincs_alarmin_connectivity

ST151_lincs_alarmin_signature_connectivity.csv

pert_iname	signature	n_sig	mean_z	se_z	z_score	p	connectivity_score	q_alarmin_family
azathioprine	alarmin_type2_genes	42	-0.09498697	0.04686141234784879	-2.0269762045873025	0.0426648462496142	0.09498697	0.11732832718643904
azithromycin	alarmin_type2_genes	42	-0.13979754	0.05141157809287738	-2.7191839602035746	0.006544320031564025	0.13979754	0.02399584011573476
baricitinib	alarmin_type2_genes	42	-0.12051914	0.0721184724830776	-1.6711271706552429	0.0946965616419668	0.12051914	0.17361036301027247
dasatinib	alarmin_type2_genes	654	0.07962565	0.019556903183603525	4.071485679138957	4.6714234767469686e-05	-0.07962565	0.0005138565824421666
everolimus	alarmin_type2_genes	180	0.07028509	0.0393938916684127	1.7841621210107754	0.07439732541293054	-0.07028509	0.1636741159084472
navitoclax	alarmin_type2_genes	42	-0.007880563	0.06463108550923675	-0.12193146462427584	0.9029532963732783	0.007880563	0.9029532963732781
quercetin	alarmin_type2_genes	42	0.07170261	0.08609166254972553	0.8328635416306158	0.4049216934935268	-0.07170261	0.6264432250607387
ruxolitinib	alarmin_type2_genes	129	-0.025178641	0.03374611205707841	-0.7461197615378325	0.4555950727714463	0.025178641	0.6264432250607387
sirolimus	alarmin_type2_genes	255	-0.021852305	0.03338088587809888	-0.6546352675831766	0.5127026117025631	0.021852305	0.6266365254142439
temsirolimus	alarmin_type2_genes	84	0.016094383	0.047330043946482415	0.3400458041094222	0.7338220342978239	-0.016094383	0.8072042377276063
tofacitinib	alarmin_type2_genes	48	0.32872936	0.1170907760018648	2.807474444361567	0.004993164149837887	-0.32872936	0.02399584011573476

Supplementary Table 152 - r18_drug_subgroup_target_trial

ST152_nhanes_drug_subgroup_target_trial.csv

comparison	outcome	n_exposed_a	n_exposed_b	OR_iptw	OR_iptw_lower	OR_iptw_upper	E_value_point	E_value_ci_limit	note
IL-6/IL-1 inhibitors vs	any_allergy	0	14						insufficient sample size
IL-6/IL-1 inhibitors vs	cancer_history	0	15						insufficient sample size
IL-6/IL-1 inhibitors vs	gallstones	0	15						insufficient sample size
IL-6/IL-1 inhibitors vs	liver_condition	0	15						insufficient sample size
JAK inhibitors vs TNF in	any_allergy	2	14						insufficient sample size
JAK inhibitors vs TNF in	cancer_history	2	15						insufficient sample size
JAK inhibitors vs TNF in	gallstones	2	15						insufficient sample size
JAK inhibitors vs TNF in	liver_condition	2	15						insufficient sample size
mTOR inhibitors vs non-i	any_allergy	3	1806	0.7248499399759903	0.0373419014289633	14.070184307103265	2.103259443246525	27.63115411492865	
mTOR inhibitors vs non-i	cancer_history	3	1921	2.441407368243772	0.2040565381114448	29.20989443847022	4.317323829462449	9.272707715224923	
mTOR inhibitors vs non-i	gallstones	3	1921	4.172314773402857	0.3714118469312544	46.87036967772831	7.810431840644219	4.827082198167197	
mTOR inhibitors vs non-i	liver_condition	3	1921	3.6973026973026974	0.1891832509730922	72.25823197966983	6.855267950148129	10.045570236050064	
Methotrexate vs non-immu	any_allergy	25	1794	1.072425418868661	0.3944665535127566	2.915573624161345	1.3511203419780748	4.507759353244038	
Methotrexate vs non-immu	cancer_history	27	1908	0.4242982717050552	0.1004342741260793	1.792505844627291	4.14507959739671	2.9843830859553178	
Methotrexate vs non-immu	gallstones	27	1908	0.3567536975266316	0.0849969808426145	1.4973849592915545	5.051177843909905	2.3603894534982857	
Methotrexate vs non-immu	liver_condition	27	1908	0.4672600127145582	0.0282185960493354	7.737164496075315	3.702200802444144	14.957036374665202	

Supplementary Table 153 - r21_t2spd_drift_model

ST153_t2spd_drift_model.csv

cycle	parameter	beta	se	lo	hi	p	n
J	drift_beta_per_decade	0.08890694125068709	0.00970441270025289	0.06988629235819142	0.10792759014318275	5.118261989619224e-20	4903
J	S0_at_age50	0.05211691248451727	0.03410719754875786	-0.014733194711048135	0.11896701968008266	0.12650442419941785	4903
J	female	-0.041212806777097176	0.043186302188297776	-0.12585795906616082	0.04343234551196646	0.3399304183502052	4903
J	weighted_r_index_age	0.16232916774811895					4903
J	drift_beta_per_decade_bm	0.0814306125318893	0.008504470833654703	0.06476184969792607	0.09809937536585252	1.0182204356315277e-21	4827
D	drift_beta_per_decade	0.056156553773467024	0.009258194731200237	0.03801049210031456	0.07430261544661948	1.3145771862885468e-09	4457
D	S0_at_age50	0.05849548949991407	0.02805459671153252	0.003508479945310332	0.11348249905451782	0.037063944874519715	4457
D	female	-0.1011869464145417	0.03268970296586506	-0.16525876422763724	-0.03711512860144618	0.0019656304822172205	4457
D	weighted_r_index_age	0.10043834192808994					4457
D	drift_beta_per_decade_bm	0.04903201118394429	0.009569069095435282	0.030276635756891136	0.06778738661099745	2.991042049981969e-07	4385
pooled	drift_beta_per_decade	0.07417142923228169	0.007085845544106978	0.060283171965832015	0.08805968649873136	1.2175578582743018e-25	9360
pooled	S0_at_age50	0.05652404850654506	0.022251167349578122	0.012911760501371945	0.10013633651171819	0.011076584481635215	9360
pooled	female	-0.06926657552275253	0.027881384116171925	-0.12391408839044951	-0.01461906265505556	0.012979524870603646	9360
pooled	weighted_r_index_age	0.13450968511322514					9360
pooled	drift_beta_per_decade_bm	0.06646119403696563	0.00672587871293645	0.05327847175961019	0.07964391631432106	5.0120466173056393e-23	9212

Supplementary Table 154 - r21_t2spd_variance_resilience

ST154_t2spd_variance_resilience.csv

cycle	analysis	index	stratum	beta	se	lo	hi	p	n
J	delta_logvar_per_decade	idx_pub	all	0.021845876057733	0.0205860557943488	-0.0185027932991907	0.0621945454146568	0.2886000424355418	4903
J	brown_forsythe_F_p	idx_pub	all	2.284050313169852				0.0438248876265223	4839
J	delta_logvar_per_decade	idx_pub	male	0.017164906987184	0.0297801085658869	-0.0412041058019543	0.0755339197763224	0.5643527319024951	2362
J	delta_logvar_per_decade	idx_pub	female	0.0269006603217759	0.0254153979135482	-0.0229135195887785	0.0767148402323305	0.289855132481304	2541
J	delta_logvar_per_decade	idx_equal	all	0.0071168605829129	0.0202851239796237	-0.0326419824171496	0.0468757035829755	0.7257073626177015	4903
D	delta_logvar_per_decade	idx_pub	all	-0.0276786142535933	0.0212084611007515	-0.0692471980110664	0.0138899695038797	0.1918676014673559	4457
D	brown_forsythe_F_p	idx_pub	all	1.3186167750981308				0.2529298682802583	4370
D	delta_logvar_per_decade	idx_pub	male	-0.0466454820254941	0.03704061516661	-0.1192450877520498	0.0259541237010615	0.2079196991711818	2146
D	delta_logvar_per_decade	idx_pub	female	-0.0091931045003643	0.0316583394219388	-0.0712434497673644	0.0528572407666357	0.7715217756449294	2311
pooled	delta_logvar_per_decade	idx_pub	all	-0.0001289568357374	0.015654078106336	-0.0308109499241561	0.0305530362526812	0.9934271757309222	9360
pooled	brown_forsythe_F_p	idx_pub	all	3.1825281898172326				0.0071310235395362	9209
pooled	delta_logvar_per_decade	idx_pub	male	-0.0089703046569887	0.0252079729670967	-0.0583779316724983	0.0404373223585207	0.721951491570324	4508
pooled	delta_logvar_per_decade	idx_pub	female	0.0096335288672548	0.0215453095953442	-0.0325952779396198	0.0518623356741295	0.6547820509626494	4852
pooled	delta_logvar_per_decade	idx_equal	all	-0.0048381562345968	0.0150739940886992	-0.0343831846484473	0.0247068721792536	0.7482403472875548	9360
J	delta_logvar_per_decade_	idx_pub	all	0.03039249139646347	0.020017505459935445	-0.00884181930501	0.06962680209793694	0.12893988465123887	4903
D	delta_logvar_per_decade_	idx_pub	all	-0.0314557838301843	0.019239663031859176	-0.06916552337262828	0.006253955712259687	0.10206065464831032	4457
pooled	delta_logvar_per_decade_	idx_pub	all	0.009973636773718868	0.014318438769444332	-0.018090503214392022	0.03803777676182976	0.486078848891501	9360

Supplementary Table 154b - r21_t2spd_residsd_by_agebin

ST154b_t2spd_residsd_by_agebin.csv

cycle	age_bin	resid_sd	n
J	20-30	0.9218660413000812	718
J	30-40	0.8571000956903945	755
J	40-50	0.8774771314918869	714
J	50-60	0.9421802028566206	831
J	60-70	0.9662764085373582	994
J	70-85	1.0023832344740755	891
D	20-30	0.9818281094681254	926
D	30-40	0.8351776187545655	756
D	40-50	0.869047520027097	776
D	50-60	0.9069842732789797	595
D	60-70	0.9651588271766081	628
D	70-85	1.0056292352366714	776
pooled	20-30	0.9518817665367347	1644
pooled	30-40	0.846998665056578	1511
pooled	40-50	0.8727742757078202	1490
pooled	50-60	0.9265375214412197	1426
pooled	60-70	0.966383070247228	1622
pooled	70-85	1.0054830380834472	1667

Supplementary Table 155 - r21_t2spd_shape_and_gain

ST155_t2spd_shape_and_gain.csv

cycle	block	model	knot	loglik	aic	n	boot_min	boot_p25	boot_median	boot_p75	boot_max	boot_boundary_frac	int_beta	int_se	int_lo	int_hi	int_p
J	drift_shape	linear		-7624.989568381	15255.979136762	4903											
J	drift_shape	quadratic		-7620.165068213	15248.330136427	4903											
J	drift_shape	hinge_best_k60	60.0	-7617.912656714	15243.825313429	4903											
J	drift_shape_boots	knot_argmin_distr				300	35.0	55.0	60.0	60.0	70.0	0.1066666666666					
D	drift_shape	linear		-6651.887282413	13309.774564827	4457											
D	drift_shape	quadratic		-6648.193113195	13304.386226390	4457											
D	drift_shape	hinge_best_k55	55.0	-6647.962846351	13303.925692703	4457											
D	drift_shape_boots	knot_argmin_distr				300	35.0	50.0	55.0	60.0	70.0	0.2133333333333					
pooled	drift_shape	linear		-14284.56890089	28575.137801781	9360											
pooled	drift_shape	quadratic		-14276.54893392	28561.097867840	9360											
pooled	drift_shape	hinge_best_k55	55.0	-14274.41231720	28556.824634408	9360											
pooled	drift_shape_boots	knot_argmin_distr				300	35.0	55.0	55.0	60.0	70.0	0.0433333333333					
J	gain_invariance	allergy ~ idx_z*a				4827							0.002159724229	0.030161456687	-0.056956730877	0.0612761793366	0.94291591459377799
D	gain_invariance	log_ige ~ idx_z*a				4379							-0.028962850794	0.019715753043	-0.067605726758	0.0096800251703	0.14182715470282192
D	gain_invariance	high_ige ~ idx_z*				4385							-0.061676750506	0.026819829253	-0.114243615842	-0.009109885170	0.021466936159418925

Supplementary Table 156 - r21_t2spd_drift_specificity

ST156_t2spd_drift_specificity.csv

block	cycle	index	role	parameter	beta	se	lo	hi	p	n
drift_specificity	J	idx_pub	reference	drift_beta_per_decade	0.088906941250687	0.0097044127002528	0.0698862923581914	0.1079275901431827	5.1182619896192244e-20	4903
drift_specificity	J	idx_equal	construction_robustness	drift_beta_per_decade	0.0679307100454738	0.0095567721860957	0.0491994365607262	0.0866619835302213	1.1763189742551802e-12	4903
drift_specificity	J	eosonly	type2_lineage	drift_beta_per_decade	0.0512043205241113	0.0098644304256309	0.0318700368898747	0.0705386041583479	2.0938845084618214e-07	5024
drift_specificity	J	neuonly	non_type2_negative_contr	drift_beta_per_decade	0.0515579428004837	0.0114467528662137	0.0291223071827048	0.0739935784182626	6.663773201802748e-06	5024
hinge_slopes	J	idx_pub	knot_60	pre_knot_slope_per_decad	0.0571050471042435	0.0145023459747422	0.0286804489937486	0.0855296452147383	8.228622610739937e-05	4903
hinge_slopes	J	idx_pub	knot_60	post_knot_increment_per_	0.130461472166033	0.0507206185038994	0.0310490598983901	0.2298738844336759	0.0101066597366131	4903
hinge_slopes	J	idx_pub	knot_60	post_knot_slope_per_deca	0.1875665192702765	0.040819464474006	0.1075603689012247	0.2675726696393283	4.326943622529194e-06	4903
drift_specificity	D	idx_pub	reference	drift_beta_per_decade	0.056156553773467	0.0092581947312002	0.0380104921003145	0.0743026154466194	1.3145771862885466e-09	4457
drift_specificity	D	idx_equal	construction_robustness	drift_beta_per_decade	0.0676164512714942	0.0100957190979625	0.0478288418394876	0.087404607035008	2.1196170693116072e-11	4457
drift_specificity	D	eosonly	type2_lineage	drift_beta_per_decade	0.0255446073683727	0.0087832280192632	0.0083294804506167	0.0427597342861287	0.0036335385114042	4486
drift_specificity	D	neuonly	non_type2_negative_contr	drift_beta_per_decade	0.0320105597568518	0.0070827643682057	0.0181283415951685	0.0458927779185351	6.198559165982184e-06	4486
hinge_slopes	D	idx_pub	knot_55	pre_knot_slope_per_decad	0.0244863617349635	0.0174238235958984	-0.0096643325129973	0.0586370559829243	0.159920788026159	4457
hinge_slopes	D	idx_pub	knot_55	post_knot_increment_per_	0.0863102448284862	0.0361829769898566	0.0153916099283673	0.1572288797286051	0.0170613656029036	4457
hinge_slopes	D	idx_pub	knot_55	post_knot_slope_per_deca	0.1107966065634497	0.0234786881526205	0.0647783777843134	0.156814835342586	2.369732238685233e-06	4457
drift_specificity	pooled	idx_pub	reference	drift_beta_per_decade	0.0741714292322816	0.0070858455441069	0.060283171965832	0.0880596864987313	1.2175578582743018e-25	9360
drift_specificity	pooled	idx_equal	construction_robustness	drift_beta_per_decade	0.0678629779937612	0.0068816680484996	0.0543749086187019	0.0813510473688204	6.118160417939337e-23	9360
drift_specificity	pooled	eosonly	type2_lineage	drift_beta_per_decade	0.0387921539451625	0.0068939131398331	0.0252800841910896	0.0523042236992354	1.8335458880080684e-08	9510
drift_specificity	pooled	neuonly	non_type2_negative_contr	drift_beta_per_decade	0.0437238901367335	0.0071331672517992	0.029742882323207	0.0577048979502601	8.806705775995308e-10	9510
hinge_slopes	pooled	idx_pub	knot_55	pre_knot_slope_per_decad	0.0388688009902838	0.0126229796238377	0.0141277609275619	0.0636098410530058	0.0020755055180168	9360
hinge_slopes	pooled	idx_pub	knot_55	post_knot_increment_per_	0.0980070123743456	0.0289731037744476	0.0412197289764282	0.154794295772263	0.0007177975114379	9360
hinge_slopes	pooled	idx_pub	knot_55	post_knot_slope_per_deca	0.1368758133646295	0.0199210497750584	0.0978305558055149	0.175921070923744	6.379198917556603e-12	9360
drift_specificity	J	lymonly	negative_weight_componen	drift_beta_per_decade	-0.09081320584758466	0.010993207598457455	-0.11235989274056127	-0.0692665189546080 6	1.4464240781560166e-16	5024
drift_specificity	D	lymonly	negative_weight_componen	drift_beta_per_decade	-0.06368118273029412	0.008221518384503357	-0.0797953587639207	-0.04756700669666754	9.507783699168336e-15	4486
drift_specificity	pooled	lymonly	negative_weight_componen	drift_beta_per_decade	-0.07690074690439803	0.007163723172710152	-0.09094164432290994	-0.06285984948588613	6.99114274466729e-27	9510
drift_specificity	J	mononly	myeloid_component	drift_beta_per_decade	0.07869153593455913	0.010489666674213739	0.058131789253100204	0.09925128261601807	6.294029405839449e-14	5024
drift_specificity	D	mononly	myeloid_component	drift_beta_per_decade	0.0801289086719984	0.012663284523468198	0.05530887100600074	0.10494894633799606	2.489132310647909e-10	4486
drift_specificity	pooled	mononly	myeloid_component	drift_beta_per_decade	0.08050888548524153	0.007959154886163647	0.06490894190836079	0.09610882906212227	4.72781678125909e-24	9510
drift_specificity	J	wbconly	total_count_control	drift_beta_per_decade	-0.004453939533981677	0.00676078898492641	-0.01770508594443744	0.008797206876474086	0.5100306906228584	5028
drift_specificity	D	wbconly	total_count_control	drift_beta_per_decade	-0.04214183287883152	0.008000685048801554	-0.05782317557448256	-0.026460490183180474	1.3846139644643869e-07	4508
drift_specificity	pooled	wbconly	total_count_control	drift_beta_per_decade	-0.013046562345230158	0.00524516715152849	-0.023327089962225996	-0.002766034728234319	0.012869900116304801	9536

Supplementary Table 157 - lung_atp_axis_colocalization

ATP01_lung_atp_axis_colocalization.csv

cell_type	n_cells	broad	compartment	IL18_pct	TSLP_pct	IL33_pct	P2RX7_pct	NLRP3_pct	CASP1_pct	GSDMD_pct	IL1B_pct	TNFSF14_pct	IL18pos_n	IL18pos_NLRP3_pct	IL18pos_CASP1_pct	IL18pos_P2RX7_pct	IL18pos_GSDMD_pct	IL18pos_IL1B_pct
macrophage	16486	myeloid leukocyt	Immune	75.25779449229	0.479194468033	0.145578066238	46.3969428606(	7.582190949896	64.93388329491	47.03991265316	40.83464757976	2.899429819240	12407.0	7.769807366809	71.67727895542	49.43177238655	51.58378334810	44.26533408559684
type i pneumocy	3116	epithelial cell of	Epithelium	71.34146341463	0.834403080872	0.256739409499	0.513478818998	0.032092426187	8.664955070603	36.64955070603	0.128369704749	0.385109114249	2223.0	0.044984255510	9.806567701304	0.449842555105	41.61043634727	0.0899685110211426
basophil	1322	granulocyte	Immune	68.83509833585	2.193645990922	0.529500756429	8.169440242057	2.193645990922	5.219364599092	19.89409984871	0.605143721633	1.210287443267	910.0	2.087912087912	5.384615384615	9.120879120879	21.42857142857	0.7692307692307693
basal cell	4015	epithelial cell	Epithelium	59.15317559153	15.41718555417	62.39103362391	1.569115815691	0.0	12.72727272727	31.23287671232	0.0	0.249066002490	2375.0	0.0	13.93684210526	1.6	36.42105263157	0.0
type ii pneumocy	11594	epithelial cell of	Epithelium	54.87321028117	3.553562187338	1.664654131447	5.476970846989	0.017250301880	18.02656546489	46.17905813351	0.129377264102	12.78247369328	6362.0	0.031436655139	21.26689720213	6.224457717698	53.75668028921	0.0943099654196797934
intermediate mo	2785	myeloid leukocyt	Immune	50.84380610412	0.107719928186	0.359066427289	32.17235188509	36.62477558348	46.60682226211	26.06822262118	22.58527827648	1.328545780969	1416.0	37.71186440677	47.24576271186	34.88700564971	27.61299435028	21.1864406779661
lung ciliated cell	1209	ciliated epithelia	Epithelium	47.80810587262	0.496277915632	9.346567411083	3.722084367245	0.248138957816	7.113316790736	41.02564102564	0.330851943755	0.248138957816	578.0	0.346020761245	9.342560553633	5.536332179930	49.30795847750	0.3460207612456747
respiratory gobl	1040	glandular epithel	Epithelium	44.23076923076	2.115384615384	30.0	1.634615384615	0.0	15.09615384615	38.75	0.0	1.057692307692	460.0	0.0	21.52173913043	2.608695652173	48.91304347826	0.0
monocyte	506	myeloid leukocyt	Immune	42.88537549407	0.197628458498	0.0	18.18181818181	24.70355731225	32.01581027667	18.18181818181	20.94861660079	3.359683794466	217.0	25.80645161290	36.40552995391	22.58064516129	21.19815668202	29.95391705069124 6
non-classical mo	527	myeloid leukocyt	Immune	34.72485768500	0.0	0.379506641366	26.94497153700	22.20113851992	44.59203036053	19.92409867172	13.28273244781	0.948766603415	183.0	25.13661202185	48.63387978142	34.97267759562	21.31147540983	12.568306010928962
classical monocy	1569	myeloid leukocyt	Immune	32.12237093690	0.254939451880	0.573613766730	22.43467176545	34.22562141491	46.78138942001	23.26322498406	25.81261950286	2.613129381771	504.0	37.10317460317	51.98412698412	24.20634920634	26.98412698412	30.753968253968253
club cell	1747	endo-epithelial c	Epithelium	19.86262163709	0.858614768174	4.922724670864	2.003434459072	0.057240984544	7.212364052661	26.96050372066	0.343445907269	1.202060675443	347.0	0.288184438040	11.81556195965	2.305475504322	38.04034582132	0.5763688760806917
adventitial cell	581	adventitial cell	Stromal	16.00688468158	0.688468158347	36.66092943201	3.270223752151	0.516351118760	6.712564543889	18.41652323580	1.376936316695	20.30981067125	93.0	0.0	6.451612903225	1.075268817204	21.50537634408	1.075268817204301
mature nk t cell	163	t cell	Immune	14.11042944785	0.0	0.0	1.840490797546	0.0	23.31288343558	25.76687116564	0.0	11.04294478527	23.0	0.0	21.73913043478	0.0	34.78260869565	0.0
natural killer cell	1019	innate lymphoid	Immune	11.08930323846	0.0	0.098135426889	6.575073601570	0.883218842001	18.15505397448	18.05691854759	0.294406280667	5.986261040235	113.0	0.884955752212	16.81415929203	10.61946902654	21.23893805309	0.0
alveolar type 2 fi	1113	fibroblast	Stromal	7.187780772686	0.988319856244	18.59838274932	3.593890386343	0.179694519317	5.660377358490	18.95777178796	0.0	5.750224618149	80.0	1.25	11.25	6.25	22.5	0.0
cd4-positive, alp	2124	t cell	Immune	3.013182674199	0.094161958568	0.188323917137	2.824858757062	1.883239171374	11.77024482109	13.41807909604	0.753295668549	4.378531073446	64.0	1.5625	20.3125	4.6875	18.75	0.0
bronchial smooth	220	contractile cell	Stromal	2.727272727272	5.909090909090	7.727272727272	1.818181818181	0.0	4.090909090909	18.18181818181	0.454545454545	0.0						
neutrophil	371	granulocyte	Immune	2.695417789757	0.0	0.539083557951	1.886792452830	35.57951482479	11.59029649595	6.469002695417	61.45552560646	12.93800539083	10.0	40.0	20.0	10.0	20.0	60.0
b cell	663	lymphocyte of b	Immune	2.413273001508	0.150829562594	0.150829562594	2.262443438914	0.150829562594	9.351432880844	10.70889894419	0.150829562594	0.150829562594	16.0	0.0	31.25	0.0	25.0	0.0
vascular associat	131	contractile cell	Stromal	2.290076335877	2.290076335877	6.106870229007	4.580152671755	0.763358778625	7.633587786259	22.13740458015	0.0	0.0						
cd8-positive, alp	1898	t cell	Immune	2.107481559536	0.052687038988	0.158061116965	3.688092729188	0.684931506849	16.96522655426	16.80716543730	0.210748155953	7.060063224446	40.0	2.5	17.5	0.0	27.5	0.0
pericyte	739	contractile cell	Stromal	2.029769959404	7.442489851150	2.571041948579	2.029769959404	0.405953991880	3.247631935047	16.10284167794	0.270635994587	0.0	15.0	0.0	0.0	0.0	6.666666666666	0.0
endothelial cell o	1754	endothelial cell	Endothelium	1.026225769669	5.302166476624	74.74344355758	2.451539338654	0.456100342075	16.02052451539	37.62827822120	0.342075256556	0.114025085518	18.0	0.0	22.22222222222	5.555555555555	55.55555555555	0.0
capillary endothe	7243	endothelial cell	Endothelium	0.704128123705	0.938837498274	12.88140273367	4.638961756178	0.041419301394	9.623084357310	27.28151318514	0.220902940770	0.069032168990	51.0	0.0	9.803921568627	9.803921568627	39.21568627450	0.0
vein endothelial	1336	endothelial cell	Endothelium	0.374251497005	5.538922155688	76.72155688622	5.763473053892	0.299401197604	19.53592814371	41.91616766467	0.149700598802	0.149700598802						
endothelial cell c	315	endothelial cell	Endothelium	0.0	0.317460317460	38.09523809523	3.492063492063	0.952380952380	10.79365079365	36.82539682539	0.317460317460	0.634920634920						
plasma cell	148	lymphocyte of b	Immune	0.0	0.675675675675	0.675675675675	6.081081081081	0.675675675675	6.756756756756	18.91891891891	0.0	0.0						

Supplementary Table 158 - atp_axis_cis_eqtl_mr

ATP02_atp_axis_cis_eqtl_mr.csv

gene	outcome	n_window_harmonized	n_instruments	instruments	lead_snp	ivw_logor	ivw_se	ivw_p	or_per_expr_sd	or_lo	or_hi	pp_h3	pp_h4	q_fdr	verdict
P2RX7	ALLERG_ASTHMA	848	10	rs11065453;rs2082	rs11065453	-0.0499	0.0604	0.40838173897870	0.9513	0.8451	1.0708	0.3695	0.0123	0.765716	null
P2RX7	ALLERG_RHINITIS	848	10	rs11065453;rs2082	rs11065453	-0.1134	0.0556	0.04134786701801	0.8928	0.8006	0.9956	0.9351	0.0047	0.155055	nominal MR only
P2RX7	L12_ATOPIC	848	10	rs11065453;rs2082	rs11065453	0.0082	0.0404	0.83844362061153	1.0083	0.9314	1.0915	0.0573	0.0112	0.956972	null
NLRP3	ALLERG_ASTHMA	743	10	rs10925027;rs1157	rs10925027	0.1074	0.0452	0.01752024216670	1.1134	1.019	1.2166	0.0233	0.017	0.087601	nominal MR only
NLRP3	ALLERG_RHINITIS	743	10	rs10925027;rs1157	rs10925027	-0.016	0.0416	0.69985731240236	0.9841	0.907	1.0677	0.1008	0.0098	0.956972	null
NLRP3	L12_ATOPIC	743	10	rs10925027;rs1157	rs10925027	0.0016	0.0303	0.95697191694256	1.0016	0.9439	1.0629	0.0332	0.0299	0.956972	null
CASP1	ALLERG_ASTHMA	554	10	rs2409065;rs74905	rs2409065	0.1263	0.0872	0.14752711513907	1.1347	0.9563	1.3462	0.0221	0.0719	0.352843	null
CASP1	ALLERG_RHINITIS	554	10	rs2409065;rs74905	rs2409065	0.0244	0.0803	0.76106487159145	1.0247	0.8755	1.1993	0.0297	0.026	0.956972	null
CASP1	L12_ATOPIC	554	10	rs2409065;rs74905	rs2409065	-0.0305	0.0584	0.60090562087401	0.9699	0.8651	1.0875	0.0084	0.0096	0.956972	null
GSDMD	ALLERG_ASTHMA	630	10	rs3889129;rs11986	rs3889129	-0.0031	0.0459	0.94530090034790	0.9969	0.9111	1.0907	0.0215	0.0083	0.956972	null
GSDMD	ALLERG_RHINITIS	630	10	rs3889129;rs11986	rs3889129	-0.134	0.0422	0.00148860539576	0.8746	0.8052	0.95	0.037	0.0183	0.022329	nominal MR only
GSDMD	L12_ATOPIC	630	10	rs3889129;rs11986	rs3889129	0.0018	0.0307	0.95209418804697	1.0018	0.9433	1.064	0.0085	0.0059	0.956972	null
IL1B	ALLERG_ASTHMA	77	4	rs6542082;rs13989	rs6542082	0.3668	0.2195	0.09470143960020	1.4431	0.9386	2.2187	0.0016	0.0163	0.284104	null
IL1B	ALLERG_RHINITIS	77	4	rs6542082;rs13989	rs6542082	-0.2804	0.2018	0.16466014872189	0.7554	0.5086	1.122	0.0009	0.009	0.352843	null
IL1B	L12_ATOPIC	77	4	rs6542082;rs13989	rs6542082	0.3954	0.1468	0.00706679927081	1.485	1.1137	1.9801	0.0063	0.008	0.053001	nominal MR only

Supplementary Table 159 - atp_axis_cis_eqtl_mr_persnp

ATP02b_atp_axis_cis_eqtl_mr_persnp.csv

gene	outcome	snp	pos	z_eqtl	n_eqtl	fstat	beta_outcome	se_outcome	wald_logor	wald_se	wald_p	or_
P2RX7	ALLERG_ASTHMA	rs11065453	121581983	-35.4265	31270	1255.0	-0.016653	0.02398	0.0831	0.1197	0.48740817749866105	1.0867
P2RX7	ALLERG_ASTHMA	rs208298	121602238	23.0096	31684	529.4	-0.00272	0.014117	-0.021	0.1092	0.8472406906910446	0.9792
P2RX7	ALLERG_ASTHMA	rs185394597	121599393	-17.8009	9446	316.9	-0.183242	0.168459	1.0005	0.9198	0.27670359349950135	2.7196
P2RX7	ALLERG_ASTHMA	rs1169809	121688026	13.9188	31684	193.7	-0.026691	0.013469	-0.3413	0.1722	0.04751132338116609	0.7108
P2RX7	ALLERG_ASTHMA	rs3751143	121622304	-13.4227	31286	180.2	0.008479	0.020633	-0.1117	0.2719	0.6811116171941786	0.8943
P2RX7	ALLERG_ASTHMA	rs1653589	121677538	12.718	28917	161.7	-0.007098	0.026057	-0.0949	0.3484	0.7853122621230704	0.9095
P2RX7	ALLERG_ASTHMA	rs117048777	121573621	12.4501	31563	155.0	-0.03217	0.03683	-0.4591	0.5256	0.3824014096731201	0.6319
P2RX7	ALLERG_ASTHMA	rs1796406	121506154	11.958	31346	143.0	-0.001839	0.013553	-0.0272	0.2007	0.8920778197703234	0.9731
P2RX7	ALLERG_ASTHMA	rs2668254	121715572	-11.8976	27279	141.6	0.006623	0.012794	-0.0919	0.1776	0.6047099963564795	0.9122
P2RX7	ALLERG_ASTHMA	rs73419624	121480803	-11.2907	20623	127.5	-0.002782	0.039901	0.0354	0.5075	0.944404925825017	1.036
P2RX7	ALLERG_RHINITIS	rs11065453	121581983	-35.4265	31270	1255.0	0.0409	0.022169	-0.2042	0.1107	0.06504938345058514	0.8153
P2RX7	ALLERG_RHINITIS	rs208298	121602238	23.0096	31684	529.4	-0.004029	0.012995	-0.0312	0.1005	0.7565336875536988	0.9693
P2RX7	ALLERG_RHINITIS	rs185394597	121599393	-17.8009	9446	316.9	-0.112527	0.151591	0.6144	0.8277	0.4579015931226701	1.8485
P2RX7	ALLERG_RHINITIS	rs1169809	121688026	13.9188	31684	193.7	-0.03233	0.012402	-0.4135	0.1586	0.009136716692697971	0.6614
P2RX7	ALLERG_RHINITIS	rs3751143	121622304	-13.4227	31286	180.2	-0.021893	0.018962	0.2885	0.2499	0.24826211773596274	1.3344
P2RX7	ALLERG_RHINITIS	rs1653589	121677538	12.718	28917	161.7	0.011927	0.023998	0.1595	0.3209	0.6191965474008438	1.1729
P2RX7	ALLERG_RHINITIS	rs117048777	121573621	12.4501	31563	155.0	0.007743	0.033681	0.1105	0.4806	0.8181837106002317	1.1168
P2RX7	ALLERG_RHINITIS	rs1796406	121506154	11.958	31346	143.0	-0.007828	0.012441	-0.1159	0.1842	0.529178276668931	0.8906
P2RX7	ALLERG_RHINITIS	rs2668254	121715572	-11.8976	27279	141.6	0.00827	0.01176	-0.1148	0.1632	0.4818914006502264	0.8915
P2RX7	ALLERG_RHINITIS	rs73419624	121480803	-11.2907	20623	127.5	0.005449	0.036769	-0.0693	0.4677	0.8821797059697679	0.933
P2RX7	L12_ATOPIC	rs11065453	121581983	-35.4265	31270	1255.0	-0.007955	0.016062	0.0397	0.0802	0.6204308909302149	1.0405
P2RX7	L12_ATOPIC	rs208298	121602238	23.0096	31684	529.4	0.005024	0.009479	0.0389	0.0733	0.5960942292260298	1.0396
P2RX7	L12_ATOPIC	rs185394597	121599393	-17.8009	9446	316.9	-0.115709	0.111631	0.6318	0.6095	0.29995445164993784	1.8809
P2RX7	L12_ATOPIC	rs1169809	121688026	13.9188	31684	193.7	0.000171	0.009008	0.0022	0.1152	0.9848763918612339	1.0022
P2RX7	L12_ATOPIC	rs3751143	121622304	-13.4227	31286	180.2	0.020629	0.013789	-0.2718	0.1817	0.13464786480111998	0.762
P2RX7	L12_ATOPIC	rs1653589	121677538	12.718	28917	161.7	0.006535	0.017483	0.0874	0.2338	0.7085369745128014	1.0913
P2RX7	L12_ATOPIC	rs117048777	121573621	12.4501	31563	155.0	-0.009622	0.024564	-0.1373	0.3505	0.6952888039062048	0.8717
P2RX7	L12_ATOPIC	rs1796406	121506154	11.958	31346	143.0	-0.000408	0.009057	-0.006	0.1341	0.9640396318963155	0.994
P2RX7	L12_ATOPIC	rs2668254	121715572	-11.8976	27279	141.6	-0.000677	0.008559	0.0094	0.1188	0.9369166331507959	1.0094
P2RX7	L12_ATOPIC	rs73419624	121480803	-11.2907	20623	127.5	0.025452	0.026698	-0.3237	0.3396	0.34042324394098533	0.7235
NLRP3	ALLERG_ASTHMA	rs10925027	247612562	33.6137	31559	1129.9	0.015677	0.013044	0.0829	0.0689	0.22940355837447546	1.0864
NLRP3	ALLERG_ASTHMA	rs11577509	247611459	-29.6632	31589	879.9	-0.032881	0.018196	0.197	0.109	0.07074813453381625	1.2178
NLRP3	ALLERG_ASTHMA	rs79796976	247553979	22.6517	31684	513.1	0.013567	0.03109	0.1066	0.2443	0.6625475908126616	1.1125
NLRP3	ALLERG_ASTHMA	rs4593864	247616733	-19.7961	31684	391.9	0.010932	0.014655	-0.0983	0.1318	0.4556789898634781	0.9064
NLRP3	ALLERG_ASTHMA	rs4925671	247622874	-16.4764	31684	271.5	0.002496	0.01349	-0.027	0.1457	0.8532234828121004	0.9734
NLRP3	ALLERG_ASTHMA	rs12137901	247581209	16.225	31684	263.3	0.041851	0.014709	0.4591	0.1614	0.004438908680011422	1.5827
NLRP3	ALLERG_ASTHMA	rs10399826	247535535	-15.4384	31684	238.3	0.009608	0.021285	-0.1108	0.2454	0.6516887136553876	0.8951
NLRP3	ALLERG_ASTHMA	rs7555997	247547334	-15.0847	31567	227.5	-0.039229	0.029909	0.462	0.3523	0.18965224808348802	1.5873
NLRP3	ALLERG_ASTHMA	rs116022280	247699511	-13.3607	28673	178.5	-0.014379	0.030863	0.1822	0.3912	0.6412807054732768	1.1999
NLRP3	ALLERG_ASTHMA	rs58287798	247605221	-12.397	29650	153.7	-0.062805	0.039085	0.8724	0.5429	0.10807723332035529	2.3925

(showing first 40 of 132 rows; full data in the accompanying CSV)

Supplementary Table 160 - instrument_loo_diagnostics

ATP02c_instrument_loo_diagnostics.csv

gene	outcome	n_instruments	ivw_logor	n_direction_positive	max_weight_share	max_weight_snp	loo_min	loo_max	loo_direction_robust	n_persnp_p005
P2RX7	ALLERG_ASTHMA	10	-0.0499	3	0.306	rs208298	-0.0953	-0.0091	True	1
P2RX7	ALLERG_RHINITIS	10	-0.1134	4	0.306	rs208298	-0.1497	-0.0714	True	1
P2RX7	L12_ATOPIC	10	0.0082	6	0.304	rs208298	-0.0051	0.0229	False	0
NLRP3	ALLERG_ASTHMA	10	0.1074	7	0.43	rs10925027	0.0775	0.1349	True	1
NLRP3	ALLERG_RHINITIS	10	-0.016	5	0.433	rs10925027	-0.0307	0.0137	False	1
NLRP3	L12_ATOPIC	10	0.0016	7	0.432	rs10925027	-0.0183	0.0678	False	0
CASP1	ALLERG_ASTHMA	10	0.1263	4	0.581	rs2409065	0.0039	0.1517	True	0
CASP1	ALLERG_RHINITIS	10	0.0244	4	0.58	rs2409065	-0.1251	0.0881	False	2
CASP1	L12_ATOPIC	10	-0.0305	3	0.581	rs2409065	-0.1063	-0.0148	True	0
GSDMD	ALLERG_ASTHMA	10	-0.0031	5	0.529	rs3889129	-0.019	0.0178	False	0
GSDMD	ALLERG_RHINITIS	10	-0.134	2	0.529	rs3889129	-0.1981	-0.1149	True	2
GSDMD	L12_ATOPIC	10	0.0018	5	0.529	rs3889129	-0.0042	0.0171	False	0
IL1B	ALLERG_ASTHMA	4	0.3668	3	0.641	rs6542082	0.3233	0.447	True	0
IL1B	ALLERG_RHINITIS	4	-0.2804	0	0.643	rs6542082	-0.5259	-0.1787	True	0
IL1B	L12_ATOPIC	4	0.3954	4	0.642	rs6542082	0.267	0.8293	True	1

Supplementary Table 161 - atp_axis_gtex_lung_sensitivity

ATP03_atp_axis_gtex_lung_sensitivity.csv

gene	outcome	n_harmonized	verdict	lead_variant	lead_tss_distance	lead_eqtl_beta	lead_eqtl_se	lead_outcome_beta	lead_wald_logor	lead_wald_se	lead_wald_p	lead_or_per_expr_sd	pp_h3	pp_h4
NLRP3	ALLERG_ASTHMA	2	uninformative (<5 ha											
NLRP3	ALLERG_RHINITIS	2	uninformative (<5 ha											
NLRP3	L12_ATOPIC	2	uninformative (<5 ha											
GSDMD	ALLERG_ASTHMA	130	null	chr8_143543623_C_T	-9584.0	-0.2838	0.0322	-0.00567	0.02	0.0536	0.709535319264615	1.0202	0.0046	0.0102
GSDMD	ALLERG_RHINITIS	130	nominal lead Wald or	chr8_143543623_C_T	-9584.0	-0.2838	0.0322	0.02831	-0.0997	0.0492	0.0427362927163575	0.9051	0.0087	0.0795
GSDMD	L12_ATOPIC	130	null	chr8_143543623_C_T	-9584.0	-0.2838	0.0322	0.00522	-0.0184	0.0358	0.607448835397347	0.9818	0.0034	0.008
P2RX7	ALLERG_ASTHMA	0	no significant Lung ci											
P2RX7	ALLERG_RHINITIS	0	no significant Lung ci											
P2RX7	L12_ATOPIC	0	no significant Lung ci											
CASP1	ALLERG_ASTHMA	0	no significant Lung ci											
CASP1	ALLERG_RHINITIS	0	no significant Lung ci											
CASP1	L12_ATOPIC	0	no significant Lung ci											
IL1B	ALLERG_ASTHMA	0	no significant Lung ci											
IL1B	ALLERG_RHINITIS	0	no significant Lung ci											
IL1B	L12_ATOPIC	0	no significant Lung ci											

Supplementary Table 162 - aging_atlas_panel_enrichment

KB01_aging_atlas_panel_enrichment.csv

[unreadable: Error tokenizing data. C error: Expected 3 fields in line 23, saw 4
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